
Econometrics I

Theory, Practice, and Applications

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Preface

Econometrics I — Theory, Practice, and Applications is the open companion textbook for the Econometrics I course at the University of Granada (UGR), Faculty of Economics and Business Studies. It consolidates three years of teaching materials — lecture notes, R tutorials, and interactive self-assessments — into a single reproducible volume.

What this book is

A unified second-year undergraduate text that walks the reader from the *idea* of an econometric model to the practical estimation, inference, and diagnostic skills required to analyse real economic data. Each chapter follows the same six-section template:

1. **Learning outcomes**
2. **Motivating empirical question**
3. **Theory**
4. **Worked example in R**
5. **Self-check** (interactive quiz)
6. **Exercises** (easy, medium, analytical)

How this book is organised

Chapter	Topic
1	Initial Concepts
2	The Simple Linear Regression Model
3	The Multiple Linear Regression Model
4	Inference
5	Prediction and Model Selection
6	Nonlinear Relationships and Qualitative Information
7	Heteroskedasticity
8	Autocorrelation
App. A	Mathematical and Statistical Prerequisites
App. B	Wooldridge–Carter–Griffiths–Lim crosswalk
App. C	Statistical tables

Editions and license

This book is released under a **Creative Commons Attribution 4.0 International (CC BY 4.0)** license. An equivalent Spanish edition is available as *Econometría I — Teoría, práctica y aplicaciones*.

How to use the source materials

The textbook is built with Quarto. Each chapter is a `.qmd` file that combines prose, executable R code, and interactive quizzes. To rebuild the book locally:

```
quarto render          # English edition (default)
quarto render --profile es  # Spanish edition
```

Required R packages

```
install.packages(c("wooldridge", "readxl", "lmtest", "sandwich", "orcutt"))
# Companion package (data + helpers)
# remotes::install_github("agaggero-ugr/econometricsUGR")
```

Acknowledgements

UGR Department of Quantitative Methods for Economics and Business; the students of three consecutive cohorts (2023/24, 2024/25, 2025/26) whose questions shaped this text; and the authors of *Introductory Econometrics* (Wooldridge) and *Principles of Econometrics* (Carter–Griffiths–Lim) on whose pedagogical traditions this book builds.

Initial Concepts

Learning outcomes

By the end of this chapter the reader should be able to:

- Explain what econometrics is and how an econometric model differs from a purely theoretical economic model.
- Interpret the two components of an econometric model: the systematic part and the random error u .
- Distinguish correlation from causation, and identify selection bias and omitted-variable bias in concrete examples.
- Explain why randomisation eliminates selection bias on average, and contrast randomised experiments with quasi-experimental and observational designs.
- Classify a dataset as cross-sectional, time-series, panel, or pooled cross-section, and recognise the research questions each structure can (and cannot) answer.
- Use basic R commands (`str()`, `summary()`, `head()`, `hist()`, `plot()`, `tapply()`, `cor()`) to explore a microeconomic dataset and compute conditional means.

Motivating empirical question

*Does going to university **cause** higher wages, or do the people who would have earned more anyway tend to go to university?*

This single question drives the rest of the course. A correlation between schooling and wages is easy to compute; turning that correlation into a credible statement about the causal return to education is hard. The tools we develop — the linear model, the assumptions on the error term, randomisation, controls, and tests — are all in the service of answering questions of this form.

1.1. What is econometrics?

ECONOMETRICS is the application of statistical methods to economic data, with the goal of giving empirical content to economic relationships (Wooldridge 2020; Hill et al. 2017). Its main objective is the **estimation of relationships between variables**. Typical pairs of interest in this course include:

- Education $\Rightarrow$ Salary
- Class attendance $\Rightarrow$ Final grade
- Tuition fees $\Rightarrow$ Enrolment

1. Initial Concepts

- Income $\Rightarrow$ Happiness
- Government spending $\Rightarrow$ Crime
- Advertising $\Rightarrow$ Sales

In general we write $X \Rightarrow Y$, where X is the **independent variable** (or regressor, explanatory variable) and Y is the **dependent variable** (or response, outcome). Two questions then arise:

1. Is there an effect of X on Y ?
2. How large is that effect?

These “how much” questions are everywhere. A city council wonders how much violent crime will fall if it spends an additional million euros on policing. A local business must estimate the relationship between advertising spending and sales. A university must estimate how much enrolment will drop if it raises tuition by 300 euros. None of these questions can be answered by economic theory alone — they require data, a statistical model, and the discipline of econometrics.

1.2. The econometric model

An econometric model has **two** components:

1. A **systematic component** — the average or systematic behaviour of many individuals, firms, or other units.
2. A **random error** u (sometimes called the *disturbance*) — the many factors that affect Y but are not included in the systematic part.

Example: salary determination

The factors associated with an individual’s salary include age, gender, experience, and education. In general functional form:

$$\text{Salary} = f(\text{Age}, \text{Gender}, \text{Experience}, \text{Education}) + u.$$

A linear specification is

$$\text{Salary} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Gender} + \beta_3 \text{Experience} + \beta_4 \text{Education} + u.$$

The coefficients $\beta_0, \beta_1, \dots, \beta_4$ are **unknown population parameters** that we will estimate from data using an econometric technique (typically Ordinary Least Squares, introduced in Chapter 2).

The functional form is itself a hypothesis about how the variables are connected. A key challenge in any applied problem is choosing a functional form that is compatible both with economic theory and with what the data look like.

Common mistake: confusing u with the residual $\hat{u}$

The error u is a *population* object: it represents every determinant of Y that is not in the model, and it is **unobservable**. The residual $\hat{u}_i = y_i - \hat{y}_i$ is its *sample* counterpart, obtained after estimating the model. Following Wooldridge’s notation, we keep u and $\hat{u}$ strictly separate throughout the book.

1.3. Uses of econometrics

We use econometric models for three broad purposes:

- **Test economic theories and hypotheses.** For example: does increasing the minimum wage reduce employment? Does class attendance improve grades?
- **Predict economic variables.** For example: forecast next quarter's GDP growth, or tomorrow's electricity demand.
- **Estimate causal relationships.** For example: does an additional year of education *cause* higher wages, or are educated workers simply different to begin with?

The third use — credible causal inference from observational data — is at the heart of the so-called **Credibility Revolution** in empirical economics. Hypothesis testing and prediction are sometimes called “Led Zeppelin” econometrics; causal inference is the modern emphasis (Hill et al. 2017).

1.4. Correlation does not imply causation

One of the most important lessons of the course — repeated, if necessary, on every page — is that **correlation does not imply causation**. The fact that two variables move together does not establish that one of them causes the other. Two reasons stand out: *selection bias* and *omitted variables*.

1.4.1. Selection bias

i Example: ventilators and deaths

A hospital wants to know whether ventilators save lives. A naive comparison shows that patients on ventilators die *more* often than patients not on ventilators. Should we conclude that ventilators *kill* patients?

Obviously not. The problem is **selection bias**: the sickest patients are precisely the ones who are placed on a ventilator. These patients would have had a higher mortality rate *regardless* of whether they received a ventilator.

The observed correlation between ventilator use and death is *positive*; the true causal effect of ventilators on survival is *negative*. A naive comparison can therefore give us not just the wrong magnitude but the **wrong sign** of the effect.

The lesson is general: whenever treatment is not assigned randomly, simple comparisons of outcomes between treated and untreated units confound the causal effect of the treatment with the characteristics of the units that selected into it.

1.4.2. Omitted variable bias

i Example: ice cream and drowning

Data show a strong positive correlation between ice-cream sales and drowning deaths. Should we ban ice cream to prevent drowning?

Of course not. The **omitted variable** is *temperature*. Hot weather causes both more

1. Initial Concepts

ice-cream sales and more swimming (hence more drowning). Without controlling for temperature, the correlation between ice cream and drowning is a **spurious correlation** — a real statistical association with no causal content.

Whenever a variable that influences Y is also correlated with X but is left out of the model, the estimate of the effect of X will be biased. We will study this formally as **omitted variable bias** in Chapter 3.

1.4.3. Why randomisation solves the problem

If we **randomly** assign individuals to a treatment group and a control group, then on average the two groups will have the same observable and unobservable characteristics *before* the treatment is administered. Any difference in outcomes *after* the treatment can therefore be attributed to the treatment itself.

In the ventilator example, if we could randomly assign ventilators (not based on severity of illness), then the treatment and control groups would have the same average severity. The difference in mortality between the two groups would be an unbiased estimate of the causal effect of ventilators.

A useful identity to keep in mind is

$$\underbrace{\text{difference in outcomes}}_{\text{what we observe}} = \underbrace{\text{causal effect}}_{\text{what we want}} + \underbrace{\text{selection bias}}_{\text{nuisance}}.$$

Randomisation forces the second term on the right to zero on average. That is why **Randomised Controlled Trials (RCTs)** are considered the gold standard for establishing causality.

i Example: private schools and test scores

A study finds that students at private schools score higher on standardised tests. Can we conclude that private schools *cause* better performance? No — students at private schools tend to come from wealthier families with more resources at home, more educated parents, and so on. The “treatment” (attending a private school) is not randomly assigned, so the comparison is contaminated by selection.

1.5. Types of experiments

1.5.1. Randomised experiments

A randomised experiment assigns participants to two or more groups by lot: one group receives the intervention (the **treatment**); another serves as the **control** and receives a placebo or standard treatment. Randomisation ensures the groups are comparable, eliminates selection bias on average, and permits causal inference.

Genuine randomised experiments are common in clinical medicine but **rare in economics and business**, for two reasons:

- *Ethical constraints* — we cannot randomly assign individuals to poverty, unemployment, or disease.
- *Cost* — well-designed field experiments are expensive and slow to run.

1.5.2. Quasi-experiments

In a **quasi-experiment**, assignment to treatment is not random but is based on some **observable** criterion — often a threshold, an eligibility rule, or a policy change that hits some units but not others.

i Example: a regional lockdown

During the COVID-19 pandemic, lockdown (the treatment) was assigned by the *Junta de Andalucía* only to municipalities with infection rates above 1{,}000 per 100{,}000 inhabitants. Municipalities just above and just below the threshold are arguably similar, so comparing their outcomes can approximate a random experiment. We return to this kind of design later in the book.

1.5.3. Nonexperimental (observational) data

With **nonexperimental** data, all variables are collected simultaneously and the values are neither fixed nor repeatable by the researcher. Survey data are the classic example — the U.S. Current Population Survey (CPS) and the Spanish *Encuesta de Población Activa* (EPA) both belong to this category. Most of the datasets we use in this book are nonexperimental, which is precisely why the issue of selection bias matters so much.

1.6. Types of economic data

Economic data come in several “flavours”. Aggregation can be *micro* (households, firms, workers) or *macro* (regions, countries); variables can be *quantitative* (numbers) or *qualitative* (categories, e.g. employed / unemployed). For econometrics what matters most is the **structure** of the dataset: do we have one snapshot, a sequence over time, or repeated observations of the same units?

i Definition: cross-sectional data

Data collected across sample units — individuals, firms, households, regions, countries — in a particular time period. Examples: income by Californian county in 2016; high-school graduation rates by U.S. state in 2015; the `wage1` dataset we use in the lab below.

i Definition: time-series data

Data collected over discrete intervals of time, with the same economic quantity recorded at a regular frequency. Examples: the annual price of wheat, daily General Electric stock prices, monthly Spanish unemployment rates from 1980 to 2024.

i Definition: panel (longitudinal) data

Observations on individual micro-units that are followed over multiple time periods. The defining feature is that *each* unit is observed in *several* periods. If every unit is observed in every period, the panel is **balanced**; otherwise it is **unbalanced**. Typically the number of time periods is small relative to the number of units, but not always.

i Definition: pooled cross-section

Independent cross-sections drawn in different periods and stacked together. Different units are sampled each time — there is *no* re-interview of the same individuals — but the periods can be compared. The Spanish EPA, which draws a fresh independent sample of households every quarter, yields a pooled cross-section when several waves are stacked.

The data structure matters because it determines the questions we can answer. With a cross-section we can study how outcomes vary *across* units at one moment in time. With a time-series we can study how a single quantity evolves *over* time. With a panel we can do both — and, crucially, we can control for time-invariant unobserved characteristics of each unit, an important tool in modern causal inference. With a pooled cross-section we can study how a relationship changes between periods (for example, before and after a policy reform).

1.7. Lab: exploring economic data

The rest of the chapter is a guided exploration of the `wage1` dataset shipped with the `wooldridge` R package. The dataset is a *cross-section* of 526 U.S. workers observed in May 1976, with information on hourly wages, education, experience, tenure, and a handful of demographic variables. (The mechanics of installing R and the `wooldridge` package are covered in Appendix A; the first formal regression with `lm()` is the opening lab of Chapter 2.)

We start by loading the package and looking at the data.

```
library(wooldridge)
data("wage1")
```

1.7.1. Structure, size, and a first look

```
str(wage1)
```

```
'data.frame':  526 obs. of  24 variables:
 $ wage      : num  3.1 3.24 3 6 5.3 ...
 $ educ      : int  11 12 11 8 12 16 18 12 12 17 ...
 $ exper     : int  2 22 2 44 7 9 15 5 26 22 ...
 $ tenure    : int  0 2 0 28 2 8 7 3 4 21 ...
 $ nonwhite  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ female    : int  1 1 0 0 0 0 0 1 1 0 ...
 $ married   : int  0 1 0 1 1 1 0 0 0 1 ...
 $ numdep    : int  2 3 2 0 1 0 0 0 2 0 ...
 $ smsa      : int  1 1 0 1 0 1 1 1 1 1 ...
 $ northcen  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ south     : int  0 0 0 0 0 0 0 0 0 0 ...
 $ west      : int  1 1 1 1 1 1 1 1 1 1 ...
 $ construc : int  0 0 0 0 0 0 0 0 0 0 ...
 $ ndurman   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ trcompu   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ trade     : int  0 0 1 0 0 0 1 0 1 0 ...
```

```

$ services: int  0 1 0 0 0 0 0 0 0 0 ...
$ profserv: int  0 0 0 0 0 1 0 0 0 0 ...
$ profocc  : int  0 0 0 0 0 1 1 1 1 1 ...
$ clerocc  : int  0 0 0 1 0 0 0 0 0 0 ...
$ servocc  : int  0 1 0 0 0 0 0 0 0 0 ...
$ lwage    : num  1.13 1.18 1.1 1.79 1.67 ...
$ expersq  : int  4 484 4 1936 49 81 225 25 676 484 ...
$ tenursq  : int  0 4 0 784 4 64 49 9 16 441 ...
- attr(*, "time.stamp")= chr "25 Jun 2011 23:03"

```

```
head(wage1)
```

```

      wage educ exper tenure nonwhite female married numdep smsa northcen south
1 3.10    11     2      0        0      1      0      2     1         0      0
2 3.24    12    22      2        0      1      1      3     1         0      0
3 3.00    11     2      0        0      0      0      2     0         0      0
4 6.00     8    44     28        0      0      1      0     1         0      0
5 5.30    12     7      2        0      0      1      1     0         0      0
6 8.75    16     9      8        0      0      1      0     1         0      0
      west construc ndurman trcommpt trade services profserv profocc clerocc
1      1          0      0        0      0      0      0      0      0
2      1          0      0        0      0      1      0      0      0
3      1          0      0        0      1      0      0      0      0
4      1          0      0        0      0      0      0      0      1
5      1          0      0        0      0      0      0      0      0
6      1          0      0        0      0      0      1      1      0
      servocc      lwage expersq tenursq
1          0 1.131402      4      0
2          1 1.175573    484      4
3          0 1.098612      4      0
4          0 1.791759   1936    784
5          0 1.667707     49      4
6          0 2.169054     81     64

```

```
nrow(wage1)
```

```
[1] 526
```

```
ncol(wage1)
```

```
[1] 24
```

The output of `str()` tells us that `wage1` is a data frame with 526 observations and 24 variables. Each row is one worker. The variables we care about most in this chapter are:

- `wage` — hourly wage, in 1976 U.S. dollars,
- `educ` — years of schooling,
- `exper` — years of potential labour-market experience,
- `tenure` — years with the current employer,
- `female` — a dummy equal to 1 for women, 0 for men,
- `married` — a dummy equal to 1 for married workers.

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1.7.2. Summary statistics and distributions

```
summary(wage1[, c("wage", "educ", "exper", "tenure")])
```

wage	educ	exper	tenure
Min. : 0.530	Min. : 0.00	Min. : 1.00	Min. : 0.000
1st Qu.: 3.330	1st Qu.:12.00	1st Qu.: 5.00	1st Qu.: 0.000
Median : 4.650	Median :12.00	Median :13.50	Median : 2.000
Mean : 5.896	Mean :12.56	Mean :17.02	Mean : 5.105
3rd Qu.: 6.880	3rd Qu.:14.00	3rd Qu.:26.00	3rd Qu.: 7.000
Max. :24.980	Max. :18.00	Max. :51.00	Max. :44.000

```
sd(wage1$wage)
```

```
[1] 3.693086
```

```
sd(wage1$educ)
```

```
[1] 2.769022
```

`summary()` gives us min, quartiles, mean, and max for each variable. The mean wage is about \$5.90 per hour and the median is about \$4.65; the mean exceeds the median, which suggests a **right-skewed** distribution — a long upper tail of high earners. A histogram makes the shape obvious:

```
hist(wage1$wage,  
     main = "Distribution of hourly wages",  
     xlab = "Wage (USD / hour)",  
     col = "lightblue",  
     breaks = 30)
```

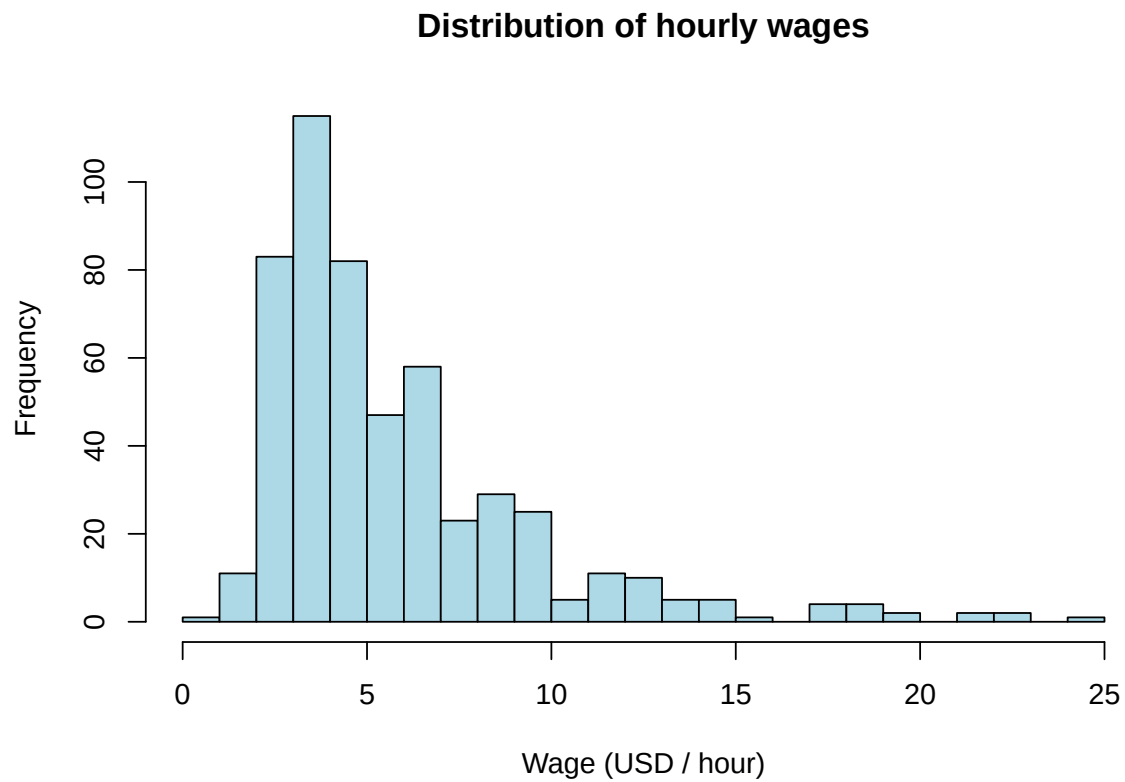



Figure 1.1.: Distribution of hourly wages in wage1 (1976 USD).

```
hist(wage1$educ,  
     main = "Distribution of years of education",  
     xlab = "Years of education",  
     col  = "lightgreen",  
     breaks = 15)
```

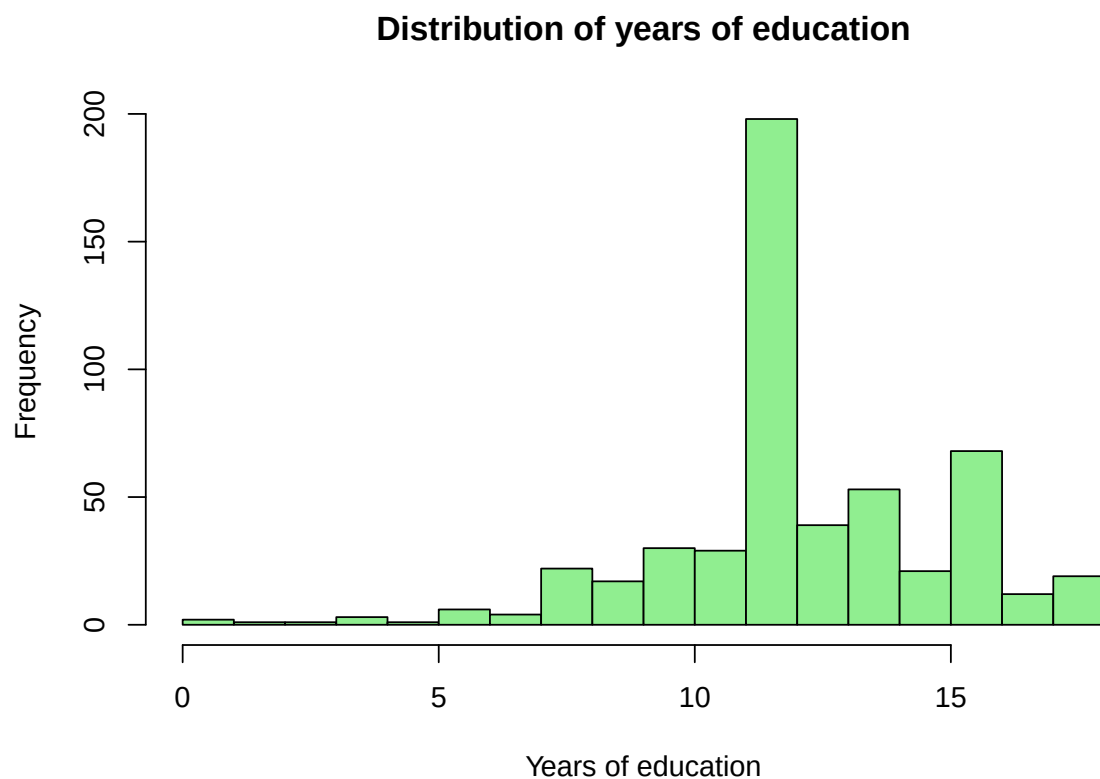


Figure 1.2.: Distribution of years of education in wage1.

The education distribution piles up at 12 years (high-school completion) and 16 years (a four-year college degree), as one would expect for U.S. data from this era.

1.7.3. Frequency tables

For categorical or count-like variables, `table()` is more useful than `summary()`:

```
table(wage1$female)
```

```
0    1
274 252
```

```
table(wage1$married)
```

```
0    1
206 320
```

```
table(wage1$female) / nrow(wage1)
```

```

0      1
0.5209125 0.4790875

```

About 48% of the sample are women and 52% are men.

1.7.4. A first scatter plot

A natural starting question is: do workers with more education earn more?

```

plot(wage1$educ, wage1$wage,
     xlab = "Years of education",
     ylab = "Hourly wage (USD)",
     main = "Wage vs. education",
     pch = 16,
     col = rgb(0, 0, 1, 0.3))

```

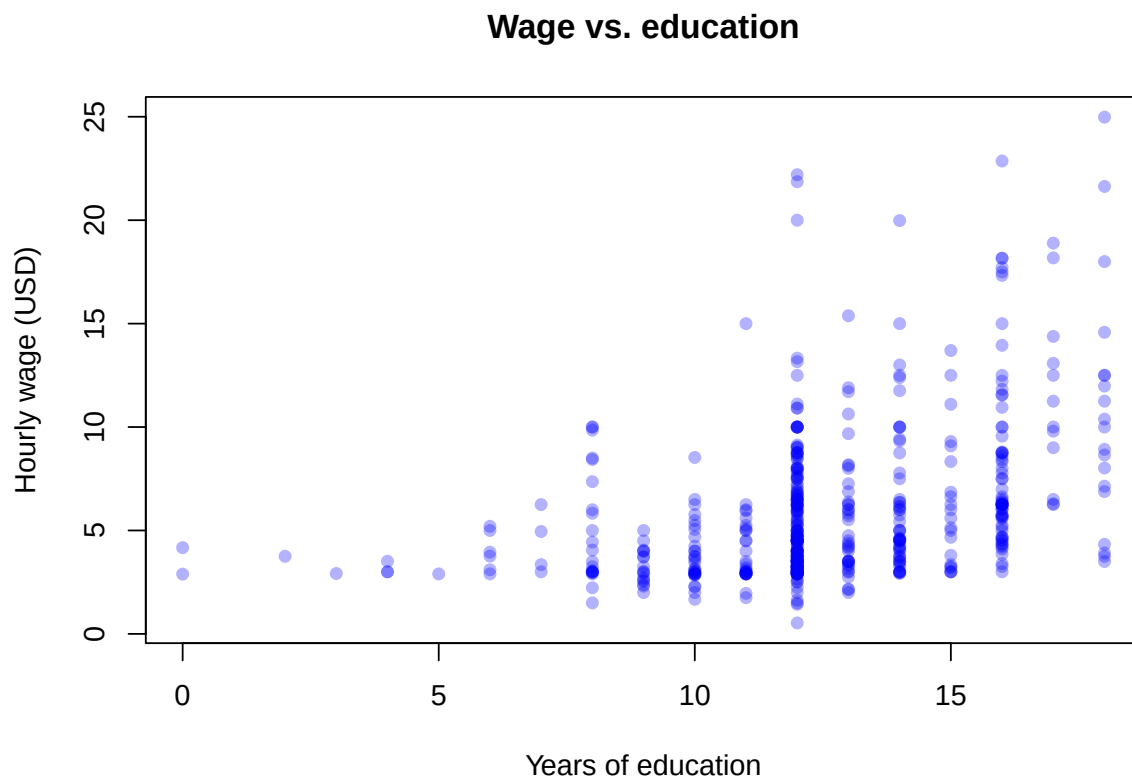


Figure 1.3.: Wage versus years of education. Each dot is one of the 526 workers.

The cloud of points slopes upward: on average, more educated workers earn higher wages. We can quantify the linear association with the **sample correlation**:

```

cor(wage1$wage, wage1$educ)

```

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```
[1] 0.4059033
```

```
cor(wage1$wage, wage1$exper)
```

```
[1] 0.1129034
```

```
cor(wage1$educ, wage1$exper)
```

```
[1] -0.2995418
```

The correlation between wage and education is about 0.41 — a clearly positive but far-from-perfect linear association. Two further patterns stand out:

- The correlation between wage and *experience* is positive but smaller than the correlation with education.
- The correlation between *education* and *experience* is **negative**. This is not because schooling destroys experience — it reflects how individuals allocate time across the life cycle: more years in school mean fewer years in the workforce, mechanically. We will see in Chapter 3 that ignoring this kind of correlation among regressors can severely bias a simple regression of wage on education.

1.7.5. Conditional means: a preview of regression

A cleaner way to look at the data is to compute the **conditional mean** of wages given another variable. `tapply()` does exactly that:

```
tapply(wage1$wage, wage1$female, mean)
```

```
      0      1  
7.099489 4.587659
```

```
tapply(wage1$wage, wage1$married, mean)
```

```
      0      1  
4.843884 6.573469
```

Men earn about \$7.10 per hour on average; women earn about \$4.59. Married workers earn about \$6.32 per hour; single workers about \$5.16. These are **raw, unconditional** comparisons.

 Common mistake: reading a conditional mean as a causal effect

The 2.51-dollar gender gap and the 1.16-dollar marriage premium are **conditional means**, not causal effects. The men and women in this sample differ in occupation, sector, experience, hours worked, and many other characteristics; married and single workers differ in age and tenure. Without holding those confounders fixed, neither gap can be interpreted as the *ceteris paribus* effect of gender or marriage on wages. Holding confounders fixed is what multiple regression (Chapter 3) is for.

A more striking selection-bias illustration uses the conditional mean of *experience* given *education*:

```
tapply(wage1$exper, wage1$educ, mean)
```

0	2	3	4	5	6	7	8
32.00000	39.00000	51.00000	41.00000	34.00000	30.50000	31.25000	31.40909
9	10	11	12	13	14	15	16
15.94118	14.83333	13.93103	18.14646	14.89744	16.13208	13.38095	12.35294
17	18						
13.33333	11.10526						

Workers with little schooling have *more* labour-market experience, and workers with a college degree have *less*. So when we compare wages across education levels, we are simultaneously comparing workers who differ in experience — the comparison is confounded. This is the same kind of selection problem we saw in the ventilator and private-school examples, now in our actual dataset. We will fix it in Chapter 3 by including both `educ` and `exper` in a multiple regression.

Why `wage1` is a cross-section

`wage1` records each worker **once**, all in 1976. There is no time index, no person identifier observed in several years. It is a clean cross-section. With this dataset alone we *cannot* track how a given worker's wage grows from one year to the next — that would require a *panel*, e.g. the U.S. Panel Study of Income Dynamics (PSID) or the Spanish *Muestra Continua de Vidas Laborales* (MCVL). Knowing what your data *can* and *cannot* tell you is half of applied econometrics.

Self-check

Eight short multiple-choice questions. Try each one before opening the answer.

Q1. What is an econometric model?

An econometric model differs from a purely economic (theoretical) model in that:

- A. It assumes the relationship between variables is exact and deterministic.
- B. It adds a stochastic error term and a functional form so that unknown parameters can be estimated from data.
- C. It does not require a sample of data because the parameters are derived from theory.
- D. It is only used in macroeconomic problems, not in microeconomic ones.

Answer: B. An econometric model couples a functional form with a random disturbance u , which makes the unknown parameters estimable from observed data.

Q2. The role of the error term

In the population regression model $y = \beta_0 + \beta_1 x + u$, the error term u :

- A. Is the difference between y_i and the OLS fitted value $\hat{y}_i$ in our sample.
- B. Is always equal to zero in the population.
- C. Captures every factor — other than x — that influences y .

1. Initial Concepts

- D. Is observable as soon as we have data on y and x .

Answer: C. u is a *population, unobservable* quantity collecting all omitted determinants of y . The residual $\hat{u}_i$ is its sample counterpart, computed only after estimation.

💡 Q3. Cross-section or not?

A dataset with information on 526 workers interviewed in May 1976 (one observation per worker, all measured at the same time) is:

- A. Cross-sectional data.
- B. Time-series data.
- C. Panel (longitudinal) data.
- D. Pooled cross-section data.

Answer: A. Many units, one time period, no repeated observation of the same unit — the textbook definition of a cross-section.

💡 Q4. Time series

Annual Spanish unemployment rates from 1980 to 2024 are an example of:

- A. Cross-sectional data.
- B. Time-series data.
- C. Panel data.
- D. Experimental data.

Answer: B. One quantity (the unemployment rate) recorded at a regular frequency over time.

💡 Q5. Panel vs pooled cross-section

A dataset follows 1{,}000 firms during the years 2010–2024, recording sales, employment and exports for each firm in every year. This is:

- A. Cross-sectional data, since each firm-year is one row.
- B. Pooled cross-section, since different units are sampled in each period.
- C. Time-series data, since it has a time dimension.
- D. Panel (longitudinal) data: the same units observed repeatedly over time.

Answer: D. The same firms are tracked over multiple periods, which is exactly the panel structure.

💡 Q6. What does correlation tell us?

A high positive correlation between two variables X and Y implies that:

- A. X causes Y .
- B. Y causes X .
- C. There exists a deterministic linear relationship between them.
- D. They tend to move together, but this association may be driven by a third variable,

by reverse causality, or by selection.

Answer: D. Correlation is silent about causal direction or confounding. The ice-cream-and-drowning example in §1.4.2 is the canonical warning.

💡 Q7. Selection bias

Selection bias arises when:

- A. The units that select into the treatment group differ systematically from those that do not, even before treatment.
- B. The researcher chooses regressors based on t -statistics.
- C. The sample size is too small to detect a true effect.
- D. The dependent variable is measured with error.

Answer: A. Selection bias is a property of the *assignment* of treatment, not of estimation or measurement. The ventilator example in §1.4.1 is the prototype.

💡 Q8. Randomisation as a remedy

In which of the following research designs does randomisation *by construction* eliminate selection bias?

- A. Comparing employed vs. unemployed individuals using cross-sectional survey data.
- B. A randomised controlled trial (RCT) where the treatment is assigned by lottery.
- C. Comparing students who chose to enrol in a private vs. a public school.
- D. Comparing countries that adopted the euro vs. countries that did not.

Answer: B. Only random assignment guarantees that, *on average*, treated and untreated units have the same pre-treatment characteristics.

Exercises

Exercise 1.1 — Classifying datasets. Classify each of the following datasets as cross-sectional, time-series, panel, or pooled cross-section.

- (a) Monthly Spanish CPI from 1980 to 2024.
- (b) `wage1`: 526 workers interviewed in May 1976, one row per worker.
- (c) The Spanish EPA: each quarter, a fresh independent sample of households is drawn (no household is revisited); several quarters are then stacked.
- (d) A dataset that follows the same 800 students from age 6 to age 18, recording their test scores every year.

💡 Show answer

- (a) Time-series. (b) Cross-sectional. (c) Pooled cross-section — new units each period.
- (d) Panel: the *same* units are followed over multiple periods.

Exercise 1.2 — Spotting selection bias. For each scenario, state whether selection bias is a likely problem, and identify which characteristic drives the selection.

1. Initial Concepts

- (a) A hospital compares mortality of patients who received a ventilator with patients who did not.
- (b) An economist compares the wages of workers who completed a job-training programme with the wages of workers who did not enrol.
- (c) Researchers compare countries that adopted the euro in 1999 with countries that did not.

 Show answer

- (a) Yes — severity of illness drives both ventilator assignment and mortality. (b) Yes — self-selection into training is driven by motivation, prior earnings, and labour-market prospects, all of which also affect wages. (c) Yes — countries that joined the euro were chosen on the basis of macroeconomic fundamentals (the Maastricht criteria), so they are not a random sample of European countries.

Exercise 1.3 — Sign of an omitted-variable bias. Suppose you regress hourly wage on years of education in a sample of currently employed adults, *without* controlling for innate ability. Assume that (i) more able workers earn more, holding education fixed, and (ii) more able workers tend to acquire more education. Will the coefficient on `educ` be biased upward (too positive) or downward (too negative)? Explain in one sentence.

A full answer is given in the Instructor Edition.

Exercise 1.4 — Designing a randomised experiment. You want to know whether a free university preparation course (the treatment) causally raises the probability that high-school students from low-income families enrol in university. Sketch the design of a randomised experiment that would identify this causal effect. Be explicit about: the population, the assignment rule, the treatment and control groups, the outcome variable, and the time horizon. Briefly discuss one ethical objection your design will have to address.

A full answer is given in the Instructor Edition.

Exercise 1.5 — A naive omitted-variable model. A worker’s wage is generated by

$$w = 5 + 2 \cdot edu + u, \quad \mathbb{E}[u] = 0, \text{ Var}(u) = 1,$$

where $edu \in \{0, 1, 2, 3\}$ are years of post-compulsory education. Suppose that ability (collected in u) is correlated with education: $\mathbb{E}[u \mid edu] = 0.5 \cdot edu$.

- (a) Compute $\mathbb{E}[w \mid edu]$.
- (b) If we naively assume $\mathbb{E}[u \mid edu] = 0$ and read off the slope, what do we get? By how much do we *miss* the true marginal effect of an extra year of education?

A full answer is given in the Instructor Edition.

Exercise 1.6 — Chocolate and Nobel prizes. It has been documented (Hill et al. 2017) that countries with higher per-capita chocolate consumption also win more Nobel prizes per capita. Is this evidence that chocolate consumption *causes* Nobel prizes? Identify (i) at least one plausible omitted variable, (ii) at least one alternative causal channel (reverse causation, selection of which countries get measured, etc.), and (iii) describe a *hypothetical* research design — experimental or quasi-experimental — that would allow you to learn the causal effect of chocolate on prize-winning. Be honest about why this design is unlikely to be feasible.

A full answer is given in the Instructor Edition.

The Simple Linear Regression Model

Learning outcomes

By the end of this chapter the reader should be able to:

- Write the population regression function (PRF) and the sample regression function (SRF) for the simple linear model, and explain what each one represents.
- Derive the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ as the minimisers of the sum of squared residuals.
- State the five Gauss–Markov assumptions for the simple linear regression (SLR) model and explain what each one guarantees.
- Prove (or follow the proof of) the unbiasedness of OLS and identify which assumptions are used at each step.
- Write down the sampling variance of $\hat{\beta}_1$, name the three factors that drive its precision, and explain what BLUE means.
- Decompose total variation into $SST = SSE + SSR$ and compute and interpret R^2 .
- Use `lm()` in R to estimate, interpret, and visualise a simple regression on the `beauty` and `bwght` datasets.

Motivating empirical question

How much higher is the wage of a worker rated “above average” in physical appearance, compared with a worker rated “average”?

The labour-economics literature (Wooldridge 2020) has long documented a “beauty premium” — workers who are rated as more physically attractive tend to earn higher wages. Whether the premium is causal (looks open doors) or driven by confounding (looks correlate with health, social skills, family background) is a separate question. Before we can even *ask* that question we need a tool that turns a cloud of points into a single number. That tool is the **simple linear regression** of wage on a measure of looks. In §2.11 we estimate exactly this regression on Hamermesh’s `beauty` dataset.

2.1. Introduction

REGRESSION analysis investigates the relationship between two or more variables. In this chapter we focus on the **simple linear regression** model (SLR), which relates a dependent variable y to a single independent variable x :

2. The Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x + u.$$

Here y is the dependent variable, x is the independent variable (also called the regressor or explanatory variable), β_0 is the **intercept**, β_1 is the **slope**, and u is the unobservable **error term** or *disturbance*, which collects every determinant of y that is not x (Wooldridge 2020). The slope parameter β_1 measures the change in y associated with a one-unit change in x , *holding all factors in u constant*:

$$\beta_1 = \frac{\Delta y}{\Delta x} \quad \text{when } \Delta u = 0.$$

The intercept β_0 is the value of y when $x = 0$. In many applications $x = 0$ is outside the range of the data (zero years of education, zero square feet, zero cigarettes per day for a non-smoker) and the intercept has no immediate economic meaning — it is still essential for the line to be in the right place.

i A small zoo of SLR examples

Each of the following is a simple linear regression of one outcome on one regressor. We will revisit several of them throughout the chapter.

- Wages on years of education: $\widehat{\text{Wage}} = 800 + 15 \cdot \text{Educ}$. Each extra year of schooling is associated with an extra 15 euros per month, on average.
- House price on square footage: $\widehat{\text{Price}} = 100,000 + 50 \cdot \text{SqFt}$.
- Car resale value on age: $\widehat{\text{Resale}} = 30,000 - 1,500 \cdot \text{YearsOld}$.
- Sales on advertising spend: $\widehat{\text{Sales}} = 50 + 2 \cdot \text{AdSpend}$.
- Blood pressure on age: $\widehat{\text{BP}} = 100 + 0.5 \cdot \text{Age}$.

2.2. The population regression function

The model $y = \beta_0 + \beta_1 x + u$ involves the unobservable disturbance u . To turn it into a statement we can in principle test, we impose a key assumption on the conditional mean of the error:

$$\mathbb{E}[u \mid x] = 0.$$

This says that whatever determinants of y are bundled into u , they are unrelated *on average* to x . Taking conditional expectations of both sides of the model and using this assumption yields the **population regression function (PRF)**:

$$\mathbb{E}[y \mid x] = \beta_0 + \beta_1 x.$$

The PRF describes the *average* value of y at each value of x in the population. It is a straight line with intercept β_0 and slope β_1 .

i Regression is a comparison

Read β_1 as a *comparison between sub-populations*. If we compare the group of individuals with $x = a$ to the group with $x = a + 1$, the latter group has a value of y that is on average

β_1 units higher. Whether that comparison reflects a causal effect depends on whether $\mathbb{E}[u \mid x] = 0$ holds — a question we keep coming back to throughout the book.

The fundamental problem with the PRF is that it **cannot be observed**. It is a statement about an entire population that we never see. The best we can do is take a sample and estimate the line that, on the available evidence, comes closest to the truth.

⚠ Common mistake: confusing the PRF with the SRF

The PRF $\mathbb{E}[y \mid x] = \beta_0 + \beta_1 x$ is a property of the *population*: β_0 and β_1 are fixed but unknown. The SRF $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ (next section) is a *random* line whose intercept and slope change from sample to sample. Confusing the two is the single most common conceptual error in this course.

2.3. The sample regression function

Let $\{(x_i, y_i) : i = 1, 2, \dots, n\}$ be a random sample of size n from the population. For each observation i the population model is

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

We define the **sample regression function (SRF)** as

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$

where $\hat{\beta}_0$ and $\hat{\beta}_1$ are **estimates** of the population parameters β_0 and β_1 , obtained from the data. The hats remind us at every step that these are sample-based quantities. Two different samples from the same population would, in general, yield two different SRFs.

2.4. Ordinary Least Squares (OLS)

Given a scatter plot of (x_i, y_i) data points, infinitely many lines could be drawn through the cloud, each with a different $(\hat{\beta}_0, \hat{\beta}_1)$. How do we choose the “best” line?

Why we need a criterion

Here is a scatter plot of 41 students from a recent UGR Econometrics I cohort, plotting their attendance rate against their final grade. Many lines look like they could “roughly” fit the cloud of points; the question is which one we choose, and why. Before deriving any formulas, it helps to *see* the problem.

```
# 41 students, UGR Econometrics I 2024-25 cohort
asistencia <- c(38.89, 94.44, 94.44, 33.33, 83.33, 0, 0, 66.67, 88.89,
               66.67, 27.78, 83.33, 0, 61.11, 0, 22.22, 0, 0, 72.22,
               88.89, 88.89, 11.11, 5.56, 33.33, 16.67, 0, 11.11,
               33.33, 100, 38.89, 5.56, 72.22, 44.44, 0, 55.56, 33.33,
```

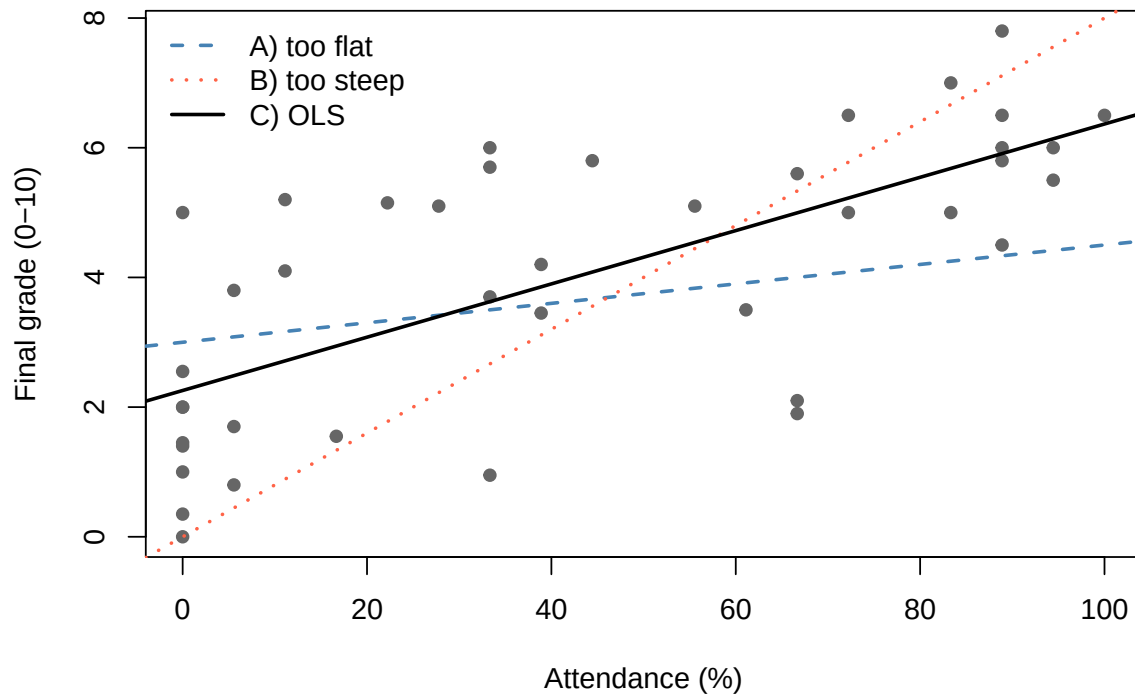
2. The Simple Linear Regression Model

```
      0, 66.67, 5.56, 88.89, 88.89)
nota    <- c(3.45, 5.5, 6, 3.7, 7, 2.55, 5, 2.1, 5.8, 1.9, 5.1, 5,
            2, 3.5, 2, 5.15, 0.35, 1.45, 5, 4.5, 7.8, 5.2, 3.8, 6,
            1.55, 1, 4.1, 5.7, 6.5, 4.2, 0.8, 6.5, 5.8, 0, 5.1,
            0.95, 1.4, 5.6, 1.7, 6.5, 6)

# Three candidate lines (we will compute their SSRs)
candidates <- data.frame(
  label = c("A) too flat", "B) too steep", "C) OLS"),
  b0     = c(3.0, 0.0, 2.256),
  b1     = c(0.015, 0.08, 0.0411)
)

# Scatter + the three candidate lines
plot(asistencia, nota,
     xlab = "Attendance (%)", ylab = "Final grade (0-10)",
     pch = 19, col = "grey40", cex = 0.9,
     main = "Three candidate lines for the same data")
cols <- c("steelblue", "tomato", "black")
ltys <- c(2, 3, 1)
for (k in seq_len(nrow(candidates))) {
  abline(a = candidates$b0[k], b = candidates$b1[k],
        col = cols[k], lwd = 2, lty = ltys[k])
}
legend("topleft", legend = candidates$label,
      col = cols, lty = ltys, lwd = 2, bty = "n")
```

Three candidate lines for the same data



```
# SSR for each candidate
ssr <- sapply(seq_len(nrow(candidates)), function(k) {
  yhat <- candidates$b0[k] + candidates$b1[k] * asistencia
  sum((nota - yhat)^2)
})
data.frame(line = candidates$label,
           beta0 = candidates$b0,
           beta1 = candidates$b1,
           SSR = round(ssr, 2))
```

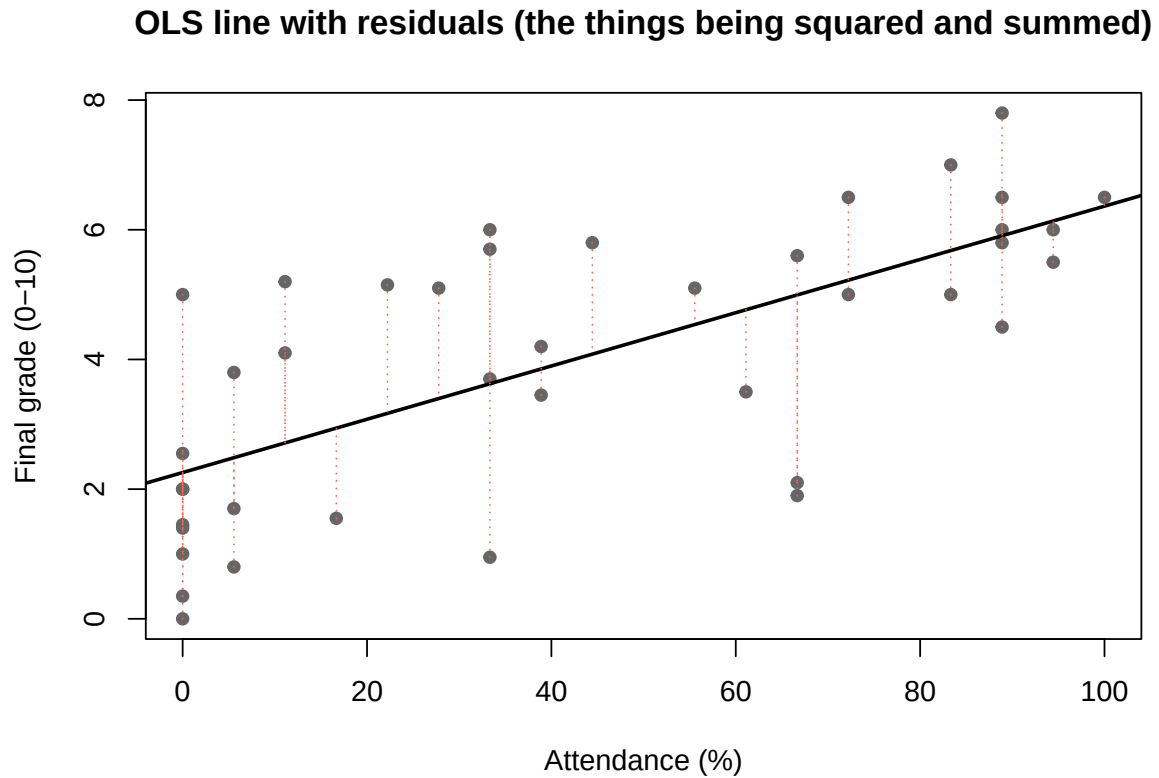
	line	beta0	beta1	SSR
1	A) too flat	3.000	0.0150	131.95
2	B) too steep	0.000	0.0800	185.12
3	C) OLS	2.256	0.0411	92.33

Each candidate line gives a different sum of squared residuals (SSR). OLS is the *unique* pair $(\hat{\beta}_0, \hat{\beta}_1)$ that minimises this sum — line C above. The next figure shows the residuals (the very quantities being squared and summed) as dashed vertical segments from each point to the OLS line.

```
m <- lm(nota ~ asistencia)
plot(asistencia, nota,
     xlab = "Attendance (%)", ylab = "Final grade (0-10)",
     pch = 19, col = "grey40", cex = 0.9,
     main = "OLS line with residuals (the things being squared and summed)")
```

2. The Simple Linear Regression Model

```
abline(m, col = "black", lwd = 2)
yhat <- predict(m)
segments(asistencia, nota, asistencia, yhat,
        col = "tomato", lty = 3)
```



The formal definition of the residual $\hat{u}_i$ follows in §2.4.1; the algebraic derivation of the minimisers $\hat{\beta}_0$ and $\hat{\beta}_1$ follows in §2.4.3. The visual intuition stays with us: OLS is the line that makes the sum of squared dashed segments as small as possible.

i Data: 41 estudiantes de Econometría I, curso 2024-25, UGR (recogidos por el autor).

2.4.1. The residual

For each observation i , the **residual** is the vertical distance between the observed y_i and the value predicted by the SRF:

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

The residual is the sample analogue of the population disturbance u_i . It is observable as soon as we have estimates $(\hat{\beta}_0, \hat{\beta}_1)$; the disturbance u_i is *never* observable.

2.4.2. The sum of squared residuals

We want the line that makes the residuals as small as possible in some overall sense. Squaring (rather than taking absolute values) penalises large deviations more heavily and yields a smooth, differentiable objective. The OLS method chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimise the **sum of squared residuals**:

$$\text{SSR} = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

2.4.3. Derivation of the OLS estimators

Differentiating SSR with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$ and setting the derivatives to zero gives the *first-order conditions* (FOCs):

$$-2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0, \quad -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

The first FOC, divided by n , gives $\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0$, hence

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Substituting this back into the second FOC and simplifying yields the celebrated formula for the slope:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\widehat{\text{Cov}}(x, y)}{\widehat{\text{Var}}(x)}.$$

The slope is the sample covariance between x and y divided by the sample variance of x . Two consequences are immediate:

- The OLS slope is a **scaled measure of association**. If x and y are positively correlated in the sample, $\hat{\beta}_1 > 0$; if uncorrelated, $\hat{\beta}_1 = 0$.
- The formula requires $\sum_{i=1}^n (x_i - \bar{x})^2 > 0$, i.e. there must be **variation in x** . If every observation has the same value of x , the denominator is zero and OLS is undefined — there is simply no way to learn how y responds to x when x never moves.

i Example: wage and education

Suppose we have a sample of $n = 1,000$ workers and we obtain

$$\widehat{\text{Wage}} = 800 + 15 \times \text{Education}.$$

We read off:

- $\hat{\beta}_0 = 800$: the predicted monthly wage for someone with zero years of education is 800 euros (extrapolation outside the data range).
- $\hat{\beta}_1 = 15$: an extra year of schooling is associated, on average, with an extra 15 euros per month.

2. The Simple Linear Regression Model

The OLS solution has three mechanical properties that follow directly from the first-order conditions and that we will use repeatedly:

1. $\sum_{i=1}^n \hat{u}_i = 0$ — the residuals sum to zero.
2. $\sum_{i=1}^n x_i \hat{u}_i = 0$ — the residuals are uncorrelated with x in the sample.
3. The OLS line passes through the point $(\bar{x}, \bar{y})$ — a direct consequence of $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

These are **algebraic** properties of OLS: they hold by construction in *every* sample, no matter how the data were generated. They are not statistical properties about the population.

2.5. The Gauss–Markov assumptions for SLR

OLS gives a number from any sample, but for that number to have desirable *statistical* properties — unbiasedness, minimum variance, sensible standard errors — we need assumptions about the population. The five **Gauss–Markov assumptions for SLR** are the bedrock of the rest of this chapter.

SLR.1 — Linearity in parameters. The population model can be written as

$$y = \beta_0 + \beta_1 x + u,$$

where β_0 and β_1 are the unknown parameters and u is the unobservable disturbance.

SLR.2 — Random sampling. We have a random sample $\{(x_i, y_i) : i = 1, \dots, n\}$ drawn from the population in SLR.1.

SLR.3 — Sample variation in x . The sample values $x_1, x_2, \dots, x_n$ are not all the same, i.e. $\sum_{i=1}^n (x_i - \bar{x})^2 > 0$.

SLR.4 — Zero conditional mean. The error has expected value zero given any value of x :

$$\mathbb{E}[u \mid x] = 0.$$

SLR.5 — Homoskedasticity. The error variance is constant in x :

$$\text{Var}(u \mid x) = \sigma^2.$$

 Common mistake: forgetting which assumption does what

SLR.1–SLR.3 are technical conditions that make the estimator well-defined. SLR.4 is what makes the estimator **unbiased** and lets us interpret β_1 as a *ceteris paribus* effect; it is also the assumption hardest to verify and most often violated in practice. SLR.5 is needed only for the **variance** formula and the Gauss–Markov efficiency result, not for unbiasedness.

2.6. Unbiasedness of OLS

Under SLR.1–SLR.4 the OLS estimators are unbiased. We sketch the argument for $\hat{\beta}_1$; the algebra for $\hat{\beta}_0$ is similar.

Start from the OLS formula and substitute the population model $y_i = \beta_0 + \beta_1 x_i + u_i$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

The second term is a weighted average of the unobservable errors, with weights $(x_i - \bar{x})/\text{SST}_x$ that sum to zero. Taking conditional expectations given $\mathbf{x} = (x_1, \dots, x_n)$ and using SLR.4 ($\mathbb{E}[u_i | x_i] = 0$ for every i),

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{x}] = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) \mathbb{E}[u_i | \mathbf{x}]}{\sum_{i=1}^n (x_i - \bar{x})^2} = \beta_1.$$

By the law of iterated expectations, $\mathbb{E}[\hat{\beta}_1] = \beta_1$. The same argument applied to $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ gives $\mathbb{E}[\hat{\beta}_0] = \beta_0$.

i Unbiasedness is a property of the estimator, not of any single estimate

The statement $\mathbb{E}[\hat{\beta}_1] = \beta_1$ does *not* say “our number is exactly right”. It says that if we drew many independent samples from the same population and computed $\hat{\beta}_1$ in each one, the long-run average of all those estimates would equal β_1 . Any individual estimate may be above or below the truth.

2.7. Sampling variances of OLS estimators

How precise are the OLS estimates? Under SLR.1–SLR.5, conditioning on the regressors:

$$\text{Var}(\hat{\beta}_1 | \mathbf{x}) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{\text{SST}_x},$$

$$\text{Var}(\hat{\beta}_0 | \mathbf{x}) = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2}.$$

Three drivers of the precision of $\hat{\beta}_1$ stand out:

1. **Error variance σ^2 .** Noisier data give less precise estimates. Adding good controls in Chapter 3 will shrink σ^2 and tighten our estimates.
2. **Variation in x (the spread of the regressor, SST_x).** A regressor that barely moves is uninformative; one that spans a wide range pins down the slope sharply.
3. **Sample size n .** A larger sample typically raises SST_x , which directly reduces $\text{Var}(\hat{\beta}_1)$.

These three drivers reappear, in similar form, in every regression model in the rest of the book.

2.8. The Gauss–Markov theorem

The single most important “why use OLS?” result is the **Gauss–Markov theorem**.

Under SLR.1–SLR.5, the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the **Best Linear Unbiased Estimators (BLUE)** of β_0 and β_1 .

“Best” means *minimum variance* in the class of all estimators that are (i) linear in y and (ii) unbiased for β . The proof considers an arbitrary linear unbiased competitor $\tilde{\beta}_1 = \sum_i w_i y_i$, writes its weights as $w_i = (x_i - \bar{x})/\text{SST}_x + d_i$ for some deviation d_i , imposes the conditions $\sum_i d_i = 0$ and $\sum_i d_i x_i = 0$ required for unbiasedness, and computes

$$\text{Var}(\tilde{\beta}_1 \mid \mathbf{x}) = \frac{\sigma^2}{\text{SST}_x} + \sigma^2 \sum_{i=1}^n d_i^2 \geq \frac{\sigma^2}{\text{SST}_x} = \text{Var}(\hat{\beta}_1 \mid \mathbf{x}),$$

with equality only when every $d_i = 0$, i.e. when $\tilde{\beta}_1 = \hat{\beta}_1$.

⚠ What Gauss–Markov does *not* say

The Gauss–Markov theorem does **not** say that OLS is the best estimator in some absolute sense. It says: among *linear, unbiased* estimators, OLS has minimum variance. There may well be *biased* estimators (ridge, LASSO) or *nonlinear* estimators (quantile regression) with lower mean squared error in particular settings.

2.9. Estimation of σ^2

The error variance σ^2 enters the variance formulas above, but it is itself unknown — we never observe the population errors u_i , only the residuals $\hat{u}_i$. The natural sample-based estimator is

$$\hat{\sigma}^2 = \frac{\text{SSR}}{n-2} = \frac{\sum_{i=1}^n \hat{u}_i^2}{n-2}.$$

Under SLR.1–SLR.5 this estimator is unbiased: $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$. We divide by $n-2$ rather than by n because two degrees of freedom have already been “used up” estimating β_0 and β_1 from the same data. The square root $\hat{\sigma}$ is sometimes called the **standard error of the regression** (R reports it as *Residual standard error*).

Combining $\hat{\sigma}^2$ with the variance formula yields the **standard error** of the slope estimate:

$$\text{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\text{SST}_x}}.$$

The standard error is the workhorse of inference in Chapter 4 — everything we will say about confidence intervals and t -tests is built on top of it.

2.10. Goodness of fit: R^2

Two regressions can have the same slope estimate and yet very different fits, depending on how much of the variation in y the regressor accounts for. To measure fit, decompose the total variation in y as

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{SSE}} + \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{\text{SSR}},$$

where SST is the *Total Sum of Squares*, SSE is the *Explained Sum of Squares*, and SSR is the *Residual Sum of Squares*. The decomposition uses the algebraic properties of OLS ($\sum_i \hat{u}_i = 0$ and $\sum_i x_i \hat{u}_i = 0$); it would not hold for an arbitrary line.

The **coefficient of determination** is then

$$R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}.$$

R^2 is the fraction of the sample variation in y that is “explained” by x . It lies between 0 and 1, with $R^2 = 1$ when every point lies on the line and $R^2 = 0$ when $\hat{\beta}_1 = 0$. In the simple linear regression, R^2 equals the squared sample correlation between y and x .

⚠ A low R^2 does *not* mean a useless model

In labour, health, and education economics, R^2 values of 5–15 % are completely normal: individual outcomes depend on many things we never measure. A small R^2 does *not* by itself invalidate the slope estimate — $\hat{\beta}_1$ can still be unbiased, statistically significant, and policy-relevant. Conversely, a high R^2 does *not* validate a model in which SLR.4 fails: $\hat{\beta}_1$ can be precisely estimated and badly biased.

2.11. Lab: simple linear regression

In this lab we estimate two simple regressions using the **wooldridge** package: the wage–looks regression from §2.4 (“Lab A”) and the birth-weight–smoking regression (“Lab B”). Both are short cross-sections, both have small R^2 , and both illustrate the warning above.

```
library(wooldridge)
data("beauty")
data("bwght")
```

2.11.1. Lab A — Wages and physical appearance (beauty)

The **beauty** dataset, used by Hamermesh and Biddle (1994) and reproduced in Wooldridge (2020), contains hourly wages and a five-point rating of physical appearance for a sample of U.S. workers. The variable **looks** takes integer values from 1 (“homely”) to 5 (“strikingly beautiful”), with 3 being “average”. The variable **wage** is the hourly wage in U.S. dollars.

We start with summary statistics and a frequency table of **looks**.

2. The Simple Linear Regression Model

```
summary(quality$wage)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.020	3.708	5.300	6.307	7.695	77.720

```
summary(quality$looks)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	3.000	3.000	3.186	4.000	5.000

```
table(quality$looks)
```

1	2	3	4	5
13	142	722	364	19

Most workers are rated 3 (“average”) — the marginal categories 1 and 5 are sparse. This already tells us that the slope of a regression of wage on looks will be driven almost entirely by the contrast among categories 2, 3, and 4.

Before any regression, let us compute the mean wage in each looks category. This is the *non-parametric* conditional mean and gives us a benchmark against which to compare the OLS fit.

```
tapply(quality$wage, quality$looks, mean)
```

1	2	3	4	5
4.621538	5.328803	6.504598	6.299341	7.388421

Mean wages rise (modestly and not entirely monotonically) with looks. The natural single-number summary of this pattern is the slope of the OLS line.

A scatter plot of the raw data, with the conditional means superimposed, makes the story visible.

```
cm <- tapply(quality$wage, quality$looks, mean)
plot(quality$looks, quality$wage,
     xlab = "Looks rating (1 = homely, 5 = strikingly beautiful)",
     ylab = "Hourly wage (USD)",
     main = "Wage vs. looks",
     pch = 16, col = rgb(0, 0, 0, 0.15))
points(as.numeric(names(cm)), cm, pch = 19, col = "blue", cex = 1.6)
```

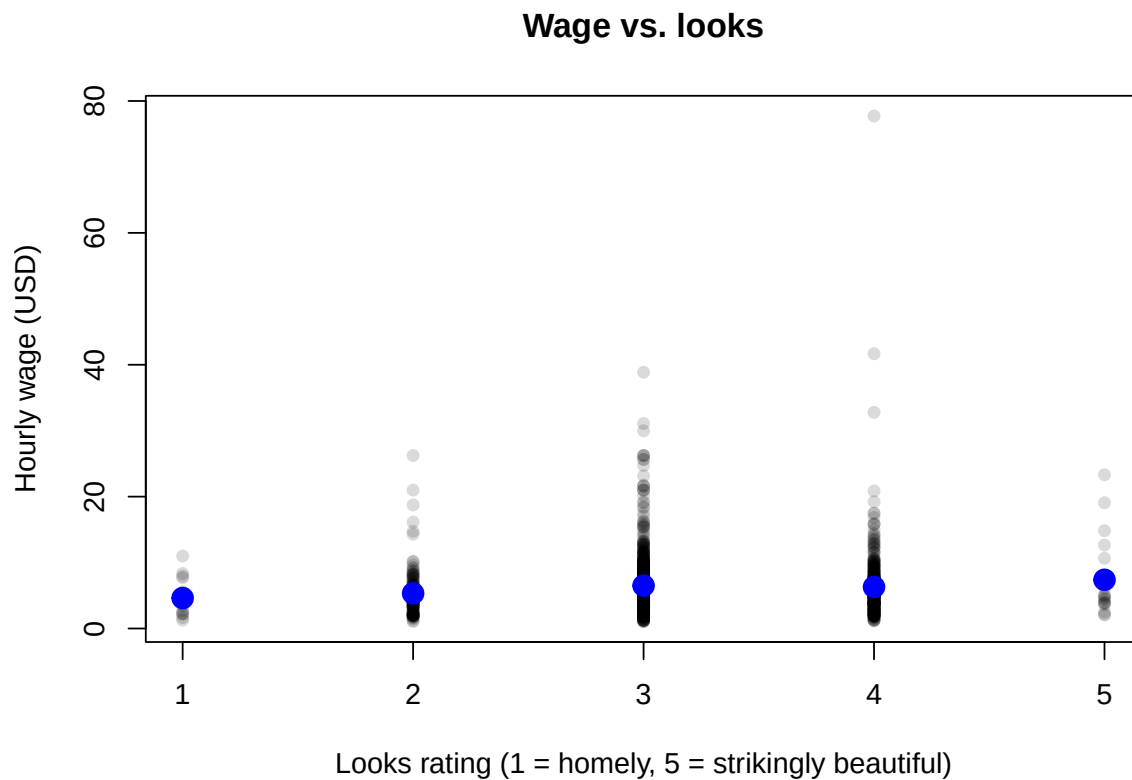


Figure 2.1.: Hourly wage vs. looks rating in the beauty dataset. Blue dots are conditional means by looks category.

Now we estimate the simple linear regression of wage on looks with `lm()`:

```
fit_beauty <- lm(wage ~ looks, data = beauty)
summary(fit_beauty)
```

Call:

```
lm(formula = wage ~ looks, data = beauty)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.452	-2.737	-0.997	1.453	71.108

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.1139	0.6242	8.192	6.24e-16 ***
looks	0.3744	0.1916	1.954	0.0509 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.655 on 1258 degrees of freedom

Multiple R-squared: 0.003027, Adjusted R-squared: 0.002235

F-statistic: 3.82 on 1 and 1258 DF, p-value: 0.05088

2. The Simple Linear Regression Model

i Reading the `summary()` output, column by column

Every column of R's `lm()` summary maps to a quantity we have already named in this chapter (or will name in Chapter 4). Use this once as a Rosetta stone — the columns recur in every regression `summary()` you will ever read.

- **Estimate** → the OLS coefficient $\hat{\beta}_j$ (intercept $\hat{\beta}_0$ in the first row, slope $\hat{\beta}_1$ in the second), defined in §2.4.3.
- **Std. Error** → the standard error $\text{se}(\hat{\beta}_j) = \hat{\sigma}/\sqrt{\text{SST}_x}$ from §2.9.
- **t value** → the ratio $\hat{\beta}_j/\text{se}(\hat{\beta}_j)$. We treat it formally in Chapter 4 §4.3; for now, read it as “how many standard errors is the estimate away from zero.”
- **Pr(>|t|)** → the two-sided p -value for $H_0 : \beta_j = 0$. We treat it formally in Chapter 4 §4.7.
- **Residual standard error** → $\hat{\sigma}$ from §2.9, i.e. the square root of $\hat{\sigma}^2 = \text{SSR}/(n-2)$.
- **Multiple R-squared** → the R^2 from §2.10 — the fraction of the variation in `wage` explained by `looks`.
- The bottom-line **F-statistic** tests $H_0 : \beta_1 = 0$ in this single-regressor setting; in Chapter 4 we generalise it to joint hypotheses on several coefficients.

Reading the output:

- The estimated intercept $\hat{\beta}_0$ is the predicted wage for a worker with `looks` = 0 — a value outside the range of the data, so the intercept has no direct economic interpretation here.
- The estimated slope $\hat{\beta}_1$ on `looks` is the change in predicted hourly wage for a one-point increase in the looks rating. A worker rated 4 (“above average”) is predicted to earn about $\hat{\beta}_1$ dollars per hour more than a worker rated 3 (“average”), on average.
- The standard error of $\hat{\beta}_1$, the t -statistic and the p -value tell us whether the slope is statistically distinguishable from zero. We treat these formally in Chapter 4; for now, note that the slope is significantly different from zero at conventional levels.
- The **Residual standard error** at the bottom of `summary()` is $\hat{\sigma}$ from §2.9.
- The **Multiple R-squared** is the R^2 from §2.10. It is small (1–2 %): `looks` alone explains only a tiny share of the variation in wages. That does not invalidate the slope — it simply reminds us that wages depend on far more than `looks` (education, experience, occupation, ...), most of which we will add as controls in Chapter 3.

To finish, overlay the OLS line on the scatter:

```
plot(beauty$looks, beauty$wage,
     xlab = "Looks rating", ylab = "Hourly wage (USD)",
     main = "Wage vs. looks with OLS line",
     pch = 16, col = rgb(0, 0, 0, 0.15))
abline(fit_beauty, col = "red", lwd = 2)
points(as.numeric(names(cm)), cm, pch = 19, col = "blue", cex = 1.6)
```

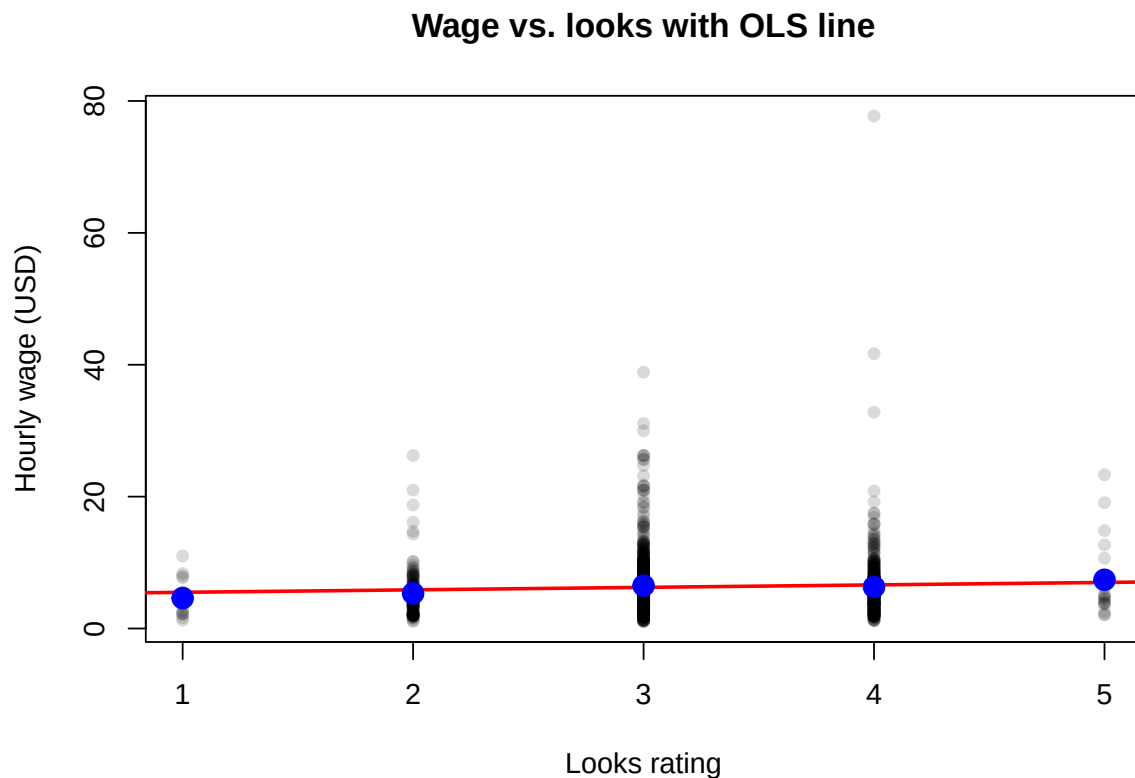


Figure 2.2.: OLS line and conditional means for the wage–looks regression.

The red OLS line approximates the trend in the blue conditional means but does not interpolate them exactly — a linear model is an approximation.

A useful sanity check is to verify the decomposition $SST = SSE + SSR$ and the identity $R^2 = 1 - SSR/SST$ from §2.10:

```
yhat <- fitted(fit_beauty)
uhat <- resid(fit_beauty)
y <- beauty$wage
SST <- sum((y - mean(y))^2)
SSE <- sum((yhat - mean(y))^2)
SSR <- sum(uhat^2)
c(SST = SST, SSE_plus_SSR = SSE + SSR)
```

```
      SST  SSE_plus_SSR
27347.44  27347.44
```

```
c(R2_manual = 1 - SSR / SST, R2_lm = summary(fit_beauty)$r.squared)
```

```
R2_manual    R2_lm
0.0030271 0.0030271
```

The two pairs of numbers match (up to numerical rounding), as the theory predicts.

2. The Simple Linear Regression Model

2.11.2. Lab B — Birth weight and smoking (bwght)

The `bwght` dataset records the birth weight (in ounces) and average daily cigarette consumption of the mother during pregnancy, among other variables, for a sample of U.S. births. Maternal smoking is one of the most studied determinants of birth weight, with clear medical plausibility for a *negative* effect.

```
summary(bwght$bwght)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23.0	107.0	120.0	118.7	132.0	271.0

```
summary(bwght$cigs)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	0.000	2.087	0.000	50.000

```
mean(bwght$cigs)
```

```
[1] 2.087176
```

Most mothers in the sample do not smoke (the median of `cigs` is zero). Among those who do, daily consumption can be as high as 50 cigarettes. The histogram makes the heavy right tail obvious:

```
hist(bwght$cigs,  
     main = "Cigarettes per day during pregnancy",  
     xlab = "Cigarettes per day",  
     col  = "lightblue",  
     breaks = 20)
```

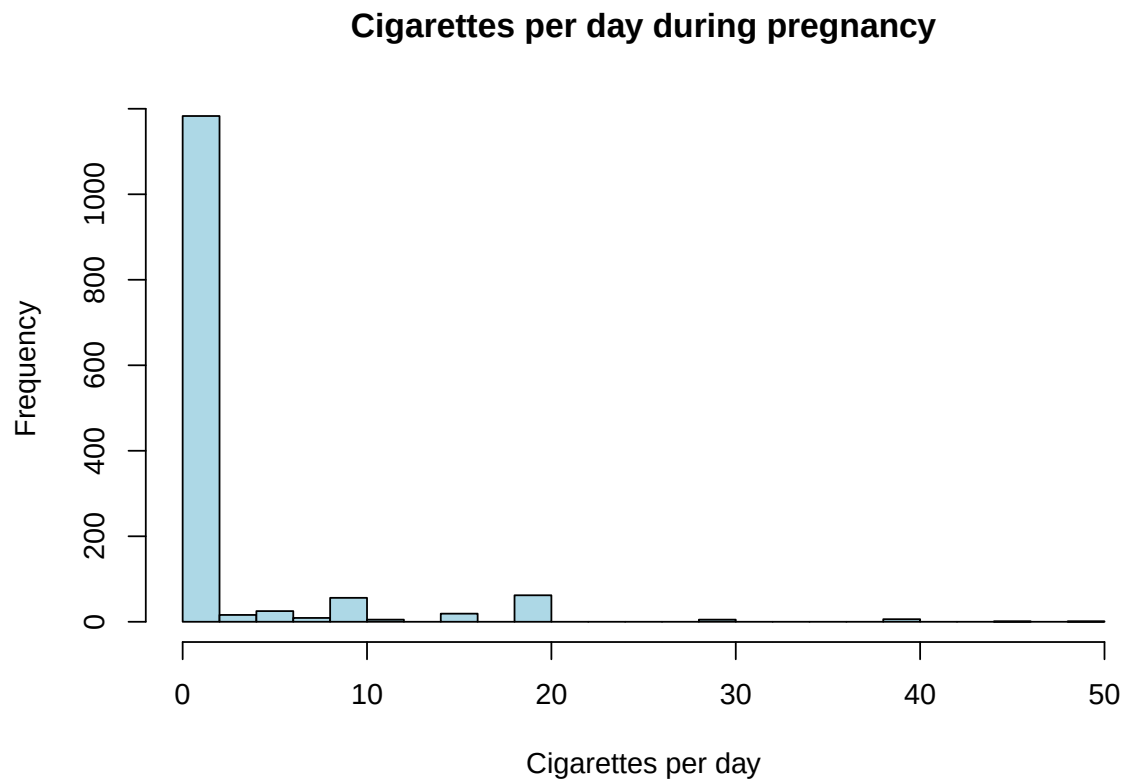



Figure 2.3.: Distribution of cigarettes smoked per day during pregnancy (bwght).

A scatter of birth weight against cigarettes:

```
plot(bwght$cigs, bwght$bwght,
     xlab = "Cigarettes per day",
     ylab = "Birth weight (oz)",
     main = "Birth weight vs. maternal smoking",
     pch = 16, col = rgb(0, 0, 0, 0.25))
```

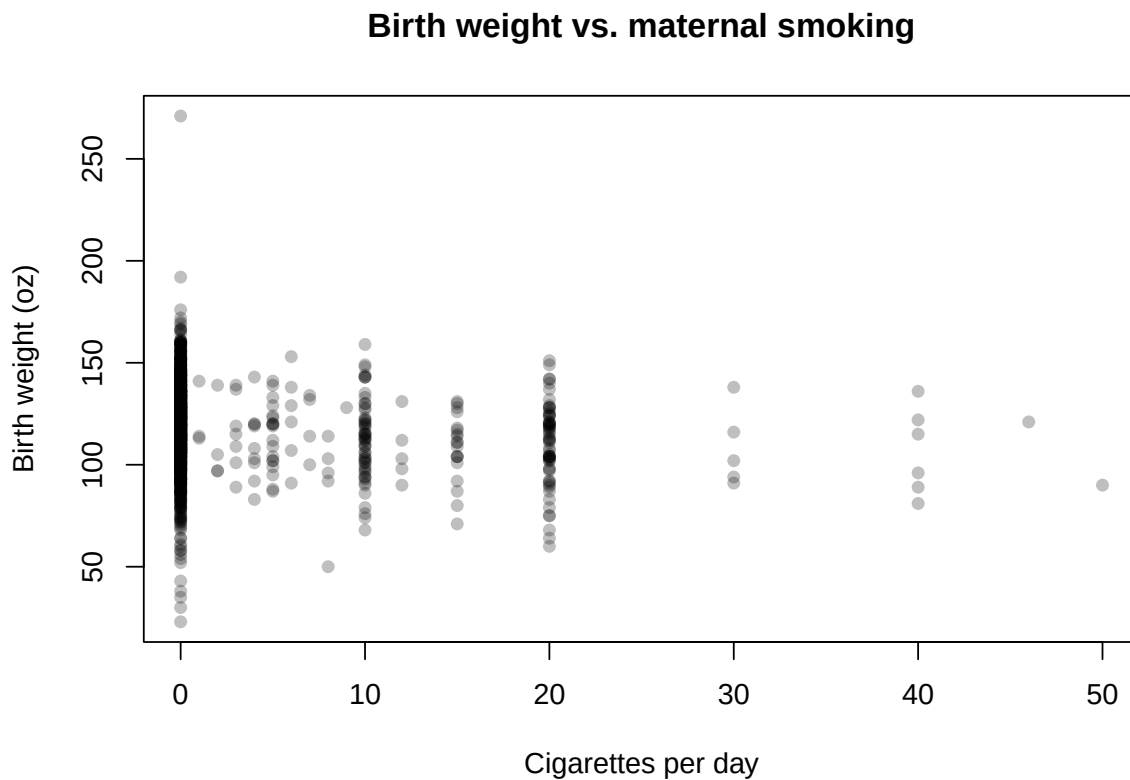


Figure 2.4.: Birth weight in ounces vs. cigarettes per day.

Now the simple linear regression of `bwght` on `cigs`:

```
fit_bwght <- lm(bwght ~ cigs, data = bwght)
summary(fit_bwght)
```

Call:

```
lm(formula = bwght ~ cigs, data = bwght)
```

Residuals:

Min	1Q	Median	3Q	Max
-96.772	-11.772	0.297	13.228	151.228

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	119.77190	0.57234	209.267	< 2e-16 ***
cigs	-0.51377	0.09049	-5.678	1.66e-08 ***

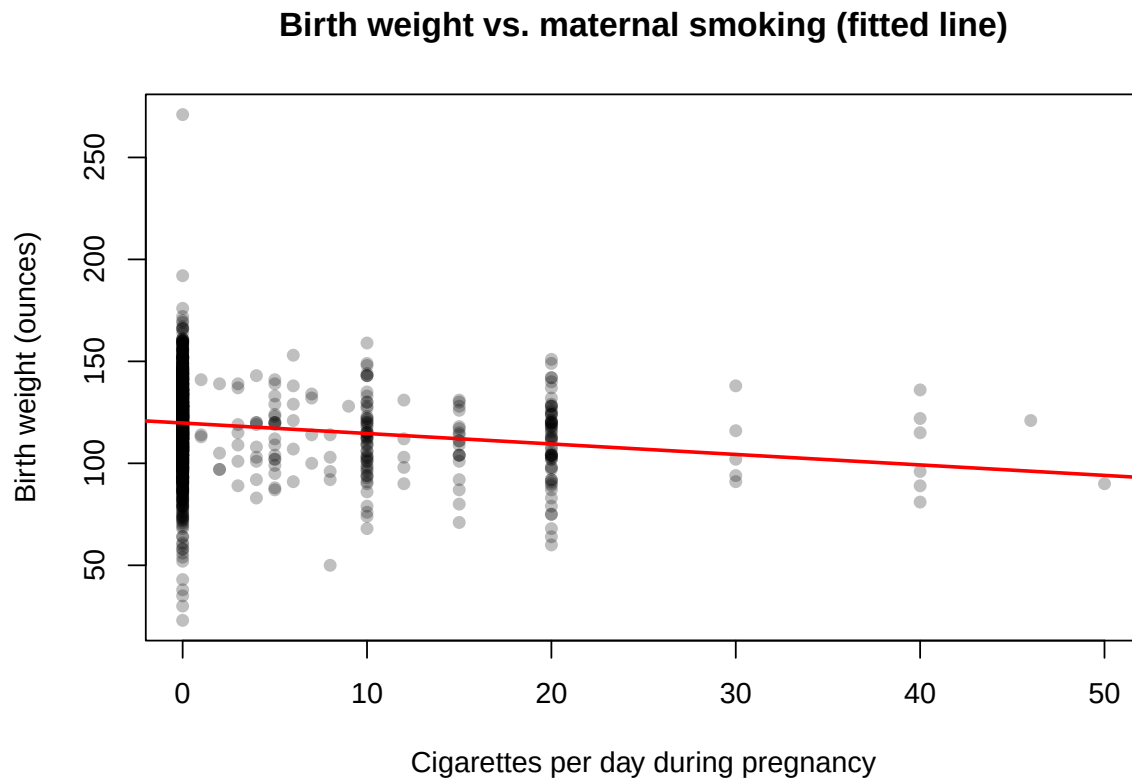
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.13 on 1386 degrees of freedom

Multiple R-squared: 0.02273, Adjusted R-squared: 0.02202

F-statistic: 32.24 on 1 and 1386 DF, p-value: 1.662e-08

```
plot(bwght$cigs, bwght$bwght,
     xlab = "Cigarettes per day during pregnancy",
     ylab = "Birth weight (ounces)",
     main = "Birth weight vs. maternal smoking (fitted line)",
     pch = 16, col = rgb(0, 0, 0, 0.25))
abline(fit_bwght, col = "red", lwd = 2)
```



Interpretation:

- $\hat{\beta}_0$ is the predicted birth weight for a baby whose mother does *not* smoke during pregnancy (`cigs` = 0). Unlike the beauty regression, this is a meaningful value because the median of `cigs` is zero.
- $\hat{\beta}_1$ is the change in predicted birth weight for one extra cigarette per day. The estimated slope is approximately -0.51 ounces per cigarette, so smoking ten more cigarettes per day is associated with roughly $10 \times (-0.51) \approx -5.1$ ounces of birth weight. The negative sign is consistent with everything we know from medicine.
- The R^2 is again small. Birth weight depends on dozens of factors — maternal nutrition, age, prior births, prenatal care — and `cigs` is only one of them. As in Lab A, a small R^2 does not invalidate the slope estimate.

Whether the estimated -0.51 is the *causal* effect of smoking on birth weight is a different question. If mothers who smoke also differ from non-smokers in income, education, and nutrition, and if those characteristics independently affect birth weight, then SLR.4 fails and $\hat{\beta}_1$ is biased. We will revisit this exact dataset in Chapter 3 once we have the multiple regression tools needed to address it.

Self-check

Seven short multiple-choice questions. Try each one before opening the answer.

💡 Q1. The slope β_1

In the simple regression model $y = \beta_0 + \beta_1 x + u$, the slope β_1 measures:

- A. The correlation between x and y .
- B. The expected change in y when x increases by one unit, holding all factors in u fixed.
- C. The expected value of y when $x = 0$.
- D. The variance of y explained by x .

Answer: B. β_1 is the *ceteris paribus* change in the conditional mean of y . C describes the intercept β_0 ; A is the correlation (a different, scale-free quantity); D is R^2 .

💡 Q2. The zero-conditional-mean assumption

The assumption $\mathbb{E}[u \mid x] = 0$ implies:

- A. The error u is normally distributed.
- B. $\text{Var}(u \mid x)$ is constant in x .
- C. $\mathbb{E}[u] = 0$ and $\text{Cov}(x, u) = 0$.
- D. x is measured without error.

Answer: C. Zero conditional mean implies both an unconditional mean of zero (by iterated expectations) and zero covariance between x and u . B is the separate homoskedasticity assumption SLR.5; A and D are unrelated.

💡 Q3. The OLS slope formula

The OLS estimator of β_1 in the simple regression $y = \beta_0 + \beta_1 x + u$ can be written as:

- A. $\hat{\beta}_1 = \bar{y}/\bar{x}$.
- B. $\hat{\beta}_1 = \widehat{\text{Var}}(y)/\widehat{\text{Cov}}(x, y)$.
- C. $\hat{\beta}_1 = \widehat{\text{Cov}}(x, y)/\widehat{\text{Var}}(x)$.
- D. $\hat{\beta}_1 = \bar{y} - \hat{\beta}_0 \bar{x}$.

Answer: C. The slope is the sample covariance of x and y over the sample variance of x . D is (a rearranged) formula for the intercept; A and B are not OLS estimators.

💡 Q4. Mechanical properties of OLS residuals

Which of the following is **NOT** an algebraic property of OLS residuals $\hat{u}_i$?

- A. $\sum_i \hat{u}_i = 0$.
- B. $\sum_i x_i \hat{u}_i = 0$.
- C. The OLS line passes through $(\bar{x}, \bar{y})$.
- D. $\hat{u}_i$ are uncorrelated with the unobserved error u_i .

Answer: D. A, B, and C follow directly from the first-order conditions and hold in every sample. D is a statement about the population disturbance u_i , which is never observable; there is no algebraic identity relating $\hat{u}_i$ to u_i .

💡 Q5. Unbiasedness vs. BLUE

Under SLR.1–SLR.4 only (no homoskedasticity), OLS is:

- A. Unbiased: $\mathbb{E}[\hat{\beta}_j] = \beta_j$.
- B. The Best Linear Unbiased Estimator (BLUE).
- C. Consistent but biased.
- D. Normally distributed.

Answer: A. Unbiasedness needs only SLR.1–SLR.4. Adding SLR.5 (homoskedasticity) is what upgrades OLS to BLUE via the Gauss–Markov theorem.

💡 Q6. Precision of $\hat{\beta}_1$

Which of the following increases the precision of $\hat{\beta}_1$ (smaller sampling variance)?

- A. A larger error variance σ^2 .
- B. A smaller sample size.
- C. Less variation in x .
- D. More variation in x (larger SST_x).

Answer: D. The variance formula is $\text{Var}(\hat{\beta}_1 | \mathbf{x}) = \sigma^2 / \text{SST}_x$. Larger SST_x shrinks the variance. A and B do the opposite; C is a re-statement of “less precision”.

💡 Q7. What R^2 does and does not tell us

A low R^2 in a simple linear regression means:

- A. OLS is biased.
- B. x has no causal effect on y .
- C. x explains a small share of the variation in y , but $\hat{\beta}_1$ can still be unbiased and policy-relevant.
- D. The Gauss–Markov assumptions are violated.

Answer: C. R^2 is a measure of fit, not of bias or causality. A small R^2 is fully compatible with an unbiased and economically meaningful slope, as the wage–looks and birth-weight–smoking labs illustrate.

Exercises

Exercise 2.1 — Reading a fitted equation. A regression on 1{,}000 workers gives the fitted line $\widehat{\text{Wage}} = 800 + 15 \times \text{Education}$, where wage is in euros per month and education is in years.

- (a) Interpret $\hat{\beta}_0$ and $\hat{\beta}_1$ in words.

2. The Simple Linear Regression Model

- (b) What is the predicted wage of a worker with 12 years of schooling?
- (c) Two workers differ by 4 years of schooling. By how much do their predicted wages differ?
- (d) Why should you be wary of the prediction at `educ = 0`?

💡 Show answer

- (a) $\hat{\beta}_0 = 800$ is the predicted wage at `educ = 0` (800 €/month). $\hat{\beta}_1 = 15$ is the predicted increase in monthly wage per extra year of schooling.
- (b) $800 + 15 \times 12 = 980$ €/month.
- (c) $4 \times 15 = 60$ €/month.
- (d) Because `educ = 0` is far below the lowest value in any realistic schooling sample — the prediction is an extrapolation and may be unreliable.

Exercise 2.2 — Algebra of OLS, by hand. Consider the toy data $x = (1, 2, 3, 4, 5)$ and $y = (3, 4, 2, 5, 6)$.

- (a) Compute $\bar{x}$ and $\bar{y}$.
- (b) Compute $\hat{\beta}_1$ using the formula in §2.4.3.
- (c) Compute $\hat{\beta}_0$.
- (d) Verify your answer in R against `coef(lm(y ~ x))`.

💡 Show answer

- (a) $\bar{x} = 3$, $\bar{y} = 4$.
- (b) Numerator: $\sum (x_i - \bar{x})(y_i - \bar{y}) = (-2)(-1) + (-1)(0) + 0(-2) + 1(1) + 2(2) = 2 + 0 + 0 + 1 + 4 = 7$. Denominator: $\sum (x_i - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 = 10$. So $\hat{\beta}_1 = 7/10 = 0.7$.
- (c) $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 4 - 0.7 \times 3 = 1.9$.
- (d) In R: `coef(lm(c(3,4,2,5,6) ~ c(1,2,3,4,5)))` returns the same numbers.

Exercise 2.3 — Variance of the OLS slope. State, in one sentence each, how the variance $\text{Var}(\hat{\beta}_1)$ changes if:

- (a) the sample size n doubles (holding the distribution of x fixed);
- (b) the error variance σ^2 doubles;
- (c) the regressor x is rescaled from minutes to hours (divided by 60).

💡 Show answer

- (a) SST_x roughly doubles, so the variance is approximately halved. (b) The variance doubles (it is linear in σ^2). (c) The variance of the *slope on the rescaled regressor* increases by a factor of $60^2 = 3600$: scaling x by c multiplies $\hat{\beta}_1$ by $1/c$, so its variance is multiplied by $1/c^2$.

Exercise 2.4 — Change of units. Show that if we redefine the regressor as $x^* = c \cdot x$ (for some constant $c \neq 0$) without changing y , the new OLS estimators are $\hat{\beta}_1^* = \hat{\beta}_1/c$ and $\hat{\beta}_0^* = \hat{\beta}_0$. Verify your derivation empirically on `sleep75`: regress `sleep` on `totwrk` (minutes per week) and then on `totwrk/60` (hours per week), and confirm that the intercept is unchanged while the slope scales by 60.

A full answer is given in the Instructor Edition.

Exercise 2.5 — Unbiasedness by simulation. Simulate 1,000 samples of size $n = 200$ from $y = 1 + 2x + u$ with $x \sim N(0, 1)$ and $u \sim N(0, 1)$. Compute $\hat{\beta}_1$ in each sample and show that its mean across replications is close to 2, while the standard deviation across replications is close to the theoretical $\sigma/\sqrt{\text{SST}_x}$ for a typical draw. Plot the histogram of $\hat{\beta}_1$.

A full answer is given in the Instructor Edition.

Exercise 2.6 — Why dropping the intercept biases the slope. Suppose the true model is $y = \beta_0 + \beta_1 x + u$ with $\beta_0 \neq 0$ and $\mathbb{E}[u | x] = 0$, but the researcher *imposes* $\beta_0 = 0$ and fits the regression through the origin $\tilde{y}_i = \tilde{\beta}_1 x_i$.

- (a) Derive the closed-form expression for $\tilde{\beta}_1 = \sum_i x_i y_i / \sum_i x_i^2$ as the minimiser of $\sum_i (y_i - \tilde{\beta}_1 x_i)^2$.
- (b) Show that $\mathbb{E}[\tilde{\beta}_1] = \beta_1 + \beta_0 \cdot \mathbb{E}[x]/\mathbb{E}[x^2]$, so $\tilde{\beta}_1$ is biased whenever $\beta_0 \neq 0$ and $\mathbb{E}[x] \neq 0$.
- (c) Under what specific condition on the population is the no-intercept slope nonetheless unbiased?

A full answer is given in the Instructor Edition.

The Multiple Linear Regression Model

Learning outcomes

By the end of this chapter the reader should be able to:

- Explain why the simple linear regression is rarely enough for credible empirical work, and read each MLR slope as a *ceteris paribus* (partial) effect.
- Write the multiple linear regression model in matrix form, $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$, and derive the OLS solution $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.
- State the Gauss–Markov assumptions MLR.1–MLR.5 and recognise which one each empirical pathology violates.
- Show, in matrix form, that OLS is unbiased under MLR.1–MLR.4 and has variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ under MLR.1–MLR.5.
- Predict the *direction* and *sign* of omitted-variable bias from an auxiliary regression, and verify it numerically on real data.
- Distinguish R^2 from adjusted R^2 and explain when adjusted R^2 may fall as regressors are added.

Motivating empirical question

Once we hold experience, tenure, gender, and marital status constant, what is the return to one additional year of education?

Chapter 1 showed that the raw wage gap by education in `wage1` confounds schooling with experience, tenure, and demographics; Chapter 2 fitted a simple regression of `wage` on `educ` and explicitly warned that the OLS slope could not be read causally. The multiple linear regression (MLR) model is the workhorse we use to take that warning seriously: by including controls, we move from raw conditional means to **partial effects** — the change in y associated with a one-unit change in one regressor, *holding all the others fixed*.

3.1. Motivation: the problem with simple regression

THE simple linear regression model

$$y = \beta_0 + \beta_1 x + u$$

3. The Multiple Linear Regression Model

has one structural weakness: every determinant of y other than x is bundled into the error term u . If any of those omitted determinants is correlated with x , then $\text{Cov}(x, u) \neq 0$, the zero-conditional-mean assumption $\mathbb{E}[u | x] = 0$ fails, and OLS is **biased** for the causal effect of x on y (Wooldridge 2020).

i Example: education and wages without controls

Suppose

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + u.$$

The error u contains *innate ability*, family background, motivation, school quality, and many other unobservables. If more able workers also acquire more education, then $\text{Cov}(\text{educ}, u) > 0$, and the simple OLS estimator $\hat{\beta}_1$ does not isolate the causal return to schooling — it conflates the return to education with the return to whatever drives ability and education together.

The fix is conceptually simple: **bring the relevant omitted factors into the model as additional regressors**. Replace the simple model by

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{tenure} + u,$$

so that experience and tenure now appear in the systematic part instead of in u . The coefficient β_1 in this longer model is interpreted as the change in expected wage for a one-year increase in education, *holding experience and tenure fixed*. That is the *ceteris paribus* statement empirical economics aims for.

Two cautions deserve emphasis from the outset:

- Adding controls is **not** the same as identifying causality. We will see in §3.9 that the bias only disappears for the variables whose relevant confounders we actually include.
- Including too many or highly correlated regressors can *inflate variance* without removing bias. There is no free lunch.

3.2. The MLR model

i Definition: the multiple linear regression model

With k regressors $x_1, \dots, x_k$, the **population MLR model** is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u.$$

Each slope β_j is the **partial effect** of x_j on y :

$$\beta_j = \frac{\partial \mathbb{E}[y | x_1, \dots, x_k]}{\partial x_j}.$$

That is, β_j measures how the conditional mean of y shifts when x_j changes by one unit and all other regressors are held constant.

In the wage example, a fitted equation such as

$$\widehat{\text{wage}} = 3.0 + 1.5 \text{educ} + 0.04 \text{age} + 0.07 \text{exper} + 0.05 \text{tenure}$$

says: across two workers with the *same* age, experience, and tenure but differing by one year of schooling, the predicted hourly wage differs by 1.50 dollars on average. The crucial qualifier is the “same age, experience, and tenure” — this is what distinguishes the MLR coefficient from the unconditional regression slope.

3.3. What does “holding fixed” actually do? Partialling out

A picture before the algebra: what does “holding the other regressors fixed” *actually do* to the data? The answer is **partialling out**. To get the MLR slope on `educ`, first scrub from `educ` whatever part of it is predictable from `exper` and `tenure` (its **residualised** version), then regress `wage` — itself residualised on the same controls — on that. The MLR slope on `educ` is the slope of the resulting scatter.

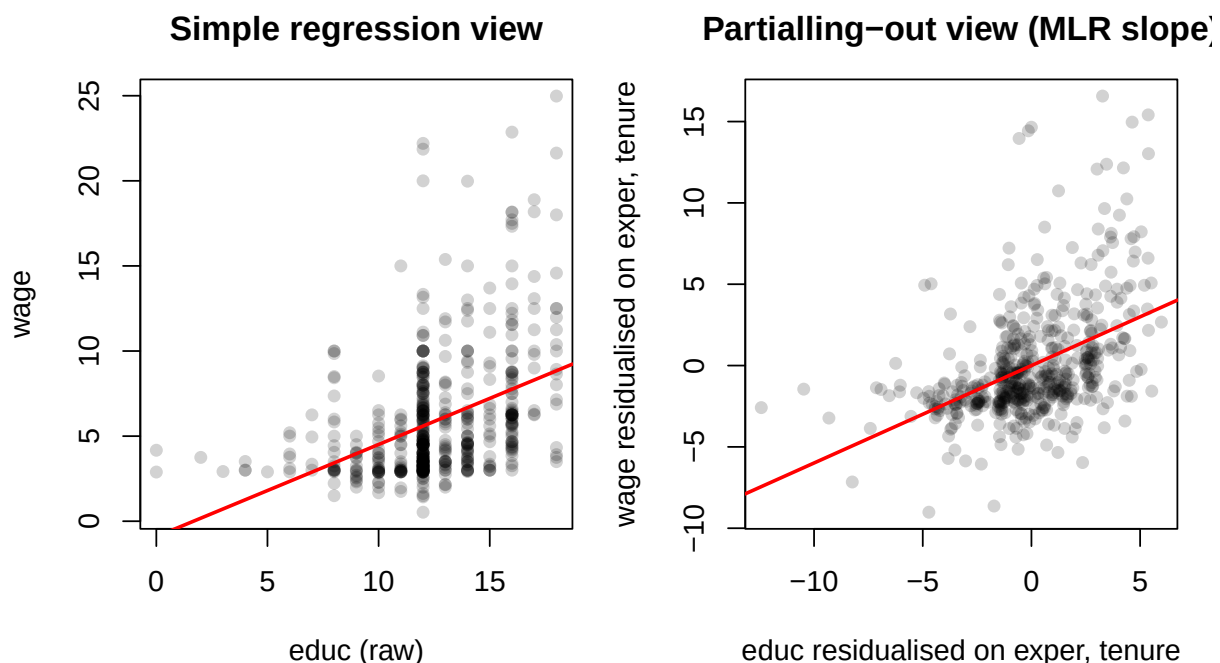


Figure 3.1.: Left: the simple-regression view — wage vs raw educ, with the simple-regression slope. Right: the multiple-regression view via partialling-out — both variables have been residualised on `exper` and `tenure`, and the slope of the resulting scatter is exactly the MLR slope on `educ`. The Frisch–Waugh–Lovell theorem (Exercise 3.6) is the algebraic statement that these two slopes coincide. The MLR slope is *not* a different estimator from OLS on a transformed scatter — it *is* OLS on a transformed scatter.

⚠ Common mistake: forgetting *ceteris paribus*

A widespread error is to read $\hat{\beta}_1$ as “the effect on y of changing x_1 ” without the conditioning clause. In MLR every slope is conditional on the other regressors in the equation. If you change which controls you include, you change what β_1 means — not just its numerical value.

3.4. Matrix notation

If matrices and the transpose are rusty, pause here and skim Appendix A.5 (three pages) before continuing.

For a sample of n observations it is far cleaner to stack the model into matrix form. Define

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}_{n \times (k+1)}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{(k+1) \times 1}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}_{n \times 1}.$$

The MLR model for the entire sample is then

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}.$$

The first column of $\mathbf{X}$ is a column of ones, which absorbs the intercept β_0 . From now on we will write proofs in matrix form; this is the standard notation in (Wooldridge 2020) and pays dividends from Chapter 4 onwards.

3.5. OLS in matrix form

The ordinary least squares (OLS) estimator chooses $\hat{\boldsymbol{\beta}}$ to minimise the sum of squared residuals

$$\text{SSR}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}.$$

Differentiating with respect to $\boldsymbol{\beta}$ and setting the result to zero gives the **normal equations**

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}.$$

If $\mathbf{X}'\mathbf{X}$ is invertible — which is exactly Assumption MLR.3 below — the unique solution is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

The fitted values are $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ and the residuals are $\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}}$. By construction the normal equations imply $\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$: residuals are orthogonal to every regressor, including the constant. In particular the sample mean of $\hat{\mathbf{u}}$ is zero whenever the model contains an intercept.

 Common mistake: confusing u with $\hat{u}$

As in Chapter 1, u is a *population, unobservable* object. The residual $\hat{u}_i = y_i - \hat{y}_i$ is the *sample* counterpart, produced *after* estimation. We will use $\hat{\mathbf{u}}$ to compute variance estimates and R^2 , but we will never observe $\mathbf{u}$ itself.

3.6. Gauss–Markov assumptions for MLR

We now collect the assumptions under which OLS has its classical properties. They generalise the simple-regression assumptions of Chapter 2.

i MLR.1 — Linearity in parameters

The population model is

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u.$$

Linearity is required *in the parameters* β_j , not in the regressors: x_j^2 or $\ln(x_j)$ are perfectly fine.

i MLR.2 — Random sampling

We observe an i.i.d. random sample $\{(y_i, x_{i1}, \dots, x_{ik}) : i = 1, \dots, n\}$ drawn from the population described by MLR.1.

i MLR.3 — No perfect collinearity

In the sample (and in the population), no regressor is constant, and there is no exact linear relationship among the regressors. Equivalently, $\mathbf{X}'\mathbf{X}$ is invertible.

i MLR.4 — Zero conditional mean

The error has mean zero given any value of the regressors:

$$\mathbb{E}[u \mid x_1, \dots, x_k] = 0, \quad \text{or, in matrix form,} \quad \mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}.$$

This is the assumption that delivers **unbiasedness** of OLS. It fails whenever a relevant variable is omitted (§3.9), the wrong functional form is used, or there is measurement error or simultaneity.

i MLR.5 — Homoskedasticity

The conditional variance of u does not depend on $\mathbf{X}$:

$$\text{Var}(u \mid x_1, \dots, x_k) = \sigma^2, \quad \text{Var}(\mathbf{u} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n.$$

Assumptions MLR.1–MLR.4 deliver **unbiasedness** (next section); add MLR.5 and OLS becomes the **best linear unbiased estimator** (BLUE) — the Gauss–Markov theorem.

⚠ Common mistake: confusing “perfect” with “high” collinearity

MLR.3 only rules out *exact* linear dependence among the regressors. Two highly correlated — but not perfectly correlated — variables (say **educ** and **IQ**) still satisfy MLR.3. They will inflate the variance of OLS estimates — a large R_j^2 (close to 1) makes $1 - R_j^2$ small, which makes the VIF $1/(1 - R_j^2)$ large — but they do not bias them and they do not break the model.

3.7. Unbiasedness of OLS (matrix form)

i Theorem: unbiasedness of OLS

Under MLR.1–MLR.4,

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta.$$

Proof sketch. Starting from the OLS solution,

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{u}) = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

Taking conditional expectations and using MLR.4 ($\mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}$),

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\mathbf{u} \mid \mathbf{X}] = \beta.$$

By the law of iterated expectations, $\mathbb{E}[\hat{\beta}] = \mathbb{E}[\mathbb{E}[\hat{\beta} \mid \mathbf{X}]] = \mathbb{E}[\beta] = \beta$ unconditionally as well — averaging the conditional-on- $\mathbf{X}$ statement over the marginal distribution of $\mathbf{X}$ does not undo it, because the conditional expectation is the constant β . $\square$

The proof makes the role of MLR.4 transparent: any correlation between $\mathbf{X}$ and $\mathbf{u}$ enters the bias term $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\mathbf{u} \mid \mathbf{X}]$. This is the matrix-form home of the omitted-variable bias formula we derive in §3.9.

3.8. Variance of OLS estimators (matrix form)

i Theorem: variance of OLS

Under MLR.1–MLR.5,

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

Proof sketch. From the proof of unbiasedness, $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}$. Hence

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{Var}(\mathbf{u} \mid \mathbf{X}) \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I}_n) \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

The diagonal entries give the variances of the individual coefficients; the off-diagonal entries give their covariances. $\square$

The variance of an individual slope $\hat{\beta}_j$ admits a useful univariate decomposition,

$$\text{Var}(\hat{\beta}_j \mid \mathbf{X}) = \frac{\sigma^2}{\text{SST}_j(1 - R_j^2)},$$

where $\text{SST}_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$ and R_j^2 is the R^2 from the auxiliary regression of x_j on **all the other regressors**. Three intuitions follow:

- More error variance (σ^2 large) $\Rightarrow$ less precise estimates.
- More sample variation in x_j (SST_j large) $\Rightarrow$ more precise estimates.
- More collinearity between x_j and the other regressors (R_j^2 close to 1) $\Rightarrow$ less precise estimates. The factor $1/(1 - R_j^2)$ is the **variance inflation factor** (VIF).

Since σ^2 is unknown, we estimate it with the residual variance,

$$\hat{\sigma}^2 = \frac{\text{SSR}}{n - k - 1} = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n - k - 1},$$

which is unbiased under MLR.1–MLR.5. The denominator $n - k - 1$ is the degrees of freedom: sample size minus number of estimated coefficients (including the intercept).

3.9. R^2 and adjusted R^2

As in simple regression, total variation in y decomposes into explained and residual variation,

$$\text{SST} = \text{SSE} + \text{SSR}, \quad R^2 = 1 - \frac{\text{SSR}}{\text{SST}} \in [0, 1].$$

There is, however, a subtlety unique to MLR: R^2 **never decreases when a regressor is added** to the model, even if that regressor is purely noise. Each new variable provides at least as much fit as no variable at all, because OLS could always set the new coefficient to zero. As a result, R^2 alone cannot adjudicate between nested specifications.

The **adjusted R^2** penalises model complexity:

$$\bar{R}^2 = 1 - \frac{\text{SSR}/(n - k - 1)}{\text{SST}/(n - 1)} = 1 - \frac{n - 1}{n - k - 1} (1 - R^2).$$

Unlike R^2 , $\bar{R}^2$ can *fall* when an irrelevant variable is added — the cost of the extra parameter exceeds the gain in fit. It is therefore a more honest summary when comparing models that differ in the number of regressors.

Common mistake: chasing R^2

A high R^2 does not mean the model is causal, nor that it predicts well out of sample. A regression of `educ` on `educ_squared` and `educ_cubed` will have an R^2 of essentially 1 and no economic content. Treat R^2 as a goodness-of-fit *summary*, not as a model-selection criterion. The AIC and BIC, introduced in Chapter 5, are better suited for the latter task.

3.10. Omitted variable bias

Omitted variable bias (OVB) is the formal name for the problem motivated in §3.1. Suppose the **true** population model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \quad \mathbb{E}[u \mid x_1, x_2] = 0,$$

but we estimate the **short** model

$$y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{u}$$

by OLS, omitting x_2 . What does $\tilde{\beta}_1$ estimate?

Theorem: omitted variable bias

Under the true model above and an OLS regression of y on x_1 alone,

$$\mathbb{E}[\tilde{\beta}_1 \mid \mathbf{x}_1] = \beta_1 + \beta_2 \tilde{\delta}_1,$$

where $\tilde{\delta}_1$ is the slope coefficient from the **auxiliary regression** of the omitted variable x_2 on the included variable x_1 :

$$x_2 = \tilde{\delta}_0 + \tilde{\delta}_1 x_1 + v.$$

Proof sketch. From the simple-regression formula,

$$\tilde{\beta}_1 = \frac{\sum_i (x_{1i} - \bar{x}_1) y_i}{\sum_i (x_{1i} - \bar{x}_1)^2}.$$

3. The Multiple Linear Regression Model

Plugging in the true model $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$ and using $\sum_i (x_{1i} - \bar{x}_1) = 0$,

$$\tilde{\beta}_1 = \beta_1 + \beta_2 \underbrace{\frac{\sum_i (x_{1i} - \bar{x}_1) x_{2i}}{\sum_i (x_{1i} - \bar{x}_1)^2}}_{=\tilde{\delta}_1} + \frac{\sum_i (x_{1i} - \bar{x}_1) u_i}{\sum_i (x_{1i} - \bar{x}_1)^2}.$$

The last term has conditional expectation zero given $\mathbf{x}_1$: MLR.4 of the *true* model gives $\mathbb{E}[u_i | x_{1i}, x_{2i}] = 0$, and the law of iterated expectations then collapses the conditioning set, $\mathbb{E}[u_i | x_{1i}] = \mathbb{E}[\mathbb{E}[u_i | x_{1i}, x_{2i}] | x_{1i}] = \mathbb{E}[0 | x_{1i}] = 0$. Substituting back leaves $\mathbb{E}[\tilde{\beta}_1 | \mathbf{x}_1] = \beta_1 + \beta_2 \tilde{\delta}_1$. $\square$

The **bias** is the product $\beta_2 \tilde{\delta}_1$. It is zero in either of two cases:

- $\beta_2 = 0$ — the omitted variable does not affect y in the true model; or
- $\tilde{\delta}_1 = 0$ — the omitted variable is uncorrelated with the included regressor.

In every other case, dropping a relevant correlated variable contaminates $\tilde{\beta}_1$.

i Extension: more than one omitted variable

The same logic extends to any number of omitted regressors. Suppose the true model has k included regressors and m omitted ones, $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \gamma_1 z_1 + \dots + \gamma_m z_m + u$, and we estimate the short regression on $(x_1, \dots, x_k)$ alone. The bias on a particular slope $\tilde{\beta}_j$ is then

$$\mathbb{E}[\tilde{\beta}_j | \mathbf{X}] - \beta_j = \sum_{\ell=1}^m \gamma_\ell \tilde{\delta}_{j\ell},$$

where $\tilde{\delta}_{j\ell}$ is the slope on x_j in the auxiliary regression of the ℓ -th omitted variable z_ℓ on **all** the included regressors. In words: each omitted variable contributes its own $\gamma_\ell \tilde{\delta}_{j\ell}$ term, and the total bias is the sum. The §3.10.2 lab applies this multivariable extension to the two omitted regressors **exper** and **tenure** simultaneously.

3.10.1. Sign of the bias: the 2×2 table

The sign of the bias is determined by the signs of β_2 and $\tilde{\delta}_1$:

	$\tilde{\delta}_1 > 0$	$\tilde{\delta}_1 < 0$
$\beta_2 > 0$	Positive bias	Negative bias
$\beta_2 < 0$	Negative bias	Positive bias

i Example: omitting ability from a wage equation

Suppose the true wage equation is $\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{ability} + u$.

- More able workers earn more, holding education fixed: $\beta_2 > 0$.
- More able workers acquire more education: $\tilde{\delta}_1 > 0$.
- The bias $\beta_2 \tilde{\delta}_1 > 0$: a simple regression of wage on education **overestimates** the causal return to schooling.

This is the canonical “ability bias” of the labour-economics literature. A naive OLS slope of, say, 0.60 dollars per year of schooling can perfectly well coexist with a true return of

0.40 once unobserved ability is netted out.

⚠ Including an irrelevant variable is not free

Adding a variable whose true coefficient is zero does **not** create bias (the OVB formula gives $\text{bias} = 0 \cdot \tilde{\delta}_1 = 0$). But it inflates the variance of every other slope through the R_j^2 term in §3.7. The empirical rule of thumb is: include controls justified by theory or by your identification strategy, not regressors that happen to “look significant”.

3.11. Lab: multiple regression

We work with `wage1` again — a cross-section of 526 U.S. workers from May 1976, shipped with the `wooldridge` package. The lab has three parts: nested wage regressions, an OVB demonstration, and a perfect-multicollinearity warning.

```
library(wooldridge)
data("wage1")
```

3.11.1. From simple to multiple regression

We first re-estimate the simple regression of `wage` on `educ` and then add `exper` and `tenure`:

```
model1 <- lm(wage ~ educ, data = wage1)
model2 <- lm(wage ~ educ + exper + tenure, data = wage1)
summary(model1)
```

Call:

```
lm(formula = wage ~ educ, data = wage1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.3396	-2.1501	-0.9674	1.1921	16.6085

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.90485	0.68497	-1.321	0.187
educ	0.54136	0.05325	10.167	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.378 on 524 degrees of freedom

Multiple R-squared: 0.1648, Adjusted R-squared: 0.1632

F-statistic: 103.4 on 1 and 524 DF, p-value: < 2.2e-16

3. The Multiple Linear Regression Model

```
summary(model2)
```

Call:

```
lm(formula = wage ~ educ + exper + tenure, data = wage1)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-7.6068 -1.7747 -0.6279  1.1969 14.6536
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.87273     0.72896  -3.941 9.22e-05 ***
educ         0.59897     0.05128  11.679 < 2e-16 ***
exper        0.02234     0.01206   1.853  0.0645 .
tenure       0.16927     0.02164   7.820 2.93e-14 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 3.084 on 522 degrees of freedom

Multiple R-squared: 0.3064, Adjusted R-squared: 0.3024

F-statistic: 76.87 on 3 and 522 DF, p-value: < 2.2e-16

i Reading `summary(model2)` — now in MLR form

The column structure is the same as in §2.11, but each cell now reads as a *partial* quantity. The mapping back to the matrix-form derivations of §3.4–§3.7:

- **Estimate** row $j \rightarrow$ the j -th entry of $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ — the partial effect of regressor j holding the other regressors fixed.
- **Std. Error** row $j \rightarrow \sqrt{\hat{\sigma}^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$, i.e. the square root of the j -th diagonal entry of the matrix variance $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$ from §3.7. Equivalently, $\hat{\sigma}/\sqrt{\text{SST}_j(1 - R_j^2)}$ — the univariate decomposition that exposes the variance inflation factor.
- **t value**, $\text{Pr}(>|t|)$ $\rightarrow$ as in Ch 2, formally meaningful from Chapter 4 onwards.
- **Residual standard error** $\rightarrow \hat{\sigma} = \sqrt{\text{SSR}/(n - k - 1)}$ with $n - k - 1$ degrees of freedom, the $k = 3$ here.
- **Multiple R-squared** $\rightarrow R^2$ from §3.8.
- **Adjusted R-squared** $\rightarrow \bar{R}^2$ from §3.8 — this is the model-selection-friendly companion that penalises adding regressors.
- **Bottom-line F-statistic** $\rightarrow$ tests the joint null $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (the *overall* significance of the regression), built from §3.8's R^2 in the form that Chapter 4 §4.8 will derive.

The point: every number in the output is one of the population objects we wrote down in §3.4–§3.8, computed from the sample.

Compare the coefficient on `educ` across the two specifications. In `model11`, an additional year of education is associated with roughly 0.54 USD/hour more; in `model12`, the same coefficient *rises* to about 0.59 once experience and tenure are held fixed. Either way, $\hat{\beta}_1$ in `model12` is the *ceteris paribus* return to education that `model11` failed to deliver.

Now add demographic controls:

```
model3 <- lm(wage ~ educ + exper + tenure + female + married, data = wage1)
summary(model3)
```

Call:

```
lm(formula = wage ~ educ + exper + tenure + female + married,
    data = wage1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.731	-1.816	-0.500	1.050	13.928

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.61815	0.72305	-2.238	0.0256 *
educ	0.55568	0.04986	11.144	< 2e-16 ***
exper	0.01874	0.01203	1.558	0.1198
tenure	0.13878	0.02114	6.566	1.25e-10 ***
female	-1.74140	0.26649	-6.535	1.52e-10 ***
married	0.55924	0.28595	1.956	0.0510 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.95 on 520 degrees of freedom

Multiple R-squared: 0.3682, Adjusted R-squared: 0.3621

F-statistic: 60.61 on 5 and 520 DF, p-value: < 2.2e-16

The coefficient on `female` is the conditional gender wage gap holding education, experience, tenure, and marital status fixed; the coefficient on `married` is the conditional marriage premium. Both are smaller in magnitude than the raw conditional means of Chapter 1 — confounders have been partly accounted for.

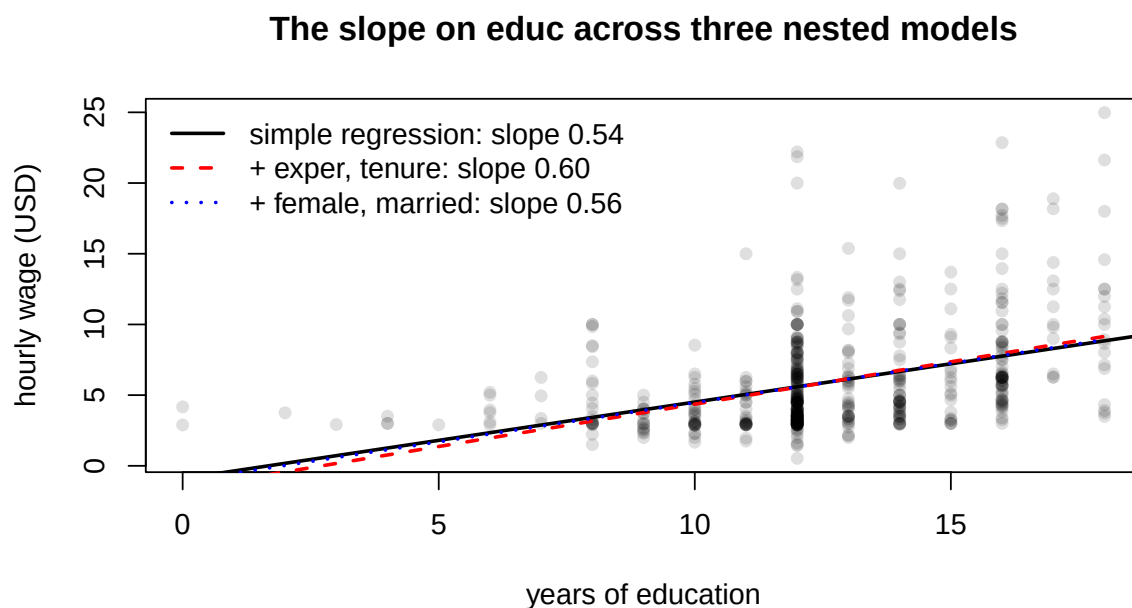


Figure 3.2.: The estimated relationship between `wage` and `educ` under three nested specifications. The black line is the simple regression. The red dashed and blue dotted lines are the multiple-regression slopes (with controls fixed at sample means) for `model2` and `model3` respectively. Adding controls *changes the slope on educ* — this is the whole reason MLR exists. The shift from black to red is the magnitude of the OVB on `educ` from omitting experience and tenure.

Compare goodness of fit across the three nested models:

```
data.frame(
  model = c("educ only", "educ+exper+tenure", "full"),
  r2     = c(summary(model1)$r.squared,
             summary(model2)$r.squared,
             summary(model3)$r.squared),
  adj_r2 = c(summary(model1)$adj.r.squared,
             summary(model2)$adj.r.squared,
             summary(model3)$adj.r.squared)
)
```

	model	r2	adj_r2
1	educ only	0.1647575	0.1631635
2	educ+exper+tenure	0.3064224	0.3024364
3	full	0.3681887	0.3621136

Both R^2 and adjusted R^2 rise as controls are added — here every added regressor is justified by labour-economics theory, so this should not surprise us.

3.11.2. Omitted variable bias on real data

Let us verify the OVB formula numerically. The “short” regression of `wage` on `educ` alone omits two regressors that we now know are correlated with education: `exper` and `tenure`. Run the

two auxiliary regressions:

```
aux_exper <- lm(exper ~ educ, data = wage1)
aux_tenure <- lm(tenure ~ educ, data = wage1)
delta_exper <- coef(aux_exper)["educ"]
delta_tenure <- coef(aux_tenure)["educ"]
c(delta_exper = delta_exper, delta_tenure = delta_tenure)
```

```
delta_exper.educ delta_tenure.educ
-1.4681823      -0.1465559
```

The slope of **exper** on **educ** is **negative**: more-schooled workers in this 1976 cross-section have less labour-market experience (this is mostly mechanical — time spent in school is time not spent working). The slope of **tenure** on **educ** is slightly positive in this sample.

Now plug the auxiliary slopes and the *long-regression* coefficients into the OVB formula and compare with the actual gap between the short and long estimates:

```
b1_short <- coef(model1)["educ"]
b1_long <- coef(model2)["educ"]
b2_long <- coef(model2)["exper"]
b3_long <- coef(model2)["tenure"]

bias_predicted <- b2_long * delta_exper + b3_long * delta_tenure
bias_observed <- b1_short - b1_long

c(predicted_bias = bias_predicted,
  observed_gap = bias_observed)
```

```
predicted_bias.exper    observed_gap.educ
-0.05760581           -0.05760581
```

The two numbers should coincide up to floating-point precision — this is the multivariable OVB extension stated in §3.9 (callout *Extension: more than one omitted variable*) at work, applied here to the two omitted regressors **exper** and **tenure** simultaneously. The simple-regression slope on **educ** differs from the multiple-regression slope by exactly $\hat{\beta}_{\text{exper}}\tilde{\delta}_{\text{exper}} + \hat{\beta}_{\text{tenure}}\tilde{\delta}_{\text{tenure}}$.

What the numerical equality actually demonstrates is the **Frisch–Waugh–Lovell** sample identity — the sample slopes from short and long regressions are related by exactly the same algebraic combination as the population OVB formula, with $(\hat{\beta}, \tilde{\delta})$ in place of (β, δ) . The population OVB statement of §3.9 is the *expected-value* version of this sample identity; Exercise 3.6 asks you to derive the underlying FWL theorem.

i Why the bias is *negative* here

Both **exper** and **tenure** have positive returns ($\hat{\beta}_2, \hat{\beta}_3 > 0$). But **exper** is *negatively* correlated with **educ** ($\tilde{\delta}_{\text{exper}} < 0$), and this dominates. The product $\hat{\beta}_2\tilde{\delta}_{\text{exper}}$ is negative; omitting experience therefore drags the short-regression slope on **educ** *downward* relative to the long regression. The 2\$×\$2 sign table above captures this exactly.

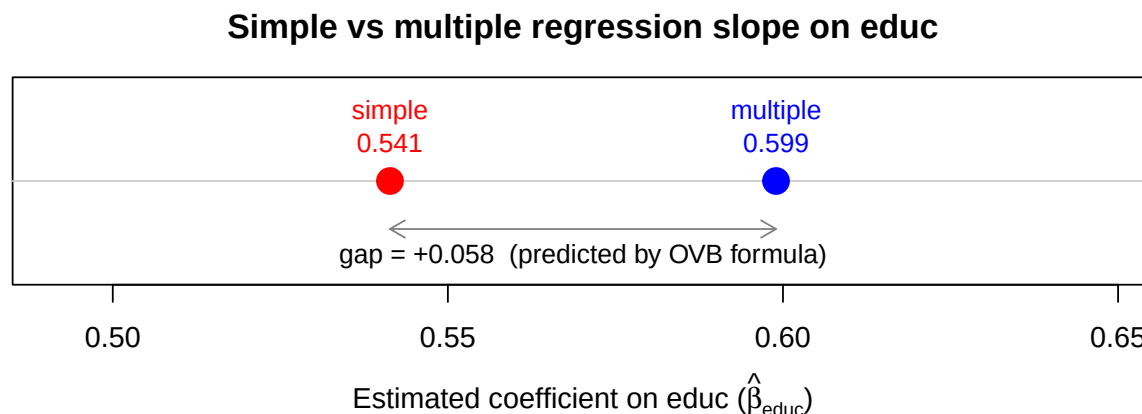


Figure 3.3.: OVB visualised on a number line. The simple-regression slope sits to the *left* of the multiple-regression slope: omitting `exper` and `tenure` from a wage regression drags the `educ` coefficient downward in this sample. The gap between the two markers is exactly $\hat{\beta}_{exper}\tilde{\delta}_{exper} + \hat{\beta}_{tenure}\tilde{\delta}_{tenure}$, the multivariable OVB formula at work.

3.11.3. Perfect multicollinearity

Now a deliberate violation of MLR.3. Suppose we measure education in years (`educ`) and additionally create a redundant copy in months (`educ_months = 12 * educ`):

```
wage1$educ_months <- 12 * wage1$educ
model_collinear <- lm(wage ~ educ + exper + tenure + educ_months, data = wage1)
summary(model_collinear)
```

Call:

```
lm(formula = wage ~ educ + exper + tenure + educ_months, data = wage1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.6068	-1.7747	-0.6279	1.1969	14.6536

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.87273	0.72896	-3.941	9.22e-05 ***
educ	0.59897	0.05128	11.679	< 2e-16 ***
exper	0.02234	0.01206	1.853	0.0645 .
tenure	0.16927	0.02164	7.820	2.93e-14 ***
educ_months	NA	NA	NA	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.084 on 522 degrees of freedom

Multiple R-squared: 0.3064, Adjusted R-squared: 0.3024

F-statistic: 76.87 on 3 and 522 DF, p-value: < 2.2e-16

R does not crash — but it does *silently drop* `educ_months` (its coefficient is reported as NA). The reason is exactly MLR.3: `educ_months` is an exact linear function of `educ`, so the design matrix $\mathbf{X}$ is rank-deficient, $\mathbf{X}'\mathbf{X}$ is singular, and $(\mathbf{X}'\mathbf{X})^{-1}$ does not exist. We cannot tell whether a one-unit change in `educ` or a 12-unit change in `educ_months` is “doing the work” — the data simply does not contain that information.

 Common mistake: the “168 minus hours” trap

A subtler version of the same problem: in a survey that records weekly hours worked (`hours`), one might be tempted to add a variable `remaining = 168 - hours` (“hours not worked in a week”, since $168 = 24 \times \$7$). The result is a perfect linear relationship between `hours`, `remaining`, and the constant: `hours + remaining = 168`. Any regression including all three will be rank-deficient. The rule of thumb: if you can write one regressor as a deterministic function of the others (and the constant), you are in violation of MLR.3.

Self-check

Six short multiple-choice questions. Try each one before opening the answer.

 Q1. Interpretation of an MLR slope

In the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the coefficient β_1 measures:

- A. The total change in y when x_1 goes up by one unit, including any indirect effect through x_2 .
- B. The expected change in y for a one-unit increase in x_1 , holding x_2 fixed.
- C. The correlation between y and x_1 .
- D. The change in y when both x_1 and x_2 rise by one unit.

Answer: B. Every MLR slope is a partial effect: it isolates the variation in y associated with x_1 once we hold the other regressors in the model constant.

 Q2. The zero-conditional-mean assumption

MLR.4 requires:

- A. $\text{Cov}(x_1, u) = 0$ only.
- B. $\text{Var}(u \mid \mathbf{X}) = \sigma^2$.
- C. $\mathbb{E}[u \mid x_1, x_2, \dots, x_k] = 0$.
- D. u to be normally distributed.

Answer: C. MLR.4 is the conditional-mean-zero condition on *all* regressors simultaneously. It is what delivers unbiasedness of OLS.

 Q3. Sign of OVB

In a wage regression we omit innate ability. Assume (i) more able workers earn more, holding education fixed, and (ii) more able workers acquire more education. The OLS coefficient on `educ` will be:

3. The Multiple Linear Regression Model

- A. Biased downward.
- B. Unbiased — OLS is always unbiased.
- C. Biased upward (overestimating the return to education).
- D. Of indeterminate sign.

Answer: C. With $\beta_{\text{ability}} > 0$ and $\tilde{\delta}_{\text{educ}} > 0$, the bias $\beta_2 \tilde{\delta}_1$ is positive: the short regression overstates the return to schooling.

💡 Q4. Which assumption rules out perfect collinearity?

- A. MLR.1 (linearity in parameters).
- B. MLR.3 (no exact linear relationship among the regressors).
- C. MLR.4 (zero conditional mean).
- D. MLR.5 (homoskedasticity).

Answer: B. MLR.3 is the assumption that $\mathbf{X}'\mathbf{X}$ is invertible. Violating it makes the OLS formula undefined; R will drop one of the offending variables and report NA.

💡 Q5. R^2 versus adjusted R^2

Which statement is true when an irrelevant regressor is added to a model?

- A. Both R^2 and adjusted R^2 always rise.
- B. R^2 always rises (weakly), but adjusted R^2 may fall.
- C. R^2 may fall, but adjusted R^2 always rises.
- D. Both always fall.

Answer: B. R^2 cannot decrease because OLS could always set the new coefficient to zero. Adjusted R^2 subtracts a penalty for parameters; if the new variable does not improve fit enough, the penalty wins.

💡 Q6. When is OVB zero?

OVB on the slope of x_1 is exactly zero when:

- A. The sample size is large.
- B. R^2 exceeds 0.5.
- C. Either the omitted variable does not affect y , or it is uncorrelated with x_1 .
- D. The errors are normal.

Answer: C. The OVB formula $\beta_2 \tilde{\delta}_1$ is zero whenever one of the two factors is zero. Large samples or high R^2 do not fix omitted-variable bias.

Exercises

Exercise 3.1 — Interpreting a fitted MLR. A researcher reports

$$\widehat{\text{wage}} = -2.87 + 0.599 \text{educ} + 0.022 \text{exper} + 0.169 \text{tenure}$$

estimated on `wage1`. (a) Interpret each slope coefficient in plain English. (b) Predict the hourly wage of a worker with 16 years of education, 5 years of experience, and 3 years of tenure. (c) Why should you *not* read $\hat{\beta}_1 = 0.599$ as the causal return to schooling?

💡 Show answer

- (a) Each slope is a partial effect: one additional year of education is associated with USD 0.60 more per hour, *holding experience and tenure fixed*; analogously for `exper` and `tenure`. (b) $\widehat{\text{wage}} = -2.87 + 0.599 \cdot 16 + 0.022 \cdot 5 + 0.169 \cdot 3 \approx 7.32$ USD/hour. (c) Because MLR.4 is unlikely to hold here — ability, family background, and school quality remain in u and are correlated with `educ`. The slope is the best within-model linear summary, not a causal coefficient.

Exercise 3.2 — Spotting perfect collinearity. For each of the following regressor sets, decide whether MLR.3 is violated. If so, identify the exact linear dependence.

- (a) `{const, age, age_squared}`.
 (b) `{const, female, male}`, where `male = 1 - female`.
 (c) `{const, exper, tenure, exper + tenure}`.
 (d) `{const, educ, log(educ)}` — assume `educ > 0`.

💡 Show answer

- (a) No violation: `age2` is not a *linear* function of `age`. (b) Violation: `female + male = 1`, the constant column. (c) Violation: the fourth regressor is the sum of the second and third. (d) No violation: `log` is nonlinear, so MLR.3 holds (though high collinearity may inflate variances).

Exercise 3.3 — Sign of OVB from theory. You want to estimate the effect of class attendance on final grades but you omit *student effort*. Suppose effort raises grades and is positively correlated with attendance. Predict the sign of the bias on the coefficient of attendance in a short regression that omits effort, and explain in one sentence.

💡 Show answer

With $\beta_{\text{effort}} > 0$ and $\tilde{\delta}_{\text{att}} > 0$, the bias $\beta_2 \tilde{\delta}_1$ is **positive**: the short regression overstates the return to attendance because part of the apparent effect is in fact the effect of the (unobserved) effort that correlates with attendance.

Exercise 3.4 — Adjusted R^2 falling. Construct a small numerical example (or argue analytically) showing that adding an irrelevant regressor can leave R^2 essentially unchanged while making adjusted R^2 smaller. Use the identity $\bar{R}^2 = 1 - (n-1)(1-R^2)/(n-k-1)$ to explain why.

💡 Show answer

With $n = 100$, suppose $R^2 = 0.30$ with $k = 3$ regressors: $\bar{R}^2 = 1 - 99 \cdot 0.70/96 \approx 0.278$. Add an irrelevant 4th regressor that lifts R^2 to 0.301: $\bar{R}^2 = 1 - 99 \cdot 0.699/95 \approx 0.272$. R^2 went up by 0.001 but $\bar{R}^2$ fell by 0.006 because $(n-1)/(n-k-1)$ grew faster than $1-R^2$ shrank. This is the formal sense in which $\bar{R}^2$ penalises complexity.

3. The Multiple Linear Regression Model

Exercise 3.5 — OVB by simulation. Generate $n = 1000$ observations from the model $y = 1 + 2x_1 + 3x_2 + u$, where $x_1 \sim \mathcal{N}(0, 1)$, $x_2 = 0.5x_1 + 0.5z$, $z \sim \mathcal{N}(0, 1)$ independent of x_1 , and $u \sim \mathcal{N}(0, 1)$ independent of everything else. (a) Estimate the short regression y on x_1 alone and the long regression y on (x_1, x_2) . (b) Use the OVB formula to predict the gap between the two slopes on x_1 and verify it numerically. Hint: $\tilde{\delta}_1 = 0.5$ by construction.

A full answer is given in the Instructor Edition.

Exercise 3.6 — Frisch–Waugh–Lovell partialling out. Let $\tilde{x}_1$ denote the residuals from regressing x_1 on the remaining regressors $(x_2, \dots, x_k)$ and the constant. Show, using the OLS normal equations, that the multiple-regression coefficient on x_1 equals the slope from a simple regression of y on $\tilde{x}_1$. Verify your derivation numerically using `wage1`: regress `educ` on `exper` and `tenure`, keep the residuals, and check that regressing `wage` on those residuals reproduces $\hat{\beta}_{\text{educ}}$ from the long regression of §3.10.

A full answer is given in the Instructor Edition.

Exercise 3.7 — The return to bwght. Chapter 2 §2.11 estimated a simple regression of birth weight on maternal smoking and warned that the slope on `cigs` might be contaminated by omitted variables (maternal income, education, nutrition). Use the multiple-regression toolkit to address that warning.

- (a) Using `bwght` from the `wooldridge` package, estimate

$$\text{bwght} = \beta_0 + \beta_1 \text{cigs} + \beta_2 \text{motheduc} + \beta_3 \text{fatheduc} + \beta_4 \text{parity} + u.$$

- Report $\hat{\beta}_1$ and compare it with the simple-regression slope -0.51 from Ch 2 §2.11. Has it moved toward zero, away from zero, or stayed essentially the same?
- (b) Apply the multivariable OVB extension from §3.9. Run the three auxiliary regressions of `motheduc`, `fatheduc`, and `parity` on `cigs`, and use the resulting $\tilde{\delta}_j$ together with the long-regression coefficients $\hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4$ to predict the gap between the simple- and multiple-regression slopes on `cigs`. Verify that your predicted gap matches the observed gap numerically (this is the FWL identity at work, exactly as in the wage example of §3.10.2).
- (c) Did including the controls change your causal interpretation of $\hat{\beta}_1$ as the effect of maternal smoking on birth weight? Which assumption of MLR.1–MLR.4 are we still relying on, and which omitted variables remain to threaten it?

A full answer is given in the Instructor Edition.

Interactive self-assessment: `LEARNR/03_multiple_linear_regression.Rmd`.

Inference

Learning outcomes

By the end of this chapter the reader should be able to:

- State the Classical Linear Model assumptions (MLR.1–MLR.6) and explain what the normality assumption MLR.6 adds to Gauss–Markov.
- Derive the sampling distribution of $\hat{\beta}_j$ under the CLM and explain why standardisation produces a t_{n-k-1} statistic rather than a standard normal.
- Construct and interpret a $(1 - \alpha) \times 100\%$ confidence interval for an individual coefficient.
- Run a four-step hypothesis test (two-sided and one-sided) for a single coefficient, and read off the corresponding p -value.
- Carry out an F -test for joint significance, distinguishing the overall F -test from a restricted-vs-unrestricted comparison.
- Reproduce all of the above in R using `summary()`, `confint()`, `qt()`, `qf()`, `pt()`, `pf()`, `anova()` and manual arithmetic on the fitted model.

Motivating empirical question

*Once we hold education, experience and age fixed, do IQ scores and schooling **jointly** matter for monthly earnings — or could the apparent contribution of cognitive ability and education have arisen by chance?*

We met IQ briefly in Chapter 3 as a candidate proxy for unobserved ability. Now we have to decide, formally, whether the coefficients we estimate are far enough from zero to take seriously. The tools of the chapter — t -statistics, p -values, confidence intervals and the F -test — are designed precisely for that decision.

4.1. From estimation to inference

CHAPTERS 2 and 3 produced **point estimates** $\hat{\beta}_j$ from a sample. These numbers are random: a different sample would have given different numbers. The question of inference is whether what we learned from *our* sample lets us say anything credible about the unknown population parameter β_j .

Statistical inference provides two complementary tools (Wooldridge 2020):

1. **Confidence intervals.** A range of plausible values for the unknown parameter, with a calibrated long-run coverage probability.

4. Inference

2. **Hypothesis tests.** A formal procedure for deciding whether the data are compatible with a specific claim about the parameter — most often, the claim that the parameter equals zero.

Both tools rely on knowing the *sampling distribution* of $\hat{\beta}_j$. Under Gauss–Markov we only know its mean (β_j) and variance; we do not know its shape. To say anything precise about probabilities — “the chance of observing a coefficient this far from zero, if the truth is zero, is less than 1%” — we need a distribution. That is what the next assumption gives us.

4.2. The Classical Linear Model (CLM) assumptions

Gauss–Markov gave us unbiasedness, a closed-form variance and the BLUE property of OLS. For exact (finite-sample) inference we need one more assumption.

i Definition: MLR.6 (Normality of the error term)

Conditional on the regressors $\mathbf{X}$, the population error u is normally distributed with mean zero and constant variance:

$$u \mid \mathbf{X} \sim \mathcal{N}(0, \sigma^2).$$

The full set MLR.1–MLR.6 is called the **Classical Linear Model (CLM) assumptions**.

Why normality, and where does it come from? In many applied settings u is the *additive* sum of a large number of small, independent omitted factors, and a central-limit heuristic suggests its distribution should be roughly bell-shaped. The qualifier “additive” matters: the heuristic does not survive if the omitted factors combine *multiplicatively*. The assumption is strong: it can fail badly for skewed outcomes such as wages or counts. We will see in Chapter 6 that taking log of the dependent variable is precisely the trick that converts multiplicative omitted factors into additive ones, which is one reason logs so often make the residuals look symmetric.

The pay-off is large. Under MLR.1–MLR.6, the OLS estimator inherits the normality of u :

$$\hat{\beta}_j \mid \mathbf{X} \sim \mathcal{N}(\beta_j, \text{Var}(\hat{\beta}_j)).$$

This is an *exact* statement that holds in any sample size, not just asymptotically. With n large, the central limit theorem delivers approximately the same conclusion even without MLR.6, but only the CLM gives us exact small-sample inference.

⚠ Common mistake: confusing the normality of u with the normality of y

MLR.6 is a statement about the *error* u , conditional on the regressors. It is *not* the claim that the marginal distribution of y is normal. Wages, for example, are right-skewed; that does not by itself contradict MLR.6, because the systematic component $\beta_0 + \beta_1 x_1 + \dots$ can absorb the skewness.

4.3. The t -distribution

Imagine for a moment that we *knew* the true error variance σ^2 . Then dividing $\hat{\beta}_j - \beta_j$ by its (known) standard deviation would give a clean standard-normal Z : one random quantity divided

by a fixed constant. In reality we do not know σ^2 ; we replace it with the sample estimate $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$. The crucial observation is that $\hat{\sigma}$ is itself a random number — different samples produce different $\hat{\sigma}$ values. So the standardised ratio is no longer a random quantity over a constant, but a *random quantity over another random quantity*, which is necessarily more variable than Z . The extra variability shows up as heavier tails: the resulting distribution is the **t -distribution**, narrower at the centre and fatter in the tails than the standard normal (see the figure below comparing t_{45} , t_5 , and $\mathcal{N}(0, 1)$).

Made formal: if σ^2 were known we would have

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{Var}(\hat{\beta}_j)}} \sim \mathcal{N}(0, 1).$$

In practice σ^2 is unknown, $\hat{\sigma}^2$ takes its place in the standard error, and the standardised statistic is no longer standard normal: it is **t -distributed** with $n - k - 1$ degrees of freedom,

$$t = \frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim t_{n-k-1}.$$

The t -distribution is symmetric, bell-shaped and centred at zero, but it has **heavier tails** than $\mathcal{N}(0, 1)$ — a reflection of the extra uncertainty from estimating σ^2 . As the degrees of freedom $n - k - 1$ grow, the tails thin out and t_{n-k-1} converges to $\mathcal{N}(0, 1)$. For $n - k - 1 \geq 120$ the two distributions are virtually indistinguishable; for small samples the t is noticeably wider.

That is the whole reason this chapter uses t critical values like 2.013 (for $\alpha = 0.05$, $df = 45$) rather than the familiar 1.96 from the normal table.

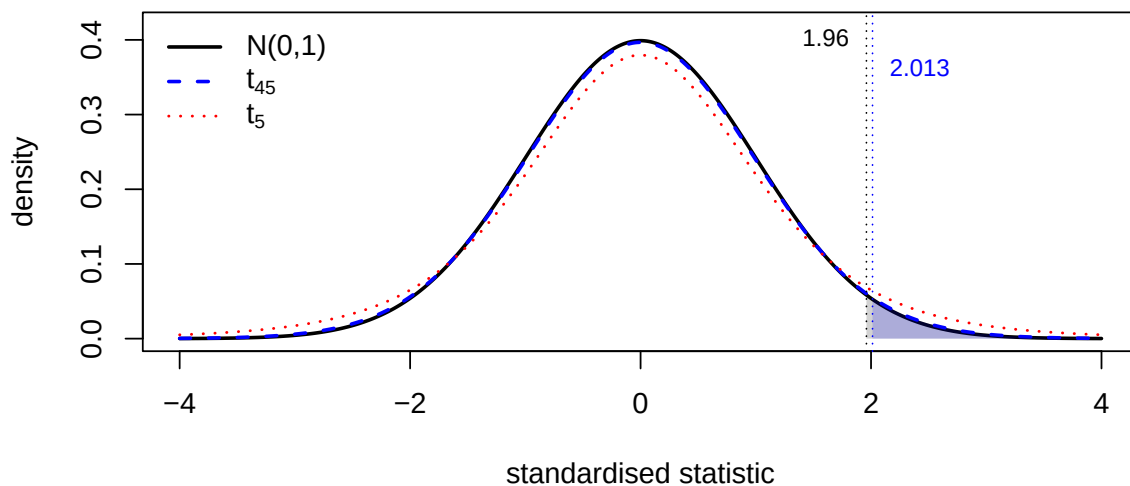


Figure 4.1.: Two-sided 5% critical values: $z_{0.025} = 1.96$ from $\mathcal{N}(0, 1)$ vs $t_{0.025, 45} = 2.013$ from the t -distribution. The t_{45} tail (blue shaded) lies slightly outside the $\mathcal{N}(0, 1)$ tail (grey shaded), so to keep the right-tail area at exactly 2.5% the t cutoff has to move outward. At $df = 5$ (red, dotted) the gap is dramatic — $t_{0.025, 5} = 2.571$.

4.4. Two ways to be wrong

A jury can convict an innocent defendant, or it can let a guilty one go free. Both mistakes are bad and the legal system is built to trade them off. A hypothesis test faces exactly the same two-by-two structure.

	H_0 is true	H_0 is false
Reject H_0	Type I error (size α)	Correct rejection (power $1 - \beta$)
Fail to reject H_0	Correct non-rejection	Type II error (probability β)

A **Type I error** is rejecting a null that is in fact true — the statistical equivalent of convicting an innocent defendant. The probability of making one, computed under H_0 , is called the **size** of the test:

$$\alpha = \Pr(\text{reject } H_0 \mid H_0 \text{ true}).$$

The conventional choices $\alpha = 0.10, 0.05, 0.01$ are precisely the Type I error rates we are willing to tolerate. Notice the **conditional**: α is *not* the probability that the null is true given the data, nor the probability that we will reject in any given study. It is the long-run frequency of false rejections *across studies where the null happens to be true*.

A **Type II error** is failing to reject a null that is in fact false — the equivalent of letting a guilty defendant go free. Its probability is denoted β . The complement, $1 - \beta$, is the **power** of the test: the probability of detecting a real effect when one exists. Unlike α , power depends on three things: how large the true effect is, how noisy the data are, and how many observations we have. Underpowered studies miss real effects routinely; this is why a “non-significant” finding in a small sample is rarely informative.

There is no free lunch. Holding n and the data-generating process fixed, lowering α (a stricter test) raises β (more missed real effects), and vice versa. The only way to reduce *both* error probabilities simultaneously is to gather more data — which makes the sampling distribution under H_0 tighter and shrinks both tails.

⚠ Common mistake: confusing α with p , and “size” with “significance”

The size α is set *before* the data are seen — it is the long-run Type I error rate we are willing to accept. The p -value is computed *after* the data are seen — it is the smallest α at which the data would lead us to reject. The two are distinct conceptual objects even though they share the same scale. Similarly, the “5% significance level” in a paper refers to a chosen α , not to a per-study probability of error: each individual study either rejected correctly or did not, and we will never know which.

We will use the vocabulary of “size” and “power” repeatedly below — particularly in §“ F -test for joint significance,” where the size of a *combined* procedure (testing several restrictions at once) is precisely what motivates the F -test in the first place.

4.5. Confidence intervals

i Definition: confidence interval for β_j

A $(1 - \alpha) \times 100\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{\alpha/2, n-k-1} \cdot \text{se}(\hat{\beta}_j),$$

where $t_{\alpha/2, n-k-1}$ is the critical value of the t -distribution that leaves probability $\alpha/2$ in each tail.

For the canonical 95% interval, $\alpha = 0.05$ and the critical value is $t_{0.025, n-k-1}$. With moderately large samples this is close to 1.96, but in small samples it can be appreciably larger.

! Common mistake: misreading the “95%”

A 95% confidence interval is a statement about the **procedure**, not about the realised interval. If we drew many independent samples and built one interval from each, about 95% of those intervals would contain the true β_j . It is **not** correct to say that there is a 95% probability that the particular interval $[L, U]$ we computed from our one sample contains β_j — β_j is a fixed (if unknown) number, and the realised interval either covers it or it does not.

Confidence intervals and two-sided hypothesis tests are two sides of the same coin: a $(1 - \alpha)$ CI for β_j contains zero **if and only if** the two-sided t -test of $H_0 : \beta_j = 0$ at level α fails to reject.

4.6. Hypothesis testing in four steps

We test claims about a single coefficient β_j using a standard four-step procedure. Let c be the hypothesised value (usually $c = 0$, “the variable does not matter”).

Step 1. State H_0 and H_1 . There are three useful forms:

- *Two-sided:* $H_0 : \beta_j = c$ versus $H_1 : \beta_j \neq c$.
- *Right one-sided:* $H_0 : \beta_j \leq c$ versus $H_1 : \beta_j > c$.
- *Left one-sided:* $H_0 : \beta_j \geq c$ versus $H_1 : \beta_j < c$.

The choice between one- and two-sided is driven by **economic theory**, not by the data: if theory tells us a priori that β_j cannot be negative, a one-sided test is appropriate.

Step 2. Choose a significance level α and read off the critical value. Standard choices are $\alpha = 0.10, 0.05, 0.01$. The critical value is

- $t_{\alpha/2, n-k-1}$ for a two-sided test,
- $t_{\alpha, n-k-1}$ for a one-sided test.

Step 3. Compute the t -statistic.

$$t = \frac{\hat{\beta}_j - c}{\text{se}(\hat{\beta}_j)}.$$

For the default $c = 0$ this reduces to $t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$, which is exactly the number R reports in the third column of `summary(lm(...))`.

4. Inference

Step 4. Compare and conclude.

- Two-sided: reject H_0 if $|t| > t_{\alpha/2, n-k-1}$.
- Right one-sided: reject H_0 if $t > t_{\alpha, n-k-1}$.
- Left one-sided: reject H_0 if $t < -t_{\alpha, n-k-1}$.

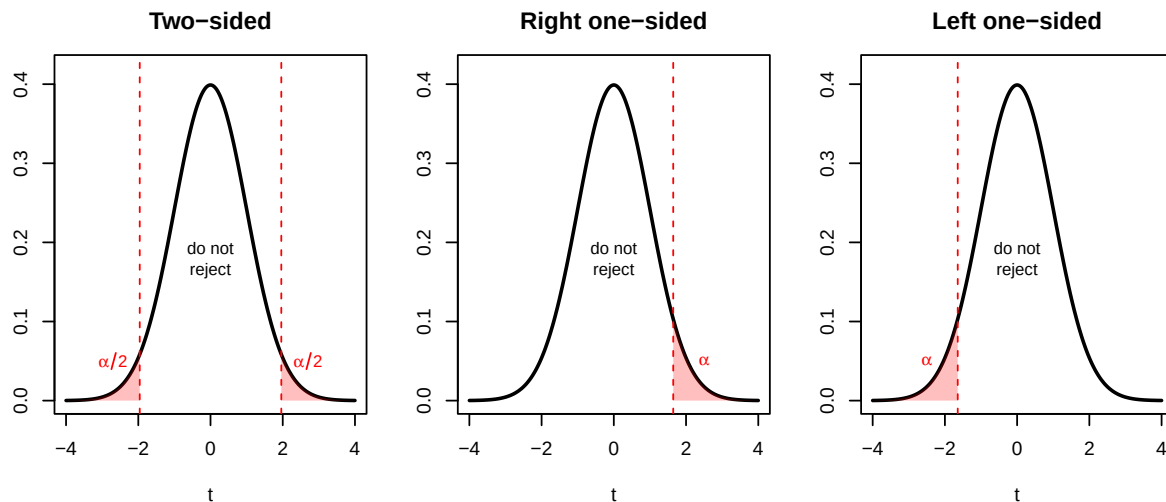


Figure 4.2.: Rejection regions of three t -tests at $\alpha = 0.05$ (using z -cutoffs from $N(0,1)$ for readability; for finite df the cutoffs move outward as shown in the previous figure). Pink-shaded tails are the rejection regions: each individual test has a total Type I error rate of α , but the *location* of that rejection mass depends on the alternative. Two-sided tests split α between the two tails; one-sided tests concentrate it in one.

If H_0 is rejected we say the coefficient is *statistically significant* at the chosen level. If we fail to reject we say the data are *consistent with* H_0 — we never “accept” H_0 , because the test was designed to detect departures from it, not to confirm it.

⚠ Common mistake: statistical significance is not economic significance

A coefficient can be statistically significant (small p -value) but economically tiny, and a coefficient can be economically large but statistically insignificant in a small sample. Always report both: the *magnitude* of $\hat{\beta}_j$ in the units of the problem, and the precision with which it is estimated. A 1% return to one extra year of education and a 10% return are both “significantly different from zero” in a large sample, but only one of them is policy-relevant.

4.7. The p -value

The four-step procedure forces us to fix α in advance. A more informative alternative is to report the p -value.

i Definition: p -value

The p -value is the probability, computed *under* H_0 , of observing a test statistic at least as extreme as the one we got. Equivalently, it is the smallest significance level α at which we would reject H_0 .

For a two-sided test of $H_0 : \beta_j = 0$ with computed statistic t ,

$$p = 2 \cdot \Pr(T_{n-k-1} > |t|) = 2[1 - F_t(|t|; n - k - 1)],$$

where F_t is the CDF of the t_{n-k-1} distribution. For a one-sided test the p -value is exactly half of this (when the sign of $\hat{\beta}_j$ agrees with the alternative).

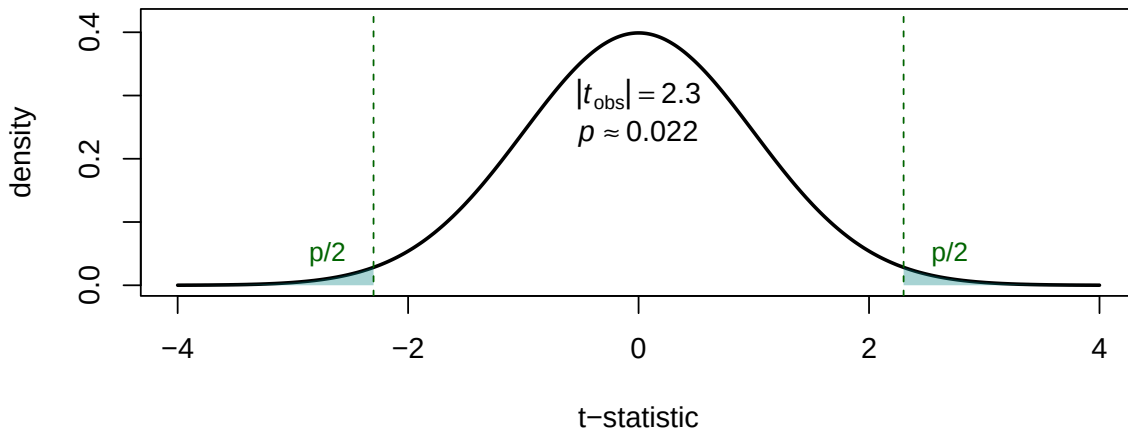


Figure 4.3.: The two-sided p -value as a tail area. The horizontal axis is the t -statistic; the curve is the t_{n-k-1} density under $H_0 : \beta_j = 0$. We observed $|t| = 2.3$. The p -value is the *total* shaded area: $p = 2 \cdot \Pr(T > 2.3) \approx 0.022$. If $p < \alpha$, the observation falls *outside* the do-not-reject zone of the previous figure; equivalently, the shaded area is smaller than the chosen size α .

Conventional rules of thumb (Wooldridge 2020):

- $p < 0.01$: strong evidence against H_0 (reject at the 1% level).
- $p < 0.05$: evidence against H_0 (reject at the 5% level).
- $p < 0.10$: weak evidence against H_0 (reject at the 10% level).
- $p \geq 0.10$: insufficient evidence to reject H_0 .

R reports two-sided p -values by default in the $\Pr(>|t|)$ column of `summary(lm(...))`, together with significance stars: ******* for $p < 0.001$, ****** for $p < 0.01$, ***** for $p < 0.05$, **.** for $p < 0.10$.

⚠ Common mistake: a small p -value is not a causal certificate

Statistical significance tells us that $\hat{\beta}_j$ is unlikely to have arisen by sampling noise if the truth were zero. It says **nothing** about whether x_j *causes* y . Causal interpretation still requires the population assumption $\mathbb{E}[u | \mathbf{X}] = 0$ (MLR.4) — no p -value, however small, can rescue a regression contaminated by omitted-variable bias or by reverse causality.

4.8. The F -test for joint significance

Recall the opening question of this chapter: do IQ and educ *jointly* matter for monthly earnings? This is precisely the kind of joint hypothesis the F -test was built for.

Single-coefficient t -tests are the right tool when we have one parameter in mind. Often, though, the question is whether **several** coefficients are jointly zero:

- “Are educ and IQ jointly irrelevant for wages once we control for hours, experience and age?”
- “Do the squared terms in a polynomial in exper add anything to the model?”
- “Do *any* of the regressors matter — is the overall regression worth running?”

These are joint hypotheses, and they require a joint test. Doing q separate t -tests does **not** answer the joint question, for two reasons. First, each individual t -test has size α , but the *combined* procedure — “reject the joint null if any individual t -test rejects” — has a Type I error rate that grows with the number of comparisons. With $q = 2$ independent tests each at size 0.05, the overall probability of rejecting at least one true null is roughly $1 - (1 - 0.05)^2 \approx 0.10$, not 0.05. Second, two coefficients can be jointly informative even when each is individually borderline; running them separately throws away the information in their *combination*.

4.8.1. Restricted vs unrestricted models

Let the **unrestricted** model be the regression with k slopes,

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u.$$

Suppose we want to test that the last q slopes are zero,

$$H_0 : \beta_{k-q+1} = \beta_{k-q+2} = \cdots = \beta_k = 0,$$

against the alternative that *at least one* of them is non-zero. The **restricted** model imposes H_0 by dropping those q regressors:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_{k-q} x_{k-q} + u.$$

Let SSR_U and SSR_R denote the sum of squared residuals of the unrestricted and restricted models. Because dropping regressors can never reduce the SSR, we always have $SSR_R \geq SSR_U$; the question is whether the *gap* is big enough to be inconsistent with H_0 .

i Definition: the F -statistic

Under H_0 and MLR.1–MLR.6,

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} \sim F_{q, n-k-1}.$$

Reject H_0 at level α if $F > F_{q, n-k-1, \alpha}$, the upper- α critical value of the F distribution with q and $n - k - 1$ degrees of freedom.

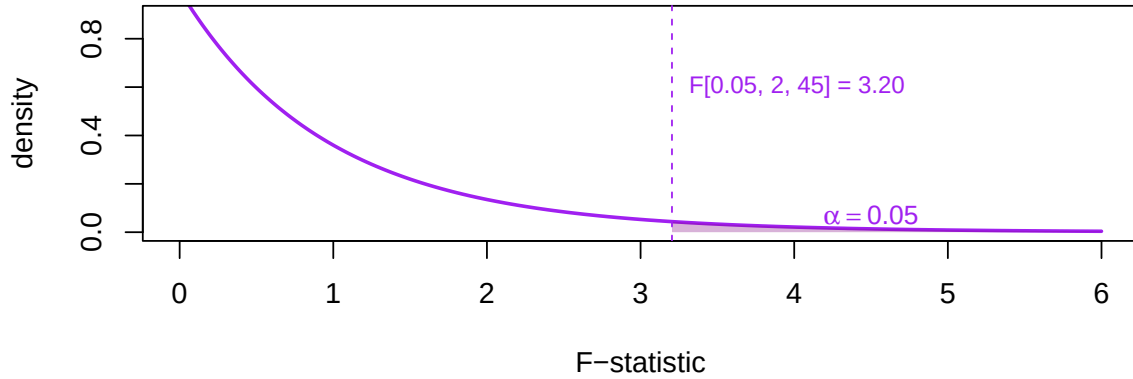


Figure 4.4.: The $F_{q,n-k-1}$ density and its 5% rejection region. Two restrictions ($q = 2$) and $n - k - 1 = 45$ are illustrated. Because the F -statistic is a ratio of *non-negative* quantities, the distribution lives on $[0, \infty)$ and is right-skewed; the rejection region is therefore a *single* upper tail, not the two-tail structure of the t -test. There is no such thing as a ‘left-sided F -test’.

An equivalent formulation uses the R^2 of each model:

$$F = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n - k - 1)}.$$

The two formulas are algebraically identical whenever y is the same in both regressions.

⚠ Common mistake: F across different dependent variables

Both the SSR-based and the R^2 -based formulas require the *same* dependent variable in the restricted and unrestricted models. If y differs between models — the classic case is testing $\log y$ against y — neither $\text{SSR}_R - \text{SSR}_U$ nor $R_U^2 - R_R^2$ is on a comparable scale, and the F -statistic computed by either formula is meaningless — not just imprecise. Tests across non-nested specifications with different left-hand-side variables require different machinery (e.g. the Vuong test or Davidson–MacKinnon J -test), which is beyond the scope of this chapter.

4.8.2. The overall F -test

A special case is the test that **every** slope is zero,

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0.$$

Here the restricted model is the regression on a constant alone, and $R_R^2 = 0$. The F -statistic collapses to

4. Inference

$$F = \frac{R^2/k}{(1-R^2)/(n-k-1)},$$

which is exactly the number `R` prints at the bottom line of `summary(lm(...))` under “F-statistic”, together with its p -value.

4.8.3. F versus t : a useful identity

For a *single* restriction ($q = 1$), the F -statistic equals the square of the corresponding t -statistic:

$$F = t^2, \quad F_{1, n-k-1} = (t_{n-k-1})^2.$$

The t - and F -tests deliver identical conclusions in this case; the F machinery is only strictly necessary when $q \geq 2$.

4.9. Lab: inference

Notation reminder

Throughout this chapter, k denotes the number of *slope* coefficients (matching the Wooldridge convention), so the unrestricted model has $k + 1$ parameters and $n - k - 1$ residual degrees of freedom. Some textbooks reserve k for the total number of parameters including the intercept; mixing the two conventions silently shifts every degrees-of-freedom count by one and gives wrong critical values. Keep Wooldridge’s convention throughout this lab.

We work with `wage2` from the **wooldridge** package: a cross-section of $n = 935$ U.S. men in the 1980 National Longitudinal Survey, with information on monthly earnings (`wage`), weekly hours (`hours`), an IQ test score (`IQ`), years of schooling (`educ`), years of work experience (`exper`), tenure with the current employer (`tenure`) and age. The goal is to translate the four-step inference machinery into R commands and then check that the built-in shortcuts give the same answers.

```
library(wooldridge)
data("wage2")
```

4.9.1. A first regression and its `summary()`

Start with a simple regression of monthly wage on weekly hours:

```
model1 <- lm(wage ~ hours, data = wage2)
summary(model1)
```

Call:

```
lm(formula = wage ~ hours, data = wage2)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

```
-839.72 -287.21 -52.38 200.46 2131.26
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  981.315      81.575   12.03  <2e-16 ***
hours        -0.532       1.832   -0.29   0.772
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 404.6 on 933 degrees of freedom

Multiple R-squared: 9.033e-05, Adjusted R-squared: -0.0009814

F-statistic: 0.08429 on 1 and 933 DF, p-value: 0.7716

The `summary()` block reports, for each coefficient, the estimate $\hat{\beta}_j$, its standard error $se(\hat{\beta}_j)$, the t -statistic $\hat{\beta}_j / se(\hat{\beta}_j)$, and the two-sided p -value. At the bottom we see the overall F -statistic and its p -value, which test $H_0 : \beta_{\text{hours}} = 0$ in this single-regressor case.

Now move to a richer specification:

```
model2 <- lm(wage ~ hours + educ + exper + IQ + age, data = wage2)
summary(model2)
```

Call:

```
lm(formula = wage ~ hours + educ + exper + IQ + age, data = wage2)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-883.49 -238.11  -47.36  190.09 2144.24
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept) -778.4403    171.2506  -4.546 6.20e-06 ***
hours        -2.5406     1.6811   -1.511 0.13104
educ         52.4505     7.2981    7.187 1.36e-12 ***
exper        10.9390     3.7196    2.941 0.00335 **
IQ           5.2917     0.9383    5.640 2.26e-08 ***
age          14.4836     4.6657    3.104 0.00197 **
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 368.9 on 929 degrees of freedom

Multiple R-squared: 0.1722, Adjusted R-squared: 0.1678

F-statistic: 38.66 on 5 and 929 DF, p-value: < 2.2e-16

i Reading the `summary()` output — columns that now carry the load

In Chapters 2 and 3 we labelled each `summary()` column but pointed the student to “we’ll come back to t and p in Ch 4.” This is that chapter. The columns that were ornamental in Ch 2 are doing the inferential work now.

- **Estimate** $\rightarrow \hat{\beta}_j$, the OLS coefficient.
- **Std. Error** $\rightarrow \text{se}(\hat{\beta}_j)$, the square root of the appropriate diagonal of $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$.
- **t value** $\rightarrow t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$, the test statistic for $H_0 : \beta_j = 0$ that we built in §4.3. Under MLR.1–MLR.6, it follows t_{n-k-1} .
- **Pr(>|t|)** $\rightarrow$ the two-sided p -value defined in §4.7. With $\alpha = 0.05$ as the conventional cutoff, anything in this column below 0.05 corresponds to a “statistically significant” coefficient.
- The **Signif. codes** line interprets **Pr(>|t|)** with stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.
- **Residual standard error: X on Y degrees of freedom** $\rightarrow \hat{\sigma}$, with $Y = n - k - 1$ residual degrees of freedom (here $n = 5 - 1$).
- **Multiple R-squared / Adjusted R-squared** $\rightarrow R^2$ and $\bar{R}^2$ from §3.8.
- **Bottom line F-statistic: X on Df1 and Df2 DF, p-value: Y** $\rightarrow$ this is the *overall F-test* of §4.8.2, which tests $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ (every slope is zero). **Df1** = k , **Df2** = $n - k - 1$, and the **p-value** is the right-tail probability under $F_{k, n-k-1}$.

Read the table line by line: **educ** and **IQ** both have large t -statistics and tiny p -values, so each is individually significant at the 1% level even after controlling for the others. **hours** is negative and significant (men who work longer hours earn slightly less *per week of hours worked*, conditional on the other covariates — a hint that hours are correlated with occupational mix). **exper** and **age** are statistically indistinguishable from zero in this specification.

4.9.2. Manual confidence intervals with `qt()`

Pull out the coefficients and standard errors programmatically, then build a 95% CI by hand.

```
beta_hat <- coef(model2)
se_hat   <- summary(model2)$coefficients[, "Std. Error"]

n <- nobs(model2)
k <- length(beta_hat) - 1      # number of slope coefficients
df <- n - k - 1

t_crit <- qt(0.975, df)
t_crit                                # critical value t_{0.025, df}
```

```
[1] 1.962521
```

The critical value is close to 1.96 because the degrees of freedom are large, but not identical. The manual 95% CI for β_{IQ} is

```
IQ_LB <- beta_hat["IQ"] - t_crit * se_hat["IQ"]
IQ_UB <- beta_hat["IQ"] + t_crit * se_hat["IQ"]
c(lower = IQ_LB, upper = IQ_UB)
```

```
lower.IQ upper.IQ
3.450211 7.133145
```

and for β_{hours} :

```
hours_LB <- beta_hat["hours"] - t_crit * se_hat["hours"]
hours_UB <- beta_hat["hours"] + t_crit * se_hat["hours"]
c(lower = hours_LB, upper = hours_UB)
```

```
lower.hours upper.hours
-5.8397528   0.7584696
```

The built-in shortcut delivers all the intervals in one call:

```
confint(model2, level = 0.95)
```

```

                2.5 %      97.5 %
(Intercept) -1114.523218 -442.3574662
hours        -5.839753   0.7584696
educ         38.127749   66.7731627
exper        3.639234   18.2387119
IQ           3.450211    7.1331452
age          5.327113   23.6400802
```

The numbers in the `IQ` and `hours` rows match what we computed by hand. Notice that the 95% interval for `IQ` excludes zero (consistent with the small p -value in the `summary()`), while the intervals for `exper` and `age` do contain zero (consistent with their non-significance).

4.9.3. Manual t -test for a single coefficient

Suppose we want to test $H_0 : \beta_{\text{educ}} = 0$ against the two-sided alternative at the 5% level. Step by step:

```
b_educ <- coef(model2)["educ"]
se_educ <- summary(model2)$coefficients["educ", "Std. Error"]

t_stat <- b_educ / se_educ
p_value <- 2 * pt(-abs(t_stat), df) # two-sided p-value

c(t = t_stat, p = p_value)
```

```

      t.educ      p.educ
7.186848e+00 1.361083e-12
```

```
summary(model2)$coefficients["educ", c("t value", "Pr(>|t|)")]
```

```

      t value      Pr(>|t|)
7.186848e+00 1.361083e-12
```

The bottom two lines are identical (up to rounding): the manual computation reproduces exactly what R prints. The t -statistic is far above the 5% critical value of roughly 1.96, so we reject H_0 . Education has a statistically significant partial effect on monthly earnings even after controlling for hours, experience, IQ and age.

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4.9.4. Manual F -test for joint significance

Now the headline question of the chapter: are `educ` and `IQ` *jointly* relevant once `hours`, `exper` and `age` are in the model? Formally,

$$H_0 : \beta_{\text{educ}} = \beta_{\text{IQ}} = 0 \quad \text{vs.} \quad H_1 : \text{at least one of them} \neq 0,$$

so $q = 2$.

The unrestricted model is `model2` above. The restricted model drops `educ` and `IQ`:

```
model3 <- lm(wage ~ hours + exper + age, data = wage2)
summary(model3)
```

Call:

```
lm(formula = wage ~ hours + exper + age, data = wage2)
```

Residuals:

Min	1Q	Median	3Q	Max
-749.69	-279.16	-48.16	203.20	2208.66

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	224.411	162.071	1.385	0.16649
hours	-1.175	1.812	-0.649	0.51665
exper	-9.430	3.443	-2.739	0.00628 **
age	27.032	4.838	5.587	3.03e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 398.4 on 931 degrees of freedom

Multiple R-squared: 0.03253, Adjusted R-squared: 0.02941

F-statistic: 10.44 on 3 and 931 DF, p-value: 9.332e-07

Compute the SSRs by hand:

```
SSR_u <- sum(residuals(model2)^2)
SSR_r <- sum(residuals(model3)^2)
c(SSR_unrestricted = SSR_u, SSR_restricted = SSR_r)
```

SSR_unrestricted	SSR_restricted
126414757	147747973

Now the F -statistic:


```

q      <- 2                # restrictions
df_u <- n - k - 1         # df of the unrestricted model

F_stat <- ((SSR_r - SSR_u) / q) / (SSR_u / df_u)
F_crit <- qf(0.95, df1 = q, df2 = df_u)
p_val  <- 1 - pf(F_stat, df1 = q, df2 = df_u)

c(F = F_stat, F_crit_5pct = F_crit, p = p_val)

```

```

      F F_crit_5pct      p
78.387043    3.005413  0.000000

```

The F -statistic is far above the 5% critical value and the p -value is essentially zero, so we strongly reject H_0 : educ and IQ are jointly significant determinants of wage.

R provides the same test in one line via `anova()`, which compares two nested models:

```
anova(model3, model2)
```

Analysis of Variance Table

```

Model 1: wage ~ hours + exper + age
Model 2: wage ~ hours + educ + exper + IQ + age
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     931 147747973
2     929 126414757  2  21333216 78.387 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Reading the anova() output — model comparison columns

`anova(restricted, unrestricted)` prints a two-row table. Row 1 is the restricted model, row 2 the unrestricted. The columns map directly to the F -formula:

- Res.Df row 2 $\rightarrow n - k - 1$ = residual degrees of freedom of the unrestricted model (the denominator df of the F in §4.8).
- Res.Df row 1 $\rightarrow n - (k - q) - 1$ for the restricted model; equivalently, Res.Df row 1 $-$ Res.Df row 2 $= q$ (the number of restrictions).
- RSS $\rightarrow$ SSR for each model (R calls it “residual sum of squares”).
- Df row 2 $\rightarrow q$, the number of restrictions being tested.
- Sum of Sq row 2 $\rightarrow \text{SSR}_R - \text{SSR}_U$, the *gain* in residual SSR that comes from dropping q regressors.
- F row 2 $\rightarrow$ the F -statistic itself: $[(\text{SSR}_R - \text{SSR}_U)/q] / [\text{SSR}_U / (n - k - 1)]$.
- Pr(>F) row 2 $\rightarrow$ the right-tail p -value under $F_{q, n-k-1}$.

The F and Pr(>F) columns reproduce the manual computation. Reporting the F -test from `anova()` is the recommended workflow; the manual derivation matters because it lets you see exactly *what is being compared*.

4.9.5. $F = t^2$ in a single-restriction test

The identity at the end of §4.7 is easy to verify numerically. Test $H_0 : \beta_{\text{educ}} = 0$ both ways:

```
t_educ <- summary(model2)$coefficients["educ", "t value"]

mR <- lm(wage ~ hours + exper + IQ + age, data = wage2) # drop educ only
ft <- anova(mR, model2)
F_single <- ft$F[2]

c(t_squared = t_educ^2, F = F_single)
```

```
t_squared      F
51.65078    51.65078
```

The two numbers agree, as the algebra predicts.

4.9.6. One-sided vs two-sided

Suppose theory tells us that an extra year of experience cannot lower monthly earnings, so the relevant alternative is *right-sided*: $H_0 : \beta_{\text{exper}} \leq 0$ vs $H_1 : \beta_{\text{exper}} > 0$. The same t -statistic feeds a different p -value:

```
co <- summary(model2)$coefficients["exper", ]
t_exper <- co["t value"]

p_one_sided <- if (t_exper > 0) 1 - pt(t_exper, df) else 1
# ^ right-tail p-value. Only meaningful when the estimate is already on the
#   side of the alternative (t_exper > 0 here). If the estimate had the
#   wrong sign for a right-sided alternative, the correct one-sided p-value
#   would be ~1 ("no evidence in the alternative's direction"), NOT the
#   naive 1 - pt(t_exper, df), which would return a value above 0.5.
p_two_sided <- 2 * pt(-abs(t_exper), df) # default in summary()

c(t = t_exper, p_one_sided = p_one_sided, p_two_sided = p_two_sided)
```

```
      t.t value p_one_sided.t value p_two_sided.t value
2.940922029      0.001676777      0.003353554
```

When $t > 0$ and the alternative is on the right, the one-sided p -value is exactly half the two-sided one. A coefficient that is borderline under the default two-sided test can become clearly significant once we are willing to commit to a sign a priori — but that commitment must come from economics, not from peeking at the data first.

Self-check

Six short multiple-choice questions. Try each one before opening the answer.

💡 Q1. What MLR.6 buys us

Adding MLR.6 (normality of u) to the Gauss–Markov assumptions allows us to:

- A. Prove that OLS is unbiased.
- B. Prove that OLS has the smallest variance among linear unbiased estimators.
- C. Derive the *exact* finite-sample distribution of $\hat{\beta}_j$ and run t - and F -tests in any sample size.
- D. Estimate σ^2 .

Answer: C. Unbiasedness (A) follows from MLR.1–MLR.4; BLUE (B) from MLR.1–MLR.5; the estimator $\hat{\sigma}^2$ (D) is defined regardless of MLR.6. Only MLR.6 gives us *exact* normality of $\hat{\beta}_j$ and therefore exact t and F inference in finite samples.

💡 Q2. Why t and not z ?

Under MLR.1–MLR.6 we use t_{n-k-1} rather than $\mathcal{N}(0, 1)$ for inference on a single coefficient because:

- A. OLS is biased in finite samples.
- B. The standard error uses an *estimated* $\hat{\sigma}^2$, and the resulting standardised statistic has heavier tails than the standard normal.
- C. The error term is heteroskedastic.
- D. The regressors are non-stochastic.

Answer: B. Replacing the unknown σ in the standard error with $\hat{\sigma}$ introduces extra variability that the t -distribution accounts for. As $n - k - 1 \rightarrow \infty$, the t converges to the standard normal.

💡 Q3. A 95% confidence interval

A 95% confidence interval for β_j that excludes zero implies:

- A. The two-sided t -test of $H_0 : \beta_j = 0$ rejects at the 5% level.
- B. The true β_j definitely is non-zero.
- C. The estimated coefficient is economically large.
- D. OLS is unbiased.

Answer: A. A CI and a two-sided test at the matching level are algebraically equivalent: the CI excludes the null value if and only if the test rejects. The CI says nothing about magnitude (C) or about OLS bias (D), and a single sample cannot certify (B).

💡 Q4. Reading a p -value

A coefficient has a two-sided p -value of 0.03. Which statement is correct?

- A. It is statistically significant at the 5% level but not at the 1% level.
- B. It is significant at every conventional level.
- C. It is not significant at the 5% level.
- D. The coefficient is economically important.

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Answer: A. Recall p is the smallest α at which we reject; $0.01 < 0.03 < 0.05$ places significance between the 1% and 5% levels. Statistical significance carries no information about economic magnitude.

💡 Q5. The F -statistic

To test $H_0 : \beta_1 = \beta_2 = 0$ jointly in a regression with k slopes and sample size n , we use:

- A. Two separate t -tests, one for each coefficient.
- B. $F = \frac{(\text{SSR}_R - \text{SSR}_U)/q}{\text{SSR}_U/(n - k - 1)}$, distributed $F_{q, n-k-1}$ under H_0 .
- C. A χ^2 -test on $\hat{\beta}_1 + \hat{\beta}_2$.
- D. A z -test, comparing the sum of the estimates with 1.96.

Answer: B. Separate t -tests do not control the **overall probability of rejecting at least one true H_0** : with q individual tests each at size α , the combined Type I error rate grows above α . The F -test is constructed so that its size on the *joint* null is exactly α . Separate t -tests also miss the case in which two regressors are individually weak but jointly informative.

💡 Q6. Significance vs causation

A regressor x_j has a coefficient with $p < 0.001$. Which of the following is true?

- A. A small p -value rules out omitted-variable bias.
- B. Statistical significance proves x_j causes y .
- C. Statistical significance and causal identification are equivalent.
- D. The p -value tells us $\hat{\beta}_j$ is unlikely to be zero by chance, but says nothing about whether x_j *causes* y — causal identification still requires $\mathbb{E}[u \mid \mathbf{X}] = 0$.

Answer: D. Section 4.6 is explicit on this point: inference is about ruling out sampling noise, not about ruling out confounding.

Exercises

Exercise 4.1 — Reading a `summary()` output. Using the `wage2` dataset, estimate the model

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{tenure} + u$$

with `lm()` and inspect `summary()`. (a) Which coefficients are significant at the 5% level? (b) Report the magnitude *and* the standard error of $\hat{\beta}_1$; what is the economic interpretation? (c) State the null hypothesis that the overall F -statistic at the bottom of `summary()` tests.

💡 Show answer

- (a) All three slopes are positive and have p -values well below 0.05 (each t -statistic is comfortably above 2 in absolute value), so `educ`, `exper` and `tenure` are individually

significant at the 5% level. (b) $\hat{\beta}_1$ is roughly 60 monthly-dollars per extra year of schooling, with a standard error of about 6 — an effect that is both statistically and economically meaningful. (c) The overall F -statistic tests $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (none of the regressors matters) against the alternative that at least one slope is non-zero.

Exercise 4.2 — Manual 95% CI. For the same model, build a 95% confidence interval for β_{educ} from scratch, using `coef()`, `summary()$coefficients[, "Std. Error"]` and `qt()`. Verify your interval against `confint(m)["educ", ]`. Does the interval contain zero? What conclusion follows for a two-sided t -test of $H_0 : \beta_{\text{educ}} = 0$ at the 5% level?

 Show answer

```
m <- lm(wage ~ educ + exper + tenure, data = wage2)
b <- coef(m)["educ"]
se <- summary(m)$coefficients["educ", "Std. Error"]
df <- nobs(m) - length(coef(m))      # n - k - 1
tc <- qt(0.975, df)
ci_manual <- c(b - tc * se, b + tc * se)
confint(m)["educ", ]                # should match
```

The interval does not contain zero, so the two-sided t -test of $H_0 : \beta_{\text{educ}} = 0$ at the 5% level rejects — consistent with the (tiny) $\Pr(>|t|)$ value in `summary()`.

Exercise 4.3 — Manual F -test. In the model $\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{tenure} + u$ on `wage2`, test $H_0 : \beta_2 = \beta_3 = 0$ by comparing the SSRs of the unrestricted and the restricted (drop `exper` and `tenure`) models. Compute the critical value with `qf()` at the 5% level and the p -value with `pf()`. Verify against `anova()`.

 Show answer

```
mU <- lm(wage ~ educ + exper + tenure, data = wage2)
mR <- lm(wage ~ educ, data = wage2)

SSR_u <- sum(resid(mU)^2)
SSR_r <- sum(resid(mR)^2)
n <- nobs(mU); k <- length(coef(mU)) - 1; q <- 2
df_u <- n - k - 1

F_stat <- ((SSR_r - SSR_u) / q) / (SSR_u / df_u)
F_crit <- qf(0.95, q, df_u)
p_val <- 1 - pf(F_stat, q, df_u)
anova(mR, mU)
```

The F -statistic is well above $F_{2, n-k-1, 0.05}$ and the p -value is essentially zero. We reject H_0 : experience and tenure are jointly significant given education.

Exercise 4.4 — One-sided test from theory. Economic theory suggests that an extra year of tenure cannot decrease monthly earnings, so the relevant alternative is right-sided: $H_0 : \beta_{\text{tenure}} \leq 0$ vs $H_1 : \beta_{\text{tenure}} > 0$. Using the regression of `wage` on `educ + exper + tenure`, compute the one-sided p -value from `summary()` output and decide at the 5% level. How does it

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compare with the default two-sided p -value that R prints?

A full answer is given in the Instructor Edition.

Exercise 4.5 — Joint vs individual significance. In the model of Exercise 4.1, add `IQ` and `age` as regressors. (a) Are `IQ` and `age` *individually* significant at the 5% level? (b) Are they *jointly* significant at the 5% level (use `anova()`)? (c) Construct an example, or explain in words, in which two regressors are individually insignificant yet jointly significant. What feature of the data drives this gap?

A full answer is given in the Instructor Edition.

Exercise 4.6 — F from R^2 . Show, starting from the SSR-based formula, that whenever the unrestricted and restricted models share the same dependent variable, the F -statistic can be written as

$$F = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n - k - 1)}.$$

Then verify the identity numerically for the F -test of $H_0 : \beta_{\text{exper}} = \beta_{\text{tenure}} = 0$ in Exercise 4.3, by pulling the two R^2 values out of `summary()` and plugging them in. Why does the formula *fail* if the dependent variable in the restricted model is `log(wage)` rather than `wage`?

A full answer is given in the Instructor Edition.

Prediction and Model Selection

Learning outcomes

By the end of this chapter the reader should be able to:

- Distinguish *causal inference* from *prediction* as two different econometric goals, and explain why a model that predicts well is not automatically a good causal model.
- Compute the point prediction $\hat{y}_0$ for a new observation with regressor values x_0 , both in the simple and multiple linear regression settings.
- Construct and interpret the two distinct interval estimates: the **confidence interval** for the conditional mean $E[y | x_0]$, and the **prediction interval** for an individual outcome y_0 .
- Explain why the prediction interval is always wider than the confidence interval for the mean, and identify the conditions under which both intervals are wider.
- Compare nested models using R^2 , adjusted R^2 , the Akaike (AIC) and Bayesian (BIC) information criteria, and the nested F -test.
- Use `predict()`, `AIC()`, `BIC()`, and `anova()` in R to predict from a fitted model and to choose among competing specifications.

Motivating empirical question

Two recruiters interview the same candidate — a 24-year-old with 16 years of education, 5 years of experience, and 2 years of tenure — and both fit the same wage regression to the same data. The first reports a 95% interval of [\$5.7, \$7.2] for the expected wage; the second reports [\$0.5, \$12.4]. Both are correct. Which interval is which?

The answer turns on a distinction that is easy to miss: the first recruiter is reporting uncertainty about the *average* wage of people like this candidate, while the second is reporting uncertainty about *this* candidate's actual wage. The same fitted model produces both numbers — they answer different questions. This chapter is about that distinction and the related question of how to choose between competing models in the first place.

5.1. Causality vs. prediction

UP to Chapter 4 we have used regression for one purpose: to estimate the *ceteris paribus* effect of x on y . The coefficient β_j was the object of interest, and the entire apparatus of Gauss–Markov assumptions, t -tests, and F -tests was in the service of getting $\hat{\beta}_j$ right — unbiased, efficient, and with valid standard errors.

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Econometrics also has a second, equally legitimate use: **prediction**. Here we do not particularly care about the value of any individual β_j ; we care about the accuracy of the fitted value $\hat{y}_0$ for a new observation.

The two goals overlap but are not the same:

- **Causal inference.** The target is $\partial \mathbb{E}[y | x] / \partial x_j$. The decisive assumption is MLR.4 (zero conditional mean): $\mathbb{E}[u | x_1, \dots, x_k] = 0$. Coefficients are interpreted; the typical statement is “a one-unit increase in x_j *causes* an expected change of β_j in y , holding other factors constant”.
- **Prediction.** The target is $\hat{y}_0$ for a new x_0 . The decisive assumption is *stability*: the new (y_0, x_0) is drawn from the same population as the sample. Coefficients need not be interpretable; we want regressors with strong predictive power and high (in-sample and out-of-sample) fit.

A model can be excellent for one purpose and poor for the other. A predictor of road accidents that includes “ice-cream sales” as a regressor may work well because ice cream proxies for hot weather; it is useless for causal policy advice about ice cream. Conversely, an instrumental-variables estimate of the return to schooling might identify a clean causal parameter from a tiny slice of variation in the data, while leaving most of the variability in wages unexplained.

 Common mistake: choosing the model with the highest R^2 and reading off causal effects

A high R^2 tells you that the fitted values track y well. It does **not** tell you that any individual coefficient measures a causal effect. The identification strategy — random assignment, an instrument, a natural experiment, or a credible argument that $\mathbb{E}[u | x] = 0$ — is what licences causal interpretation. Fit-based selection criteria such as adjusted R^2 , AIC, and BIC are tools for prediction; they cannot rescue a model that suffers from omitted-variable bias.

5.2. Point prediction

Consider, for clarity, the simple linear regression (SLR) model:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad \mathbb{E}[u_i | x_i] = 0.$$

Given the OLS estimates $(\hat{\beta}_0, \hat{\beta}_1)$ and a new value x_0 of the regressor, the **point prediction** is just the fitted value of the regression at x_0 :

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

The same construction extends, term by term, to multiple linear regression (MLR). If $x_0 = (1, x_{01}, x_{02}, \dots, x_{0k})$ is the row vector of regressor values for the new observation, then

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \hat{\beta}_2 x_{02} + \dots + \hat{\beta}_k x_{0k}.$$

How good is this prediction? The natural quantity is the **forecast error**:

$$f \equiv y_0 - \hat{y}_0 = (\beta_0 + \beta_1 x_0 + u_0) - (\hat{\beta}_0 + \hat{\beta}_1 x_0).$$

Taking the conditional expectation under MLR.1–MLR.4,

$$\mathbb{E}[f \mid x_0] = (\beta_0 + \beta_1 x_0) - \mathbb{E}[\hat{\beta}_0 + \hat{\beta}_1 x_0] = 0.$$

So $\hat{y}_0$ is an **unbiased predictor** of y_0 . In fact, under SLR.1–SLR.5 (and their MLR counterparts), one can show that $\hat{y}_0$ has the smallest variance among all linear unbiased predictors: it is the **Best Linear Unbiased Predictor (BLUP)**.

⚠ $\hat{y}_0$ plays a dual role

The same number $\hat{y}_0$ is *both* the point prediction of the individual outcome y_0 *and* the estimate of the conditional mean $\mathbb{E}[y \mid x_0]$. The two interpretations of $\hat{y}_0$ are numerically identical, but they have different *uncertainty*. The next section makes this concrete.

5.3. Forecast variance and prediction intervals

A point estimate is rarely enough. We need an interval that quantifies uncertainty — and there are two distinct uncertainties to quantify.

5.3.1. Confidence interval for the conditional mean $\mathbb{E}[y \mid x_0]$

The conditional mean $\mathbb{E}[y \mid x_0] = \beta_0 + \beta_1 x_0$ is a fixed (non-random) population quantity. Its estimator is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$, which is random because the OLS estimates are random. Its variance is

$$\text{Var}(\hat{y}_0 \mid x) = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\text{SST}_x} \right],$$

where $\text{SST}_x = \sum_{i=1}^n (x_i - \bar{x})^2$ is the total sum of squares of the regressor. Replacing σ^2 by the estimator $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$, the $100(1 - \alpha)\%$ confidence interval for the *mean* prediction is

$$\hat{y}_0 \pm t_c \cdot \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\text{SST}_x}},$$

with t_c the appropriate critical value of a t -distribution with $n - k - 1$ degrees of freedom.

5.3.2. Prediction interval for an individual outcome y_0

The new outcome $y_0 = \beta_0 + \beta_1 x_0 + u_0$ is random because *both* the OLS estimates *and* the new error u_0 are random. Assuming u_0 is independent of the sample,

$$\text{Var}(y_0 - \hat{y}_0 \mid x) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\text{SST}_x} \right].$$

The extra “1” inside the brackets is the contribution of u_0 — the irreducible part of the new draw that no amount of data could ever predict. The $100(1 - \alpha)\%$ **prediction interval** for y_0 is

5. Prediction and Model Selection

$$\hat{y}_0 \pm t_c \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\text{SST}_x}}.$$

Imagine two questions about the candidate. Question A: “across all women like this in the population, what is their *average* wage?” Question B: “what wage will *this particular* woman earn?” Even with infinite data we could pin down Question A exactly — but Question B still depends on this woman’s specific ability, employer, and luck, bundled into u_0 . That irreducible per-person variation is the “+1”.

In MLR the same logic gives $\text{Var}(y_0 - \hat{y}_0 \mid X) = \sigma^2[1 + x_0'(X'X)^{-1}x_0]$, with the CI for the mean obtained by dropping the “1” in the brackets.

i The two intervals at a glance

Quantity targeted	Variance ingredient	Interval	Width
Mean $\mathbb{E}[y \mid x_0]$	$1/n + (x_0 - \bar{x})^2/\text{SST}_x$	Confidence interval	Narrower
Individual y_0	$1 + 1/n + (x_0 - \bar{x})^2/\text{SST}_x$	Prediction interval	Wider

The PI is always wider than the CI. The gap is governed by the “1”: the irreducible variance of the new draw u_0 .

A few more observations on the formulas:

- Both variances shrink as n grows — with infinite data the CI for the mean would collapse to a point, but the PI would not, because the “1” never goes away.
- Both variances grow as x_0 moves *away from the sample mean* $\bar{x}$. Predictions are most precise near the centre of the data; predicting at the edges (or extrapolating beyond them) is dangerous.
- Both variances shrink as SST_x grows. More spread-out regressors give more leverage to pin down the slope, which in turn tightens prediction.

5.3.3. Worked numerical example

Suppose we have estimated the SLR

$$\hat{y} = 2.0 + 0.7x,$$

with sample size $n = 100$, $\bar{x} = 12$, $\text{SST}_x = 200$, and $\hat{\sigma}^2 = 4$ (so $\hat{\sigma} = 2$). We want to predict at $x_0 = 16$. Using $t_c \approx 1.98$ for a 95% interval with 98 degrees of freedom:

- **Point prediction.** $\hat{y}_0 = 2.0 + 0.7 \times 16 = 13.2$.
- **CI for the mean** $\mathbb{E}[y \mid x_0]$.

$$\text{se}(\hat{y}_0) = \hat{\sigma} \sqrt{\frac{1}{100} + \frac{(16-12)^2}{200}} = 2\sqrt{0.01 + 0.08} = 0.6.$$

$$95\% \text{ CI: } 13.2 \pm 1.98 \times 0.6 = [12.01, 14.39].$$

- **Prediction interval** for y_0 .

$$\text{se}(f) = \hat{\sigma} \sqrt{1 + \frac{1}{100} + \frac{(16-12)^2}{200}} = 2\sqrt{1.09} \approx 2.088.$$

$$95\% \text{ PI: } 13.2 \pm 1.98 \times 2.088 = [9.06, 17.34].$$

The CI is roughly ± 1.20 wide around the point prediction; the PI is roughly ± 4.14 wide. The ratio is set by the relative size of the “1” against $1/n + (x_0 - \bar{x})^2 / \text{SST}_x = 0.09$. Here the new-draw uncertainty dwarfs the parameter uncertainty by an order of magnitude.

5.4. Prediction in MLR

Real wage models include more than one regressor. Nothing essential changes in the construction of $\hat{y}_0$ and the two intervals — the formulas just become vector-matrix versions of the SLR formulas.

For a row vector $x_0 = (1, x_{01}, \dots, x_{0k})$,

$$\hat{y}_0 = x'_0 \hat{\beta}, \quad \text{Var}(y_0 - \hat{y}_0 \mid X) = \sigma^2 [1 + x'_0 (X'X)^{-1} x_0].$$

Dropping the “1” gives the variance of $\hat{y}_0$ as an estimator of $\mathbb{E}[y \mid x_0]$. In R, the function `predict()` does all of this automatically with `interval = "confidence"` or `interval = "prediction"`. The user supplies the new x_0 as a data frame; the function returns the fit and the bounds. We try this on `wage1` in the lab.

Don't extrapolate

The prediction formulas assume that (y_0, x_0) comes from the *same population* as the sample. If x_0 is far outside the range of the regressors actually observed — predicting wages for someone with 30 years of education and 80 years of experience, say — both intervals can be technically computed, but neither is trustworthy. There is no information in the sample about what happens at those values. Stay inside the support of the data.

5.5. Model selection: R^2 , adjusted R^2 , AIC, BIC

Suppose we have several candidate regression models for the same outcome. How do we choose? Four criteria are commonly used.

5.5.1. R^2 and the garbage-variable problem

The coefficient of determination

$$R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

measures the proportion of the variation of y explained by the model. It has one fatal flaw as a model-selection device: **adding any regressor cannot reduce R^2** . Even a pure noise regressor

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will, almost surely, slightly increase R^2 , because OLS can always find a tiny linear combination of the noise that fits some residual variation by chance.

If we picked our model by maximising R^2 , we would include every regressor we could lay our hands on — including dozens of irrelevant ones. The model would fit the sample beautifully and predict horribly out of sample. This is the classic *overfitting* problem. The slides illustrate it by simulating a model with two genuine regressors X_1, X_2 and then adding 196 “garbage” variables of pure noise: R^2 climbs steadily towards 1 as garbage piles up, even though none of the new variables has any predictive content.

5.5.2. Adjusted R^2

A simple fix is to penalise complexity. The **adjusted R^2** is

$$\bar{R}^2 = 1 - \frac{\text{SSR}/(n - k - 1)}{\text{SST}/(n - 1)},$$

where k is the number of regressors (excluding the intercept). The numerator $\text{SSR}/(n - k - 1)$ is $\hat{\sigma}^2$, the unbiased estimator of the error variance. Adding a regressor reduces SSR but also reduces $n - k - 1$; $\bar{R}^2$ goes up only if the gain in fit beats the cost of the extra parameter.

A few useful facts:

- $\bar{R}^2 \leq R^2$ always, with equality only when $k = 0$.
- Unlike R^2 , $\bar{R}^2$ can *decrease* when an irrelevant regressor is added — and in the garbage simulation, it does.
- $\bar{R}^2$ can in principle be negative if the model is so poor that $\hat{\sigma}^2$ exceeds the sample variance of y .

5.5.3. AIC and BIC

The **Akaike Information Criterion** and the **Bayesian Information Criterion** combine model fit (a log-likelihood) with a complexity penalty:

$$\text{AIC} = -2 \ln \hat{L} + 2(k + 1), \quad \text{BIC} = -2 \ln \hat{L} + (k + 1) \ln n.$$

For a linear model with normally distributed errors the log-likelihood is a monotone transformation of $-\text{SSR}/n$, so up to additive constants

$$\text{AIC} \approx n \ln \left(\frac{\text{SSR}}{n} \right) + 2(k + 1), \quad \text{BIC} \approx n \ln \left(\frac{\text{SSR}}{n} \right) + (k + 1) \ln n.$$

Lower AIC and BIC indicate better models. Both criteria balance fit (the SSR term) against complexity (the second term). The only difference is the size of the penalty: AIC charges 2 per extra parameter, BIC charges $\ln n$. For $n \geq 8$, $\ln n > 2$, so BIC penalises complexity more strongly and tends to select more parsimonious models. AIC is built to minimise prediction error; BIC is built to identify the “true” model when one exists.

i How big is a “big” AIC difference?

AIC and BIC are unitless and only meaningful in *differences* between models fit to the same data with the same dependent variable. Rule of thumb (Burnham & Anderson): $\Delta\text{AIC} < 2$ between two models means they are essentially equivalent; $\Delta\text{AIC} > 10$ is strong evidence for the lower-AIC model. The same rule applies approximately to BIC.

! AIC and BIC require the same dataset and the same dependent variable

If two models use different dependent variables (e.g. `wage` and `log(wage)`), or are estimated on different sub-samples, their AIC and BIC are *not* comparable — the log-likelihoods are on different scales. The same warning applies to R^2 and $\bar{R}^2$.

5.5.4. Nested F -test

When one model is nested inside another — the smaller model is obtained by setting some coefficients of the larger model to zero — the **F -test of joint restrictions** (Chapter 4) is the formal test of “are these extra regressors jointly informative?”. In R, `anova(small, big)` does it. The test answers a different question from AIC/BIC: it asks whether the data *reject* the smaller model in favour of the larger, not which model is “best” by some predictive criterion.

5.5.5. Putting it together

In practice we report all four criteria side by side. They usually agree on the broad ranking; when they disagree, the disagreement is informative:

- If R^2 rises but $\bar{R}^2$, AIC, and BIC all worsen, the new regressors are noise.
- If $\bar{R}^2$ and AIC favour the larger model but BIC favours the smaller, the extra regressors are weakly predictive; with a large sample BIC may be over-penalising. In a *purely predictive* exercise, lean towards AIC; for *parsimony* or for picking a “true” model, lean towards BIC.

And, always: model selection criteria choose the model that *fits* best on the criterion at hand. They do not choose the model that *identifies a causal parameter* best. That choice belongs to the identification strategy.

5.5.6. In-sample versus out-of-sample

A final word on what these criteria are actually doing. $\bar{R}^2$, AIC, and BIC are all *in-sample* statistics that approximate out-of-sample predictive accuracy. They exist because splitting your sample in half was historically wasteful: with $n = 50$ observations, dedicating 25 to a test set throws away half the information available for estimation. With modern data, **train/test splitting** (and *cross-validation*, beyond the scope of this chapter) gives a more direct answer to the question “how well will this model predict on data it has not seen?”. We demonstrate the train/test idea in Step 9 of the lab.

5.6. Lab: prediction and model selection

We work with the `wage1` dataset of 526 U.S. workers (Chapter 1). The goal is to (i) fit a sequence of nested wage models, (ii) compare them using $\bar{R}^2$, AIC, BIC, and an F -test, (iii) produce point predictions and the two interval estimates for a specific worker, and (iv) visualise the difference between the confidence band for the mean and the prediction band for an individual.

```
library(wooldridge)
data("wage1")
```

5.6.1. Step 1 — Fit four nested wage models

We progressively enrich the specification.

```
model1 <- lm(wage ~ educ, data = wage1)
model2 <- lm(wage ~ educ + exper, data = wage1)
model3 <- lm(wage ~ educ + exper + tenure, data = wage1)
model4 <- lm(wage ~ educ + exper + tenure + female + married, data = wage1)
```

5.6.2. Step 2 — Compare the four models

Collect R^2 , $\bar{R}^2$, AIC, and BIC in a single table.

```
tab <- sapply(list(m1 = model1, m2 = model2, m3 = model3, m4 = model4),
  function(m) c(
    R2      = summary(m)$r.squared,
    AdjR2   = summary(m)$adj.r.squared,
    AIC     = AIC(m),
    BIC     = BIC(m)))
round(tab, 3)
```

	m1	m2	m3	m4
R2	0.165	0.225	0.306	0.368
AdjR2	0.163	0.222	0.302	0.362
AIC	2777.423	2739.937	2683.662	2638.600
BIC	2790.219	2756.999	2704.988	2668.457

A few things to read off this table:

- R^2 rises *monotonically* from `model1` to `model4`. That is mechanical: adding regressors cannot decrease R^2 .
- $\bar{R}^2$ also rises across the four models, indicating that each newly added regressor contributes more fit than its complexity cost.
- AIC and BIC both *decrease* as we go from `model1` to `model4` (lower is better). All four criteria agree: of these four specifications, `model4` is preferred.

5.6.3. Step 3 — A nested F -test

Are `female` and `married` *jointly* informative once `educ`, `exper`, and `tenure` are in the model?

```
anova(model3, model4)
```

Analysis of Variance Table

```
Model 1: wage ~ educ + exper + tenure
Model 2: wage ~ educ + exper + tenure + female + married
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     522 4966.3
2     520 4524.0  2     442.27 25.418 2.938e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The F -statistic compares the two SSRs; if its p -value is small, we reject the joint null $H_0 : \beta_{\text{female}} = \beta_{\text{married}} = 0$. In `wage1` it is very small, confirming the AIC/BIC preference for `model4` with a formal test.

5.6.4. Step 4 — The garbage-variable demonstration

Now we deliberately add three pure-noise regressors to `model4` and watch what happens to each criterion.

```
set.seed(42)
wage1$noise1 <- rnorm(nrow(wage1))
wage1$noise2 <- rnorm(nrow(wage1))
wage1$noise3 <- rnorm(nrow(wage1))

model_garbage <- lm(wage ~ educ + exper + tenure + female + married +
  noise1 + noise2 + noise3, data = wage1)

rbind(
  model4 = c(R2 = summary(model4)$r.squared,
    AdjR2 = summary(model4)$adj.r.squared,
    AIC = AIC(model4), BIC = BIC(model4)),
  model_garbage = c(R2 = summary(model_garbage)$r.squared,
    AdjR2 = summary(model_garbage)$adj.r.squared,
    AIC = AIC(model_garbage), BIC = BIC(model_garbage)))
```

	R2	AdjR2	AIC	BIC
model4	0.3681887	0.3621136	2638.600	2668.457
model_garbage	0.3708921	0.3611573	2642.345	2684.998

Inspect the output: R^2 has risen slightly (as it must), but $\bar{R}^2$ has *fallen*, and both AIC and BIC have *increased*. The three pure-noise regressors look superficially helpful to plain R^2 ; every penalised criterion sees through them. This is the entire point of using $\bar{R}^2$, AIC, or BIC rather than raw R^2 to choose a model.

5.6.5. Step 5 — Point prediction for a specific worker

Take the candidate from the chapter opening: 16 years of education, 5 years of experience, 2 years of tenure, female, unmarried. We use `model4`.

```
new_worker <- data.frame(educ = 16, exper = 5, tenure = 2,
                        female = 1, married = 0)
predict(model4, newdata = new_worker)
```

```
1
5.90268
```

This is $\hat{y}_0$ — the recruiter’s single-number wage prediction.

5.6.6. Step 6 — The two intervals

The same `predict()` function delivers the CI for the mean and the PI for the individual.

```
predict(model4, newdata = new_worker,
        interval = "confidence", level = 0.95)
```

```
      fit      lwr      upr
1 5.90268 5.323082 6.482277
```

```
predict(model4, newdata = new_worker,
        interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr
1 5.90268 0.07919504 11.72616
```

The first interval is the recruiter’s answer to “what is the average wage of women with these characteristics?”. The second is the answer to “what wage might *this* woman actually earn?”. The PI is much wider — in this dataset, an order of magnitude wider — because it includes the irreducible variance of the new draw u_0 .

5.6.7. Step 7 — A manual sanity check

To make the formulas concrete, compute the CI by hand and compare:

```
yhat <- predict(model4, newdata = new_worker, se.fit = TRUE)
tc    <- qt(0.975, df = model4$df.residual)
ci_manual <- c(yhat$fit - tc * yhat$se.fit,
              yhat$fit + tc * yhat$se.fit)
ci_manual
```

```
1      1
5.323082 6.482277
```



```
# Now the prediction interval, using  $SE(f)^2 = SE(\hat{y})^2 + \sigma_{\hat{y}}^2$ 
sig2 <- summary(model4)$sigma^2
se_pred <- sqrt(yhat$se.fit^2 + sig2)
pi_manual <- c(yhat$fit - tc * se_pred,
              yhat$fit + tc * se_pred)
pi_manual
```

```
      1      1
0.07919504 11.72616433
```

These should match (to rounding) the bounds returned by `predict()` in Step 6. The decomposition $\text{Var}(y_0 - \hat{y}_0) = \text{Var}(\hat{y}_0) + \sigma^2$ is the operational version of “two sources of uncertainty: the parameters, and the new draw”.

5.6.8. Step 8 — Visualising CI vs. PI across educ

The two-interval distinction is easiest to see in a picture. We compute both bands across a grid of education values (holding the other regressors fixed) and plot them on top of the fitted line.

```
grid <- data.frame(educ = seq(0, 20, by = 0.5),
                  exper = 5, tenure = 3,
                  female = 0, married = 1)
ci_band <- predict(model4, newdata = grid, interval = "confidence")
pi_band <- predict(model4, newdata = grid, interval = "prediction")

plot(grid$educ, ci_band[, "fit"], type = "l", lwd = 2, col = "blue",
     xlab = "Years of education",
     ylab = "Predicted wage (USD/hour)",
     main = "CI for the mean vs. PI for an individual",
     ylim = range(pi_band))
lines(grid$educ, ci_band[, "lwr"], lty = 2, col = "blue")
lines(grid$educ, ci_band[, "upr"], lty = 2, col = "blue")
lines(grid$educ, pi_band[, "lwr"], lty = 2, col = "red")
lines(grid$educ, pi_band[, "upr"], lty = 2, col = "red")
legend("topleft",
     legend = c("Fitted", "95% CI (mean)", "95% PI (individual)"),
     col = c("blue", "blue", "red"),
     lty = c(1, 2, 2), lwd = c(2, 1, 1))
```

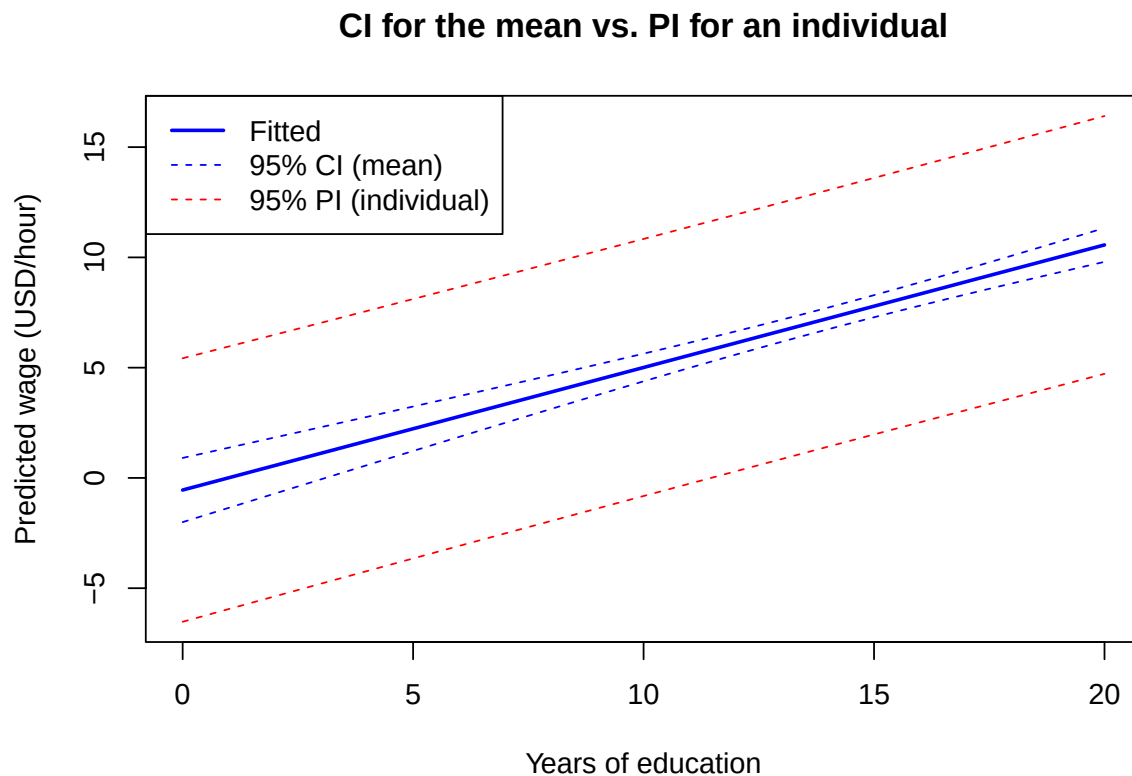


Figure 5.1.: Fitted wage (solid blue) with the 95% confidence band for the conditional mean (dashed blue) and the 95% prediction band for individuals (dashed red), across years of education. Other regressors fixed at `exper=5`, `tenure=3`, `female=0`, `married=1`.

Two features of the plot are worth pointing out:

- The PI (red) is much wider than the CI (blue) at every value of `educ`. The vertical gap between the two bands is, essentially, $\pm t_c \hat{\sigma}$ — the contribution of u_0 .
- Both bands widen at the *extremes* of `educ`, away from the sample mean. Predicting wages for a worker with no schooling, or with 20 years of schooling, is intrinsically less precise than predicting near the sample average. This is the $(x_0 - \bar{x})^2 / \text{SST}_x$ term in the variance formula doing its work.

We can now answer the opening question. The narrow interval is the CI for the *mean* wage of workers like the candidate; the wide interval is the PI for *this* candidate's wage. Both are produced by the same model, from the same data, with one keystroke of difference in the call to `predict()`.

5.6.9. Step 9 — In-sample versus out-of-sample MSE

So far every fit statistic we have computed was evaluated on the same 526 observations used to estimate the model. The criteria of the previous section ($\bar{R}^2$, AIC, BIC) all *approximate* the predictive error we would see on new data, but the most direct way to measure that error is to simply hold some data aside.

We split `wage1` into a *training set* (the first 400 observations) and a *test set* (the remaining 126). We fit `model4` — the chapter’s preferred specification — on the training set alone, then evaluate its squared-error loss on both sets.

```
set.seed(42)
n      <- nrow(wage1)
train_idx <- 1:400
test_idx  <- 401:n

model4_train <- lm(wage ~ educ + exper + tenure + female + married,
                  data = wage1[train_idx, ])

# In-sample MSE: residuals on the training set
mse_in  <- mean(residuals(model4_train)^2)

# Out-of-sample MSE: predict on the test set, then compute squared errors
yhat_test <- predict(model4_train, newdata = wage1[test_idx, ])
mse_out   <- mean((wage1$wage[test_idx] - yhat_test)^2)

c(in_sample_MSE = mse_in, out_of_sample_MSE = mse_out)
```

```
in_sample_MSE out_of_sample_MSE
      8.920853       7.846613
```

Two observations on the comparison:

- The out-of-sample MSE is *larger* than the in-sample MSE. This is the rule, not the exception: OLS is fit to minimise the sum of squared residuals on the training data, so the in-sample loss is by construction the smallest the model can achieve on those points. On *new* data the model has no such home-field advantage, and the squared errors are typically larger.
- The gap between the two is what $\bar{R}^2$, AIC, and BIC were trying to estimate when we computed them on the full sample. Train/test splitting measures the same thing more directly — at the cost of using only part of the data to estimate $\hat{\beta}$.

A single train/test split is noisy: a different split (or a different seed) can give a noticeably different out-of-sample MSE. **Cross-validation** averages over many splits to reduce that noise, and is the standard tool when predictive accuracy is the primary goal. We do not pursue it here, but every modern statistical-learning text covers it in its first chapter.

Self-check

Six short questions. Try each one before opening the answer.

💡 Q1. Which interval is which?

A recruiter wants to know the *average* wage of all workers in the population who share the candidate’s profile. The relevant interval is:

- A. The 95% prediction interval for y_0 .
- B. The 95% confidence interval for $\mathbb{E}[y \mid x_0]$.

5. Prediction and Model Selection

- C. The 95% confidence interval for the slope β_1 .
- D. The OLS residual.

Answer: B. The CI for $\mathbb{E}[y \mid x_0]$ targets the conditional *mean* and is the narrower of the two intervals. The PI in option A would be the right choice for “what might *this* candidate actually earn?”.

💡 Q2. Why is the PI wider than the CI?

The 95% prediction interval for y_0 is wider than the 95% confidence interval for $\mathbb{E}[y \mid x_0]$ because:

- A. The prediction interval uses a larger critical value t_c .
- B. The prediction interval includes the variance of the new error u_0 , the irreducible part of the new draw.
- C. The CI ignores the variance of the OLS estimates entirely.
- D. The PI is built with a smaller sample size.

Answer: B. Both intervals use the same t_c . The PI adds a “1” inside the bracket in the variance formula, which captures $\text{Var}(u_0) = \sigma^2$ — something the data can never predict.

💡 Q3. Where are the intervals widest?

Holding the sample fixed, both the CI for the mean and the PI for an individual are *widest* when:

- A. x_0 equals the sample mean of the regressor.
- B. The sample size is very large.
- C. x_0 is far from the sample mean of the regressor.
- D. The residual variance is zero.

Answer: C. Both variances contain the term $(x_0 - \bar{x})^2 / \text{SST}_x$, which grows as x_0 moves away from $\bar{x}$. Predictions are most precise near the centre of the data and least precise at the edges; extrapolating beyond the support of the data is dangerous.

💡 Q4. Adjusted R^2 and a noise regressor

Adding a pure-noise regressor to an OLS model:

- A. Strictly increases R^2 and strictly increases $\bar{R}^2$.
- B. Cannot decrease R^2 but can decrease $\bar{R}^2$.
- C. Always lowers R^2 because the new regressor is uninformative.
- D. Has no effect on either criterion.

Answer: B. R^2 never falls when a regressor is added. $\bar{R}^2$ penalises complexity through the degrees-of-freedom correction, so an irrelevant regressor can lower it. AIC and BIC will also worsen.

💡 Q5. Comparing AIC across models

The Akaike Information Criterion can legitimately be used to compare:

- A. Two models estimated on different sub-samples of the data.
- B. A regression of `wage` on `educ` against a regression of `log(wage)` on `educ`.
- C. Two models with the same dependent variable, the same data, and different sets of regressors.
- D. Models estimated by different methods (OLS, IV, GMM) on the same data.

Answer: C. AIC (and BIC) are only comparable when the dependent variable, the sample, and the estimation method are the same; otherwise the log-likelihoods are on different scales. The same warning applies to R^2 and $\bar{R}^2$.

💡 Q6. Causality vs. prediction

The model with the smallest AIC is necessarily the best for:

- A. Both prediction *and* causal identification of individual coefficients.
- B. Prediction of y given x — but not necessarily for identifying the causal effect of a specific regressor.
- C. Causal identification only.
- D. Neither, since AIC is always misleading.

Answer: B. AIC is a predictive criterion. Causal identification depends on the assumption $\mathbb{E}[u | x] = 0$, which is a statement about the design (random assignment, an instrument, a credible argument), not about model fit. A predictively excellent model can still suffer from omitted-variable bias.

Exercises

Exercise 5.1 — The two intervals by hand. Suppose you have fit $\hat{y} = 2.0 + 0.7x$ on a sample with $n = 100$, $\bar{x} = 12$, $\text{SST}_x = 200$, and $\hat{\sigma}^2 = 4$. Take $x_0 = 16$ and use $t_c = 1.98$.

- (a) Compute the point prediction $\hat{y}_0$.
- (b) Compute the 95% confidence interval for $\mathbb{E}[y | x_0]$.
- (c) Compute the 95% prediction interval for y_0 .
- (d) Comment briefly on which interval a *recruiter* would quote and why.

💡 Show answer

- (a) $\hat{y}_0 = 2.0 + 0.7 \times 16 = 13.2$.
- (b) $\text{se}(\hat{y}_0) = 2\sqrt{1/100 + 16/200} = 2\sqrt{0.09} = 0.6$, so $\text{CI} = 13.2 \pm 1.98 \times 0.6 = [12.01, 14.39]$.
- (c) $\text{se}(f) = 2\sqrt{1 + 0.09} = 2\sqrt{1.09} \approx 2.088$, so $\text{PI} = 13.2 \pm 1.98 \times 2.088 = [9.06, 17.34]$.
- (d) Both. The CI answers “what is the *average* wage of workers like the candidate?” — useful for planning aggregate compensation. The PI answers “what might *this* candidate actually earn?” — useful for an individual job offer. Reporting only one of the two is misleading.

5. Prediction and Model Selection

Exercise 5.2 — Reading a selection table. Using `wage1`, estimate the four nested models from the lab and present a 4-by-4 table of R^2 , $\bar{R}^2$, AIC, and BIC. Which model wins under each criterion? Do all four criteria agree?

💡 Show answer

In `wage1` the four criteria agree on the ranking: R^2 and $\bar{R}^2$ are highest for `model4`; AIC and BIC are lowest for `model4`. Whenever all four agree the choice is easy. When BIC selects a smaller model than AIC (which can happen with larger samples), the gap reflects BIC's stronger penalty $\ln n > 2$; the choice then depends on whether you prioritise predictive accuracy (AIC) or parsimony (BIC).

Exercise 5.3 — A 12-year worker and a 20-year worker. Using `model4` from the lab, compute the 95% prediction interval for two profiles: (i) `educ = 12`, `exper = 10`, `tenure = 5`, `female = 0`, `married = 1` and (ii) `educ = 20`, `exper = 10`, `tenure = 5`, `female = 0`, `married = 1`. Is the second interval wider, narrower, or the same width as the first? Why?

💡 Show answer

The second interval is wider. Twenty years of education is far from the sample mean of `educ` in `wage1` (around 12.5 years), so the $(x_0 - \bar{x})'(X'X)^{-1}(x_0 - \bar{x})$ term in the variance formula is larger. This is the warning against extrapolation: the further x_0 is from the centre of the data, the larger both intervals become, and at some point they become so wide that the prediction is uninformative.

Exercise 5.4 — A garbage variable in your own data. Take any dataset of your choice (real or simulated). Fit a baseline model with one or two genuine regressors and report R^2 , $\bar{R}^2$, AIC, BIC. Add five regressors of pure Gaussian noise (`rnrm(n)`), refit, and report the same four numbers. By how much does each change? Which criterion is fooled, and which is not?

A full answer is given in the Instructor Edition.

Exercise 5.5 — Causality vs. prediction with a confound. Suppose y is true grade, x is hours studied, and z is innate ability. Hours and ability are positively correlated. You fit two models on a large sample: (a) $y \sim x$, (b) $y \sim x + z$. Argue, informally, that:

- (i) Model (b) has the higher R^2 and higher $\bar{R}^2$ — it is the better *predictor*.
- (ii) Model (a) gives a *biased* causal estimate of the effect of hours on grades; model (b) does not.
- (iii) AIC and BIC will both prefer model (b).

What conclusion do you draw about using AIC/BIC to choose a *causal* model?

A full answer is given in the Instructor Edition.

Exercise 5.6 — Out-of-sample prediction across specifications. Step 9 of the lab applied the 400/126 train/test split to `model4`. Repeat the exercise for `model11`, `model12`, and `model13` (use the same `train_idx` and `test_idx`). For each of the four models, report (i) the in-sample MSE on the training set, and (ii) the out-of-sample MSE on the test set. Which model wins on in-sample MSE? Which wins on out-of-sample MSE? Comment on the relationship between $\bar{R}^2$ /AIC/BIC computed on the *training* sample and the *out-of-sample* MSE.

 Show answer

Fit each model on `wage1[train_idx, ]` and compute the two MSEs exactly as in Step 9. The in-sample MSE decreases *monotonically* from `model11` to `model14` — adding regressors cannot increase the in-sample sum of squared residuals. The out-of-sample MSE is *not* monotone: typically it also falls from `model11` through `model14` in `wage1` (the added regressors carry real predictive content, as $\bar{R}^2$ /AIC/BIC indicated on the full sample), but the gain shrinks at each step and on a different random split one of the smaller models can occasionally win. The general lesson: $\bar{R}^2$, AIC, and BIC computed on the training sample track the *ranking* of out-of-sample MSE well, but they do not give an unbiased estimate of its *level*. A larger gap between in-sample and out-of-sample MSE for a richer model is the signature of mild overfitting.

Nonlinear Relationships and Qualitative Information

Learning outcomes

By the end of this chapter the reader should be able to:

- Recognise when a linear-in-parameters model can accommodate a nonlinear relationship in the variables, and choose between quadratic and logarithmic specifications.
- Compute partial effects and turning points in quadratic models, and decide whether a turning point is economically meaningful.
- Interpret coefficients in level–level, log–level, level–log, and log–log models in terms of slopes, semi-elasticities, and elasticities.
- Build and interpret interaction terms, including interactions between two continuous variables and between a continuous variable and a dummy (slope dummies).
- Use dummy variables for binary, nominal, and ordinal qualitative information, and avoid the dummy variable trap.
- Test for a structural break across two subsamples using the Chow F -test, computed both by hand and via `anova()`.

Motivating empirical question

*Does an additional year of experience always increase wages by the same amount?
And is the return to experience — or to looks — the same for women as it is for men?*

A linear regression of wages on experience implicitly assumes that the tenth year of experience pays exactly the same as the first. Common sense and decades of labour-economics evidence both suggest otherwise: the wage–experience profile is *concave*. Likewise, a single coefficient on `female` captures an average gap in intercepts but says nothing about whether the *return* to experience differs by gender. The tools in this chapter — quadratics, logs, interactions, dummies, and the Chow test — let us answer questions of this form without abandoning OLS.

6.1. Nonlinear relationships in a linear model

MANY economic relationships are not straight lines. Diminishing returns to experience, inverted-U Kuznets curves, U-shaped average cost curves, and constant-elasticity

6. Nonlinear Relationships and Qualitative Information

demand curves all violate strict linearity in the variables. Fortunately the workhorse model of this course requires linearity only in the **parameters**. A specification such as

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

is *nonlinear* in x but *linear* in $\beta_0, \beta_1, \beta_2$, so it can be estimated by OLS exactly like the multiple regression of Chapter 3, just with x^2 entered as an additional regressor. The same is true of $\ln(y)$, $\ln(x)$, $x_1 \cdot x_2$, $1/x$, and any other transformation of the variables that you can compute before running the regression.

Two practical takeaways follow.

- *Specification is a modelling choice.* Theory, prior empirical work, and a scatter plot of the data should guide whether to use a quadratic, a log, an interaction, or some combination.
- *Interpretation changes.* Once you transform variables, the coefficients no longer have the simple “one-unit increase in x ” reading. The bulk of this chapter is about getting those interpretations right.

i Definition: linear in parameters

A regression model is *linear in parameters* if it can be written as $y = \beta_0 + \beta_1 g_1(\mathbf{x}) + \dots + \beta_k g_k(\mathbf{x}) + u$, where $g_1, \dots, g_k$ are known functions of the regressors. The functions g_j may be highly nonlinear (squares, logs, interactions, ratios), but each enters multiplied by a single unknown coefficient. OLS estimation, the Gauss–Markov theorem, and the inference machinery of Chapters 3–5 all apply.

6.2. Quadratic models

Wages rise quickly in the first years of a career but level off — and may even fall — as workers approach retirement. To let our model bend, add the square of experience as a second regressor:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u.$$

Now the partial effect of x on y is no longer constant:

$$\frac{\partial y}{\partial x} = \beta_1 + 2\beta_2 x.$$

The marginal effect depends on *where* on the x -axis we evaluate it. The signs of β_1 and β_2 determine the shape of the relationship:

- $\beta_1 > 0$, $\beta_2 < 0$: increasing then decreasing — an inverted U with a maximum at $x^* = -\beta_1/(2\beta_2)$.
- $\beta_1 < 0$, $\beta_2 > 0$: decreasing then increasing — a U-shape with a minimum at x^* .
- $\beta_1 > 0$, $\beta_2 > 0$: increasing at an increasing rate (convex and upward-sloping).
- $\beta_1 < 0$, $\beta_2 < 0$: decreasing at an increasing rate.

Setting the partial effect to zero and solving gives the **turning point**

$$x^* = -\frac{\beta_1}{2\beta_2}.$$

Whether x^* is a maximum or a minimum is read off the sign of β_2 (negative for a maximum, positive for a minimum).

i Example: wage and age

A study estimates

$$\widehat{\text{wage}} = -10 + 1.5 \text{ age} - 0.02 \text{ age}^2.$$

The marginal effect of age is $1.5 - 0.04 \text{ age}$.

- At age 20: $1.5 - 0.04(20) = 0.7$ euros per year of age.
- At age 30: $1.5 - 0.04(30) = 0.3$ euros per year.
- At age 40: $1.5 - 0.04(40) = -0.1$ euros per year.

The turning point is $\text{age}^* = 1.5/(2 \cdot 0.02) = 37.5$ years. Before 37.5, an extra year of age raises predicted wages; after 37.5, it lowers them.

⚠ Common mistake: reporting a turning point outside the data

The formula $x^* = -\beta_1/(2\beta_2)$ always exists, but the turning point is only economically meaningful if it lies inside the range of x observed in the sample. If x^* is, say, 75 years of experience and nobody in the sample has more than 50, the “turning point” is an extrapolation and should not be sold as evidence of decreasing returns at experienced workers — the data simply do not cover that region.

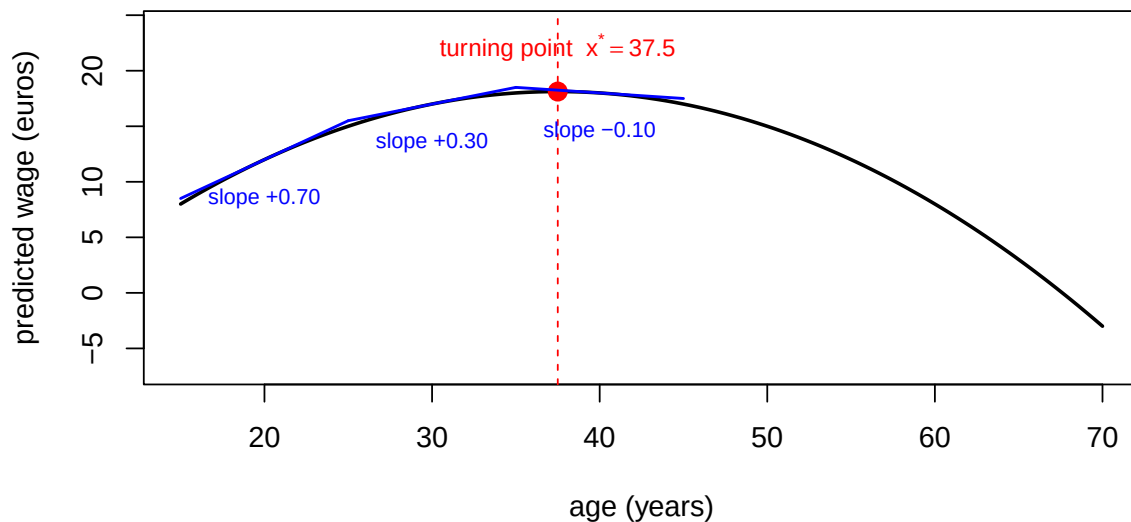


Figure 6.1.: The wage–age parabola $\widehat{\text{wage}} = -10 + 1.5 \text{ age} - 0.02 \text{ age}^2$ from the example above, with the turning point $\text{age}^* = 37.5$ marked (vertical dashed line). The blue tangent segments at ages 20, 30, 40 show the partial effect $\partial y/\partial x$ flattening out as age grows and then turning negative past the peak. A single number ($\hat{\beta}_3$, the slope on the linear term) would have to summarise *all* three tangents at once — which is precisely why a linear specification cannot fit a concave profile.

In R, the squared term is included with `I(exper^2)`. The wrapper `I()` (for “as is”) protects the `^` operator from being parsed by R’s formula language as the symbol for crossed effects.

6.3. Logarithmic transformations

Taking the natural logarithm of one or both sides of the regression equation changes how the coefficient is interpreted. The four cases in common use are summarised below, with β_1 denoting the slope on the (possibly transformed) regressor.

Model	Dependent	Independent	Interpretation of β_1
Level–level	y	x	One-unit $\uparrow$ in $x \Rightarrow \Delta y \approx \beta_1$ units
Log–level	$\ln(y)$	x	One-unit $\uparrow$ in $x \Rightarrow y$ changes by $\approx 100 \beta_1 \%$ (semi-elasticity)
Level–log	y	$\ln(x)$	1% $\uparrow$ in $x \Rightarrow \Delta y \approx \beta_1/100$ units
Log–log	$\ln(y)$	$\ln(x)$	1% $\uparrow$ in $x \Rightarrow y$ changes by $\approx \beta_1 \%$ (elasticity)

Three practical reasons to log a strictly positive variable like wage or income are: to reduce right-skew, to stabilise variance (a partial remedy for some kinds of heteroskedasticity, see Chapter 7), and to give coefficients a percentage reading. The log–log specification is especially common when economic theory delivers a constant-elasticity prediction (e.g. Cobb–Douglas production functions, isoelastic demand curves).

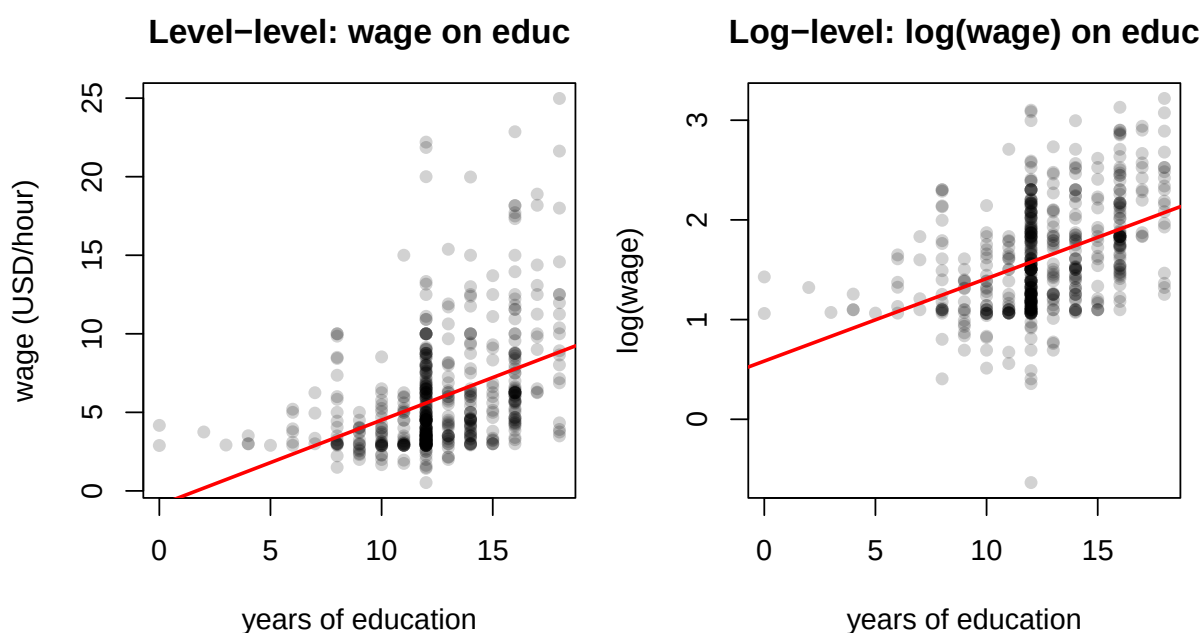


Figure 6.2.: Why logging the dependent variable is so common in wage equations. Left: hourly wages plotted directly — the long right tail of high earners pulls the fitted line and inflates residuals at high education levels. Right: the same fit on $\log(\text{wage})$ — the cloud of points becomes much more symmetric around the line, the slope is read as a *semi-elasticity* (one extra year of education $\Rightarrow$ approximately $100\hat{\beta}_1\%$ change in wage), and the variance of the residuals is much more nearly constant across education levels.

i Example: log-level wage equation

If the estimated equation is

$$\ln(\widehat{\text{wage}}) = 1.5 + 0.02 \text{ age},$$

then an additional year of age is associated with approximately a $0.02 \times 100 = 2\%$ increase in the wage, holding everything else fixed. The approximation $\% \Delta y \approx 100 \beta_1$ is accurate for small β_1 ; for large coefficients the exact percentage change is $100(e^{\beta_1} - 1)$.

! Common mistake: logging zeros or negatives

The natural log is undefined at zero and negative numbers. Before logging a variable, check that all observations are strictly positive. For variables that are zero for many units (e.g. hours worked, donations, exports), $\ln$ is not an option without further care — alternatives include $\ln(1 + x)$, the inverse hyperbolic sine, or a two-part model. We stick to strictly positive variables in this chapter.

6.4. Interaction terms

Does an extra year of schooling pay off the same for men and for women? If we suspect not, we cannot capture that with separate dummies — we need a variable that lets the slope on education itself change with gender. That variable is an **interaction term**: a regressor that lets the partial effect of one variable depend on another. With two continuous regressors x_1 and x_2 ,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 \cdot x_2) + u,$$

the partial effect of x_1 on y is

$$\frac{\partial y}{\partial x_1} = \beta_1 + \beta_3 x_2.$$

It now depends on the level of x_2 . Symmetrically, the partial effect of x_2 depends on x_1 . The coefficient β_3 on the product term tells us how the slope of x_1 changes when x_2 moves by one unit (and vice versa).

The same idea applies when one of the two variables is a dummy (Section 6.5.2) and when both are dummies. A significant interaction means that the effect of one variable on y is *heterogeneous* in the other — a property that matters enormously for policy: a treatment with a positive *average* effect can still hurt particular subgroups if the interaction is large enough in the wrong direction.

! Common mistake: dropping the main effects when including an interaction

A model that contains $x_1 \cdot x_2$ but omits x_1 and/or x_2 on their own forces the partial effect of the omitted main effect through zero. Unless you have a very strong theoretical reason to do so, *always include the main effects whenever you include their interaction*. In R, the syntax `x1*x2` automatically expands to `x1 + x2 + x1:x2`, which prevents this mistake.

⚠ Common mistake: a significant interaction is not heterogeneous causation

A significant $\hat{\beta}_3$ tells you the *conditional mean* of y shifts heterogeneously with the two regressors — this is a statement about description, not about causation. It does *not* by itself certify that the **causal** effect of one regressor varies across groups defined by the other. That stronger interpretation still requires MLR.4 ($\mathbb{E}[u \mid \mathbf{X}] = 0$) to hold for *both* main effects **and** the interaction term. Functional flexibility is not identification: a flexible specification can describe a more nuanced pattern in the data without ruling out the possibility that omitted variables are doing the work. The “correlation is not causation” warning from Chapter 1 applies to interactions exactly as it does to simple slopes.

6.5. Dummy variables

Qualitative information — gender, marital status, region, sector, season — enters a regression via **dummy variables** (also called *indicator* or *binary* variables). A dummy takes the value 1 when the characteristic of interest is present and 0 otherwise:

$$D = \begin{cases} 1 & \text{if characteristic present,} \\ 0 & \text{otherwise.} \end{cases}$$

Three kinds of qualitative variable are typical in applied work.

- *Binary*: two categories, encoded by a single dummy. Example: `female` = 1 for women, 0 for men.
- *Nominal*: more than two, unordered. Example: a variable `platform` with values Instagram, TikTok, X. There is no natural numeric ordering and no notion of “more” or “less”.
- *Ordinal*: more than two, ordered. Example: the `looks` variable in the `beauty` dataset, coded from 1 (below average) to 5 (above average).

For an ordinal variable you have a choice. You can treat it as continuous and include a single slope coefficient (assuming the effect of moving from 2 to 3 equals the effect of moving from 4 to 5), or you can convert it into a set of dummies and let the data tell you whether the steps are equally spaced. The lab in §6.8 does both for comparison.

6.5.1. The dummy variable trap

If a qualitative variable has J categories, you can include at most $J - 1$ dummies in a regression that also contains an intercept. Including all J would create perfect multicollinearity with the constant: the J dummies sum to a column of ones, which is exactly the intercept column. OLS cannot then identify the individual coefficients — the matrix $X'X$ is singular — and any software will either drop one of the dummies automatically or refuse to compute. The dropped category is called the **reference** or **base** category, and every other dummy coefficient is interpreted as a *difference relative to it*.

i Example: intercept dummy for gender

Suppose

$$\widehat{\text{wage}} = 20 - 5 \text{ female} + 2 \text{ age}.$$

- For men (`female` = 0): $\widehat{\text{wage}} = 20 + 2 \text{ age}$.

- For women (female = 1): $\widehat{\text{wage}} = 15 + 2 \text{ age}$.

The dummy shifts the regression line *down* by 5 euros for women but leaves the *slope* on age unchanged. Men and women are predicted to gain the same 2 euros per year of age; only the intercept differs.

6.5.2. Slope dummies (interactions with a dummy)

To allow *both* the intercept and the slope to differ between groups, interact the dummy with a continuous regressor. The model

$$y = \beta_0 + \beta_1 D + \beta_2 x + \beta_3 (D \cdot x) + u$$

has partial effect $\partial y / \partial x = \beta_2 + \beta_3 D$, which equals β_2 for the reference group ($D = 0$) and $\beta_2 + \beta_3$ for the indicator group ($D = 1$). The coefficient β_3 on the interaction is exactly the *difference in slopes* between the two groups, and a *t*-test on β_3 is a test of equal slopes.

i Example: different returns to age by gender

Estimated equation:

$$\widehat{\text{wage}} = 20 - 5 \text{ female} + 2 \text{ age} - 1 (\text{female} \cdot \text{age}).$$

- Men: $\widehat{\text{wage}} = 20 + 2 \text{ age}$. Each year of age adds 2 euros.
- Women: $\widehat{\text{wage}} = 15 + 1 \text{ age}$. Each year of age adds 1 euro.

The intercept dummy (−5) and the slope dummy (−1) capture two different forms of inequality: women start out lower *and* gain less from ageing.

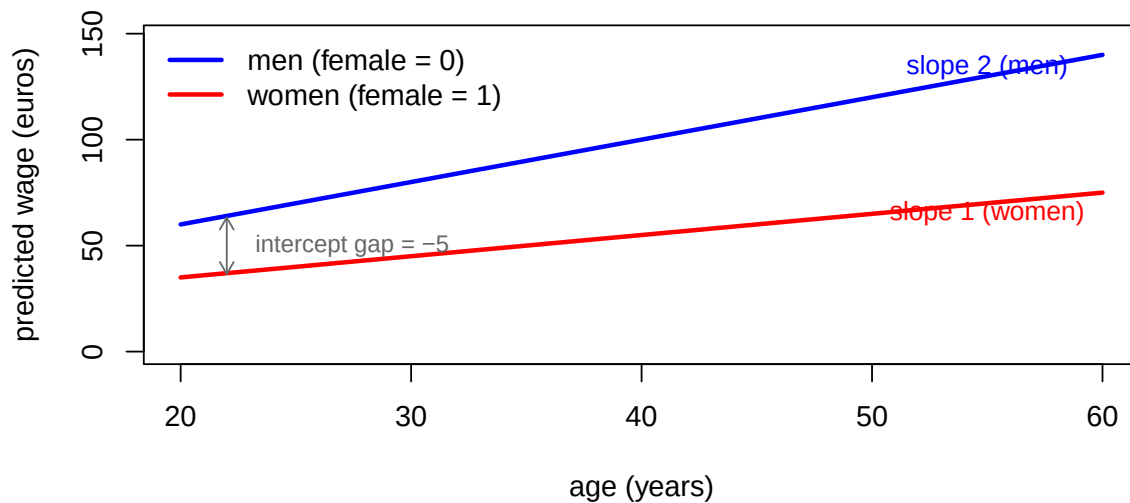


Figure 6.3.: The slope-dummy specification produces *two different lines*, one per group. The intercept dummy (-5) is the vertical gap between the two intercepts; the slope dummy (-1) is the difference in slopes (2 for men vs 1 for women). A t -test on the slope-dummy coefficient is a test of H_0 : equal slopes. Note that the gap between the two lines *grows* with age — this is the heterogeneity that a single $\hat{\beta}_{\text{female}}$ in a pure intercept-dummy regression would miss.

6.5.3. Interactions between two dummies

The same logic extends to the case in which *both* interacted variables are dummies. Consider

$$\ln(\widehat{\text{wage}}) = \hat{\beta}_0 + \hat{\beta}_1 \text{female} + \hat{\beta}_2 \text{married} + \hat{\beta}_3 (\text{female} \cdot \text{married}).$$

With two binary regressors there are only four possible cells, one for each combination of (**female**, **married**). Reading the table of fitted values off the equation:

- *Unmarried men* (female = 0, married = 0): $\hat{\beta}_0$.
- *Unmarried women* (female = 1, married = 0): $\hat{\beta}_0 + \hat{\beta}_1$.
- *Married men* (female = 0, married = 1): $\hat{\beta}_0 + \hat{\beta}_2$.
- *Married women* (female = 1, married = 1): $\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3$.

The female–male gap among the *unmarried* is $\hat{\beta}_1$; among the *married* it is $\hat{\beta}_1 + \hat{\beta}_3$. The coefficient $\hat{\beta}_3$ on the product term is therefore the **difference** between the female–male gap among the married and the female–male gap among the unmarried — equivalently, how the *marriage premium itself* differs by gender. This is the dummy $\times$ dummy reading of a difference-in-differences: a single number summarising the interaction of two binary characteristics.

```
library(wooldridge)
data("wage1")
model_dd <- lm(log(wage) ~ female * married, data = wage1)
summary(model_dd)
```



```

Call:
lm(formula = log(wage) ~ female * married, data = wage1)

Residuals:
    Min       1Q   Median       3Q      Max
-2.0240 -0.3245 -0.0800  0.3155  1.6849

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.52081    0.05099   29.827  < 2e-16 ***
female        -0.13164    0.06680   -1.971   0.0493 *
married         0.42669    0.06155    6.932 1.23e-11 ***
female:married -0.37479    0.08571   -4.373 1.48e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4728 on 522 degrees of freedom
Multiple R-squared:  0.2132,    Adjusted R-squared:  0.2087
F-statistic: 47.15 on 3 and 522 DF,  p-value: < 2.2e-16

```

In this `wage1` regression the coefficient on `female:married` is the estimated *difference* between the female–male log-wage gap among married workers and the same gap among unmarried workers; a *t*-test on the `female:married` row is precisely the test of whether the marriage premium for women equals the marriage premium for men. A negative and statistically significant interaction would say that being female is associated with a *larger* wage penalty among the married than among the unmarried — the marriage premium accrues disproportionately to men. With all three main effects and the interaction included, the four fitted cells above are reproduced exactly by `lm()`; the interaction simply repackages those four group means into a more interpretable set of contrasts.

6.6. The Chow test

Suppose we suspect that the *entire* regression — intercept and every slope — differs between two groups (men vs women, before vs after a reform, treated vs control regions). The **Chow test** is the *F*-test of the joint null

$$H_0 : \text{all coefficients are equal across the two groups.}$$

It compares the residual sum of squares from a **pooled** regression (one equation on the full sample) with the sum of residual sums of squares from **two separate** regressions (one per group).

Let SSR_P denote the pooled SSR, and SSR_1, SSR_2 the SSRs from the two group-specific regressions. With k slope coefficients in the (common) specification and $n = n_1 + n_2$ observations,

$$F = \frac{(SSR_P - SSR_1 - SSR_2)/(k+1)}{(SSR_1 + SSR_2)/(n - 2(k+1))} \sim F_{k+1, n-2(k+1)} \text{ under } H_0.$$

The numerator degrees of freedom $k+1$ counts the coefficients that are restricted to be equal under the null (the intercept plus the k slopes). The denominator degrees of freedom $n - 2(k+1)$ comes from running two separate regressions, each with $k+1$ parameters.

6. Nonlinear Relationships and Qualitative Information

Algebraically, the same test can be run by pooling the data, fully interacting the group dummy with every regressor, and using `anova()` to compare the restricted and unrestricted models. The lab in §6.8 verifies that the two approaches give identical F -statistics and p -values.

⚠ Common mistake: forgetting that Chow assumes equal error variances

The Chow F -statistic is derived under the assumption that the error variance is the same in the two groups. If the two subsamples have very different residual variances, the test can be misleading. The cleanest fix is a Wald test with heteroskedasticity-robust standard errors on the fully-interacted regression (Chapter 7). Logging the dependent variable sometimes also reduces variance heterogeneity, but it is *not* a substitute: it changes the question being asked — you are now testing equality of *log-wage* equations, not *wage* equations, and the two test results need not agree.

6.7. Lab A: nonlinear forms in the beauty dataset (Tutorial 6)

The `beauty` dataset from the `wooldridge` package contains hourly wages, years of education and experience, an ordinal `looks` score from 1 to 5, and dummies for gender and marital status. We use it to fit a linear baseline, add a quadratic term in experience, log the wage, and finally introduce an interaction between looks and education.

```
library(wooldridge)
data("beauty")
```

6.7.1. A linear baseline

```
model_lin <- lm(wage ~ looks + educ + exper, data = beauty)
summary(model_lin)
```

Call:

```
lm(formula = wage ~ looks + educ + exper, data = beauty)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.248	-2.263	-0.821	1.272	71.925

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.82166	0.85684	-3.293	0.00102 **
looks	0.41300	0.18318	2.255	0.02433 *
educ	0.45703	0.04807	9.508	< 2e-16 ***
exper	0.11374	0.01055	10.785	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.361 on 1256 degrees of freedom

Multiple R-squared: 0.1265, Adjusted R-squared: 0.1244
 F-statistic: 60.62 on 3 and 1256 DF, p-value: < 2.2e-16

The baseline shows the usual signs: education and looks are associated with higher wages, experience too. But the linear specification forces the marginal return to experience to be the same at year 1 and year 30 — a strong restriction.

6.7.2. Adding a quadratic term in experience

```
model_quad <- lm(wage ~ looks + educ + exper + I(exper^2), data = beauty)
summary(model_quad)
```

Call:

```
lm(formula = wage ~ looks + educ + exper + I(exper^2), data = beauty)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.164	-2.276	-0.694	1.200	71.927

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-3.795520	0.872656	-4.349	1.48e-05	***
looks	0.436807	0.181628	2.405	0.0163	*
educ	0.422904	0.048156	8.782	< 2e-16	***
exper	0.299390	0.039670	7.547	8.53e-14	***
I(exper^2)	-0.004327	0.000892	-4.851	1.38e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.323 on 1255 degrees of freedom

Multiple R-squared: 0.1426, Adjusted R-squared: 0.1398

F-statistic: 52.17 on 4 and 1255 DF, p-value: < 2.2e-16

Reading the quadratic regression output

The column structure is the same Rosetta stone as in Chs 2–4. What is new here is the `I(exper^2)` row. Map the rows of `summary(model_quad)` to the parameters of the quadratic specification $wage = \beta_0 + \beta_1 \text{looks} + \beta_2 \text{educ} + \beta_3 \text{exper} + \beta_4 \text{exper}^2 + u$ as follows:

- `(Intercept)` $\rightarrow \hat{\beta}_0$.
- `looks` $\rightarrow \hat{\beta}_1$.
- `educ` $\rightarrow \hat{\beta}_2$.
- `exper` $\rightarrow \hat{\beta}_3$.
- `I(exper^2)` $\rightarrow \hat{\beta}_4$. The `I(...)` wrapper is R-syntax, not part of the variable's name; R is just protecting `^` from being parsed as a formula operator (it normally means “all interactions up to order N”). The coefficient sitting on this row is the slope on `exper`².

Reading the partial effect for `exper` requires *both* $\hat{\beta}_3$ and $\hat{\beta}_4$ at once — the partial effect

is $\hat{\beta}_3 + 2\hat{\beta}_4 \text{exper}$, evaluated at whatever level of experience interests you. This is the operational sense in which a quadratic captures diminishing (or accelerating) returns: the slope on each row, taken alone, is half of the story.

The coefficient on `exper` should be positive and the coefficient on `I(exper^2)` negative — the classic concave wage–experience profile. The marginal effect $\partial \text{wage} / \partial \text{exper} = \hat{\beta}_3 + 2\hat{\beta}_4 \text{exper}$ can be evaluated at several experience levels:

```
b3 <- coef(model_quad)["exper"]
b4 <- coef(model_quad)["I(exper^2)"]

# Range of experience in the sample
range(beauty$exper)
```

```
[1] 0 48
```

```
# Marginal effect at exper = 10, 20, 30, 40
sapply(c(10, 20, 30, 40), function(x) b3 + 2 * b4 * x)
```

```
      exper      exper      exper      exper
0.21284197 0.12629376 0.03974555 -0.04680266
```

The marginal return falls as experience accumulates. The **turning point** is

```
exp_star <- -b3 / (2 * b4)
exp_star
```

```
      exper
34.5923
```

Provided `exp_star` falls inside the observed range of `exper`, the parabola has a meaningful maximum: beyond that point the predicted wage starts to fall. If it falls outside the range, we can only say that returns are diminishing throughout the observed sample.

6.7.3. Logging the wage

To shift to a semi-elasticity interpretation, take logs on the left-hand side:

```
model_logq <- lm(log(wage) ~ looks + educ + exper + I(exper^2), data = beauty)
summary(model_logq)
```

Call:

```
lm(formula = log(wage) ~ looks + educ + exper + I(exper^2), data = beauty)
```

Residuals:

```
      Min      1Q   Median      3Q      Max
```

```
-1.62220 -0.31622  0.02116  0.31609  2.78775
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.0462774  0.1053330   0.439  0.66049
looks        0.0620681  0.0219232   2.831  0.00471 **
educ         0.0683689  0.0058127  11.762 < 2e-16 ***
exper        0.0487326  0.0047884  10.177 < 2e-16 ***
I(exper^2)   -0.0006985  0.0001077  -6.487 1.26e-10 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5217 on 1255 degrees of freedom

Multiple R-squared: 0.2323, Adjusted R-squared: 0.2298

F-statistic: 94.91 on 4 and 1255 DF, p-value: < 2.2e-16

The coefficient on `educ` is now read as “an extra year of education is associated with a $100 \cdot \hat{\beta}_{\text{educ}}$ per cent increase in hourly wages, holding looks and experience fixed”. The marginal **percentage** effect of one extra year of experience at a given level x is approximately $100 (\hat{\beta}_{\text{exper}} + 2\hat{\beta}_{\text{exper}^2} x)$:

```
b3l <- coef(model_logq)["exper"]
b4l <- coef(model_logq)["I(exper^2)"]
sapply(c(10, 30), function(x) 100 * (b3l + 2 * b4l * x))
```

```
      exper      exper
3.4763529 0.6825305
```

A worker with 10 years of experience gains a different *percentage* per additional year than a worker with 30.

6.7.4. Interaction: looks $\times$ education

Does the wage premium associated with looking better depend on how much schooling a worker has? The `looks * educ` shorthand expands to `looks + educ + looks:educ`:

```
model_int <- lm(log(wage) ~ looks * educ + exper + I(exper^2), data = beauty)
summary(model_int)
```

Call:

```
lm(formula = log(wage) ~ looks * educ + exper + I(exper^2), data = beauty)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-1.62325 -0.31741  0.02058  0.31601  2.78849
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0870703  0.3103984  -0.281  0.77913
```

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```
looks      0.1053823  0.0973400   1.083  0.27918
educ       0.0790930  0.0241898   3.270  0.00111 **
exper      0.0487614  0.0047903  10.179 < 2e-16 ***
I(exper^2) -0.0006988  0.0001077  -6.488 1.25e-10 ***
looks:educ -0.0034672  0.0075915  -0.457  0.64795
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5219 on 1254 degrees of freedom

Multiple R-squared: 0.2324, Adjusted R-squared: 0.2293

F-statistic: 75.92 on 5 and 1254 DF, p-value: < 2.2e-16

The marginal effect of `looks` on $\ln(\text{wage})$ is $\hat{\beta}_{\text{looks}} + \hat{\beta}_{\text{looks:educ}} \cdot \text{educ}$. Multiplied by 100 it is the percentage wage change associated with a one-unit increase in `looks`:

```
b_looks <- coef(model_int)["looks"]
b_int   <- coef(model_int)["looks:educ"]
me <- function(e) 100 * (b_looks + b_int * e)
sapply(c(12, 16), me)
```

```
      looks      looks
6.377586 4.990705
```

A simple t -test on the interaction coefficient (the line `looks:educ` in the summary table) tells us whether the premium genuinely varies with education or is statistically indistinguishable from a constant.

6.8. Lab B: dummies and the Chow test (Tutorial 7)

This lab returns to the gender question that opened the chapter. We use the same `beauty` data, build categorical dummies from the ordinal `looks` variable, and finish with a manual Chow test.

```
library(wooldridge)
data("beauty")
table(beauty$female)
```

```
 0    1
824 436
```

```
table(beauty$married)
```

```
 0    1
389 871
```

```
table(beauty$looks)
```

```
 1   2   3   4   5
13 142 722 364  19
```

6.8.1. Intercept dummies for gender and marriage

```
model_d <- lm(wage ~ looks + educ + exper + female + married, data = beauty)
summary(model_d)
```

Call:

```
lm(formula = wage ~ looks + educ + exper + female + married,
    data = beauty)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-7.520 -2.190 -0.599   1.149 72.998
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.66350	0.86433	-1.925	0.0545 .
looks	0.39694	0.17618	2.253	0.0244 *
educ	0.44112	0.04623	9.542	< 2e-16 ***
exper	0.08280	0.01065	7.778	1.53e-14 ***
female	-2.37281	0.26662	-8.899	< 2e-16 ***
married	0.69044	0.27502	2.511	0.0122 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.192 on 1254 degrees of freedom

Multiple R-squared: 0.1944, Adjusted R-squared: 0.1912

F-statistic: 60.52 on 5 and 1254 DF, p-value: < 2.2e-16

The coefficient on `female` is the estimated wage gap between women and men, holding looks, education, and experience fixed; the coefficient on `married` is the marriage premium. Both shifts apply only to the intercept — the slopes on the continuous regressors are constrained to be the same for both groups.

6.8.2. Slope dummy: does the return to looks differ by gender?

```
model_slope <- lm(wage ~ looks * female + educ + exper + married, data = beauty)
summary(model_slope)
```

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Call:

```
lm(formula = wage ~ looks * female + educ + exper + married,
    data = beauty)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-7.483 -2.200 -0.582  1.135  72.936
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.52087    0.95946  -1.585   0.1132
looks         0.35024    0.22272   1.573   0.1161
female       -2.76205    1.16595  -2.369   0.0180 *
educ          0.44150    0.04626   9.544 < 2e-16 ***
exper         0.08270    0.01065   7.764 1.71e-14 ***
married       0.69404    0.27531   2.521   0.0118 *
looks:female  0.12209    0.35602   0.343   0.7317
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.193 on 1253 degrees of freedom

Multiple R-squared: 0.1945, Adjusted R-squared: 0.1906

F-statistic: 50.41 on 6 and 1253 DF, p-value: < 2.2e-16

```
b_looks <- coef(model_slope)["looks"]
b_int   <- coef(model_slope)["looks:female"]
me_men  <- b_looks
me_women <- b_looks + b_int
c(men = me_men, women = me_women)
```

```
men.looks women.looks
0.3502386  0.4723292
```

The `looks:female` coefficient is the difference in the slope of `looks` between women and men. A *t*-test on that single coefficient is precisely the test of equal slopes; if it is small in absolute value, the marginal return to looks is statistically the same for both groups.

6.8.3. Categorical dummies from the ordinal looks

Treating `looks` as continuous imposes equal spacing between adjacent categories. A more flexible approach is to build dummies for the categories and let the data speak. We collapse `looks` into three bins — below average (≤ 2), average ($= 3$), above average (≥ 4) — and use `average` as the omitted reference:

```
beauty$below_av <- ifelse(beauty$looks <= 2, 1, 0)
beauty$ab_av    <- ifelse(beauty$looks >= 4, 1, 0)
# 'average' (looks == 3) is the reference category and is omitted.
```



```
model_cat <- lm(wage ~ below_av + ab_av + educ + exper + female + married,
               data = beauty)
summary(model_cat)
```

Call:

```
lm(formula = wage ~ below_av + ab_av + educ + exper + female +
    married, data = beauty)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-7.351 -2.198 -0.551  1.188 73.141
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.31216	0.69931	-0.446	0.6554
below_av	-0.87991	0.37197	-2.366	0.0182 *
ab_av	0.07682	0.26956	0.285	0.7757
educ	0.44377	0.04618	9.610	< 2e-16 ***
exper	0.08113	0.01066	7.614	5.21e-14 ***
female	-2.36363	0.26669	-8.863	< 2e-16 ***
married	0.67890	0.27499	2.469	0.0137 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.191 on 1253 degrees of freedom

Multiple R-squared: 0.1952, Adjusted R-squared: 0.1914

F-statistic: 50.65 on 6 and 1253 DF, p-value: < 2.2e-16

The coefficient on `below_av` is the wage gap (in 1976 USD per hour) between “below average looks” workers and the omitted “average looks” group, holding all controls fixed. The coefficient on `ab_av` is the analogous gap for “above average”. If the two gaps are roughly symmetric around zero — $\hat{\beta}_{\text{below_av}} \approx -\hat{\beta}_{\text{ab_av}}$ — the continuous specification was a reasonable simplification; if not, the dummies are revealing asymmetry.

6.8.4. Chow test for a structural break by gender

We now ask whether the *whole* wage equation differs between men and women, not just the intercept or the slope on a single regressor. The Chow test compares one pooled regression on the full sample with two separate regressions, one per gender. Note that `female` is dropped from the subgroup regressions because it is constant within each subsample:

```
# Pooled regression (full sample)
model_p <- lm(wage ~ below_av + ab_av + educ + exper + female + married,
             data = beauty)

# Group-specific regressions
model_f <- lm(wage ~ below_av + ab_av + educ + exper + married,
             data = beauty, subset = female == 1)
```

```

model_m <- lm(wage ~ below_av + ab_av + educ + exper + married,
              data = beauty, subset = female == 0)

SSR_P <- sum(residuals(model_p)^2)
SSR_F <- sum(residuals(model_f)^2)
SSR_M <- sum(residuals(model_m)^2)

n <- nrow(beauty)
k <- length(coef(model_p)) - 1 # number of slope coefficients in pooled model

F_chow <- ((SSR_P - (SSR_F + SSR_M)) / (k + 1)) /
          ((SSR_F + SSR_M) / (n - 2 * (k + 1)))
p_value <- 1 - pf(F_chow, df1 = k + 1, df2 = n - 2 * (k + 1))

c(F = F_chow, df1 = k + 1, df2 = n - 2 * (k + 1), p_value = p_value)

```

	F	df1	df2	p_value
	1.4767802	7.0000000	1246.0000000	0.1713911

If `F_chow` exceeds the critical value (or equivalently if `p_value` is below your chosen significance level), reject H_0 and conclude that the wage equation is not the same for men and women — some combination of intercept and slopes differs.

6.8.5. Chow via `anova()`

The same test can be run on the pooled sample by interacting `female` with every other regressor and comparing the restricted and unrestricted models with `anova()`:

```

mR <- lm(wage ~ below_av + ab_av + educ + exper + married + female,
          data = beauty)
mU <- lm(wage ~ female * (below_av + ab_av + educ + exper + married),
          data = beauty)
anova(mR, mU)

```

Analysis of Variance Table

```

Model 1: wage ~ below_av + ab_av + educ + exper + married + female
Model 2: wage ~ female * (below_av + ab_av + educ + exper + married)
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1   1253 22009
2   1248 21828  5    181.09 2.0708 0.06657 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

The F -statistic, degrees of freedom and p -value match the manual computation. This second form is often more convenient because it lends itself naturally to robust standard errors (Chapter 7) and to tests of partial structural breaks (only some coefficients allowed to differ).

Self-check

Six short multiple-choice questions. Try each one before opening the answer.

💡 Q1. Marginal effect in a quadratic model

In the model $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$, the marginal effect of x on y is:

- A. β_1 , constant.
- B. β_2 , constant.
- C. $\beta_1 + 2\beta_2 x$ — it depends on x .
- D. Always positive.

Answer: C. Differentiating the model with respect to x gives $\partial y / \partial x = \beta_1 + 2\beta_2 x$. The marginal effect varies with the level at which it is evaluated, which is exactly why a quadratic specification captures diminishing (or accelerating) returns.

💡 Q2. Interpreting a log-level coefficient

In a regression $\ln(y) = \beta_0 + \beta_1 x + u$, the coefficient β_1 is approximately:

- A. The change in y for a one-unit change in x .
- B. The percentage change in y for a one-unit change in x (about $100\beta_1$ per cent).
- C. The elasticity of y with respect to x .
- D. The level change in y when x doubles.

Answer: B. A log-level coefficient is a **semi-elasticity**: a one-unit increase in x is associated with approximately a $100\beta_1$ per cent change in y . Elasticities (option C) come from log-log specifications.

💡 Q3. Why is logging often useful?

A common reason for taking the log of a strictly positive variable like wage is:

- A. To violate the Gauss–Markov assumptions on purpose.
- B. Because R^2 is always higher in log models.
- C. To reduce right-skew, partially stabilise the variance, and obtain coefficients with a percentage interpretation.
- D. To make the regressor normally distributed.

Answer: C. All three motivations matter in applied work. Logging often turns a skewed wage distribution into something closer to symmetric, dampens variance that grows with the level, and gives semi-elasticities or elasticities that are easy to communicate.

💡 Q4. Interpreting an interaction

If $\hat{\beta}_3$ in the interaction model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 \cdot x_2) + u$ is statistically significant, then:

- A. The effect of x_1 on y is the same regardless of x_2 .
- B. x_1 and x_2 have no separate effects.

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- C. The effect of x_1 on y is *not* constant: it changes with x_2 .
- D. OLS becomes biased.

Answer: C. A significant interaction means heterogeneous slopes. The partial effect of x_1 is $\beta_1 + \beta_3 x_2$, which varies with x_2 . Causally, this is the language of *heterogeneous treatment effects*.

💡 Q5. The dummy variable trap

A categorical variable has 4 categories. Including an intercept, you should add:

- A. 4 dummies, one per category.
- B. 3 dummies; the fourth category is the omitted reference.
- C. 2 dummies.
- D. 1 dummy.

Answer: B. With $J = 4$ categories, including all four dummies plus the intercept creates perfect multicollinearity: the four dummies sum to the column of ones. Drop one to serve as the reference category, and interpret the remaining three coefficients as differences relative to it.

💡 Q6. What the Chow test tests

The Chow test for a structural break across two groups tests the null hypothesis:

- A. All coefficients (intercept and slopes) are equal across the two groups.
- B. Only the intercept differs.
- C. Only the slope on one regressor differs.
- D. The two error variances are equal.

Answer: A. The Chow F -statistic compares the pooled-sample SSR with the sum of the two group-specific SSRs. Under H_0 every coefficient — the intercept and all k slopes — is the same in both groups. Rejecting H_0 says only that *some* coefficient differs; it does not by itself tell you which one.

The interactive `learnr` version with twelve questions and live R code is in `LEARNR/06_nonlinear_and_qualitati`

Exercises

Exercise 6.1 — Marginal effect and turning point. Using `wage1` from the `wooldridge` package, estimate

$$\ln(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + u.$$

- Report $\hat{\beta}_2$, $\hat{\beta}_3$, and their standard errors.
- Compute the marginal *percentage* return to experience at `exper = 5, 15, 25`.
- Find the experience level x^* that maximises predicted log wage and check that it lies inside the data range.

A full answer is given in the Instructor Edition.

Exercise 6.2 — How many dummies? A categorical variable `region` takes the values North, South, East, West.

- (a) How many dummies should you include in a regression that contains an intercept? Why?
- (b) Write the model with the intercept and $J - 1$ dummies and explain how to interpret each coefficient.
- (c) Describe in one sentence what would change if you instead dropped the intercept and included all four dummies.

A full answer is given in the Instructor Edition.

Exercise 6.3 — Chow test for married vs unmarried. Using `wage1`, perform a Chow test for whether the wage equation

$$\ln(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{tenure} + u$$

differs between married and unmarried workers. Report (i) the three SSRs, (ii) the F -statistic with its two degrees of freedom, and (iii) the p -value. Verify your manual computation against `anova()` on the equivalent fully-interacted model.

A full answer is given in the Instructor Edition.

Exercise 6.4 — Slope dummy by gender. Using `wage1`, estimate

$$\ln(\text{wage}) = \beta_0 + \beta_1 \text{female} + \beta_2 \text{educ} + \beta_3 (\text{female} \cdot \text{educ}) + u.$$

- (a) Interpret $\hat{\beta}_3$ in plain English.
- (b) What is the estimated percentage return to one extra year of education for men? For women?
- (c) Test $H_0 : \beta_3 = 0$ at the 5% level and state the economic conclusion.

A full answer is given in the Instructor Edition.

Exercise 6.5 — Ordinal as continuous vs as dummies. Using `beauty`:

- (a) Estimate $\text{wage} = \beta_0 + \beta_1 \text{looks} + \beta_2 \text{educ} + \beta_3 \text{exper} + u$, treating `looks` as continuous.
- (b) Re-estimate the same model after replacing `looks` with the dummies `below_av` and `ab_av` defined in §6.8.
- (c) Compare the implied wage gaps (below average vs average, above average vs average). Are they roughly symmetric around zero? What does that tell you about the linearity assumption embedded in (a)?

A full answer is given in the Instructor Edition.

Exercise 6.6 — Why the two Chow forms agree. Show algebraically that running two separate regressions on the male and female subsamples gives exactly the same fitted values, residuals, and SSR as running the pooled regression with `female` fully interacted with every regressor. Use that fact to argue that the manual three-SSR Chow F -statistic and the `anova()` F -statistic must be numerically identical.

A full answer is given in the Instructor Edition.

Heteroskedasticity

Learning outcomes

By the end of this chapter the reader should be able to:

- State the homoskedasticity assumption MLR.5 in Wooldridge’s notation and recognise when it is likely to fail in cross-sectional data.
- Explain why heteroskedasticity leaves OLS unbiased and consistent but destroys efficiency and invalidates the usual standard errors, t -tests, and F -tests.
- Detect heteroskedasticity informally with residual plots and formally with the Goldfeld–Quandt and Breusch–Pagan tests.
- Correct heteroskedasticity by Weighted Least Squares (WLS) when the form of the conditional variance is known.
- Compute and report heteroskedasticity-robust (White / HC1) standard errors when the form is unknown.
- Build a side-by-side table comparing OLS standard errors with robust standard errors and interpret the ratio.

Motivating empirical question

Why do conventional OLS confidence intervals understate the true uncertainty in a wage regression with very dispersed top earners — and what should we do about it?

A regression of hourly wages on years of education delivers a clean point estimate of the return to schooling. But the spread of wages around the regression line is **not** the same for high-school dropouts and for PhDs: graduates fan out across a much wider range of salaries. The textbook OLS formula for the standard error of $\hat{\beta}_1$ assumes the spread is constant. When it is not, the reported standard errors are wrong, the t -statistics are wrong, and the policy recommendations built on them are wrong. This chapter shows how to diagnose the problem and how to fix it.

A separation of concerns up front: this chapter is about **inference given identification**, not about identification itself. OLS coefficient estimates remain meaningful as estimates of conditional means even when standard errors are wrong; we fix the standard errors here, not the causal question. The “correlation is not causation” thread from Chapters 1–5 still applies — nothing in this chapter rescues a regression with omitted-variable bias. We are repairing the inferential machinery on top of OLS, not the OLS slope itself.

7.1. Introduction

IN Chapter 4 we listed the multiple linear regression assumptions MLR.1–MLR.6 that justify OLS. The **homoskedasticity** assumption MLR.5 states that the error variance is the same for every value of the regressors:

$$\text{Var}(u_i | \mathbf{x}_i) = \sigma^2, \quad i = 1, \dots, n.$$

When MLR.5 fails, the conditional variance depends on $\mathbf{x}_i$:

$$\text{Var}(u_i | \mathbf{x}_i) = \sigma_i^2,$$

and the error is said to be **heteroskedastic**. The Greek root is helpful: *homo-skedastic* means “same spread”; *hetero-skedastic* means “different spread”.

i Example: wage variability and education

Workers with only primary education in a typical European labour market earn between roughly 15{,}000 and 25{,}000 euros per year — a narrow range. University graduates earn anywhere from 20{,}000 to 200{,}000 euros — a much wider range. The conditional variance of wages clearly increases with education. The error term in a wage equation $\text{wage}_i = \beta_0 + \beta_1 \text{educ}_i + u_i$ therefore exhibits heteroskedasticity: $\text{Var}(u_i | \text{educ}_i)$ grows with educ_i .

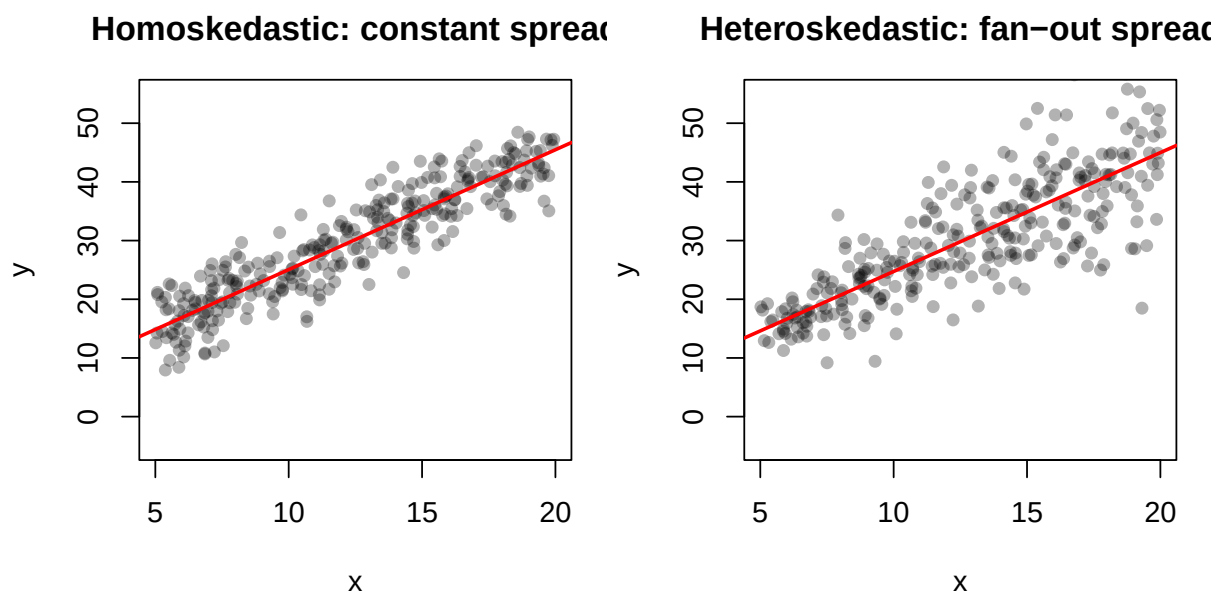


Figure 7.1.: Two simulated regression scatters with the *same* slope and intercept — only the variance structure differs. Left: $\text{Var}(u_i | x_i) = \sigma^2$ is constant (the homoskedastic ideal of MLR.5). The vertical spread of the cloud is uniform across the horizontal axis. Right: $\text{Var}(u_i | x_i) \propto x_i$ — the wedge or fan shape that is the *visual fingerprint* of heteroskedasticity. The two fitted red lines are essentially identical (heteroskedasticity does not bias the slope); what differs is *how informative* each part of the scatter is about that slope.

Heteroskedasticity is **not** a problem of identification. As long as $\mathbb{E}[u_i | \mathbf{x}_i] = 0$ (assumption MLR.4) holds, the *meaning* of β_j as a *ceteris paribus* effect survives. What heteroskedasticity breaks is **inference** — the machinery that turns a point estimate $\hat{\beta}_j$ into a confidence interval or a hypothesis test.

⚠ Common mistake: confusing heteroskedasticity with omitted variable bias

Heteroskedasticity is about $\text{Var}(u_i | \mathbf{x}_i)$; omitted variable bias is about $\mathbb{E}[u_i | \mathbf{x}_i]$. The first leaves $\hat{\beta}_j$ unbiased and only damages the standard errors. The second biases $\hat{\beta}_j$ itself. The two problems require different tools, and the same dataset can suffer from one, the other, both, or neither.

i Where heteroskedasticity typically appears

Heteroskedasticity is the **rule rather than the exception** in cross-sectional microeconomic data, for at least three reasons:

- **Scale effects.** Larger units have more room to vary. Spending by large firms varies more in absolute terms than spending by small firms.
- **Discrete and bounded outcomes.** A binary dependent variable mechanically has $\text{Var}(y | \mathbf{x}) = p(\mathbf{x})[1 - p(\mathbf{x})]$, which is not constant.
- **Aggregation.** Means computed over groups of different sizes have variances proportional to $1/n_g$, generating heteroskedasticity by construction.

These are precisely the data structures applied economists work with, which is why robust standard errors have become the default in modern practice.

7.2. Consequences of heteroskedasticity

Assume MLR.1–MLR.4 hold and only MLR.5 fails. Three statements summarise the consequences:

1. **OLS is still unbiased and consistent.** Unbiasedness of $\hat{\beta}_j$ requires only MLR.1–MLR.4. Heteroskedasticity does not enter the proof.
2. **OLS is no longer BLUE.** The Gauss–Markov theorem requires homoskedasticity. Under heteroskedasticity, *other* linear unbiased estimators can have a smaller variance than OLS. We will see one such estimator, WLS, in §7.4.
3. **The usual OLS standard errors are wrong.** The textbook formula $\widehat{\text{Var}}(\hat{\beta}_j) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})_{jj}^{-1}$ assumes the error variance-covariance matrix is $\sigma^2\mathbf{I}_n$. When it is not, this formula is biased. In the most common case — variance that *increases* with the regressors, the typical “fan” shape illustrated above — OLS standard errors are *too small* and *t*-statistics *too large*, so confidence intervals are too narrow and we reject the null more often than we should. The opposite direction is possible too: when high-variance observations happen to have *low leverage* on the slope, robust standard errors can come out *smaller* than OLS. This is less common in practice but worth knowing so the diagnosis is “the OLS formula is wrong” rather than “the OLS formula is always optimistic”.

i The correct (sandwich) variance under heteroskedasticity

If we collect the conditional variances on the diagonal of a matrix $\Omega = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, the *correct* conditional variance of the OLS estimator is

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\Omega\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}.$$

The “bread” $(\mathbf{X}'\mathbf{X})^{-1}$ surrounds the “meat” $\mathbf{X}'\Omega\mathbf{X}$ — which is why this form is universally called a **sandwich** estimator. The usual OLS formula simply replaces Ω by $\sigma^2\mathbf{I}_n$ and collapses the sandwich to a single slice. The White / HC1 estimator we use in §7.4 estimates Ω by $\text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2)$, with a small-sample degrees-of-freedom correction.

In short: heteroskedasticity is a *standard-error* problem, not a *coefficient* problem. The fix is to repair the inferential machinery (robust SEs) or, if we can afford the extra modelling, to use an estimator that exploits the variance structure (WLS).

7.3. Detection of heteroskedasticity

We describe one informal diagnostic and two formal tests. The informal plot is cheap and should be the first thing you look at; the formal tests give a structured decision rule.

7.3.1. Informal detection: residual plots

After estimating the OLS model, compute the residuals $\hat{u}_i = y_i - \hat{y}_i$ and plot them against either the fitted values $\hat{y}_i$ or against each regressor x_j . The visual fingerprint of heteroskedasticity is a **fan** or **wedge** shape: the vertical spread of the residual cloud widens (or narrows) as you move along the horizontal axis. Squared residuals $\hat{u}_i^2$ against x_j make the pattern even more obvious because the sign of the residual no longer matters.

If the cloud looks like a horizontal band of roughly constant thickness, homoskedasticity is plausible. If it spreads out — as it typically does when x_j is income, firm size, or any other “scale” variable — heteroskedasticity is the natural suspect.

7.3.2. The Goldfeld–Quandt test

The Goldfeld–Quandt (GQ) test is appropriate when you suspect heteroskedasticity is driven by **one specific** regressor. The idea is simple: split the sample into a low- x group and a high- x group and compare the residual variances of the two subsamples.

Steps.

1. Order the observations by the suspected variable x .
2. Optionally drop the middle m observations (often $m \approx n/3$) to accentuate the contrast.
3. Estimate the model separately on the low group (subsample 1) and on the high group (subsample 2). Obtain the sums of squared residuals SSR_1 and SSR_2 , with degrees of freedom df_1 and df_2 .
4. Compute

$$F = \frac{\text{SSR}_2 / \text{df}_2}{\text{SSR}_1 / \text{df}_1} \sim F_{\text{df}_2, \text{df}_1} \quad \text{under } H_0.$$

Hypotheses. H_0 : homoskedasticity. H_1 : $\text{Var}(u)$ increases with x (the most common alternative). Reject H_0 when F exceeds the upper-tail critical value of the F distribution.

The GQ test was historically popular in small samples and is intuitive, but it requires the analyst to *commit* to a single sorting variable and to a sample-splitting rule. The Breusch–Pagan test, introduced next, drops both of these commitments.

7.3.3. The Breusch–Pagan test

Start from the picture, not the formula. If heteroskedasticity exists, then $\text{Var}(u_i | \mathbf{x}_i)$ moves with $\mathbf{x}_i$ — and because the squared residual $\hat{u}_i^2$ is our sample window onto the unobserved u_i^2 , large squared residuals should *cluster* at certain values of $\mathbf{x}$ rather than scatter evenly. So the natural diagnostic is: regress $\hat{u}_i^2$ on $\mathbf{x}_i$ and see if anything explains it. If the regressors explain a non-trivial share of the variation in $\hat{u}_i^2$, then $\text{Var}(u | \mathbf{x})$ depends on $\mathbf{x}$ — which is the definition of heteroskedasticity. If they explain essentially nothing, the squared residuals look like noise around a constant, and homoskedasticity is plausible.

That is the whole Breusch–Pagan (BP) idea. Unlike Goldfeld–Quandt, it detects heteroskedasticity related to **any** of the regressors, without committing to a single sorting variable. It is the workhorse formal test in modern applied work, and the test statistic below is just a way of asking “is the auxiliary R^2 large enough to take seriously?”

Steps.

1. Estimate the original model by OLS and store the residuals $\hat{u}_i$.
2. Run the **auxiliary regression** of the squared residuals on all the original regressors:

$$\hat{u}_i^2 = \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \cdots + \gamma_k x_{ki} + v_i.$$

3. Compute the BP statistic

$$\text{BP} = n \cdot R_{\text{aux}}^2 \sim \chi_k^2 \quad \text{under } H_0,$$

where R_{aux}^2 is the coefficient of determination of the auxiliary regression and k is the number of slope coefficients in it.

Hypotheses. $H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_k = 0$ (homoskedasticity). Reject when $\text{BP} > \chi_{k, \alpha}^2$.

A close cousin — the **White test** — adds squares and cross-products of the regressors to the auxiliary regression and so detects more general forms of heteroskedasticity, at the cost of degrees of freedom.

 **Common mistake:** chaining tests until one rejects

Do not run GQ, then BP, then White, and report whichever rejects “first”. This is a form of *p*-hacking. Pick one test based on what you suspect *before* looking at the residuals, or — following modern practice — skip the testing step altogether and report robust standard errors by default.

7.4. Remedies

There are two main remedies, suited to two different states of knowledge.

7.4.1. Weighted Least Squares (WLS)

Suppose we know (or are willing to model) the form of the conditional variance:

$$\text{Var}(u_i | \mathbf{x}_i) = \sigma^2 h(\mathbf{x}_i),$$

where $h(\mathbf{x}_i) > 0$ is a known function of the regressors. A typical example is $h(\mathbf{x}_i) = x_i$ when the variance is proportional to a scale variable such as income.

Divide every term in the regression by $\sqrt{h(\mathbf{x}_i)}$:

$$\frac{y_i}{\sqrt{h(\mathbf{x}_i)}} = \beta_0 \frac{1}{\sqrt{h(\mathbf{x}_i)}} + \beta_1 \frac{x_i}{\sqrt{h(\mathbf{x}_i)}} + \frac{u_i}{\sqrt{h(\mathbf{x}_i)}}.$$

The new error $u_i^* = u_i / \sqrt{h(\mathbf{x}_i)}$ has constant variance:

$$\text{Var}(u_i^*) = \frac{\sigma^2 h(\mathbf{x}_i)}{h(\mathbf{x}_i)} = \sigma^2.$$

OLS on the transformed model is equivalent to **weighted least squares**: minimise

$$\sum_{i=1}^n \frac{1}{h(\mathbf{x}_i)} (y_i - \beta_0 - \beta_1 x_i - \dots)^2.$$

Each observation receives weight $w_i = 1/h(\mathbf{x}_i)$. Observations with high variance receive *less* weight; observations with low variance receive *more*. When the weights are correctly specified, WLS is BLUE: it has lower variance than OLS while remaining unbiased.

 Common mistake: forcing WLS when you don't know $h(\mathbf{x})$

WLS is only BLUE under *correctly specified* weights. With wrong weights it can be *less* efficient than OLS, and its standard errors are again incorrect. If you have only a vague idea of the variance structure, prefer robust standard errors.

7.4.2. Heteroskedasticity-robust standard errors

When the form of $h(\mathbf{x})$ is unknown, the safer remedy is to keep OLS for the point estimates and to compute **heteroskedasticity-consistent (HC) standard errors**, also called **robust** or **White standard errors** after (White 1980).

In the simple regression case with $\hat{u}_i$ denoting the OLS residual,

$$se_{\text{robust}}(\hat{\beta}_1) = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2 \hat{u}_i^2}{\left[\sum_{i=1}^n (x_i - \bar{x})^2 \right]^2}}.$$

The general matrix expression is the sandwich estimator of §7.2 with Ω replaced by $\text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2)$. Several variants exist (HC0, HC1, HC2, HC3) that differ in their finite-sample degrees-of-freedom correction; **HC1** is the default reported by Stata's `, robust` option and is the standard in published applied work.

Robust SEs leave the point estimates $\hat{\beta}_j$ unchanged. Only the variance — and therefore the t -statistics, p -values, and confidence intervals — is corrected.

i When to use what

- **Variance structure known and trustworthy** $\Rightarrow$ WLS (BLUE, smaller SEs).
- **Variance structure unknown** $\Rightarrow$ OLS with robust SEs (unbiased coefficients, valid inference, larger SEs).
- **In doubt** $\Rightarrow$ report robust SEs. In modern applied econometrics, robust SEs have become the default rather than a backup. Many journals expect them by construction.

7.5. Lab: heteroskedasticity

This lab follows Tutorial 8 of the course. The dataset `het_data.xlsx` is a custom cross-section of household-level consumption and income, divided into three income groups (`group` $\in \{1, 2, 3\}$, low to high). We fit a simple Engel-curve-style regression of consumption on income, detect heteroskedasticity, and then fix it twice — first by WLS and then by robust standard errors.

We use three packages: `readxl` to read the Excel file, `lmtest` for the Breusch–Pagan and Goldfeld–Quandt tests and for `coeftest()`, and `sandwich` for the robust variance–covariance matrix.

```
library(readxl)
library(lmtest)
library(sandwich)
```

7.5.1. Loading the data

The original tutorial uses a relative path (`read_excel("het_data.xlsx")`) after running `setwd()` into the Tutorial 8 folder. For portability in this book we keep that style but document the absolute path used at compile time. Adapt the path to your machine.

```
het_path <- file.path(
  "..", "..", "..", "Practicas", "2025", "Tutorial 8", "het_data.xlsx"
)
het_data <- read_excel(het_path)

# Construct a tercile-of-income indicator for the Goldfeld-Quandt split:
```

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```
# group == 1 = low-income third, 2 = middle third, 3 = high-income third.
het_data$group <- as.integer(cut(het_data$income,
                                breaks = quantile(het_data$income,
                                                    probs = c(0, 1/3, 2/3, 1)),
                                include.lowest = TRUE))
str(het_data)
```

```
tibble [30 x 4] (S3: tbl_df/tbl/data.frame)
 $ id      : num [1:30] 1 2 3 4 5 6 7 8 9 10 ...
 $ consumption: num [1:30] 433 663 690 727 800 ...
 $ income    : num [1:30] 649 939 956 1067 1071 ...
 $ group     : int [1:30] 1 1 1 1 1 1 1 1 1 1 ...
```

```
head(het_data)
```

```
# A tibble: 6 x 4
   id consumption income group
  <dbl>      <dbl>  <dbl> <int>
1     1         433.   649.     1
2     2         663.   939.     1
3     3         690.   956.     1
4     4         727.  1067.     1
5     5         800.  1071.     1
6     6         777.  1215.     1
```

The original xlsx ships with `income` and `consumption`. We add a derived `group` variable equal to the income tercile (1 = low, 2 = middle, 3 = high) so the Goldfeld–Quandt test below can subset the sample.

7.5.2. Step 1. OLS baseline and residual plot

```
model1 <- lm(consumption ~ income, data = het_data)
summary(model1)
```

Call:

```
lm(formula = consumption ~ income, data = het_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-280.684	-52.408	2.257	58.517	234.525

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.94848	70.77718	0.197	0.845
income	0.65532	0.03864	16.961	2.91e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 117.5 on 28 degrees of freedom

Multiple R-squared: 0.9113, Adjusted R-squared: 0.9081

F-statistic: 287.7 on 1 and 28 DF, p-value: 2.91e-16

The slope $\hat{\beta}_1$ is the average marginal propensity to consume out of income. To check homoskedasticity informally, plot the OLS residuals against income:

```
plot(het_data$income, resid(model1),
     xlab = "Income",
     ylab = "Residuals",
     main = "Residuals vs income",
     pch = 16,
     col = rgb(0, 0, 1, 0.4))
abline(h = 0, lty = 2, col = "red")
```

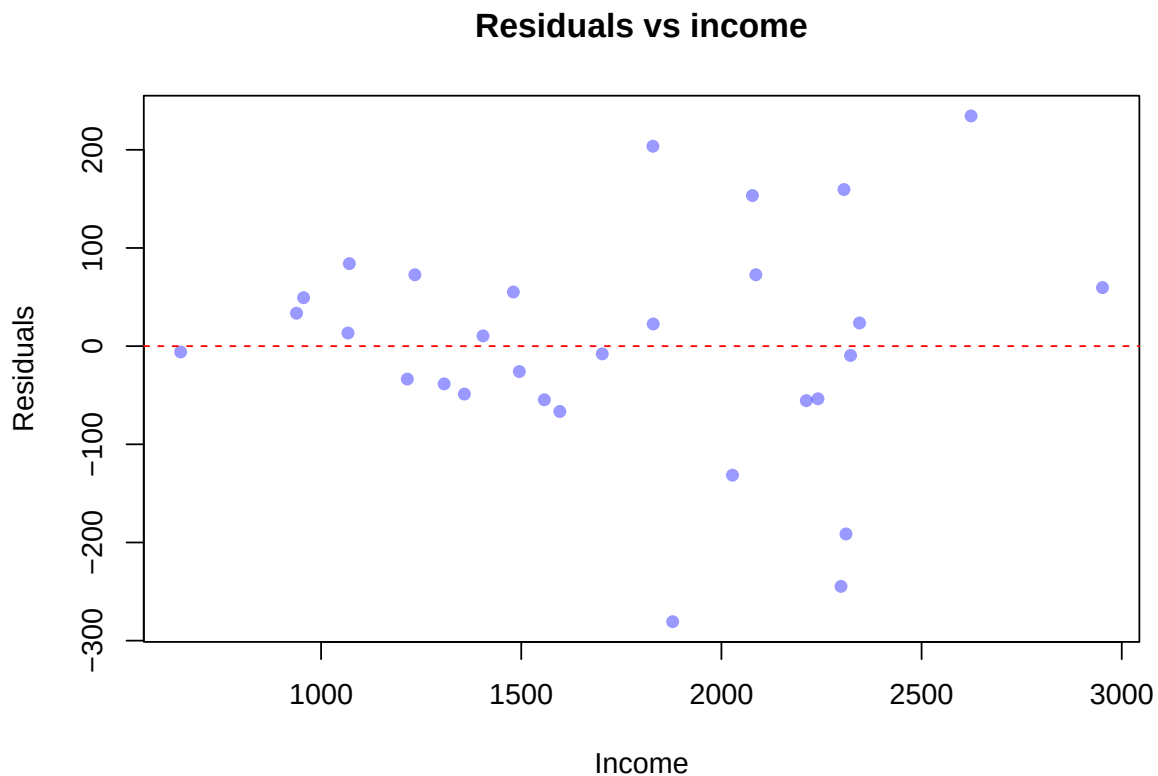


Figure 7.2.: Residuals from consumption $\sim$ income against income. A widening cloud suggests heteroskedasticity.

The cloud opens up as income grows: the residuals fan out. This is exactly the wedge shape we warned about in §7.3.1.

7.5.3. Step 2. Goldfeld–Quandt test by hand

Estimate the model separately on the low-income (`group == 1`) and high-income (`group == 3`) subsamples, then compare the two residual variances.

```
model_low <- lm(consumption ~ income, data = het_data, subset = group == 1)
model_high <- lm(consumption ~ income, data = het_data, subset = group == 3)

SSR_low <- sum(residuals(model_low)^2)
SSR_high <- sum(residuals(model_high)^2)

n_low <- nobs(model_low)
n_high <- nobs(model_high)
k <- length(coef(model_low)) # intercept + slope

df_low <- n_low - k
df_high <- n_high - k

F_gq <- (SSR_high / df_high) / (SSR_low / df_low)
F_crit <- qf(0.95, df_high, df_low)
p_gq <- 1 - pf(F_gq, df_high, df_low)

c(F = F_gq, F_crit_5pct = F_crit, p_value = p_gq)
```

```
      F F_crit_5pct      p_value
9.582976952 3.438101233 0.002205927
```

If F_{gq} is well above F_{crit_5pct} — equivalently, if the p -value is below 0.05 — we reject homoskedasticity in favour of “variance increases with income”.

7.5.4. Step 3. Breusch–Pagan test (by hand and with `bptest()`)

The auxiliary regression of squared residuals on income gives us R^2_{aux} ; the BP statistic is $n \cdot R^2_{aux}$ and follows a χ^2_1 under H_0 (we have one regressor, `income`).

```
res <- residuals(model1)
res_sq <- res^2
aux <- lm(res_sq ~ income, data = het_data)

n <- nobs(model1)
R2_aux <- summary(aux)$r.squared
BP_stat <- n * R2_aux
chi_crit <- qchisq(0.95, df = 1)
p_bp <- 1 - pchisq(BP_stat, df = 1)

c(BP_manual = BP_stat, chi2_1_5pct = chi_crit, p_value = p_bp)
```

```
BP_manual chi2_1_5pct      p_value
4.86609184 3.84145882 0.02738946
```


The package version delivers the same answer in one line:

```
bptest(model1)
```

studentized Breusch-Pagan test

data: model1

BP = 4.8661, df = 1, p-value = 0.02739

i Reading the bptest() output

bptest() prints four lines and a model line. The map to §7.3.3 is:

- **studentized Breusch-Pagan test** → name of the procedure.
- **data: model1** → the OLS fit the test is being applied to.
- **BP** → the test statistic, $n \cdot R_{\text{aux}}^2$ from the auxiliary regression of $\hat{u}_i^2$ on the regressors.
- **df** → the degrees of freedom of the asymptotic χ^2 reference distribution; equals the number of slope regressors in the auxiliary regression (one here, **income**).
- **p-value** → the right-tail probability under χ_{df}^2 . Reject H_0 at the 5% level when this is below 0.05.

A small p -value (below 0.05) means we reject homoskedasticity. The two computations should agree up to rounding.

7.5.5. Step 4. WLS correction

Assume $\text{Var}(u_i \mid \text{income}_i) \propto \text{income}_i$, so $h(\text{income}_i) = \text{income}_i$. The WLS weights are $w_i = 1/\text{income}_i$.

```
wls <- lm(consumption ~ income, data = het_data, weights = 1 / income)
summary(wls)
```

Call:

```
lm(formula = consumption ~ income, data = het_data, weights = 1/income)
```

Weighted Residuals:

Min	1Q	Median	3Q	Max
-6.4405	-1.1894	0.0547	1.3568	4.7847

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	34.45170	55.08024	0.625	0.537
income	0.64357	0.03354	19.185	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.607 on 28 degrees of freedom

Multiple R-squared: 0.9293, Adjusted R-squared: 0.9268

F-statistic: 368.1 on 1 and 28 DF, p-value: < 2.2e-16

7. Heteroskedasticity

To check whether the weights were well chosen, re-test for heteroskedasticity in the WLS fit:

```
bptest(wls)
```

```
studentized Breusch-Pagan test
```

```
data: wls
```

```
BP = 0.0056423, df = 1, p-value = 0.9401
```

If the BP test no longer rejects (large p -value), the assumed variance structure was approximately correct and WLS has done its job.

7.5.6. Step 5. Robust (HC1) standard errors

The safer remedy — valid even when we have no idea what $h(\mathbf{x})$ looks like — is to keep the OLS point estimates and recompute the standard errors with `vcovHC()`:

```
coeftest(model1, vcov = vcovHC(model1, type = "HC1"))
```

```
t test of coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.948477	49.970518	0.2791	0.7822
income	0.655317	0.035729	18.3411	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

i Reading the `coeftest()` output — the column journals expect

`coeftest()` from `lmtest` prints a familiar-looking coefficient table, but the **Std. Error** column is computed from the matrix you pass to `vcov` — here the HC1 sandwich estimator from `vcovHC()`. The map:

- **Estimate** → identical to `summary(model1)$coefficients[, "Estimate"]`. Robust standard errors do not change point estimates.
- **Std. Error** → the *robust* standard error: $\sqrt{[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\hat{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$, with $\hat{\Omega} = \text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2)$ scaled by the HC1 small-sample correction. **This is the column most journals will ask for.**
- **t value** → **Estimate** / **Std. Error**, with the robust SE in the denominator. Compare to the OLS **t value** from `summary(model1)`: the numerator is the same, the denominator has changed.
- **Pr(>|t|)** → two-sided p -value computed from the same t -distribution as in §4.3; the difference from the OLS column is driven entirely by the change in the standard error.

The point estimates are identical to `summary(model1)`. Only the standard errors, t -statistics, and p -values change.

7.5.7. Step 6. Comparison table: OLS SEs vs robust SEs

A clean way to communicate the result is to put the two standard errors side by side and report their ratio.

```
se_ols    <- summary(model1)$coefficients[, "Std. Error"]
se_robust <- sqrt(diag(vcovHC(model1, type = "HC1")))

comparison <- data.frame(
  OLS_SE    = round(se_ols, 4),
  Robust_SE = round(se_robust, 4),
  Ratio     = round(se_robust / se_ols, 3)
)
comparison
```

	OLS_SE	Robust_SE	Ratio
(Intercept)	70.7772	49.9705	0.706
income	0.0386	0.0357	0.925

If the ratio in the last column is materially larger than 1, the OLS standard errors were understating uncertainty — exactly the diagnosis we made in §7.2.

Reading the comparison table

The point of this table is not to “choose” one column over the other — the coefficients are the same in both columns. The table tells you *how much* the OLS standard error has been distorted by heteroskedasticity. A ratio close to 1.0 means OLS was fine even though MLR.5 technically failed; a ratio above 1.5 means OLS would have led you to over-reject the null. Either way, robust SEs are now the standard column to report.

Self-check

Six short multiple-choice questions. Try each one before opening the answer.

Q1. Definition

Heteroskedasticity in a linear regression model means:

- A. $E[u \mid x] \neq 0$.
- B. $\text{Var}(u \mid x)$ depends on x — it is not constant.
- C. The error u is correlated across observations.
- D. x is collinear with another regressor.

Answer: B. Heteroskedasticity is, by definition, a failure of MLR.5: the conditional variance of u varies with the regressors. The other options describe MLR.4, autocorrelation, and multicollinearity respectively.

💡 Q2. Consequences

Under heteroskedasticity (with MLR.1–MLR.4 still holding), OLS is:

- A. Biased in finite samples.
- B. Inconsistent.
- C. Still unbiased and consistent, but no longer BLUE, and the usual standard errors are wrong.
- D. Still BLUE.

Answer: C. Unbiasedness and consistency of $\hat{\beta}_j$ do not require MLR.5. Efficiency (BLUE) and the validity of the textbook standard error formula do.

💡 Q3. Identification versus inference

Why do we say heteroskedasticity is an *inference* problem, not an *identification* problem?

- A. Because heteroskedasticity biases $\hat{\beta}_j$ but leaves the standard errors intact.
- B. Because $\hat{\beta}_j$ is no longer consistent for the population slope.
- C. Because, if MLR.4 still holds, $\hat{\beta}_j$ still estimates the *ceteris paribus* effect; only the standard errors are wrong.
- D. Because heteroskedasticity is the same thing as omitted variable bias.

Answer: C. The causal interpretation of $\hat{\beta}_j$ survives heteroskedasticity. What fails is the machinery that turns the estimate into a *t*-test or a confidence interval.

💡 Q4. Breusch–Pagan

The Breusch–Pagan test runs an auxiliary regression of:

- A. $\hat{u}$ on the regressors and uses the resulting F statistic.
- B. $\hat{u}^2$ on the regressors and uses $n \cdot R^2$ as a χ^2 statistic.
- C. y on $\hat{u}^2$ and uses a t statistic.
- D. x on $\hat{u}$ and uses the slope as a test statistic.

Answer: B. The BP test regresses the squared residuals on the regressors; under H_0 of homoskedasticity, nR_{aux}^2 is asymptotically χ_k^2 .

💡 Q5. Goldfeld–Quandt

The Goldfeld–Quandt test compares:

- A. The two estimated slopes.
- B. The residual variances of two subsamples (typically low vs high values of a regressor).
- C. The R^2 of two models.
- D. Two intercepts.

Answer: B. GQ splits the sample by the suspected variable, fits the regression on each half, and uses the ratio of mean squared residuals as an F statistic.

💡 Q6. WLS versus robust SEs

When should you prefer Weighted Least Squares to OLS with robust standard errors?

- A. When the form of $\text{Var}(u \mid \mathbf{x})$ is correctly known and you want efficiency gains.
- B. Whenever the Breusch–Pagan test rejects, regardless of what you know.
- C. When the sample is small and you cannot compute robust SEs.
- D. Never — robust SEs always dominate WLS.

Answer: A. WLS is BLUE only under correctly specified weights. With unknown variance structure, OLS plus robust SEs is the safer default; with a credible model for $h(\mathbf{x})$, WLS buys efficiency.

Exercises

Exercise 7.1 — Spotting heteroskedasticity. For each of the following regressions, predict whether the error term is likely to be heteroskedastic and explain why in one sentence.

- (a) Annual food expenditure regressed on household income across a sample of 4{,}000 households.
- (b) Number of children regressed on mother's age, in a sample restricted to women aged 30–35.
- (c) Firm-level R&D spending regressed on firm sales in a sample mixing small startups and Fortune 500 companies.
- (d) A linear-probability model where the dependent variable is a 0/1 indicator for employment.

💡 Show answer

- (a) Yes — richer households have more discretionary spending and therefore wider absolute variation in food spending. (b) Probably not strongly — the sample is narrow in age, so scale effects are small. (c) Yes, severely — firm size spans several orders of magnitude. (d) Yes, by construction — $\text{Var}(y \mid \mathbf{x}) = p(\mathbf{x})[1 - p(\mathbf{x})]$ depends on $\mathbf{x}$ whenever $p(\mathbf{x})$ does.

Exercise 7.2 — Reading a BP output. A student runs `bptest(lm(consumption ~ income, data = het_data))` and obtains a test statistic of about 9.8 on 1 degree of freedom, with p -value ≈ 0.002 .

- (a) State the null and alternative hypotheses of the test.
- (b) What is the conclusion at the 5% level?
- (c) Should the student now (i) keep OLS standard errors, (ii) switch to robust standard errors, or (iii) re-estimate by WLS? Justify briefly.

💡 Show answer

- (a) H_0 : homoskedasticity ($\gamma_1 = 0$ in the auxiliary regression of $\hat{u}^2$ on income); H_1 : $\text{Var}(u \mid \text{income})$ depends on income. (b) Reject H_0 at the 5% level: the p -value is well below 0.05. (c) At minimum, switch to robust SEs (option ii). WLS (option iii) is an option *if* the student is confident the variance is proportional to income; otherwise the robust route is safer.

7. Heteroskedasticity

Exercise 7.3 — Goldfeld–Quandt by hand. Suppose for a regression of consumption on income you obtain $SSR_{\text{low}} = 120$ on $df_{\text{low}} = 28$ observations in the low-income subsample, and $SSR_{\text{high}} = 540$ on $df_{\text{high}} = 28$ in the high-income subsample. The 5% critical value of $F_{28,28}$ is approximately 1.88.

- (a) Compute the Goldfeld–Quandt F statistic.
- (b) State your conclusion at the 5% level.
- (c) What economic story would explain a result of this kind?

A full answer is given in the Instructor Edition.

Exercise 7.4 — Why the sandwich. Show that under homoskedasticity ($\Omega = \sigma^2 \mathbf{I}_n$), the sandwich variance

$$(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$$

reduces to the usual OLS variance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. Use this to argue, in one paragraph, that the robust standard error is *never worse asymptotically* than the OLS standard error.

A full answer is given in the Instructor Edition.

Exercise 7.5 — WLS as transformed OLS. Take the SLR model $y_i = \beta_0 + \beta_1 x_i + u_i$ with $\text{Var}(u_i | x_i) = \sigma^2 x_i$ (so $x_i > 0$).

- (a) Write the transformed model after dividing by $\sqrt{x_i}$.
- (b) Show that the error of the transformed model has constant variance equal to σ^2 .
- (c) Conclude that OLS on the transformed model is BLUE for (β_0, β_1) , and write down the equivalent WLS criterion in terms of weights w_i .

A full answer is given in the Instructor Edition.

Exercise 7.6 — Lab extension. Using `het_data.xlsx` from Tutorial 8:

- (a) Reproduce the comparison table of OLS standard errors and HC1 robust standard errors. Report the ratio for the slope on `income`.
- (b) Re-fit the model using WLS with weights $w_i = 1/\text{income}_i^2$ (variance proportional to income_i^2 rather than income_i). Run the BP test on the resulting WLS fit. Does this second weighting scheme outperform the $w_i = 1/\text{income}_i$ scheme of §7.5?
- (c) Suppose the BP test still rejects after WLS. Which of the two remedies — WLS or robust SEs — should you trust more, and why?

A full answer is given in the Instructor Edition.

Autocorrelation

Learning outcomes

By the end of this chapter the reader should be able to:

- Explain why autocorrelation is the time-series analogue of heteroskedasticity, and articulate what it does (and does not do) to OLS unbiasedness, efficiency, and standard errors.
- Write down the AR(1) error model $u_t = \rho u_{t-1} + \varepsilon_t$, state the stationarity condition $|\rho| < 1$, and derive the stationary mean, variance, and autocovariances.
- Detect first-order serial correlation visually (time plot of residuals, lag plot of $\hat{u}_t$ vs $\hat{u}_{t-1}$) and formally with the Durbin–Watson statistic, using the lower/upper bounds d_L, d_U .
- Recognise when the DW test is invalid (lagged dependent variable) and apply Durbin’s h -test instead.
- Apply the Cochrane–Orcutt iterative procedure to obtain efficient FGLS estimates under AR(1) errors, and explain the role of the Prais–Winsten correction for the first observation.
- Implement the full workflow in R using `lmtest::dwtest()` and `orcutt::cochrane.orcutt()`, and compare OLS to AR(1)-corrected estimates on a simulated quarterly sales-vs-advertising regression.

Motivating empirical question

Why do quarterly sales-vs-advertising regressions almost always produce optimistic t -statistics?

A marketing analyst regresses quarterly sales on advertising and a time trend, gets a t -statistic of 7.4 on advertising, and proudly reports a “highly significant” effect. The board approves a 30 % budget increase. Next year sales disappoint. What went wrong? Very often, the answer is **autocorrelation**: the errors at quarter t are correlated with the errors at $t - 1$, OLS standard errors are too small, and the t -statistic is overstated. This chapter is about diagnosing and fixing that problem.

8.1. From cross-section to time series

IN Chapters 1–7 every observation was a *unit* (a worker, a firm, a household) sampled at a single point in time. We indexed observations by $i = 1, \dots, n$. Time series are different: there is one unit — a country, a firm, a household, a market — and we observe it repeatedly across discrete periods. We replace the index i with $t = 1, \dots, n$, and the population model becomes

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$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t.$$

In matrix form, $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$, exactly as before — only the index has changed. A typical example is monthly company-level data:

Month	Sales y_t	Advertising x_t
Jan 2020	120	15
Feb 2020	135	18
Mar 2020	128	10
$\vdots$	$\vdots$	$\vdots$
Nov 2020	175	24
Dec 2020	190	30

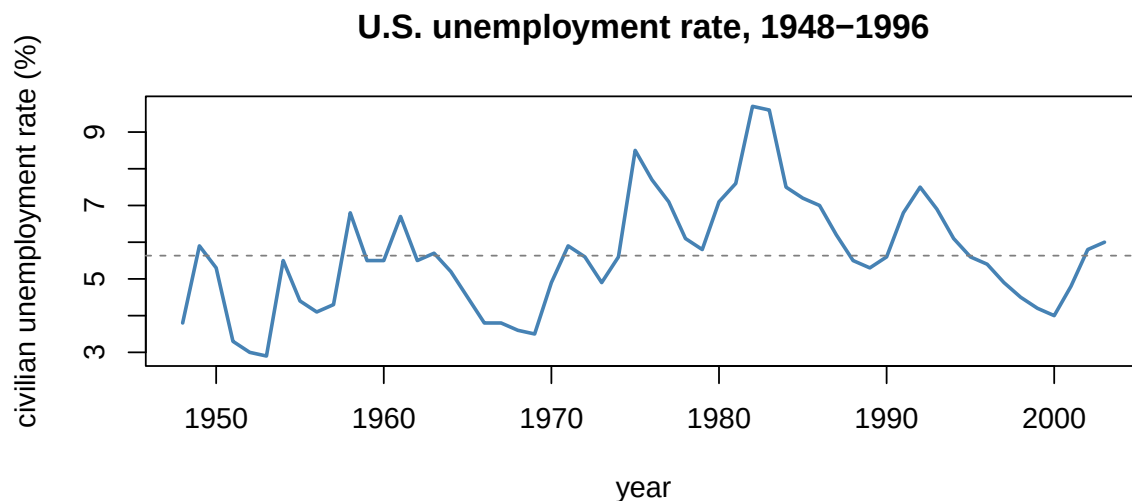


Figure 8.1.: A real macroeconomic time series: U.S. quarterly unemployment, 1948–1996 (Wooldridge `phillips` dataset). Notice that consecutive observations are obviously *not* independent draws from a population — there are long expansions and long recessions, and a low unemployment year is almost always followed by another low year. That persistence is the same phenomenon we will call **autocorrelation** below. The cross-sectional ideal of MLR.2 (independent draws across i) cannot survive contact with data of this shape, and the rest of the chapter is about what to do about it.

Mechanically, almost everything from Chapters 2–6 carries over: OLS still solves $\min_{\beta} \sum_t (y_t - \mathbf{x}'_t \beta)^2$; the partialling-out interpretation is unchanged; nonlinear and qualitative regressors behave as before. What changes is the **error structure**. Two errors u_t and u_s refer to the *same* economic unit observed at *different* points in time, and there is no reason in general for them to be uncorrelated.

i Recap of MLR.2 (random sampling) in time-series context

Assumption MLR.2 in cross-sections required the sample to consist of n *independent* draws — for any pair $i \neq j$, $\mathbb{E}[u_i u_j] = 0$. The mechanism that delivered MLR.2 was random sampling from a fixed population: knowing u_i told you nothing about u_j .

In a time series there is no random sampling across units. Successive errors u_t and u_{t+1} are generated by the same process unfolding over time, and the natural default is that they *are* related. Whenever $\mathbb{E}[u_t u_s] \neq 0$ for some $t \neq s$, we say the errors are **autocorrelated** (or **serially correlated**). Will January’s unobserved demand shock influence February’s? Almost certainly yes — that is the working definition of autocorrelation.

The contrast with Chapter 7 is useful. Heteroskedasticity is a cross-sectional pathology in which $\text{Var}(u_i)$ varies with the regressors but errors across units are independent. Autocorrelation is its time-series cousin: the off-diagonal elements of the error variance–covariance matrix are nonzero, while the diagonal may or may not stay constant. Both leave OLS unbiased but invalidate its standard errors.

8.2. Definition, AR(1) structure, and consequences

8.2.1. Definition

i Definition: autocorrelation

The errors u_t exhibit **autocorrelation** (or **serial correlation**) when

$$\text{Cov}(u_t, u_s) \neq 0 \quad \text{for some } t \neq s.$$

The variance–covariance matrix of $\mathbf{u} = (u_1, \dots, u_n)'$ is

$$\text{Var}(\mathbf{u}) = \begin{pmatrix} \sigma^2 & \gamma_1 & \gamma_2 & \cdots & \gamma_{n-1} \\ \gamma_1 & \sigma^2 & \gamma_1 & \cdots & \gamma_{n-2} \\ \gamma_2 & \gamma_1 & \sigma^2 & \cdots & \gamma_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1} & \gamma_{n-2} & \gamma_{n-3} & \cdots & \sigma^2 \end{pmatrix},$$

with at least one off-diagonal element $\gamma_k = \text{Cov}(u_t, u_{t-k}) \neq 0$.

Under the classical cross-sectional assumptions, the off-diagonal entries are all zero and $\text{Var}(\mathbf{u}) = \sigma^2 \mathbf{I}_n$. Autocorrelation drops that simplification.

8.2.2. The AR(1) workhorse

Before any algebra, the picture. Think of ρ as the fraction of yesterday’s shock that carries over into today: if yesterday’s error was +2 and $\rho = 0.7$, then today already starts with 1.4 baked in before any new disturbance arrives. The closer ρ is to 1, the longer shocks live; at $\rho = 0$ each period is a fresh draw.

Almost all of the chapter — and the vast majority of applied undergraduate work — uses one specific model for serial dependence: the **first-order autoregressive** process, or AR(1):

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$$u_t = \rho u_{t-1} + \varepsilon_t, \quad |\rho| < 1, \quad (\text{AR}(1))$$

where ε_t is **white noise**: $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma_\varepsilon^2$ constant, $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$, and (often) $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.

The single parameter ρ summarises the entire dependence structure. Under stationarity ($|\rho| < 1$) one can show:

$$\mathbb{E}[u_t] = 0, \quad \text{Var}(u_t) = \frac{\sigma_\varepsilon^2}{1 - \rho^2}, \quad \text{Cov}(u_t, u_{t-k}) = \rho^k \cdot \frac{\sigma_\varepsilon^2}{1 - \rho^2}.$$

So $\gamma_k = \rho^k \sigma^2$ with $\sigma^2 = \sigma_\varepsilon^2 / (1 - \rho^2)$ — the autocovariances **decay geometrically** in the lag k . The sign of ρ classifies the pattern:

- $\rho > 0$ (**positive autocorrelation**). Errors of the same sign cluster. A positive shock today persists, decaying smoothly: long runs of positive residuals, then long runs of negative residuals. This is by far the most common case in macroeconomic and business data.
- $\rho < 0$ (**negative autocorrelation**). Errors alternate in sign: each positive shock is followed disproportionately by a negative one. Less common, but it does appear in inventory series and in models where adjustment overshoots.
- $\rho = 0$ (**no autocorrelation**). $u_t = \varepsilon_t$ and the errors are white noise: the cross-sectional default is restored.

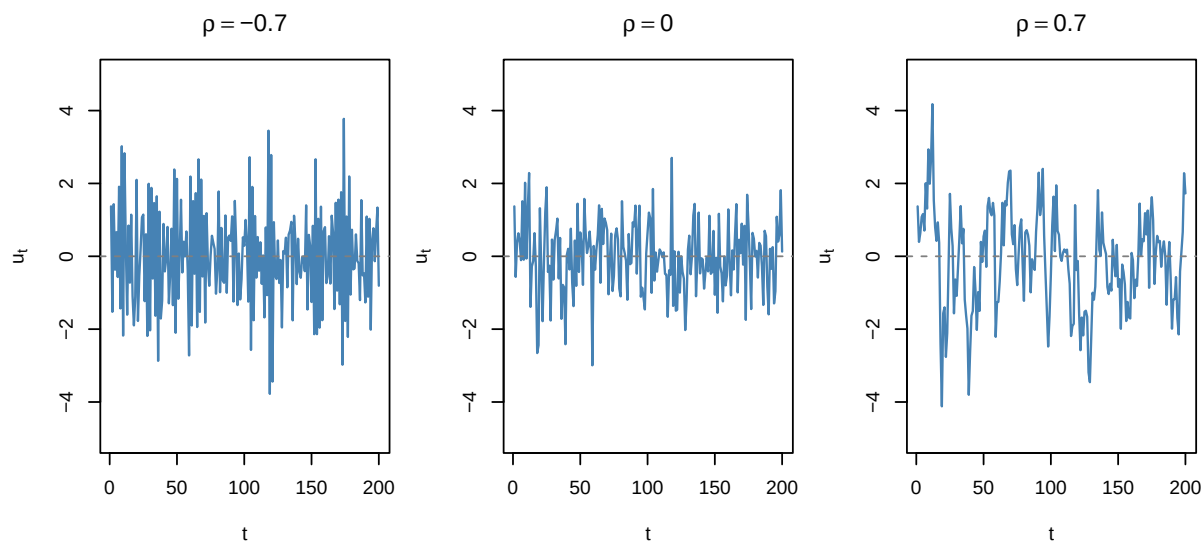


Figure 8.2.: Three simulated AR(1) error series of length 200, sharing the same innovation variance $\sigma_\varepsilon^2 = 1$ and the same random seed but different values of ρ . Left ($\rho = -0.7$): residuals alternate sign in a sawtooth pattern. Middle ($\rho = 0$): white noise, no visible pattern. Right ($\rho = 0.7$): long, smooth excursions above and below zero — the cumulative-shock pattern of positive autocorrelation. Reading off the visual fingerprint is most of what graphical diagnostics do.

Higher-order processes (AR(2), AR(p), ARMA, ...) exist and are studied in Econometrics II; for our purposes AR(1) is enough to fix ideas and to motivate the diagnostics and corrections that follow.

8.2.3. Consequences for OLS

What does autocorrelation do to the OLS estimator $\hat{\beta}$?

1. **OLS is still unbiased and consistent** — provided the conditional mean is correctly specified, i.e. $\mathbb{E}[u_t | \mathbf{x}_t] = 0$. Autocorrelation is a property of the *off-diagonal* elements of $\text{Var}(\mathbf{u})$ and does not push the average estimate away from β . The point estimates are not the problem.
2. **OLS is no longer BLUE**. It is no longer the minimum-variance linear unbiased estimator; an alternative estimator (Generalised Least Squares, see §8.4) is more efficient.
3. **The conventional OLS standard errors are wrong**. The textbook formula $\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ presupposes $\text{Var}(\mathbf{u}) = \sigma^2\mathbf{I}_n$, which fails. With **positive** autocorrelation the OLS-reported standard errors typically *understate* the true sampling variability, *t*-statistics are too large, *p*-values too small, and confidence intervals too narrow. The “optimistic *t*-statistics” of the motivating question are exactly this artefact.

⚠ Causality vs efficiency: the lagged-dependent-variable trap

There is one case in which autocorrelation does more than just inflate *t*-statistics: it destroys identification. If the regression includes a **lagged dependent variable** y_{t-1} and the errors are AR(1),

$$y_t = \alpha y_{t-1} + \beta_0 + \beta_1 x_t + u_t, \quad u_t = \rho u_{t-1} + \varepsilon_t,$$

then y_{t-1} depends on u_{t-1} , and u_{t-1} is correlated with u_t through ρ . The regressor is correlated with the error, $\mathbb{E}[u_t | y_{t-1}] \neq 0$, and OLS is no longer just inefficient — it becomes **biased and inconsistent**. The causal coefficient β_1 is no longer identified. This is the time-series cousin of the omitted-variable bias from Chapter 3.

The intuition for why positive autocorrelation makes OLS standard errors too small is worth dwelling on: if successive errors carry information about one another, then n correlated observations contain less independent information than n uncorrelated ones. The effective sample size is smaller than the nominal n , but the OLS formula treats every observation as a fresh draw. The reported precision is therefore exaggerated.

8.3. Detection

We have two complementary toolkits: *graphical* (quick, intuitive) and *analytical* (formal hypothesis tests). The recommended workflow is to look first and test second.

8.3.1. Graphical methods

Time plot of residuals. Plot $\hat{u}_t$ against t . If the errors are uncorrelated, the residuals should scatter randomly around zero — no visible pattern, no clustering, no obvious cycles. If you see **long runs** of same-sign residuals (many consecutive positives, then many consecutive negatives), that is the visual signature of $\rho > 0$. If residuals alternate in a sawtooth pattern, that is $\rho < 0$.

Lag plot of residuals. Plot $\hat{u}_t$ on the vertical axis against $\hat{u}_{t-1}$ on the horizontal axis. Under $\rho = 0$ the cloud has no slope; under $\rho > 0$ it tilts up-right (positive linear association); under $\rho < 0$ it tilts down. Mechanically this is the picture of the OLS regression $\hat{u}_t = \rho \hat{u}_{t-1} + (\text{noise})$, whose slope is exactly the estimate of ρ that we will use below.

8. Autocorrelation

The three benchmark shapes are: a featureless cloud ($\rho \approx 0$), an upward-sloping line ($\rho > 0$), and a downward-sloping line ($\rho < 0$). In the lab below you will see all three.

8.3.2. The Durbin–Watson test

The Durbin–Watson (DW) statistic is the workhorse formal test for first-order autocorrelation.

i Definition: Durbin–Watson statistic

$$DW = \frac{\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^n \hat{u}_t^2}.$$

The test is of $H_0 : \rho = 0$ (no autocorrelation) against $H_1 : \rho \neq 0$ (or a one-sided alternative).

Why $DW \approx 2(1 - \hat{\rho})$. Expand the numerator:

$$\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2 = \sum_{t=2}^n \hat{u}_t^2 - 2 \sum_{t=2}^n \hat{u}_t \hat{u}_{t-1} + \sum_{t=2}^n \hat{u}_{t-1}^2.$$

For large n the two outer sums are each approximately $\sum_{t=1}^n \hat{u}_t^2$ (they differ by a single observation). Dividing through gives

$$DW \approx \frac{2 \sum_{t=2}^n \hat{u}_t^2 - 2 \sum_{t=2}^n \hat{u}_t \hat{u}_{t-1}}{\sum_{t=1}^n \hat{u}_t^2} = 2 \left(1 - \frac{\sum_{t=2}^n \hat{u}_t \hat{u}_{t-1}}{\sum_{t=1}^n \hat{u}_t^2} \right) = 2(1 - \hat{\rho}),$$

where $\hat{\rho} = \sum_{t=2}^n \hat{u}_t \hat{u}_{t-1} / \sum_{t=1}^n \hat{u}_t^2$ is the sample first-order autocorrelation of the residuals. Since $\rho \in [-1, 1]$, the statistic lives in $[0, 4]$:

- $DW \approx 0 \Leftrightarrow \hat{\rho} \approx +1$ (strong positive autocorrelation).
- $DW \approx 2 \Leftrightarrow \hat{\rho} \approx 0$ (no autocorrelation).
- $DW \approx 4 \Leftrightarrow \hat{\rho} \approx -1$ (strong negative autocorrelation).

Bounds and decision rule. The exact distribution of DW depends on the design matrix $\mathbf{X}$, so Durbin and Watson tabulated a **lower bound** d_L and an **upper bound** d_U (functions of n , the number of regressors k , and the significance level). The decision rule, for the one-sided test $H_1 : \rho > 0$, is:

Region	Conclusion
$0 < DW < d_L$	Reject H_0 : positive autocorrelation.
$d_L < DW < d_U$	Inconclusive.
$d_U < DW < 4 - d_U$	Do not reject H_0 : no autocorrelation.
$4 - d_U < DW < 4 - d_L$	Inconclusive.
$4 - d_L < DW < 4$	Reject H_0 : negative autocorrelation.

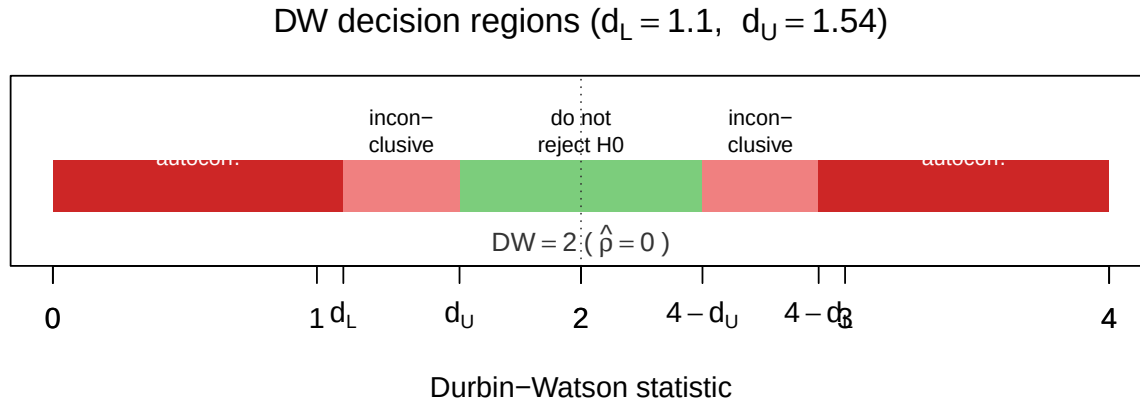


Figure 8.3.: The DW decision regions as a number line (illustrative bounds $d_L = 1.10$, $d_U = 1.54$, e.g. $n \approx 30$, $k = 2$). The statistic lives in $[0, 4]$, with $DW = 2$ corresponding to $\hat{\rho} = 0$ (the dashed grey vertical). Red bands flag rejection; pink bands flag inconclusive regions; green flags ‘no evidence against H_0 ’. Note the *symmetry*: positive autocorrelation pushes DW down towards 0, negative autocorrelation pushes it up towards 4.

The two inconclusive regions are an unavoidable feature of the test — a price one pays for not having to assume a particular design matrix. In modern software (e.g. `lmtest::dwtest()` in R) the test reports an exact p -value computed for the actual $\mathbf{X}$, sidestepping the table-bound mechanics. We use the bounds for hand exercises; we read the p -value in the lab.

💡 Worked example (small-sample DW)

A regression of small road-accident counts Y on registered cars X over 10 years yields $\hat{Y}_t = 2.5676 + 0.04937 X_t$ with $R^2 = 0.942$. From the table of residuals one computes

$$\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2 = 18.796, \quad \sum_{t=1}^n \hat{u}_t^2 = 22.844,$$

so

$$DW = \frac{18.796}{22.844} \approx 0.823.$$

For $n = 10$ and $k = 1$ regressors, the 5 % bounds are $d_L = 0.879$, $d_U = 1.320$. Since $DW < d_L$, we reject H_0 and conclude that the residuals exhibit **positive** first-order autocorrelation. (We will use this same example to illustrate Cochrane–Orcutt in §8.4.)

8.3.3. When DW fails: Durbin’s h -test

The DW test rests on two assumptions that often go unstated: the error process is AR(1), and the regression does **not** include a lagged dependent variable. The lagged-dependent-variable case — discussed in the warning box of §8.2.3 — is precisely the case in which OLS becomes inconsistent, and where DW itself becomes badly biased toward 2 (toward the null of no autocorrelation), which is the worst possible behaviour for a diagnostic test.

For models of the form

$$y_t = \alpha y_{t-1} + \beta_0 + \beta_1 x_{1t} + \cdots + \beta_k x_{kt} + u_t, \quad u_t = \rho u_{t-1} + \varepsilon_t,$$

8. Autocorrelation

Durbin proposed the h **statistic**

$$h = \hat{\rho} \sqrt{\frac{n}{1 - n \widehat{\text{Var}}(\hat{\alpha})}} \sim \mathcal{N}(0, 1) \quad \text{under } H_0 : \rho = 0,$$

where $\hat{\alpha}$ is the OLS coefficient on y_{t-1} and $\widehat{\text{Var}}(\hat{\alpha})$ is the usual (incorrect) OLS variance estimate. One often plugs in $\hat{\rho} \approx 1 - DW/2$. Reject H_0 at level α if $|h| > z_{1-\alpha/2}$. If $n \cdot \widehat{\text{Var}}(\hat{\alpha}) \geq 1$ the h statistic is undefined (the radicand is non-positive); in that case use Breusch–Godfrey instead.

💡 Worked example (Durbin's h)

Consider an OLS fit on $n = 21$ observations:

$$\hat{Y}_t = 1.3 + 0.97 Y_{t-1} + 2.31 X_t, \quad \text{se}(\hat{\alpha}) = 0.07, \quad DW = 1.21.$$

Then $\hat{\rho} \approx 1 - 1.21/2 = 0.395$, $\widehat{\text{Var}}(\hat{\alpha}) = 0.07^2 = 0.0049$, and

$$h = 0.395 \cdot \sqrt{\frac{21}{1 - 21 \cdot 0.0049}} = 0.395 \cdot 4.838 \approx 1.91.$$

At the 5 % level the critical value is $z_{0.975} = 1.96$. Since $|h| = 1.91 < 1.96$, we do **not** reject H_0 : there is no statistically significant evidence of first-order autocorrelation. (A naive DW reading of 1.21 might have suggested otherwise, but DW is biased here.)

Other tests exist — the Breusch–Godfrey LM test, which handles higher-order processes; the Ljung–Box portmanteau test on residual autocorrelations — and are covered in Econometrics II. For this course, DW (or Durbin's h when needed) is the standard tool.

8.4. The robust shortcut: HAC standard errors

Before turning to a full correction, it is worth noting the time-series parallel of Chapter 7's robust standard errors. If all you want is honest standard errors and you don't care about the efficiency gain, **HAC** (Heteroskedasticity- and Autocorrelation-Consistent) standard errors are the one-line fix. They leave the OLS coefficients alone — which, recall, are still unbiased under autocorrelation — and just correct the variance, replacing the wrong $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ with an estimator that accounts for the off-diagonal entries of $\text{Var}(\mathbf{u})$. The Newey–West (1987) variance estimator is the standard implementation: it weights residual cross-products $\hat{u}_t \hat{u}_{t-k}$ by a kernel that down-weights distant lags, so the analyst does not have to commit to a specific error model like AR(1).

The trade-off is the same as for heteroskedasticity-robust SEs. HAC buys you valid inference under very weak assumptions, but it does not deliver an efficient estimator: with strong autocorrelation, GLS-style methods (next section) typically produce tighter confidence intervals. In modern applied work HAC is the default for time-series regressions where the error model is uncertain; Cochrane–Orcutt is reached for when one is willing to commit to AR(1) in exchange for efficiency. We show the modern one-line workflow first — on a small simulated regression with AR(1) errors, paralleling the lab's full setup — and then develop the GLS approach.

```

library(lmtest)
library(sandwich)

set.seed(1)
n_demo <- 100
x_demo <- 1:n_demo + rnorm(n_demo)
u_demo <- numeric(n_demo); u_demo[1] <- rnorm(1)
for (t in 2:n_demo) u_demo[t] <- 0.7 * u_demo[t-1] + rnorm(1)
y_demo <- 1 + 0.5 * x_demo + u_demo
fit_demo <- lm(y_demo ~ x_demo)

# OLS standard errors (wrong under autocorrelation):
coeftest(fit_demo)

```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.7338669	0.2798236	2.6226	0.01012 *
x_demo	0.5030121	0.0048028	104.7335	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

# HAC (Newey-West) standard errors --- the one-line fix:
coeftest(fit_demo, vcov = NeweyWest(fit_demo))

```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.7338669	0.5836579	1.2574	0.2116
x_demo	0.5030121	0.0092992	54.0921	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The coefficients are identical in the two outputs — HAC does not touch them — but the standard errors in the second block are larger, the t -statistics smaller, and the p -values more honest. If you also want efficiency, read on for Cochrane–Orcutt.

8.5. Correction: from GLS to Cochrane–Orcutt

If OLS is inefficient under AR(1), what should we do? The answer is **Generalised Least Squares (GLS)**: transform the model so that the transformed errors are white noise, then run OLS on the transformed model.

8.5.1. The GLS quasi-difference

The trick: if today's error is part-yesterday's-error-plus-noise, then today minus ρ times yesterday cancels out the dependence and leaves only the noise. That's the entire idea of GLS for AR(1).

Start from the model and its ρ -multiplied lag:

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad \rho y_{t-1} = \rho \beta_0 + \rho \beta_1 x_{t-1} + \rho u_{t-1}.$$

Subtract the second from the first:

$$y_t - \rho y_{t-1} = \beta_0(1 - \rho) + \beta_1(x_t - \rho x_{t-1}) + (u_t - \rho u_{t-1}).$$

By the AR(1) assumption $u_t - \rho u_{t-1} = \varepsilon_t$, which is white noise. Define the **quasi-differenced** variables

$$y_t^* = y_t - \rho y_{t-1}, \quad x_t^* = x_t - \rho x_{t-1}, \quad \beta_0^* = \beta_0(1 - \rho).$$

The transformed regression

$$y_t^* = \beta_0^* + \beta_1 x_t^* + \varepsilon_t$$

has homoskedastic, uncorrelated errors — the classical regression assumptions are restored, and OLS on the transformed model is the **GLS estimator**, which is BLUE.

The first observation. The transformation as written loses observation $t = 1$. To keep it, the **Prais–Winsten** modification rescales it differently:

$$y_1^* = \sqrt{1 - \rho^2} y_1, \quad x_1^* = \sqrt{1 - \rho^2} x_1, \quad \text{constant}_1^* = \sqrt{1 - \rho^2}.$$

The factor $\sqrt{1 - \rho^2}$ is exactly what makes $\text{Var}(u_1^*) = \sigma_\varepsilon^2$, matching the other observations. In samples where n is small, this matters; in large samples, dropping observation 1 (the “two-step Cochrane–Orcutt”) is essentially equivalent.

In matrix form, the GLS transformation can be written compactly as

$$\mathbf{Y}^* = \begin{pmatrix} \sqrt{1 - \rho^2} y_1 \\ y_2 - \rho y_1 \\ \vdots \\ y_n - \rho y_{n-1} \end{pmatrix}, \quad \mathbf{X}^* = \begin{pmatrix} \sqrt{1 - \rho^2} & \sqrt{1 - \rho^2} x_{21} & \cdots & \sqrt{1 - \rho^2} x_{k1} \\ 1 - \rho & x_{22} - \rho x_{21} & \cdots & x_{k2} - \rho x_{k1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 - \rho & x_{2n} - \rho x_{2,n-1} & \cdots & x_{kn} - \rho x_{k,n-1} \end{pmatrix}.$$

Running OLS on $(\mathbf{Y}^*, \mathbf{X}^*)$ delivers $\hat{\beta}_{\text{GLS}}$.

8.5.2. The Cochrane–Orcutt iterative procedure

The catch: ρ is unknown. We need to estimate it. The **Cochrane–Orcutt (CO)** procedure is the simplest feasible-GLS algorithm: alternate between estimating ρ from residuals and re-estimating β on the quasi-differenced data, until ρ stabilises.

i Algorithm: Cochrane–Orcutt iterative procedure

1. Estimate the original model $y_t = \beta_0 + \beta_1 x_t + \dots + u_t$ by OLS; collect residuals $\hat{u}_t$.
2. Estimate ρ by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ (no intercept):

$$\hat{\rho} = \frac{\sum_{t=2}^n \hat{u}_t \hat{u}_{t-1}}{\sum_{t=2}^n \hat{u}_{t-1}^2}.$$

3. Form the quasi-differenced variables $y_t^* = y_t - \hat{\rho} y_{t-1}$, $x_t^* = x_t - \hat{\rho} x_{t-1}$ (and, if Prais–Winsten, the rescaled first observation).
4. Run OLS on (y_t^*, x_t^*) to get new $\hat{\beta}^{(2)}$; compute new residuals $\hat{u}_t^{(2)}$.
5. Plug $\hat{u}_t^{(2)}$ back into step 2 to update $\hat{\rho}$.
6. Iterate until $\hat{\rho}$ converges (changes negligibly between iterations).

The final coefficient vector is the Cochrane–Orcutt FGLS estimate; reported standard errors are the OLS standard errors *from the transformed regression*, which are valid under AR(1).

There are alternatives — the two-step CO method (one iteration only), the Hildreth–Lu grid search over ρ , Durbin’s two-step method, Prais–Winsten with restricted least squares — but they all share the same underlying idea: estimate ρ , quasi-difference, and apply OLS to recover an efficient estimator.

💡 Worked example (Cochrane–Orcutt on the accidents data)

Returning to the accidents-vs-cars example from §8.3.2: with $DW = 0.823$ and a small $n = 10$, we apply the Prais–Winsten variant.

Step 1 (OLS): $\hat{Y}_t = 2.568 + 0.0494 X_t$.

Step 2 ($\hat{\rho}$ from residuals): regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ yields $\hat{\rho} = 0.4226$.

Step 3 (transform): for $t > 1$, $Y_t^* = Y_t - 0.4226 Y_{t-1}$ and $X_t^* = X_t - 0.4226 X_{t-1}$; for $t = 1$, multiply by $\sqrt{1 - 0.4226^2} \approx 0.906$.

Step 4 (OLS on transformed): $\hat{Y}_t^* = 2.111 + 0.0498 X_t^*$.

The slope barely moves — OLS was unbiased to start with — but the new DW computed on the transformed residuals is 1.52, comfortably inside the no-autocorrelation region. The standard errors that come out of step 4 are the ones to report.

The takeaway: in the AR(1) world, OLS gives you the right *point* estimates (on average) but the wrong *uncertainty*; Cochrane–Orcutt fixes the uncertainty without much changing the point.

8.6. Lab: autocorrelation

The lab follows the workflow of Tutorial 9 (`autocorr_demo.xlsx`): regress quarterly sales on advertising and a time trend, diagnose autocorrelation visually and with Durbin–Watson, and correct it with Cochrane–Orcutt. To keep the chapter self-contained and reproducible without the external `.xlsx`, we generate a synthetic version of the tutorial dataset that has the same structure (sales depending on advertising and a trend, with AR(1) errors of known ρ). The interactive tutorial `LEARNR/08_autocorrelation.Rmd` uses an identical simulation.

```
library(lmtest)
library(orcutt)
```

8. Autocorrelation

```
set.seed(2026)
n          <- 120                                # quarterly observations, ~30 years
time       <- 1:n
advertising <- 5 + 0.5 * time + rnorm(n, 0, 3)      # trending ad spend
rho_true   <- 0.7                                # AR(1) parameter
u          <- numeric(n); u[1] <- rnorm(1, 0, 2)
for (t in 2:n) u[t] <- rho_true * u[t-1] + rnorm(1, 0, 2)
sales      <- 10 + 0.8 * advertising + 0.3 * time + u
autocorr_demo <- data.frame(time = time,
                             advertising = advertising,
                             sales = sales)
head(autocorr_demo)
```

	time	advertising	sales
1	1	7.0617672	19.76182
2	2	2.7609277	19.74102
3	3	6.9177143	21.74968
4	4	6.7457536	22.11135
5	5	5.5000812	16.28280
6	6	0.4517329	13.36700

The true data-generating process has slope 0.8 on advertising, a time-trend coefficient 0.3, and AR(1) errors with $\rho = 0.7$ (strong positive autocorrelation). A real analyst would not know any of these numbers.

8.6.1. OLS and a first look at the residuals

```
fit <- lm(sales ~ advertising + time, data = autocorr_demo)
summary(fit)
```

Call:

```
lm(formula = sales ~ advertising + time, data = autocorr_demo)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.9456	-1.6830	0.1018	1.4430	7.2580

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.03424	0.55212	21.796	< 2e-16 ***
advertising	0.79301	0.07210	10.999	< 2e-16 ***
time	0.28555	0.03669	7.782	3.13e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.399 on 117 degrees of freedom

Multiple R-squared: 0.9902, Adjusted R-squared: 0.99

F-statistic: 5895 on 2 and 117 DF, p-value: < 2.2e-16

The coefficient on `advertising` is in the right ballpark (around 0.8), the trend is in the right ballpark (around 0.3), and the t -statistics are spectacular. Are they real? Plot the residuals over time:

```
plot(ts(resid(fit)),
      ylab = "Residuals",
      main = "Residuals over time")
abline(h = 0, lty = 2)
```

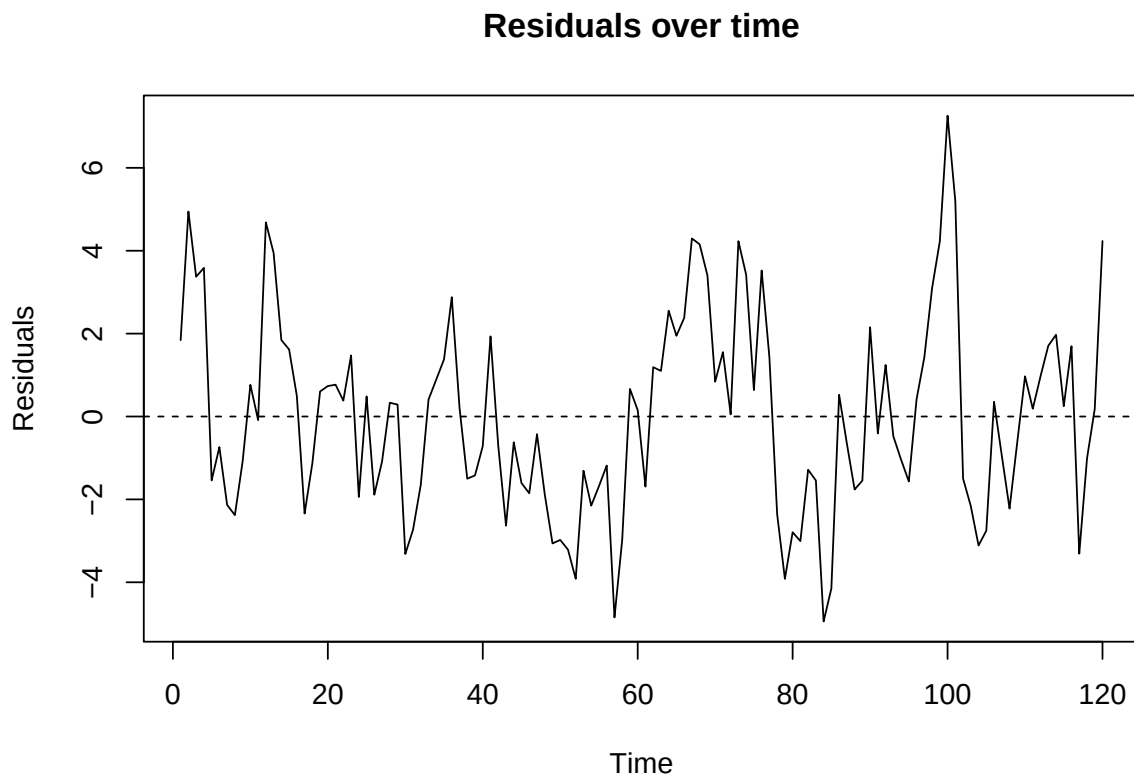


Figure 8.4.: OLS residuals plotted over time. Long runs of same-sign residuals are the visual signature of positive autocorrelation.

The residuals trace out long, smooth excursions above and below zero: this is exactly the picture of $\rho > 0$. Compare with what an uncorrelated residual series should look like (a featureless cloud around zero) and the difference is obvious.

8.6.2. The lag plot

```
e <- resid(fit)
plot(head(e, -1), tail(e, -1),
     xlab = expression(hat(u)[t-1]),
     ylab = expression(hat(u)[t]),
     main = "Residuals vs. lagged residuals",
     pch = 16,
```

8. Autocorrelation

```
col = rgb(0, 0, 1, 0.4))
abline(0, 1, lty = 3)
```

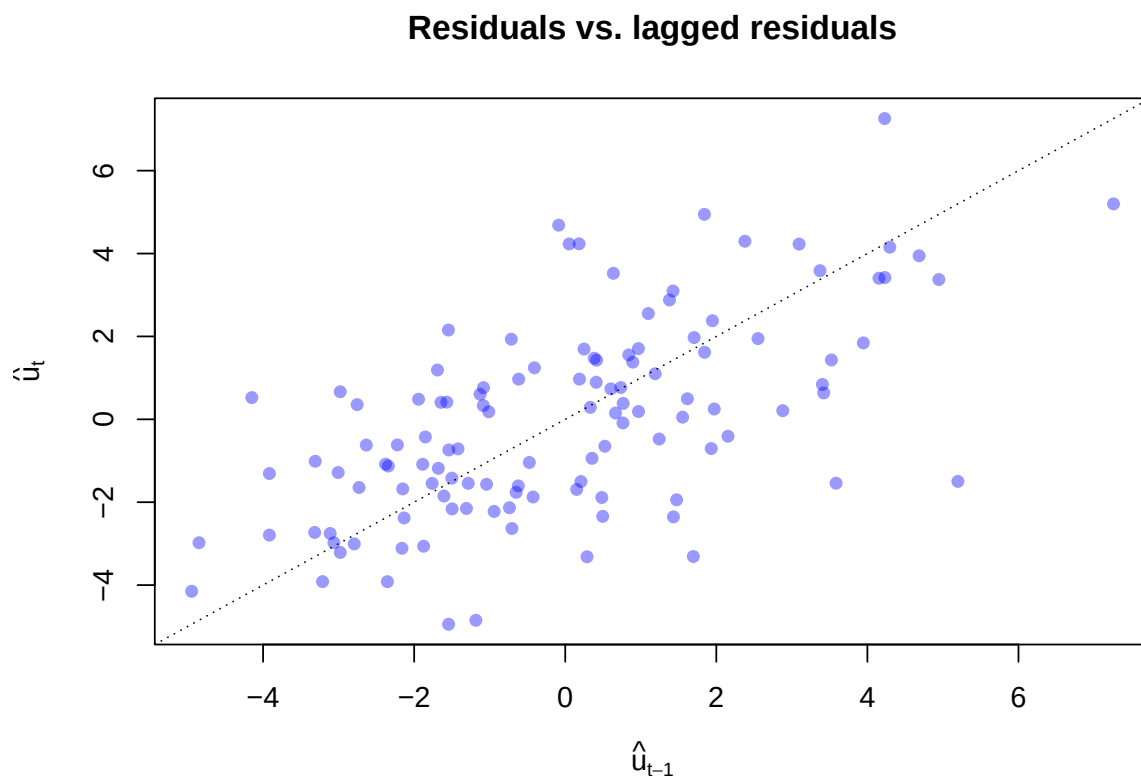


Figure 8.5.: Lag plot of OLS residuals: $\hat{u}_t$ versus $\hat{u}_{t-1}$. The upward tilt confirms positive first-order autocorrelation.

The slope of the cloud is the estimate of ρ . We can compute it directly:

```
# Textbook sample autocorrelation, matching the algebra in section 8.3.2.
# An OLS slope of e on e_lag would be very close but not identical
# in finite samples; here we use the formula consistent with DW = 2(1 - rho_hat).
rho_hat <- sum(e[-1] * e[-n]) / sum(e^2)
rho_hat
```

```
[1] 0.6072128
```

```
2 * (1 - rho_hat) # back-of-envelope DW
```

```
[1] 0.7855744
```

The residual autocorrelation $\hat{\rho}$ should be close to the true 0.7, and $2(1 - \hat{\rho})$ gives a quick approximation to what the Durbin–Watson statistic will report.

8.6.3. Durbin–Watson

```
dwtest(fit)
```

Durbin-Watson test

```
data: fit
DW = 0.75392, p-value = 1.519e-12
alternative hypothesis: true autocorrelation is greater than 0
```

The DW statistic is well below 2, the p -value is tiny, and we reject $H_0 : \rho = 0$ in favour of positive autocorrelation. (Compare DW to $2(1 - \hat{\rho})$ from the previous chunk: the two should agree to within rounding.)

8.6.4. Cochrane–Orcutt correction

```
co <- cochrane.orcutt(fit)
summary(co)
```

Call:

```
lm(formula = sales ~ advertising + time, data = autocorr_demo)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.660805	0.984502	11.844	< 2.2e-16 ***
advertising	0.808566	0.045719	17.686	< 2.2e-16 ***
time	0.282872	0.026587	10.639	< 2.2e-16 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Residual standard error: 1.8894 on 116 degrees of freedom
Multiple R-squared: 0.9618 , Adjusted R-squared: 0.9612
F-statistic: 1462.1 on 2 and 116 DF, p-value: < 5.37e-83
```

Durbin-Watson statistic

```
(original): 0.75392 , p-value: 1.519e-12
(transformed): 1.88504 , p-value: 2.5e-01
```

`cochrane.orcutt()` iterates the algorithm of §8.4.2 until convergence and prints the corrected coefficients, the iterated $\hat{\rho}$, and a DW statistic computed on the transformed residuals.

i A note on `cochrane.orcutt()` vs Prais–Winsten

The algebra in §8.4.1 derived a Prais–Winsten transformation that *rescales* the first observation by $\sqrt{1 - \rho^2}$ so it can be retained. The `cochrane.orcutt()` function from the **orcutt** package implements the two-step Cochrane–Orcutt variant, which *drops* the first observation instead. With $n = 120$ as in our sample the two are essentially equivalent; in

small samples the rescaling matters and you would want to use `prais::prais_winsten()` for a true Prais–Winsten fit. We stay with the dropped-first-observation variant here for compatibility with the §8.4.1 algorithm description.

Compare the OLS and CO results side by side:

```
cat("--- OLS coefficients ---\n")
```

```
--- OLS coefficients ---
```

```
print(round(coef(fit), 4))
```

```
(Intercept) advertising      time
      12.0342      0.7930      0.2855
```

```
cat("\n--- Cochrane-Orcutt coefficients ---\n")
```

```
--- Cochrane-Orcutt coefficients ---
```

```
print(round(co$coefficients, 4))
```

```
(Intercept) advertising      time
      11.6608      0.8086      0.2829
```

```
cat("\nEstimated rho:      ", round(co$rho, 4),
    "(true rho = ", rho_true, ")\n")
```

```
Estimated rho:      0.6256 (true rho = 0.7 )
```

```
cat("DW after correction:  ", round(co$DW[1], 4), "\n")
```

```
DW after correction:  0.7539
```

Two observations are typical of this kind of correction:

1. The **point estimates** of the slope coefficients barely move. OLS was unbiased, after all — the only thing that was wrong with it was the uncertainty.
2. The **standard errors** in the CO output are larger than the OLS standard errors. The OLS *t*-statistics were inflated; the corrected ones are more honest. Hypotheses that were “highly significant” under OLS may become merely significant, or even fail to reject.
3. The DW statistic on the transformed residuals is close to 2, indicating that the AR(1) correction has done its job. (If it has not — if DW is still far from 2 — the residuals may follow a richer process such as AR(2), which is beyond the scope of this chapter.)

8.6.5. A manual one-step Cochrane–Orcutt

To see exactly what `cochrane.orcutt()` does internally, here is a one-step version of the algorithm:

```
y <- autocorr_demo$sales
x1 <- autocorr_demo$advertising
x2 <- autocorr_demo$time

# Step 2: rho from residuals (already computed above as rho_hat)
# Step 3: quasi-differenced variables
y_q <- y[-1] - rho_hat * y[-n]
x1_q <- x1[-1] - rho_hat * x1[-n]
x2_q <- x2[-1] - rho_hat * x2[-n]

# Step 4: OLS on transformed model
fit_q <- lm(y_q ~ x1_q + x2_q)
summary(fit_q)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.5880640	0.36909986	12.43041	4.561547e-23
x1_q	0.8082269	0.04613918	17.51715	2.273774e-34
x2_q	0.2827161	0.02646062	10.68441	5.735566e-19

```
# Recover the original-scale intercept
beta0_original_scale <- coef(fit_q)["(Intercept)"] / (1 - rho_hat)
beta0_original_scale
```

```
(Intercept)
  11.68079
```

```
dwtest(fit_q) # DW should now be close to 2
```

Durbin-Watson test

```
data: fit_q
DW = 1.8507, p-value = 0.1938
alternative hypothesis: true autocorrelation is greater than 0
```

One iteration already gets us most of the benefit. The fully iterated `cochrane.orcutt()` polishes $\hat{\rho}$ further until it converges; in practice a few iterations suffice.

 Bias is not the worry — uncertainty is

The OLS coefficient on **advertising** was unbiased, even with $\rho = 0.7$ errors. The problem was that its standard error was *too small*. If you had taken the OLS *t*-statistic at face value and built a forecast confidence interval, your interval would have undercovered the truth — exactly the cautionary tale of the motivating question.

A second cautionary tale belongs here too. If your residuals look autocorrelated, the *first* question is not “should I run Cochrane–Orcutt?” but “have I omitted a slowly-moving covariate that would explain the persistence?” Persistent residuals are sometimes a flag for missing controls — a violation of MLR.4 that an AR(1) correction cannot fix. The “correlation is not causation” warning of Chapter 1 still applies in time series: HAC standard errors and Cochrane–Orcutt repair the inferential machinery, but they do not rescue a regression whose identifying assumption was wrong to begin with.

Self-check

Seven short multiple-choice questions. Try each one before opening the answer.

💡 Q1. What autocorrelation is

Autocorrelation (serial correlation) means:

- A. $\text{Var}(u_t)$ depends on the regressors x_t .
- B. $\text{Cov}(u_t, u_s) \neq 0$ for some $t \neq s$ — the errors are correlated across time periods.
- C. x_t is correlated with u_t .
- D. $\mathbb{E}[u_t] \neq 0$.

Answer: B. Autocorrelation is a property of the *off-diagonal* elements of the error variance–covariance matrix. (A) is heteroskedasticity, (C) is endogeneity, (D) is a violation of zero-mean errors.

💡 Q2. Consequences for OLS

Under AR(1) errors but with $\mathbb{E}[u_t | x_t] = 0$, OLS is:

- A. Always biased.
- B. Always inconsistent.
- C. Still unbiased and consistent, but no longer BLUE, and the usual standard errors are wrong.
- D. Still BLUE.

Answer: C. OLS keeps its point-estimate properties but loses efficiency and produces invalid standard errors. (B) becomes correct only in the special case of a lagged dependent variable.

💡 Q3. The AR(1) process

In the model $u_t = \rho u_{t-1} + \varepsilon_t$ with $|\rho| < 1$ and ε_t white noise, autocorrelation is:

- A. Positive when $\rho > 0$, negative when $\rho < 0$; the autocovariances decay geometrically.
- B. Always positive.
- C. Independent of ρ .
- D. Equivalent to heteroskedasticity.

Answer: A. The sign of ρ pins down the pattern, and $\text{Cov}(u_t, u_{t-k}) = \rho^k \sigma^2$ decays geometrically with the lag.

💡 Q4. Reading the residual time plot

A plot of OLS residuals against time shows long runs of positive residuals followed by long runs of negative residuals. This suggests:

- A. Positive autocorrelation.
- B. Negative autocorrelation.
- C. Heteroskedasticity.
- D. No systematic pattern.

Answer: A. Persistent same-sign runs are the visual signature of $\rho > 0$. Sawtooth patterns (alternating signs) suggest $\rho < 0$.

💡 Q5. The DW statistic

The Durbin–Watson statistic satisfies, for large n :

- A. $DW = \hat{\rho}$.
- B. $DW = 1 + \hat{\rho}$.
- C. $DW \approx 2(1 - \hat{\rho})$.
- D. $DW = 2\hat{\rho}$.

Answer: C. Expanding the squared differences in the numerator and using $\sum \hat{u}_t^2 \approx \sum \hat{u}_{t-1}^2$ for large n gives $DW \approx 2(1 - \hat{\rho})$.

💡 Q6. When DW is not valid

The Durbin–Watson test is **not** valid when:

- A. The errors are normally distributed.
- B. The sample size is large.
- C. There are many regressors.
- D. The model includes a lagged dependent variable.

Answer: D. A lagged dependent variable creates correlation between regressor and error, biases DW toward 2, and ruins the test. Use Durbin's h instead.

💡 Q7. After Cochrane–Orcutt

After a successful Cochrane–Orcutt correction, the DW statistic computed on the transformed regression should:

- A. Be close to 0.
- B. Be close to 4.
- C. Equal $\hat{\rho}$.
- D. Be close to 2.

Answer: D. A successful AR(1) correction whitens the residuals, so $\hat{\rho}^*$ on the transformed model is close to 0 and $DW^* \approx 2(1 - 0) = 2$. If DW after correction is still far from 2, the AR(1) assumption is wrong.

The fuller interactive set (twelve items including simulation exercises) lives in `LEARNR/08_autocorrelation.Rmd`.

Exercises

Exercise 8.1 — Reading a Durbin–Watson statistic. A regression of quarterly retail sales on advertising spending and a time trend uses $n = 80$ quarters and $k = 2$ regressors. The OLS output reports $DW = 0.94$. The relevant 5 % bounds are $d_L = 1.59$, $d_U = 1.69$. State your conclusion and explain what it implies for the OLS t -statistics in the regression.

💡 Show answer

Since $0 < DW < d_L$ (i.e. $0.94 < 1.59$), we reject $H_0 : \rho = 0$ in favour of $H_1 : \rho > 0$: there is positive first-order autocorrelation. The implied $\hat{\rho} \approx 1 - DW/2 = 0.53$. The OLS standard errors are therefore too small, the t -statistics inflated, and any “significant” coefficients should be re-tested using Cochrane–Orcutt or a Newey–West HAC variance estimator.

Exercise 8.2 — Sign of autocorrelation from a lag plot. A scatter of $\hat{u}_t$ against $\hat{u}_{t-1}$ shows a clear *downward*-sloping cloud, with most points in the upper-left and lower-right quadrants. What is the approximate sign of ρ ? What would the DW statistic be approximately?

💡 Show answer

Downward slope $\Rightarrow \hat{\rho} < 0$: **negative** autocorrelation. With (say) $\hat{\rho} \approx -0.5$ we have $DW \approx 2(1 - (-0.5)) = 3$, comfortably above 2.

Exercise 8.3 — DW from a residual table. Using the small-sample example of §8.3.2, $\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2 = 18.796$ and $\sum_{t=1}^n \hat{u}_t^2 = 22.844$.

- Compute the DW statistic and the implied $\hat{\rho}$.
- For $n = 10$, $k = 1$, the 5 % bounds are $d_L = 0.879$, $d_U = 1.320$. What do you conclude?
- Briefly explain why the bounds depend on n and k .

💡 Show answer

- $DW = 18.796/22.844 \approx 0.823$, so $\hat{\rho} \approx 1 - DW/2 \approx 0.589$. (b) Since $DW < d_L$, reject H_0 : positive autocorrelation. (c) The exact distribution of DW depends on $\mathbf{X}'\mathbf{X}$; Durbin and Watson bracketed it with two bounds that depend only on (n, k) , leaving an inconclusive region.

Exercise 8.4 — Durbin’s h . An OLS regression with $n = 21$ produces

$$\widehat{Y}_t = 1.3 + 0.97 Y_{t-1} + 2.31 X_t, \quad \text{se}(\hat{\alpha}) = 0.07, \quad DW = 1.21.$$

Compute Durbin’s h statistic and decide whether the residuals are autocorrelated at the 5 % level. Why is DW alone misleading here?

💡 Show answer

$$\hat{\rho} \approx 1 - 1.21/2 = 0.395; \quad \widehat{\text{Var}}(\hat{\alpha}) = 0.07^2 = 0.0049;$$

$$h = 0.395 \sqrt{21/(1 - 21 \cdot 0.0049)} \approx 1.91.$$

Critical value $z_{0.975} = 1.96$; $|h| < 1.96$, do **not** reject H_0 . The plain DW reading of 1.21 is

biased toward 2 (the null) because the regression contains Y_{t-1} ; trusting DW would have falsely flagged autocorrelation.

Exercise 8.5 — Why positive autocorrelation deflates OLS standard errors (sketch).

In one short paragraph, explain — using the idea of *effective sample size* — why positive serial correlation makes the conventional OLS standard error of $\hat{\beta}_1$ in $y_t = \beta_0 + \beta_1 x_t + u_t$ too small. You do not need to derive a formula; an intuition based on the information content of n correlated observations is enough.

A full answer is given in the Instructor Edition.

Exercise 8.6 — One-step Cochrane–Orcutt by hand. A small dataset has $\hat{\rho} = 0.4226$ obtained from the OLS residuals of $Y_t = \beta_0 + \beta_1 X_t + u_t$ on $n = 10$ observations.

- (a) Write down the formulas for the quasi-differenced variables Y_t^* , X_t^* and the transformed constant term, *including* the Prais–Winsten treatment of $t = 1$.
- (b) Explain why the standard error of $\hat{\beta}_1$ computed on (Y_t^*, X_t^*) is the one to report, while the OLS standard error from the *original* regression is not.
- (c) After running OLS on (Y_t^*, X_t^*) the new *DW* statistic is 1.52. With $n = 9$, $k = 1$, $d_L = 0.824$, $d_U = 1.320$, would you iterate again or stop?

A full answer is given in the Instructor Edition.

Mathematical and Statistical Prerequisites

This appendix collects the mathematical and statistical tools used throughout the book. It is meant as a reference: skim it once at the start of the course, then come back to individual sections as they are invoked. The level is deliberately operational — enough to follow the derivations in the main text without turning the book into a course on real analysis (Wooldridge 2020). We adopt Wooldridge’s notation everywhere: population parameters are $\beta_0, \beta_1, \dots$; the population error is u ; the sample residual is $\hat{u}$; fitted values are $\hat{y}$; and the usual sums of squares are SST, SSE, and SSR.

Learning outcomes

By the end of this appendix the reader should be able to:

- Manipulate exponents and logarithms confidently, and read off elasticities and semi-elasticities from log-transformed equations.
- Compute single-variable derivatives and partial derivatives, and use first-order conditions to solve a small optimisation problem.
- State and apply the linearity of expectation, the formulas for variance and covariance, and the conditional-expectation properties used by Wooldridge.
- Recall the shapes and uses of the normal, t , χ^2 , and F distributions, and locate critical values in standard tables.
- Read and manipulate a vector or matrix, including transpose, inverse, and rank, well enough to follow the matrix form of OLS in §3.4.
- Install R and RStudio, load the `wooldridge` package, and run a first inspection of the `wage1` dataset.

A.1. Algebra refresher

A.1.1. Functions and notation

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ assigns to each real number x a real number $f(x)$. The graphs we draw in this book are almost always of functions of one or two variables. A linear function takes the form

$$f(x) = \beta_0 + \beta_1 x,$$

with intercept β_0 and slope β_1 . The slope β_1 tells us by how many units $f(x)$ changes when x increases by one unit. This is exactly how we will read every estimated coefficient in the book.

A.1.2. Exponents and powers

Every interest-rate, growth, and elasticity calculation you will do this term either *is* an exponentiation or hides one inside `exp(...)` or `log(...)`, so the rules below are the algebraic plumbing of the entire course.

For any $a > 0$ and any real numbers m, n :

$$a^m \cdot a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad a^0 = 1.$$

(The same identities hold for any real $a \neq 0$ when m, n are integers; the $a > 0$ restriction matters as soon as we allow fractional exponents — for instance, the square root, $\sqrt{a} = a^{1/2}$, is defined for $a \geq 0$ only.)

A.1.3. Logarithms

Logging a regressor lets us read coefficients as percentage effects, which is half of what makes regressions communicable to non-economists — so the log rules below are not optional.

The natural logarithm $\ln(x)$ is the inverse of the exponential function e^x : $\ln(e^x) = x$ and $e^{\ln(x)} = x$ for $x > 0$. The properties we use over and over are:

$$\ln(xy) = \ln(x) + \ln(y), \quad \ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y), \quad \ln(x^a) = a \ln(x), \quad \ln(1) = 0.$$

A useful approximation, exploited throughout the book, is that for small changes,

$$\ln(x + \Delta x) - \ln(x) \approx \frac{\Delta x}{x}.$$

That is, a difference of logs is approximately a *percentage change*. This is why log-transformed regressions are so popular in applied work — they let us read coefficients as elasticities or percentage effects.

i Definition: elasticity and semi-elasticity

In the **log-log** specification $\ln(y) = \beta_0 + \beta_1 \ln(x) + u$, the coefficient β_1 is an **elasticity**: a 1% increase in x raises y by approximately $\beta_1\%$.

In the **log-level** specification $\ln(y) = \beta_0 + \beta_1 x + u$, the coefficient β_1 is a **semi-elasticity**: a one-unit increase in x raises y by approximately $100 \beta_1\%$.

Forward reference: these interpretations are used systematically from Chapter 2 onwards, and become central in the dummy-variables chapter where we read β_1 as a “log-wage gap” of $100 \beta_1\%$.

A.1.4. Summations

Every estimator in the book is a sum — $\hat{\beta}_1$, $\bar{y}$, SST_x — so once you can manipulate $\sum$ cleanly, you can manipulate OLS.

The summation symbol $\sum$ is shorthand we cannot live without:

$$\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n.$$

Two properties used constantly:

$$\sum_{i=1}^n (ax_i + by_i) = a \sum_{i=1}^n x_i + b \sum_{i=1}^n y_i, \quad \sum_{i=1}^n c = n c.$$

The sample mean is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, and a property we will use in the OLS derivation is $\sum_{i=1}^n (x_i - \bar{x}) = 0$.

⚠ Common mistake: $\ln(x + y) \neq \ln(x) + \ln(y)$

The log of a *sum* is not the sum of the logs. This trips up many first-time users when they try to “simplify” terms like $\ln(\beta_0 + \beta_1 x)$ — they cannot be split. Logarithms split products, not sums.

A.2. A.2 Calculus refresher

A.2.1. Derivatives of a single variable

OLS is the algorithm that sets the derivative of the sum of squared residuals to zero; you cannot derive a single coefficient in this book without this tool.

The derivative $f'(x)$ measures the instantaneous rate of change of f at x :

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Visually, the derivative is the **slope of the tangent** to the graph of f at x . The fraction inside the limit is the slope of a **secant** between two nearby points x and $x+h$; as h shrinks, the secant pivots towards the tangent.

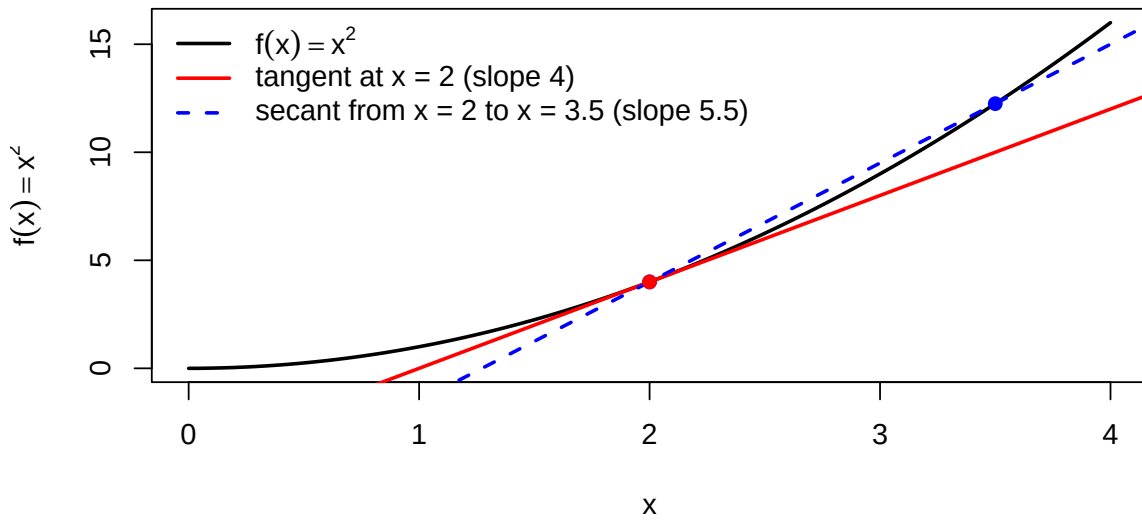


Figure A.1.: Derivative as the slope of the tangent. The dashed blue line is a secant from $x = 2$ to $x = 3.5$; as the second point slides toward $x = 2$, the secant rotates onto the solid red tangent line. Its slope, $f'(2) = 4$, is the derivative of $f(x) = x^2$ at $x = 2$.

The rules we need are short:

$$\frac{d}{dx}(c) = 0, \quad \frac{d}{dx}(x^n) = n x^{n-1}, \quad \frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(\ln x) = \frac{1}{x}.$$

Linearity, product, quotient, and chain rules:

$$(af + bg)' = af' + bg', \quad (fg)' = f'g + fg',$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad (f \circ g)'(x) = f'(g(x)) g'(x).$$

A.2.2. Partial derivatives

This *is* the *ceteris paribus* interpretation of β_j : the partial derivative of $E[y \mid \mathbf{x}]$ with respect to x_j , holding the other regressors fixed. Everything Chapter 3 says about “controlling for” is the operator below in disguise.

For a function of several variables $f(x_1, x_2, \dots, x_k)$, the partial derivative with respect to x_j holds the other variables fixed:

$$\frac{\partial f}{\partial x_j} = \lim_{h \rightarrow 0} \frac{f(\dots, x_j + h, \dots) - f(\dots, x_j, \dots)}{h}.$$

This is precisely the *ceteris paribus* reading we want from a multiple regression: for the population model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

we have $\partial y / \partial x_1 = \beta_1$ holding x_2 fixed — the so-called *partial effect*. Chapter 3 makes this interpretation precise.

A.2.3. First-order conditions and optimisation

Both OLS (minimising SSR) and Maximum Likelihood (maximising the log-likelihood) reduce to the same three-step recipe: take derivatives, set them to zero, solve.

To find an interior minimum or maximum of a smooth function, set the derivative to zero (the **first-order condition**, FOC) and check the second-order condition. A *sufficient* condition for a strict local minimum is $f''(x^*) > 0$; the *necessary* second-order condition is the weaker $f''(x^*) \geq 0$. The distinction matters because edge cases like $f(x) = x^4$ have a minimum at $x = 0$ with $f''(0) = 0$. For functions of several variables, set every partial derivative to zero simultaneously.

i Example: minimising a simple quadratic

Consider $f(\beta) = \sum_{i=1}^n (y_i - \beta)^2$. The FOC is

$$\frac{df}{d\beta} = -2 \sum_{i=1}^n (y_i - \beta) = 0 \iff \beta = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}.$$

The minimiser is the sample mean. This is exactly the structure of OLS: OLS minimises a sum of squared residuals by setting partial derivatives equal to zero. We carry out this derivation in detail in §2.4.

A.3. A.3 Probability refresher

A.3.1. Random variables

A **random variable** X is a numerical summary of the outcome of a random experiment. It is *discrete* if it can take only a countable number of values (e.g. the number of children in a household), and *continuous* if it can take any value in an interval (e.g. an hourly wage). Throughout the book we write the (unknown) population mean as $\mu = \mathbb{E}[X]$ and the population variance as $\sigma^2 = \text{Var}(X)$.

A.3.2. Expectation

The expected value of X , denoted $\mathbb{E}[X]$, is a population average. The properties below are used on essentially every page of the book.

$$\begin{aligned} \mathbb{E}[c] &= c, & \mathbb{E}[aX + b] &= a \mathbb{E}[X] + b, \\ \mathbb{E}[aX_1 + bX_2] &= a \mathbb{E}[X_1] + b \mathbb{E}[X_2], & \mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] &= \sum_{i=1}^n a_i \mathbb{E}[X_i]. \end{aligned}$$

The last identity is **linearity of expectation**: it holds whether or not the X_i are independent.

A.3.3. Variance

The variance measures dispersion around the mean:

$$\text{Var}(X) = \sigma^2 = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The standard deviation is $\sigma = \sqrt{\text{Var}(X)}$ and has the same units as X . Useful properties:

$$\text{Var}(c) = 0, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

Notice that an additive constant b shifts the distribution but does *not* affect its dispersion. A multiplicative constant a scales the variance by a^2 , not by a .

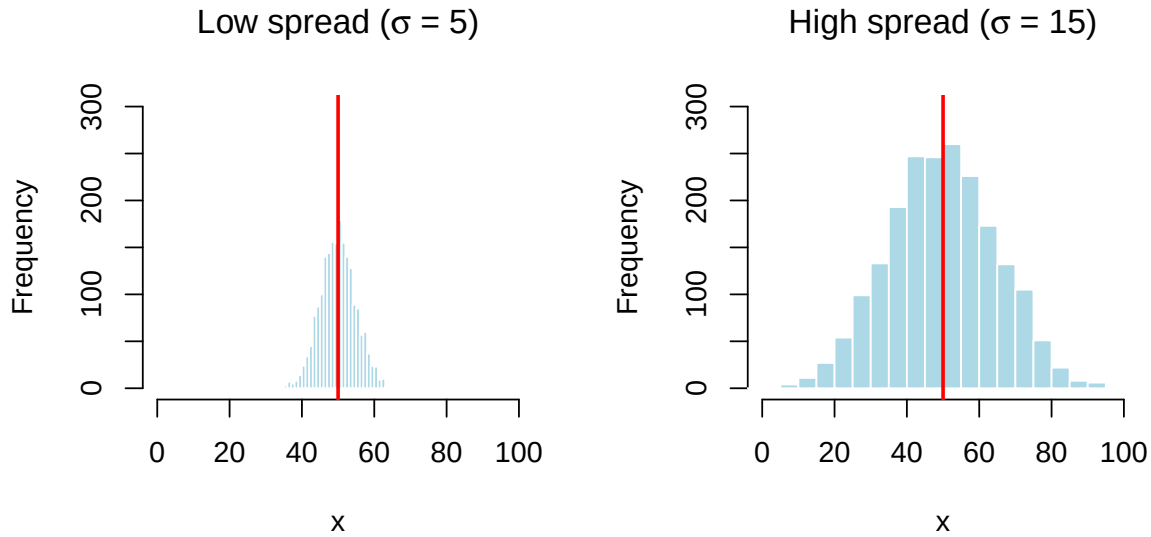


Figure A.2.: Variance as spread, not location. Both samples have the same mean (50), but the right panel has three times the standard deviation. The variance captures the dispersion around the mean — *not* where the mean is. Adding a constant b would shift each red line left or right without changing the shape; multiplying by a would stretch each histogram horizontally by $|a|$ and the variance by a^2 .

A.3.4. Covariance and correlation

The covariance between two random variables measures the direction of their linear co-movement:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y].$$

If either $\mathbb{E}[X] = 0$ or $\mathbb{E}[Y] = 0$, the second expression collapses to $\text{Cov}(X, Y) = \mathbb{E}[XY]$. Properties:

$$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y), \quad \text{Cov}(X, X) = \text{Var}(X).$$

If X and Y are independent, then $\text{Cov}(X, Y) = 0$. The converse is false in general: zero covariance only rules out *linear* dependence. The **correlation coefficient** rescales the covariance to lie in $[-1, 1]$:

$$\text{Corr}(X, Y) = \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}.$$

A.3.5. Variance of a sum

A formula we will use many times in the book is

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

When $\text{Cov}(X, Y) = 0$ this simplifies to $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. The general n -variable version,

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j),$$

is invoked when we derive the sampling variance of the OLS estimator.

A.3.6. Conditional expectation

The **conditional expectation** $\mathbb{E}[Y | X]$ is the mean of Y given knowledge of X . It is itself a function of X . The properties we use most often are:

$$\mathbb{E}[X | X] = X, \quad \mathbb{E}[g(X) | X] = g(X),$$

$$\mathbb{E}[aY + b(X) | X] = a \mathbb{E}[Y | X] + b(X).$$

If X and Y are independent, $\mathbb{E}[Y | X] = \mathbb{E}[Y]$. Finally, the **Law of Iterated Expectations** (also called the tower property) is the single most useful tool in the book:

$$\mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[Y].$$

The zero conditional mean assumption $\mathbb{E}[u | x] = 0$ that underpins OLS unbiasedness in §3.2 is a statement about conditional expectation in exactly this sense.

 Common mistake: $\text{Cov}(X, Y) = 0$ does not mean independence

Zero covariance only rules out *linear* association. Two variables can have $\text{Cov}(X, Y) = 0$ and still be deterministically related — the textbook example is $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$, which satisfy $\text{Cov}(X, Y) = 0$ even though Y is a function of X .

A.4. A.4 Statistical inference refresher

A.4.1. Sampling distributions

We will repeatedly contrast a population quantity (e.g. β_1) with its estimator computed from a sample (e.g. $\hat{\beta}_1$). The estimator is itself a random variable — it varies from sample to sample — and its distribution is called the **sampling distribution**. Two features of the sampling distribution determine the quality of the estimator:

- The **mean**: an estimator $\hat{\theta}$ is *unbiased* for θ if $\mathbb{E}[\hat{\theta}] = \theta$.
- The **variance**: among *linear* unbiased estimators, the one with the smallest variance is the most *efficient*. The Gauss-Markov theorem in Chapter 3 makes this comparison formal for OLS. The “linear” qualifier is load-bearing: nonlinear estimators (ridge, LASSO, quantile regression) can sometimes do strictly better in mean-squared-error terms by accepting a small bias in exchange for a large variance reduction.

A.4.2. The Central Limit Theorem

If $X_1, X_2, \dots, X_n$ are independent and identically distributed (i.i.d.) with mean μ and variance $\sigma^2 < \infty$, then

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

The remarkable point is that the *shape* of the underlying distribution of X_i does not matter: the sample mean is approximately normal in large samples, whatever X looks like. This is why t - and z -statistics built from OLS estimators are approximately normal in large samples, even when the errors u are not.

A.4.3. The normal distribution

The standard normal $\mathcal{N}(0, 1)$ has density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right),$$

is symmetric about 0, and has $\mathbb{E}[Z] = 0$, $\text{Var}(Z) = 1$. Two-sided critical values you will memorise: $z_{0.975} = 1.96$ (5% level) and $z_{0.995} = 2.576$ (1% level). A general normal $X \sim \mathcal{N}(\mu, \sigma^2)$ can always be standardised to $Z = (X - \mu)/\sigma$.

A.4.4. The t , χ^2 , and F distributions

These three distributions appear naturally when we test hypotheses about regression coefficients.

- The χ_k^2 distribution is the distribution of a sum of k squared independent standard normals. It is non-negative, right-skewed, and has mean k .
- The t_k distribution is the ratio $Z/\sqrt{V/k}$ where $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_k^2$ are independent. It is symmetric and bell-shaped, with heavier tails than the standard normal; as $k \rightarrow \infty$ it converges to $\mathcal{N}(0, 1)$.

- The F_{k_1, k_2} distribution is the ratio $(V_1/k_1)/(V_2/k_2)$ of two independent χ^2 random variables divided by their respective degrees of freedom. It is non-negative and right-skewed.

In §4.3 we will use t -statistics for single-coefficient tests, F -statistics for joint tests, and χ^2 variants for large-sample tests. Critical values are tabulated in any statistical software; in R you will use `qt()`, `qf()`, and `qchisq()`.

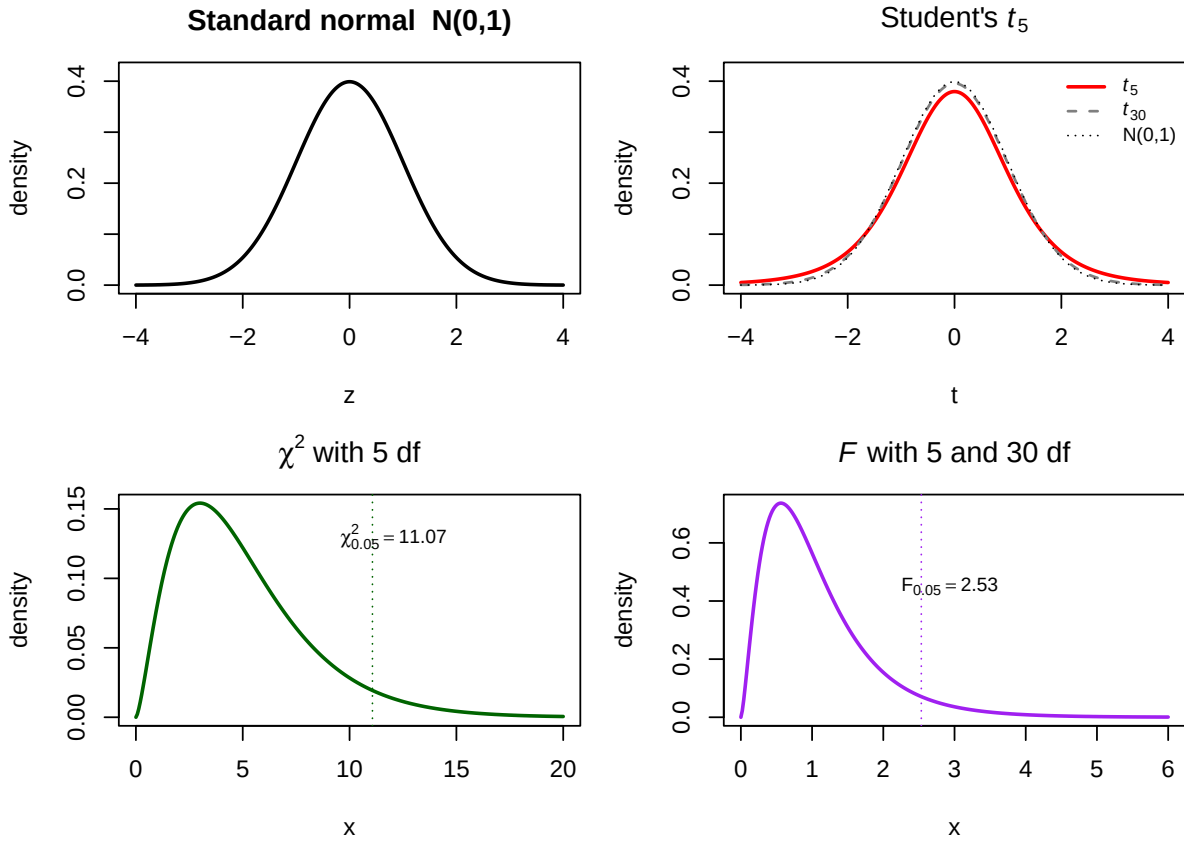


Figure A.3.: The four reference distributions. Normal and t_5 live on the real line and are symmetric, with the t noticeably heavier-tailed at low degrees of freedom; the t_{30} overlay (grey dashed) in the second panel shows how quickly the t approaches $N(0,1)$ as df grows. The χ^2_5 and $F_{5,30}$ live on the non-negative half-line and are right-skewed — which is why their critical values come from the *upper* tail.

i Definition: p -value

The p -value of a test is the probability, under the null hypothesis, of observing a test statistic at least as extreme as the one we got. Small p -values are evidence against the null. The conventional thresholds are 1%, 5%, and 10%, but the p -value itself is what you should report — not just “rejected” or “not rejected”.

A.4.5. Confidence intervals

A $(1 - \alpha)$ confidence interval for a parameter θ is a random interval that, in repeated samples, contains the true θ with probability $1 - \alpha$. For an estimator $\hat{\theta}$ with approximate normal sampling distribution and standard error $\text{se}(\hat{\theta})$, the standard 95% interval is $\hat{\theta} \pm 1.96 \cdot \text{se}(\hat{\theta})$. Chapter 4 builds confidence intervals for individual OLS coefficients in exactly this form.

A.5. A.5 Linear algebra refresher

The matrix form of OLS in §3.4 needs a small bag of linear-algebra tools. We collect them here.

A.5.1. Vectors and matrices

Wooldridge writes the multiple-regression model as $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$, and the rest of his Chapter 3 is one long matrix calculation — you need to be able to follow it.

A column **vector** of length n is a stack of n real numbers:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

A **matrix** of dimension $n \times k$ is a rectangular array of n rows and k columns:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nk} \end{pmatrix}.$$

Throughout this book, the **data matrix** $\mathbf{X}$ has one row per observation and one column per regressor (including a leading column of ones for the intercept). The outcome vector $\mathbf{y}$ has length n . With this convention, the multiple regression model is written in matrix form as $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$.

A.5.2. Transpose

The **transpose** $\mathbf{A}'$ swaps rows and columns: if $\mathbf{A}$ is $n \times k$, then $\mathbf{A}'$ is $k \times n$, with $(\mathbf{A}')_{ij} = a_{ji}$. Key properties:

$$(\mathbf{A}')' = \mathbf{A}, \quad (\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}', \quad (\mathbf{AB})' = \mathbf{B}'\mathbf{A}'.$$

A.5.3. Matrix multiplication

The product $\mathbf{AB}$ is defined when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$. If $\mathbf{A}$ is $n \times k$ and $\mathbf{B}$ is $k \times m$, then $\mathbf{AB}$ is $n \times m$ with entries

$$(\mathbf{AB})_{ij} = \sum_{\ell=1}^k a_{i\ell} b_{\ell j}.$$

Matrix multiplication is associative ($(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$) and distributive over addition, but **not commutative**: in general $\mathbf{AB} \neq \mathbf{BA}$.

A.5.4. Inverse and identity

The $k \times k$ **identity matrix** $\mathbf{I}_k$ has ones on the diagonal and zeros elsewhere. It satisfies $\mathbf{I}_k \mathbf{A} = \mathbf{A} \mathbf{I}_k = \mathbf{A}$ for any conformable $\mathbf{A}$.

A square matrix $\mathbf{A}$ is **invertible** (or non-singular) if there exists a matrix $\mathbf{A}^{-1}$ with $\mathbf{A} \mathbf{A}^{-1} = \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}_k$. Properties:

$$(\mathbf{A}^{-1})^{-1} = \mathbf{A}, \quad (\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}, \quad (\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'.$$

The matrix-form OLS estimator in §3.4 is $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, which requires $\mathbf{X}'\mathbf{X}$ to be invertible.

A.5.5. Rank and the no-multicollinearity assumption

This is the technical version of “no perfectly redundant regressor” (Wooldridge’s MLR.3): full rank k on the $n \times k$ design matrix is exactly what makes the OLS solution unique.

The **rank** of a matrix is the number of linearly independent columns (equivalently, of linearly independent rows). A square $k \times k$ matrix is invertible if and only if it has full rank k . For the data matrix $\mathbf{X}$ of dimension $n \times k$, the condition $\text{rank}(\mathbf{X}) = k$ is the **no perfect multicollinearity** assumption (Wooldridge’s MLR.3): no column of $\mathbf{X}$ is an exact linear combination of the others. When this fails, $\mathbf{X}'\mathbf{X}$ is singular and OLS cannot be computed.

 Common mistake: confusing perfect and near-perfect multicollinearity

Perfect multicollinearity (e.g. including both a dummy for male and a dummy for female *plus* an intercept — the “dummy-variable trap”) makes OLS literally impossible to compute. Near multicollinearity (highly but not perfectly correlated regressors) is a different beast: the estimator still exists, but its variance becomes large. We discuss both, and the diagnostic VIF, in Chapter 3.

A.6. A.6 Setting up R and RStudio

The empirical labs in every chapter assume a working installation of R (4.2) and RStudio. If you have never installed them, follow these steps once:

1. Download and install **R** from <https://cran.r-project.org/>.
2. Download and install **RStudio Desktop** (free edition) from <https://posit.co/download/rstudio-desktop/>.
3. Open RStudio. The “Console” pane on the left is where you type R commands.

We will use a handful of contributed packages throughout the book. Install them once, then in each session load only the ones you need.

```
install.packages(c(
  "wooldridge", # textbook datasets
  "readxl",     # read Excel files
  "lmtest",     # tests after lm() (Chs. 4, 7)
  "sandwich",  # robust standard errors (Ch. 7)
```

```
"orcutt"          # Cochrane-Orcutt procedure (Ch. 8)
))
```

The `wooldridge` package ships the datasets that Wooldridge uses in his textbook, including `wage1`, which we work with in Chapter 1.

```
library(wooldridge)
data("wage1")
```

A first sanity check on any dataset is to look at its structure, its head, and a summary.

```
str(wage1)
```

```
'data.frame':  526 obs. of  24 variables:
 $ wage      : num  3.1 3.24 3 6 5.3 ...
 $ educ      : int  11 12 11 8 12 16 18 12 12 17 ...
 $ exper     : int  2 22 2 44 7 9 15 5 26 22 ...
 $ tenure    : int  0 2 0 28 2 8 7 3 4 21 ...
 $ nonwhite  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ female    : int  1 1 0 0 0 0 0 1 1 0 ...
 $ married   : int  0 1 0 1 1 1 0 0 0 1 ...
 $ numdep    : int  2 3 2 0 1 0 0 0 2 0 ...
 $ smsa      : int  1 1 0 1 0 1 1 1 1 1 ...
 $ northcen  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ south     : int  0 0 0 0 0 0 0 0 0 0 ...
 $ west      : int  1 1 1 1 1 1 1 1 1 1 ...
 $ construc : int  0 0 0 0 0 0 0 0 0 0 ...
 $ ndurman   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ trcompu   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ trade     : int  0 0 1 0 0 0 1 0 1 0 ...
 $ services  : int  0 1 0 0 0 0 0 0 0 0 ...
 $ profserv  : int  0 0 0 0 0 1 0 0 0 0 ...
 $ profocc   : int  0 0 0 0 0 1 1 1 1 1 ...
 $ clerocc   : int  0 0 0 1 0 0 0 0 0 0 ...
 $ servocc   : int  0 1 0 0 0 0 0 0 0 0 ...
 $ lwage     : num  1.13 1.18 1.1 1.79 1.67 ...
 $ expersq   : int  4 484 4 1936 49 81 225 25 676 484 ...
 $ tenursq   : int  0 4 0 784 4 64 49 9 16 441 ...
 - attr(*, "time.stamp")= chr "25 Jun 2011 23:03"
```

```
head(wage1)
```

	wage	educ	exper	tenure	nonwhite	female	married	numdep	smsa	northcen	south
1	3.10	11	2	0	0	1	0	2	1	0	0
2	3.24	12	22	2	0	1	1	3	1	0	0
3	3.00	11	2	0	0	0	0	2	0	0	0
4	6.00	8	44	28	0	0	1	0	1	0	0
5	5.30	12	7	2	0	0	1	1	0	0	0
6	8.75	16	9	8	0	0	1	0	1	0	0


```

west construc ndurman trcommpu trade services profserv profocc clerocc
1      1         0       0         0      0         0         0         0
2      1         0       0         0      0         1         0         0
3      1         0       0         0      1         0         0         0
4      1         0       0         0      0         0         0         1
5      1         0       0         0      0         0         0         0
6      1         0       0         0      0         0         1         1
servocc    lwage expersq tenursq
1          0 1.131402      4      0
2          1 1.175573    484      4
3          0 1.098612      4      0
4          0 1.791759   1936    784
5          0 1.667707     49      4
6          0 2.169054     81     64

```

```
summary(wage1[, c("wage", "educ", "exper", "tenure")])
```

wage	educ	exper	tenure
Min. : 0.530	Min. : 0.00	Min. : 1.00	Min. : 0.000
1st Qu.: 3.330	1st Qu.:12.00	1st Qu.: 5.00	1st Qu.: 0.000
Median : 4.650	Median :12.00	Median :13.50	Median : 2.000
Mean : 5.896	Mean :12.56	Mean :17.02	Mean : 5.105
3rd Qu.: 6.880	3rd Qu.:14.00	3rd Qu.:26.00	3rd Qu.: 7.000
Max. :24.980	Max. :18.00	Max. :51.00	Max. :44.000

The output of `str()` tells us that `wage1` is a data frame with 526 observations and 24 variables; `head()` prints the first six rows; and `summary()` gives the five-number summary plus the mean for each variable.

i Quick R glossary

- `<-` is the **assignment** operator: `x <- 5` stores 5 in the object `x`.
- `c()` **combines** values into a vector: `c(1, 2, 3)`.
- `data.frame` is the standard table structure: one column per variable, one row per observation.
- `$` extracts a column by name: `wage1$wage`.
- `?function_name` opens the help file for `function_name`.

A.7. A.7 First plots

A picture is almost always the right place to start. The two functions we use most often in this book are `hist()` for a single variable and `plot()` for a scatterplot of two variables.

```

hist(wage1$wage,
     main = "Distribution of hourly wages",
     xlab = "Wage (USD/hour)",
     col  = "lightblue",
     breaks = 30)

```



Figure A.4.: Distribution of hourly wages in wage1 (1976 USD). Note the long right tail.

```
hist(wage1$educ,  
     main = "Distribution of years of education",  
     xlab = "Years of education",  
     col  = "lightgreen",  
     breaks = 15)
```

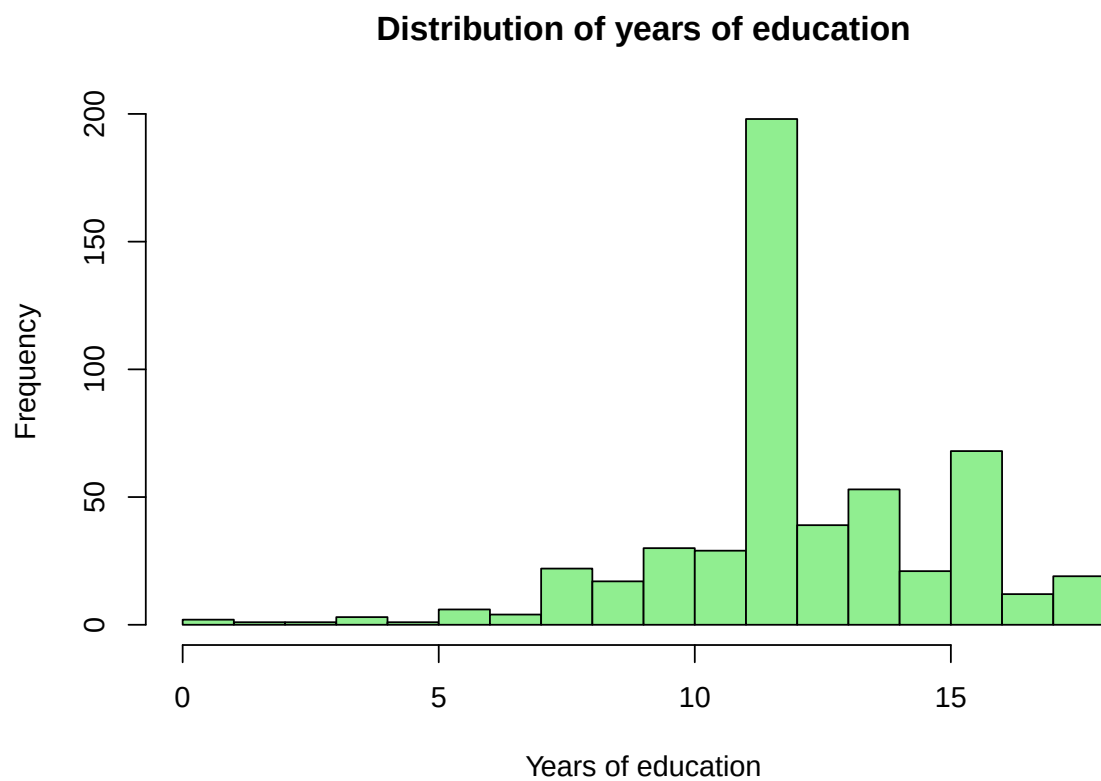


Figure A.5.: Distribution of years of education in wage1, with visible piles at 12 and 16 years.

A scatterplot reveals bivariate structure:

```
plot(wage1$educ, wage1$wage,
     xlab = "Years of education",
     ylab = "Hourly wage (USD)",
     main = "Wage vs. education",
     pch = 16,
     col = rgb(0, 0, 1, 0.3))
```

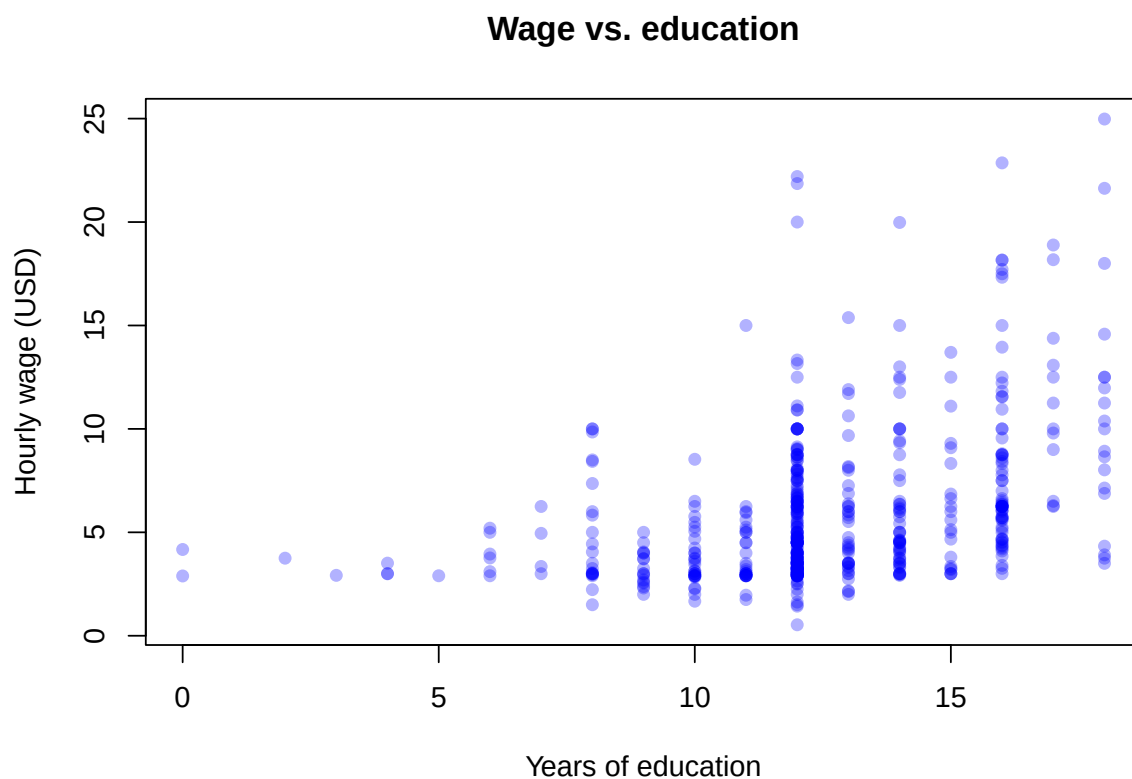


Figure A.6.: Wage versus years of education in wage1.

Conditional means — the average of one variable within bins of another — are a non-parametric preview of regression. In base R, `tapply()` does the job:

```
tapply(wage1$wage, wage1$educ, mean)
```

0	2	3	4	5	6	7	8
3.530000	3.750000	2.920000	3.170000	2.900000	3.985000	4.387500	5.038182
9	10	11	12	13	14	15	16
3.275882	3.835667	4.185517	5.371364	5.598974	6.231698	6.321429	8.041618
17	18						
11.343333	10.678947						

```
tapply(wage1$wage, wage1$female, mean)
```

0	1
7.099489	4.587659

The first table is the average wage at each level of education; the second is the average wage by gender. Chapter 1 discusses the interpretation of these numbers at length (and why they are *not* causal effects). The mechanics of computing them belong here, in the prerequisites.

i Where to go next

Readers who feel rusty on any block above should skim a standard reference before tackling Chapter 2: Wooldridge's own Appendices A–D are an excellent and notation-consistent companion (Wooldridge 2020). Linear algebra at the level of §A.5 is covered in any first-year text; for a quick refresher tailored to econometrics, see Wooldridge's Appendix D.

Wooldridge Carter–Griffiths–Lim Crosswalk

B.1. B.1 Why this crosswalk exists

Many UGR students arrive at this course with a copy of Carter, Griffiths and Lim’s *Principles of Econometrics* 5e (CGL) on the shelf (Hill et al. 2017), because that is the blue book the library stocks in bulk. This textbook, however, follows the notation of Wooldridge’s *Introductory Econometrics* 7e (Wooldridge 2020), in line with decision D9. The two presentations are pedagogically compatible — they cover the same OLS theory, the same Gauss–Markov theorem, the same heteroskedasticity and autocorrelation diagnostics — but they label the same objects with different letters and, in one notorious case (sums of squares), with the *same* letters in a different order. The purpose of this appendix is not to re-teach any of that material but to give the reader a one-page dictionary so that flipping between the two books does not become a source of confusion. Read §B.2 once before Chapter 2, glance at §B.3 when you start a new chapter, and consult §B.4–B.5 only if you mean to use CGL as your main reading companion.

B.2. B.2 Notation crosswalk (extended)

The table below collects every notational difference that has bitten a student in tutorial during the last three editions of this course. Wooldridge notation is what appears in the main text of this book; the CGL column is what you will see if you open the blue book to the corresponding chapter.

Concept	Wooldridge (this book)	Carter–Griffiths–Lim 5e
Population intercept	β_0	β_1
Population slope on x_1	β_1	β_2
Population slope on x_j (MLR)	β_j	β_{j+1}
Error term	u	e
Fitted value	$\hat{y}$	$\hat{y}$
Residual	$\hat{u}$	$\hat{e}$
Total sum of squares	$SST = \sum (y_i - \bar{y})^2$	$SST = \sum (y_i - \bar{y})^2$
Sum of squared residuals	$SSR = \sum \hat{u}_i^2$	$SSE = \sum \hat{e}_i^2$
Explained sum of squares	$SSE = \sum (\hat{y}_i - \bar{y})^2$	$SSR = \sum (\hat{y}_i - \bar{y})^2$
Coefficient of determination	$R^2 = 1 - SSR/SST$	$R^2 = 1 - SSE/SST$
Sample size	n	N (sometimes n)
Number of regressors (excl. intercept)	k	$K - 1$

Concept	Wooldridge (this book)	Carter–Griffiths–Lim 5e
Number of parameters (incl. intercept)	$k + 1$	K
Estimator of β_j	$\hat{\beta}_j$	b_{j+1}
Standard error of $\hat{\beta}_j$	$\text{se}(\hat{\beta}_j)$	$\widehat{\text{sd}}(b_{j+1})$ or $\text{se}(b_{j+1})$
Error variance	σ^2	σ^2
Estimator of σ^2	$\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$	$\hat{\sigma}^2 = \text{SSE}/(N - K)$
t -statistic for $H_0: \beta_j = 0$	$t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$	$t = b_{j+1} / \text{se}(b_{j+1})$
F -statistic, q restrictions	F with df $(q, n - k - 1)$	F with df $(J, N - K)$
95% confidence interval	$\hat{\beta}_j \pm t_{0.975, n-k-1} \cdot \text{se}(\hat{\beta}_j)$	$b_{j+1} \pm t_c \cdot \text{se}(b_{j+1})$
Heteroskedasticity-robust SE	“heteroskedasticity-robust” / HC0–HC3 (via <code>sandwich::vcovHC</code> , <code>lmtest::coeftest</code>)	“White” standard errors
AR(1) error process	$u_t = \rho u_{t-1} + \varepsilon_t$	$e_t = \rho e_{t-1} + v_t$
Dummy / indicator variable	dummy variable, $d \in \{0, 1\}$	indicator variable, $D \in \{0, 1\}$
Interaction	$\beta_3 x_1 x_2$	$\beta_4 x_2 x_3$
Quadratic	$\beta_1 x + \beta_2 x^2$	$\beta_2 x + \beta_3 x^2$
Log-level	$\ln(y) = \beta_0 + \beta_1 x + u$	$\ln(y) = \beta_1 + \beta_2 x + e$
Level-log	$y = \beta_0 + \beta_1 \ln(x) + u$	$y = \beta_1 + \beta_2 \ln(x) + e$
Log-log (elasticity)	$\ln(y) = \beta_0 + \beta_1 \ln(x) + u$	$\ln(y) = \beta_1 + \beta_2 \ln(x) + e$

⚠ The SSR/SSE swap

The names “SSR” and “SSE” are **swapped** between the two textbook traditions. In Wooldridge, **SSR** is the sum of *squared residuals* and **SSE** is the *explained* sum of squares. In CGL, **SSE** is the sum of squared *errors* (i.e. residuals) and **SSR** is the *regression* (explained) sum of squares. Both books define $R^2 = 1 - (\text{residual SS})/\text{SST}$ — they only disagree on which three letters spell “residual SS”. When you read software output, note that R’s `summary(lm())` and `anova(lm())` follow Wooldridge’s convention: the row labelled “Residuals” is SSR in this book’s notation.

A second subtlety hides in the indexing convention. CGL writes the multiple regression model with K parameters in total, indexed $\beta_1, \beta_2, \dots, \beta_K$, so the intercept is β_1 and the slope on the j -th regressor is β_{j+1} . Wooldridge writes the same model with k regressors *plus* an intercept, indexed $\beta_0, \beta_1, \dots, \beta_k$, so the slope on x_j is simply β_j . The residual degrees-of-freedom expression therefore reads $N - K$ in CGL and $n - k - 1$ in Wooldridge — numerically the same thing, written differently.

B.3. B.3 Chapter mapping (extended)

The table below maps each chapter of this book to the corresponding chapters in Wooldridge 7e and CGL 5e, with approximate page ranges in CGL for students using the blue book as a secondary reference. Page numbers refer to the 5th edition in English (Wiley, 2017); they shift by 2–3 pages in the international edition.

Chapter in this book	Wooldridge 7e	CGL 5e (chapter)	CGL 5e (approx. pp.)
Ch. 1 Initial Concepts	Ch. 1	Ch. 1 An Introduction to Econometrics	1–40
Ch. 2 Simple Linear Regression	Ch. 2	Ch. 2 The Simple Linear Regression Model	41–110
Ch. 3 Multiple Linear Regression	Ch. 3	Ch. 5 The Multiple Regression Model	187–240
Ch. 4 Inference	Ch. 4	Ch. 3 Interval Estimation and Hypothesis Testing; Ch. 6 Further Inference in the Multiple Regression Model	111–150; 241–290
Ch. 5 Prediction	parts of Ch. 6	Ch. 4 Prediction, Goodness-of-Fit, and Modeling Issues	151–186
Ch. 6 Nonlinear forms & dummies	Ch. 6 (functional form), Ch. 7 (dummies)	Ch. 7 Using Indicator Variables; portions of Ch. 4 (nonlinear functional forms)	291–340; 162–180
Ch. 7 Heteroskedasticity	Ch. 8	Ch. 8 Heteroskedasticity	341–380
Ch. 8 Autocorrelation / time series	Ch. 10–12	Ch. 9 Regression with Time-Series Data: Stationary Variables	381–440
Appendix A Prerequisites	Apps. A–D	Probability Primer; Apps. A–C	xxiii–xl; 581–640

A reader who follows CGL chapter-by-chapter will notice that CGL covers MLR (its Ch. 5) only *after* interval estimation and hypothesis testing (its Ch. 3) and prediction (its Ch. 4) in the simple-regression setting. Wooldridge inverts this: he finishes MLR mechanics in Ch. 3 before opening inference in Ch. 4. This textbook follows Wooldridge — so if your CGL is the main reading, expect to flip back and forth between its Ch. 2–3 and its Ch. 5–6 as you work through our Chapters 2–4.

B.4. B.4 Where the two diverge pedagogically

The two books are close cousins, but they make four pedagogical choices differently. None of these are right-or-wrong; they reflect the authors' background and audience.

Matrix algebra. Wooldridge introduces the matrix form $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$ as an appendix to Chapter 3 and uses index notation in the main text. CGL relegates matrix algebra to a late chapter and an appendix, preferring scalar notation throughout the body of the book. This book follows Wooldridge: the matrix form appears in §3.4. Readers whose linear algebra is rusty should consult Appendix A.5 before tackling that section.

Running examples. CGL’s signature dataset is `food`, the food expenditure / income example used to motivate the simple linear regression model. Across its later chapters CGL also leans on capital-asset-pricing-model regressions and on US wage data. Wooldridge — and therefore this book — builds the running narrative around `wage1` (hourly wages and education), `hprice1` and `hprice2` (housing prices), and various education-and-earnings datasets that ship in the `wooldridge` R package. Both authors use the data ethically and consistently; only the *headline* example differs.

Flavour. CGL has a slightly more applied flavour: the chapters open with practical motivation and the derivations are kept to a minimum. Wooldridge is slightly more rigorous: each result is stated as a numbered theorem, with the assumptions (MLR.1 through MLR.6) named and reused. Neither book pretends to be a graduate text, but a student who plans to continue with Econometrics II at UGR will find Wooldridge’s habit of naming assumptions a useful preparation.

Indicator vs. dummy. CGL says “indicator variable” everywhere; Wooldridge says “dummy variable” (and uses “indicator” only as a synonym in passing). The two words refer to the same object — a regressor taking values in $\{0, 1\}$ — and we use “dummy” in the main text of this book to match Wooldridge. CGL’s “Chow test for structural change” is exactly Wooldridge’s “Chow test”; CGL’s “interaction term between indicator and continuous variable” is Wooldridge’s “interaction with a dummy”. Same idea, two costumes.

Heteroskedasticity tests. Both books teach the Breusch–Pagan test and the Goldfeld–Quandt test in their heteroskedasticity chapters. CGL additionally develops the **White general test** (which Wooldridge treats more briefly as the “special case of Breusch–Pagan with squares and cross-products”). Chapter 7 of this book sides with Wooldridge but mentions the White test by name; readers interested in the longer treatment should consult CGL §8.6.

B.5. B.5 If you only have CGL: a reading map for this book

The following pairing lets a CGL-only reader follow this book chapter-by-chapter. Each bullet pairs one of our chapters with the CGL chapter and section to read alongside it.

- **Our Ch. 1 (Initial Concepts)** CGL Ch. 1 (especially §1.1–1.5 on what econometrics is, types of data, and the simple regression idea).
- **Our Ch. 2 (Simple Linear Regression)** CGL Ch. 2 (entire chapter), translating $\beta_1, \beta_2 \rightarrow \beta_0, \beta_1$ and $e \rightarrow u$ as you read.
- **Our Ch. 3 (Multiple Linear Regression)** CGL Ch. 5 §5.1–5.6 for the mechanics and the Gauss–Markov theorem; the matrix form in §3.4 of this book does not appear in CGL until its Appendix.
- **Our Ch. 4 (Inference)** CGL Ch. 3 §3.1–3.6 for t - and confidence-interval logic in the simple regression setting, then CGL Ch. 6 §6.1–6.3 for the F -test and joint hypotheses in MLR. Be aware that CGL splits this material in two; this book treats it together.
- **Our Ch. 5 (Prediction)** CGL Ch. 4 §4.1–4.3 (point and interval prediction) and §4.5–4.6 (functional form, residual diagnostics).
- **Our Ch. 6 (Nonlinear forms & dummies)** CGL Ch. 7 (entire chapter) for indicator variables and Chow tests; supplement with CGL §4.5 for the log/quadratic functional-form material.
- **Our Ch. 7 (Heteroskedasticity)** CGL Ch. 8 (entire chapter). The White test is in CGL §8.6 and is worth reading even though this book treats it briefly.

- **Our Ch. 8 (Autocorrelation)** CGL Ch. 9 §9.1–9.5 for AR(1) errors, the Durbin–Watson test, and feasible GLS / Cochrane–Orcutt. CGL’s notation $e_t = \rho e_{t-1} + v_t$ is what this book writes as $u_t = \rho u_{t-1} + \varepsilon_t$.
- **Our Appendix A (Prerequisites)** CGL Probability Primer and Appendices A–C. The conditional-expectation material in §A.3 of this book is in CGL’s Probability Primer §P.4; the linear-algebra material in §A.5 is in CGL’s Appendix B.

A CGL-only reader who works through the book in this order will have no trouble keeping up; the only sustained effort required is the mechanical translation of indices and the constant attention to the SSR/SSE swap flagged in §B.2.

Statistical Tables

This appendix gathers the critical-value tables we use most often in the book: the t , F , and χ^2 distributions for the inferential procedures of Chapter 4, plus the Durbin–Watson bounds used in the serial-correlation diagnostic of Chapter 8. Because the entries below are produced inside R by the quantile functions `qt()`, `qf()`, and `qchisq()`, they are exact to machine precision rather than rounded as in printed appendices. A reader who wants a value not listed here can simply re-run the corresponding chunk with extra rows or columns.

How to read these tables

A **critical value** c_α is the threshold a test statistic must cross to reject the null at significance level α : for a one-sided t -test of $H_0 : \beta_j = 0$ against $H_1 : \beta_j > 0$ at the 5% level, we reject when $t > c_{0.05}$. A **p -value** turns the question around: it is the smallest α at which the null would have been rejected given the observed statistic. The two views are equivalent — given a p -value you know which critical values you would have crossed — but the tables here are organised around critical values because that is still how most printed inference works through.

For the t -distribution, the columns below report **one-tailed** upper-tail critical values: c such that $\Pr(T_{df} > c) = \alpha$. For a **two-tailed** test at level α , look up the $\alpha/2$ column instead — so the 5% two-tailed critical value is the entry in the $\alpha = 0.025$ column. The F and χ^2 tables are intrinsically one-tailed (their statistics are non-negative) and need no such adjustment. The t table is invoked in §4.5 for single-coefficient hypothesis testing; the F table is used in §4.7 for joint tests and again in §7.3 for the Breusch–Pagan heteroskedasticity test; the χ^2 table covers the large-sample variants of those tests; and the Durbin–Watson bounds in §C.4 are the reference for the serial-correlation diagnostic of §8.3.

C.1. C.1 Critical values of the t -distribution

The entries below are one-tailed upper critical values c such that $\Pr(T_{df} > c) = \alpha$. The last row, $df = \infty$, coincides with the standard-normal quantiles $z_{1-\alpha}$ — the bridge formalised by the CLT in §A.4.

```
df_t      <- c(1:30, 40, 60, 90, 120, Inf)
alpha_t   <- c(0.10, 0.05, 0.025, 0.01, 0.005, 0.001)

t_tab     <- sapply(alpha_t, function(a) qt(1 - a, df = df_t))
t_tab     <- as.data.frame(t_tab)
colnames(t_tab) <- paste0(" = ", format(alpha_t, scientific = FALSE))
```

```

t_tab <- cbind(df = ifelse(is.infinite(df_t), "∞", as.character(df_t)),
              t_tab)

knitr::kable(
  t_tab,
  digits = 3,
  align = c("l", rep("r", length(alpha_t))),
  caption = "One-tailed critical values of the  $t$ -distribution. For a two-tailed test at level  $\alpha$ , read the  $\alpha/2$  column."
)

```

Table C.1.: One-tailed critical values of the t -distribution. For a two-tailed test at level α , read the $\alpha/2$ column.

df	= 0.100	= 0.050	= 0.025	= 0.010	= 0.005	= 0.001
1	3.078	6.314	12.706	31.821	63.657	318.309
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.610
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
60	1.296	1.671	2.000	2.390	2.660	3.232
90	1.291	1.662	1.987	2.368	2.632	3.183
120	1.289	1.658	1.980	2.358	2.617	3.160
∞	1.282	1.645	1.960	2.326	2.576	3.090

A handful of entries are worth memorising. With $df = \infty$ the table reproduces $z_{0.975} = 1.96$ (column $\alpha = 0.025$) and $z_{0.995} = 2.576$ (column $\alpha = 0.005$), the standard-normal critical values for two-sided tests at 5% and 1%. For $df \geq 30$ the rule of thumb that t is “close to normal” already pays off: the 5% two-sided critical value (column $\alpha = 0.025$) sits within 5% of 1.96.

C.2. C.2 Critical values of the F -distribution

The F -tables tabulate the upper- α critical value of F_{k_1, k_2} , where k_1 is the **numerator** degrees of freedom (the number of restrictions in a joint test) and k_2 is the **denominator** degrees of freedom (typically $n - k - 1$ in a k -regressor model with an intercept). We tabulate the two conventional significance levels separately.

```
num_df <- c(1, 2, 3, 4, 5, 6, 7, 8, 10, 15, 20, 30)
den_df <- c(10, 15, 20, 25, 30, 40, 60, 120, Inf)

f_tab_05 <- outer(den_df, num_df,
  function(d, n) qf(0.95, df1 = n, df2 = d))
f_tab_05 <- as.data.frame(f_tab_05)
colnames(f_tab_05) <- paste0("k1 = ", num_df)
f_tab_05 <- cbind(`k2 (den. df)` = ifelse(is.infinite(den_df), "∞",
  as.character(den_df)),
  f_tab_05)

knitr::kable(
  f_tab_05,
  digits = 3,
  align = c("l", rep("r", length(num_df))),
  caption = "Critical values of the  $F$ -distribution at  $\alpha = 0.05$  (upper tail). Rows:
  )
```

Table C.2.: Critical values of the F -distribution at $\alpha = 0.05$ (upper tail). Rows: denominator df k_2 . Columns: numerator df k_1 .

k2 (den. df)	k1 = 1	k1 = 2	k1 = 3	k1 = 4	k1 = 5	k1 = 6	k1 = 7	k1 = 8	k1 = 10	k1 = 15	k1 = 20	k1 = 30
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	2.978	2.845	2.774	2.700
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.544	2.403	2.328	2.247
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.348	2.203	2.124	2.039
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.236	2.089	2.007	1.919
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.165	2.015	1.932	1.841
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.077	1.924	1.839	1.744
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	1.993	1.836	1.748	1.649
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.910	1.750	1.659	1.554
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.831	1.666	1.571	1.459

```
f_tab_01 <- outer(den_df, num_df,
  function(d, n) qf(0.99, df1 = n, df2 = d))
f_tab_01 <- as.data.frame(f_tab_01)
colnames(f_tab_01) <- paste0("k1 = ", num_df)
```

```
f_tab_01 <- cbind(`k2 (den. df)` = ifelse(is.infinite(den_df), "∞",
                                           as.character(den_df)),
                  f_tab_01)

knitr::kable(
  f_tab_01,
  digits = 3,
  align = c("l", rep("r", length(num_df))),
  caption = "Critical values of the $F$-distribution at $\alpha = 0.01$ (upper tail). Rows:
)

```

Table C.3.: Critical values of the F -distribution at $\alpha = 0.01$ (upper tail). Rows: denominator df k_2 . Columns: numerator df k_1 .

k2 (den. df)	k1 = 1	k1 = 2	k1 = 3	k1 = 4	k1 = 5	k1 = 6	k1 = 7	k1 = 8	k1 = 10	k1 = 15	k1 = 20	k1 = 30
10	10.044	7.559	6.552	5.994	5.636	5.386	5.200	5.057	4.849	4.558	4.405	4.247
15	8.683	6.359	5.417	4.893	4.556	4.318	4.142	4.004	3.805	3.522	3.372	3.214
20	8.096	5.849	4.938	4.431	4.103	3.871	3.699	3.564	3.368	3.088	2.938	2.778
25	7.770	5.568	4.675	4.177	3.855	3.627	3.457	3.324	3.129	2.850	2.699	2.538
30	7.562	5.390	4.510	4.018	3.699	3.473	3.304	3.173	2.979	2.700	2.549	2.386
40	7.314	5.179	4.313	3.828	3.514	3.291	3.124	2.993	2.801	2.522	2.369	2.203
60	7.077	4.977	4.126	3.649	3.339	3.119	2.953	2.823	2.632	2.352	2.198	2.028
120	6.851	4.787	3.949	3.480	3.174	2.956	2.792	2.663	2.472	2.192	2.035	1.860
∞	6.635	4.605	3.782	3.319	3.017	2.802	2.639	2.511	2.321	2.039	1.878	1.696

The bottom row ($k_2 = \infty$) is a useful sanity check: the $F_{k_1, \infty}$ distribution coincides (after scaling) with $\chi_{k_1}^2/k_1$, so its 5% critical value equals $\chi_{k_1, 0.95}^2/k_1$. Compare, for instance, the ($k_1 = 4, k_2 = \infty$) entry of the 5% F table with $\chi_{4, 0.95}^2/4$ from the next section.

C.3. C.3 Critical values of the χ^2 -distribution

```
df_chi    <- c(1:10, 15, 20, 30, 50, 100)
alpha_chi <- c(0.10, 0.05, 0.025, 0.01, 0.005)

chi_tab <- sapply(alpha_chi, function(a) qchisq(1 - a, df = df_chi))
chi_tab <- as.data.frame(chi_tab)
colnames(chi_tab) <- paste0(" = ", format(alpha_chi, scientific = FALSE))
chi_tab <- cbind(df = as.character(df_chi), chi_tab)

knitr::kable(
  chi_tab,
  digits = 3,
  align = c("l", rep("r", length(alpha_chi))),
  caption = "Upper-tail critical values of the $\chi^2$-distribution: $c$ such that $\Pr
)

```


Table C.4.: Upper-tail critical values of the χ^2 -distribution: c such that $\Pr(\chi_{df}^2 > c) = \alpha$.

df	= 0.100	= 0.050	= 0.025	= 0.010	= 0.005
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.070	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589
10	15.987	18.307	20.483	23.209	25.188
15	22.307	24.996	27.488	30.578	32.801
20	28.412	31.410	34.170	37.566	39.997
30	40.256	43.773	46.979	50.892	53.672
50	63.167	67.505	71.420	76.154	79.490
100	118.498	124.342	129.561	135.807	140.169

The χ^2 table is the natural reference for large-sample versions of the joint test (the **Wald statistic** is asymptotically χ_q^2 where q is the number of restrictions) and for the Breusch–Pagan test of §7.3, whose *LM* statistic is χ_k^2 under the null of homoskedasticity.

C.4. C.4 Durbin–Watson bounds

The Durbin–Watson statistic $d = \sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2 / \sum_{t=1}^n \hat{u}_t^2$ does not have a single critical value: its exact null distribution depends on the design matrix $\mathbf{X}$, not just on n and the number of regressors. Durbin and Watson (1951) sidestepped this by tabulating two bounds, d_L and d_U , between which the *true* critical value must lie regardless of $\mathbf{X}$. The bounds below are taken from Durbin, J. and Watson, G. S. (1951), “Testing for serial correlation in least squares regression. II,” *Biometrika* **38**(1), 159–178, for $\alpha = 0.05$ and k' regressors excluding the constant. A representative subset is reproduced; consult the original tables for values of n not listed.

```
dw_05 <- data.frame(
  n      = c(15, 20, 25, 30, 40, 50, 75, 100, 150, 200),
  dL_k1  = c(1.08, 1.20, 1.29, 1.35, 1.44, 1.50, 1.60, 1.65, 1.72, 1.76),
  dU_k1  = c(1.36, 1.41, 1.45, 1.49, 1.54, 1.59, 1.65, 1.69, 1.75, 1.78),
  dL_k2  = c(0.95, 1.10, 1.21, 1.28, 1.39, 1.46, 1.57, 1.63, 1.71, 1.75),
  dU_k2  = c(1.54, 1.54, 1.55, 1.57, 1.60, 1.63, 1.68, 1.72, 1.76, 1.79),
  dL_k3  = c(0.82, 1.00, 1.12, 1.21, 1.34, 1.42, 1.54, 1.61, 1.69, 1.74),
  dU_k3  = c(1.75, 1.68, 1.66, 1.65, 1.66, 1.67, 1.71, 1.74, 1.77, 1.80),
  dL_k4  = c(0.69, 0.90, 1.04, 1.14, 1.29, 1.38, 1.51, 1.59, 1.68, 1.73),
  dU_k4  = c(1.97, 1.83, 1.77, 1.74, 1.72, 1.72, 1.74, 1.76, 1.79, 1.81),
  dL_k5  = c(0.56, 0.79, 0.95, 1.07, 1.23, 1.34, 1.49, 1.57, 1.67, 1.72),
  dU_k5  = c(2.21, 1.99, 1.89, 1.83, 1.79, 1.77, 1.77, 1.78, 1.80, 1.82)
)

colnames(dw_05) <- c(
  "n",
```

```

"k'=1 dL", "k'=1 dU",
"k'=2 dL", "k'=2 dU",
"k'=3 dL", "k'=3 dU",
"k'=4 dL", "k'=4 dU",
"k'=5 dL", "k'=5 dU"
)

knitr::kable(
  dw_05,
  digits = 2,
  align = c("l", rep("r", 10)),
  caption = "Durbin-Watson 5% bounds  $d_L$  and  $d_U$  by sample size  $n$  and number of regressors  $k'$  excluding the constant."
)

```

Table C.5.: Durbin–Watson 5% bounds d_L and d_U by sample size n and number of regressors k' excluding the constant. Source: Durbin & Watson (1951).

n	k'=1 dL	k'=1 dU	k'=2 dL	k'=2 dU	k'=3 dL	k'=3 dU	k'=4 dL	k'=4 dU	k'=5 dL	k'=5 dU
15	1.08	1.36	0.95	1.54	0.82	1.75	0.69	1.97	0.56	2.21
20	1.20	1.41	1.10	1.54	1.00	1.68	0.90	1.83	0.79	1.99
25	1.29	1.45	1.21	1.55	1.12	1.66	1.04	1.77	0.95	1.89
30	1.35	1.49	1.28	1.57	1.21	1.65	1.14	1.74	1.07	1.83
40	1.44	1.54	1.39	1.60	1.34	1.66	1.29	1.72	1.23	1.79
50	1.50	1.59	1.46	1.63	1.42	1.67	1.38	1.72	1.34	1.77
75	1.60	1.65	1.57	1.68	1.54	1.71	1.51	1.74	1.49	1.77
100	1.65	1.69	1.63	1.72	1.61	1.74	1.59	1.76	1.57	1.78
150	1.72	1.75	1.71	1.76	1.69	1.77	1.68	1.79	1.67	1.80
200	1.76	1.78	1.75	1.79	1.74	1.80	1.73	1.81	1.72	1.82

i The four-region decision rule

Given an observed Durbin–Watson statistic d , the test for **positive** first-order autocorrelation at level α partitions the unit-line into four regions:

- $d < d_L$: **reject** H_0 in favour of positive autocorrelation.
- $d_L \leq d \leq d_U$: **inconclusive** — the test cannot decide.
- $d_U < d < 4 - d_U$: **do not reject** H_0 of no autocorrelation.
- $4 - d_U \leq d \leq 4 - d_L$: **inconclusive** (negative side).
- $d > 4 - d_L$: **reject** H_0 in favour of **negative** autocorrelation.

Inconclusive regions are the price paid for the design-free bounds: when d falls into one of them, switch to a test that does not depend on the design matrix (e.g. the Breusch–Godfrey LM test of §8.3), which is now standard in applied work.

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White, Halbert. 1980. “A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity.” *Econometrica* 48 (4): 817–38. <https://doi.org/10.2307/>

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